PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL AUTOMATED INVENTORY MANAGEMENT SYSTEM FOR PHARMACY AND CONVENIENCE STORE

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Chapter1
INTRODUCTION

1.1 Background of the Study

The retail pharmacy and convenience store sector plays an important role in providing medicines, healthcare products, and daily necessities to the local community. These establishments are expected to maintain accurate inventory records and efficient customer service to meet the growing demands of consumers. However, many pharmacies and convenience stores still rely on manual methods for inventory tracking and sales recording, which often result in delays, inaccurate records, and operational inefficiencies. Manual processes such as handwritten inventory logs and manual stock counting are time-consuming and prone to human error, leading to inventory discrepancies, stock shortages, overstocking, and customer dissatisfaction.

In addition, outdated scanning systems and manual transaction processing can slow down operations and affect the overall customer experience. Employees may also experience difficulty in monitoring stock levels and updating inventory records in real time, which can reduce productivity and business efficiency. As businesses continue to grow, traditional inventory management methods become less effective in handling increasing numbers of products and transactions.

With the continuous advancement of information technology, businesses are increasingly adopting automated systems to improve operational performance and simplify daily processes. Automated inventory management systems provide organizations with more accurate and efficient methods of tracking inventory, recording sales, and monitoring product movement in real time. These systems reduce human intervention in repetitive tasks, minimize operational errors, and improve the overall reliability of business data. Through automation, businesses are able to maintain updated inventory records, generate reports quickly, monitor stock levels efficiently, and make informed decisions regarding purchasing and inventory replenishment.

To address these challenges, this capstone project proposes the development of an Automated Inventory Management System (AIMS) specifically designed for pharmacies and convenience stores. The proposed system aims to streamline inventory and sales operations by integrating automated inventory tracking, real-time stock monitoring, sales recording, and barcode scanning into a single system. The system is expected to improve inventory accuracy, reduce transaction delays, enhance operational efficiency, and provide better customer service. Additionally, the proposed system seeks to minimize manual workload and support employees in performing inventory-related tasks more effectively and efficiently.

The development of the proposed Automated Inventory Management System is expected to provide significant benefits to both business owners and customers. For business owners, the system can improve operational productivity, reduce inventory-related losses, and support better decision-making through accurate and updated information. For employees, the system can simplify inventory management tasks and reduce the stress associated with manual processes. For customers, faster transactions and improved product availability can contribute to a more convenient and satisfying shopping experience. Through the implementation of automation and modern inventory management practices, the proposed system aims to help pharmacies and convenience stores adapt to the growing demands of modern retail operations while improving the quality and efficiency of their services.

1.2 Statement of the Problem

General Problem

Pharmacies and convenience stores still face problems in managing their daily operations. Many stores still use manual methods to manage inventory and record sales, which can lead to mistakes, inaccurate records, and delays. Some stores also use old barcode scanners that make transactions slower and reduce customer satisfaction.

Because of these problems, this study aims to develop an Automated Inventory Management System (AIMS) for pharmacies and convenience stores.

Specific Problems

Specifically, this study seeks to answer the following questions:

- How can a digital inventory management system be designed and developed to replace manual inventory tracking processes?
- How can the proposed system improve the accuracy and monitoring of inventory records in pharmacies and convenience stores?
- How can the system automate sales recording to reduce inventory discrepancies and prevent loss of revenue?
- How can the integration of an updated barcode scanning system improve transaction speed and customer service efficiency?
- How can the proposed Automated Inventory Management System improve the overall efficiency and organization of pharmacy and convenience store operations?
- How acceptable is the proposed Automated Inventory Management System in terms of:
    - Functionality
    - Usability
    - Efficiency
    - Reliability

1.2.1 General Problem

Pharmacies and convenience stores play an important role in providing medicines, food, and other daily needs to customers. Because they serve many customers every day, they need a good system to manage their products and sales. A well-organized inventory system helps store owners know which products are available, which items need to be restocked, and which products are about to expire. It also helps employees work faster and provide better service to customers.

However, many pharmacies and convenience stores still use manual methods to manage their inventory and sales. Employees often write records on paper or enter data manually into a computer. This process takes time and increases the chance of making mistakes. Incorrect records can make it difficult to know the actual number of products available in the store. As a result, some products may become out of stock without the staff noticing, while others may be overstocked.

Manual inventory management can also affect the sales process. If product information is not updated immediately after every sale, the inventory records may become inaccurate. This can cause differences between the actual stock and the recorded stock. These mistakes can lead to problems in ordering products, tracking inventory, and preparing reports. Store owners may also lose income because of missing or incorrect inventory records.

Another problem is the use of old or slow barcode scanning systems. Some barcode scanners take longer to read product codes, especially if the barcode is damaged or the scanner is outdated. Slow scanning can increase the waiting time at the cashier, especially during busy hours. Long waiting lines may cause customers to become impatient and dissatisfied with the service. Fast and accurate transactions are important to improve the shopping experience.

Managing inventory manually also makes it difficult for store owners to monitor their products. Employees need to spend more time checking the shelves and counting products by hand. This process requires extra effort and may still result in errors. It is also harder to identify products that are running low or are close to their expiration date. Without proper monitoring, stores may experience product shortages, expired items, or unnecessary waste.

Technology can help solve these problems by making inventory management and sales recording easier and more accurate. A digital inventory system can automatically update stock whenever a product is sold or added to the inventory. It can also generate reports, monitor stock levels, and provide real-time information about available products. An updated barcode scanning system can speed up transactions and reduce errors during checkout.

Because of these challenges, there is a need for a better and more reliable inventory management system for pharmacies and convenience stores. This study aims to develop an Automated Inventory Management System (AIMS) that can replace manual inventory tracking, improve the accuracy of inventory records, automate sales recording, and use an updated barcode scanning system. The proposed system is expected to improve the efficiency of daily operations, help employees perform their tasks more easily, and provide faster and better service to customers.

1.2.2 Specific Problem

The study sought to identify and address the specific operational problems encountered by pharmacies and convenience stores that affected the efficiency, accuracy, and reliability of their daily inventory and sales operations. These operational concerns were identified through an assessment of existing business processes and were found to significantly influence inventory control, transaction processing, product monitoring, and managerial decision-making. Specifically, the study focused on the following operational problems:

Inventory Monitoring

Inventory monitoring in many pharmacies and convenience stores was primarily performed using manual recording methods, paper-based inventory logs, or spreadsheet applications. Employees were required to manually update stock quantities whenever products were received, sold, returned, or adjusted. While these methods allowed businesses to maintain inventory records, they required continuous manual effort and became increasingly difficult to manage as the number of products and daily transactions increased.

The reliance on manual inventory management increased the possibility of inaccurate stock records, duplicate entries, delayed inventory updates, and inventory discrepancies. Management also experienced difficulties in monitoring real-time stock availability, identifying low-stock products, determining reorder schedules, and maintaining accurate inventory records. These operational limitations often resulted in stock shortages, overstocking, unnecessary purchasing costs, and reduced operational efficiency.

Sales Transaction Management

Sales transactions were processed through conventional point-of-sale procedures that required employees to manually encode product information or search for items before completing customer purchases. Although these procedures enabled stores to process sales, they often required additional time and employee effort, particularly during periods of high customer demand.

Manual transaction processing increased the likelihood of pricing errors, incorrect product selection, incomplete sales records, and delayed transaction processing. In addition, inventory records were not always updated immediately after every sale, creating discrepancies between actual stock levels and recorded inventory. These issues affected transaction accuracy, delayed customer service, and reduced the reliability of sales records required for daily business operations.

Sales Analytics

Although pharmacies and convenience stores generated large volumes of sales data through daily transactions, much of this information was used only for record-keeping purposes rather than business analysis. Existing systems provided limited capability for automatically organizing sales information into meaningful reports that could support management decisions.

Without an integrated sales analytics feature, store managers experienced difficulties in identifying best-selling and slow-moving products, evaluating sales performance, monitoring daily and monthly revenue trends, analyzing customer purchasing patterns, and forecasting future inventory requirements. As a result, important business decisions regarding product replenishment, promotional activities, and inventory planning were often based on manual observations instead of accurate analytical information.

Stock and Expiration Monitoring

Monitoring stock levels and product expiration dates was commonly performed through periodic manual inspection of shelves and storage areas. Employees were required to regularly check product quantities and expiration dates individually, making the monitoring process time-consuming and labor-intensive.

Because these monitoring activities relied heavily on manual checking, products approaching their expiration dates could be overlooked, while low-stock items might not be replenished promptly. These operational limitations increased the risk of expired products remaining on display, unnecessary product wastage, stock shortages, and lost sales opportunities. The absence of an automated notification system also reduced the ability of management to respond proactively to inventory-related issues.

Barcode-Based Transaction Processing

Many pharmacies and convenience stores continued to utilize outdated barcode scanning systems or relied on manual product searches during sales transactions. These existing methods slowed transaction processing and required employees to spend additional time locating products and verifying pricing information.

The lack of an updated barcode scanning system increased the possibility of scanning errors, incorrect product identification, delayed customer transactions, and reduced cashier productivity. During busy business hours, slow transaction processing contributed to longer customer waiting times and decreased overall service efficiency.

Report Generation

Sales and inventory reports were commonly prepared through manual consolidation of transaction records and inventory documents. Employees were required to collect daily sales information, compute totals, reconcile inventory records, and prepare reports before management could evaluate business performance.

The manual preparation of reports required considerable administrative effort and frequently delayed the availability of essential operational information. As a result, management experienced difficulties in monitoring inventory movement, evaluating sales performance, identifying inventory discrepancies, and making timely business decisions based on accurate and up-to-date reports.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to design and develop an Automated Inventory Management System (AIMS) that will improve inventory tracking, sales recording, and overall operational efficiency for pharmacies and convenience stores. The proposed system aims to minimize manual processes, reduce human errors, enhance transaction speed, and provide a more organized and reliable method of managing products, sales, and inventory data. Through automation and digital monitoring, the system also seeks to help store staff perform daily operations more efficiently while improving customer service and business productivity.

1.3.2 Specific Objectives

Specifically, this study aims to:

- Analyze the current inventory management practices, user requirements, and existing limitations of manual inventory and sales processes in pharmacies and convenience stores.
- Design and develop an integrated Point of Sale (POS) system that can automatically record sales transactions and update inventory data in real time.
- Implement a real-time inventory tracking feature that allows staff to monitor product availability, stock movement, and inventory levels accurately.
- Develop a low-stock alert and notification system to help prevent product shortages and improve inventory replenishment management.
- Enhance and integrate bar code scanning functionality to reduce transaction delays, improve scanning accuracy, and speed up the checkout process.
- Create a user-friendly and intuitive interface that enables staff users to easily navigate, manage inventory records, and process transactions efficiently.
- Improve the accuracy, organization, and security of inventory and sales records through automated digital data management.
- Reduce human errors commonly associated with manual inventory counting, handwritten sales recording, and outdated monitoring systems.
- Evaluate the proposed Automated Inventory Management System in terms of functionality, usability, efficiency, reliability, and overall performance based on user feedback and system testing.

1.4 Scope and Delimitation

This study focuses on the design and development of an Automated Inventory Management System (AIMS) intended for pharmacies and convenience stores. The proposed system aims to improve inventory monitoring, sales recording, and transaction processing through automation and digital data management. The system is designed to assist store owners and staff in managing daily operations more efficiently and accurately.

The scope of the study includes the development of the following features and functionalities:

- Automated Inventory Tracking – The system will automatically monitor product quantities, stock movement, and inventory updates whenever products are added, sold, or restocked.
- Integrated Point of Sale (POS) System – The system will include a POS feature that records sales transactions automatically and updates inventory data in real time.
- Real-Time Stock Updates – The inventory database will provide updated stock information to help staff monitor product availability and prevent stock discrepancies.
- Low Stock Alert Notifications – The system will notify users when product quantities reach a minimum threshold to support timely restocking and inventory management.

- Bar code Scanning Integration – The system will support bar code scanning to improve transaction speed, reduce manual encoding, and enhance accuracy during checkout operations.
- User Management Dashboard – The system will include a dashboard for authorized users to manage products, monitor inventory, access sales records, and oversee system activities.
- ***Sales and Inventory Reports – The system will generate basic reports related to sales transactions, inventory levels, and product movement for monitoring and documentation purposes.***
- User-Friendly Interface – The proposed system will provide an intuitive and easy-to-use interface suitable for pharmacy and convenience store staff with basic computer knowledge.

The study is limited only to the development and testing of a prototype Automated Inventory Management System. The project will focus primarily on inventory and sales management functionalities and will not include advanced features such as:

- predictive analytics,
- artificial intelligence–based forecasting,
- online ordering systems,
- multi-branch synchronization,
- customer loyalty programs,
- accounting integration, and
- large-scale cloud deployment.

In addition, the study will only evaluate the system in terms of functionality, usability, efficiency, and reliability within the selected testing environment. The system is intended for academic and prototype purposes only and is not designed for full commercial implementation at this stage.

1.5 Significance of the Study

The proposed Automated Inventory Management System (AIMS) is expected to provide significant benefits to pharmacies and convenience stores by improving inventory management, sales recording, and transaction processing. Through automation and real- time monitoring, the system aims to reduce manual work, minimize errors, and enhance the overall efficiency of store operations.

The study will benefit the following:

Users

The system will benefit users, particularly customers and store staff, by providing more accurate inventory tracking and faster transaction processing. Customers will experience improved service due to reduced waiting time during purchases and better product availability monitoring. Staff users will also benefit from a more organized and efficient method of recording sales, checking stock levels, and managing inventory information.

Store owners

Store owners will benefit from the proposed system through improved operational efficiency and better inventory management. The automation of inventory tracking and sales recording can help reduce human errors, prevent inventory discrepancies, and minimize financial losses caused by inaccurate stock monitoring. The system can also assist owners in making informed decisions by providing updated inventory and sales records.

Employees and Staff

The proposed system will help employees and staff perform their daily tasks more efficiently. By automating manual processes such as stock counting, sales recording, and barcode scanning, the system can reduce workload, improve productivity, and simplify transaction processing. The user-friendly interface will also make the system easier to learn and operate.

BSIT Students

This study will serve as a practical application of the knowledge and skills acquired by Bachelor of Science in Information Technology (BSIT) students in the fields of software development, database management, system analysis, and system integration. It will also provide students with experience in designing and developing real-world information systems that address actual business problems.

Future Researchers

Future researchers may use this study as a reference for further research related to automated inventory systems, retail management systems, and Point of Sale (POS) technologies. The findings, methodologies, and system design presented in this study may help future studies improve and expand similar systems by incorporating additional features and technologies.

Pharmacies and Convenience Stores

Pharmacies and convenience stores may benefit from this study by gaining a more modern and efficient approach to inventory and sales management. The implementation of automated systems can improve accuracy, organization, and overall business operations, helping establishments provide better service to customers and maintain more reliable inventory records.

1.6 Definition of Terms

Automated Inventory Management System (AIMS)

An Automated Inventory Management System is a computerized system designed to monitor, organize, and manage inventory levels, product movement, and sales transactions automatically. In this study, the system is used to reduce manual inventory processes, improve accuracy, and enhance operational efficiency in pharmacies and convenience stores.

Inventory Tracking

Inventory tracking refers to the process of monitoring the quantity, availability, and movement of products within a store or business establishment. This includes recording incoming stocks, sold items, and remaining inventory levels to maintain accurate product records.

Point of Sale (POS)

Point of Sale (POS) refers to a computerized system used to process customer purchases and sales transactions. In this study, the POS system automatically records sales, updates inventory levels, and helps improve transaction speed and accuracy.

Real-Time Updates

Real-time updates refer to the immediate and automatic updating of inventory and sales data whenever transactions or stock movements occur. This feature allows users to access current and accurate information without delays.

Barcode Scanning System

A bar code scanning system is a technology used to read product bar codes for faster and more accurate item identification during transactions and inventory management. In this study, bar code scanning helps reduce manual encoding and speeds up the checkout process.

Operational Efficiency

Operational efficiency refers to the ability of a business or system to perform tasks accurately, quickly, and with minimal wasted time, effort, or resources. The proposed system aims to improve operational efficiency by automating inventory and sales management processes.

Low Stock Alert

A low stock alert is a system-generated notification that informs users when the quantity of a product reaches a minimum or critical level. This feature helps prevent stock shortages and supports timely restocking of products.

User Management

User management refers to the process of controlling and organizing user access within the system. It includes managing user accounts, roles, permissions, and login credentials to ensure proper system security and authorized access.

Database

A database is an organized collection of digital information stored electronically for easy access, management, and updating. In this study, the database stores inventory records, sales transactions, product details, and user information.

CHAPTER 2
REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

2.1.1 Automated inventory management systems and its impact on supply chain risk management in manufacturing firms of Pakistan. International Journal of Management

According to Saleem (2020), inventory management serves as one of the most essential functions within an organization because it directly influences operational efficiency, customer satisfaction, profitability, and long-term business sustainability. The study emphasized that organizations operating in retail, manufacturing, distribution, and service industries rely heavily on effective inventory control to ensure that products and materials are available when needed without maintaining excessive stock levels. As business environments become increasingly competitive and customer expectations continue to rise, organizations are compelled to adopt modern inventory management practices that improve accuracy, efficiency, and responsiveness. The author highlighted that traditional inventory management methods are no longer sufficient for many organizations because manual processes often fail to provide the speed and precision required in modern business operations. The literature explained that many organizations historically relied on manual inventory management practices such as handwritten records, spreadsheets, physical stock counting, and paper-based reporting systems. Although these methods were suitable for smaller operations in the past, they present numerous challenges as businesses expand and inventory volumes increase. Manual inventory management consumes substantial amounts of time and labor, requiring employees to regularly update records, conduct physical stock verification, and prepare inventory reports. These activities not only increase operational costs but also expose organizations to a higher risk of human error. Mistakes such as incorrect data entry, misplaced records, inaccurate stock counts, and delayed inventory updates can result in inventory discrepancies that negatively affect business performance. The study noted that these problems often lead to stock shortages, overstocking, delayed deliveries, and customer dissatisfaction.

Saleem further discussed how advancements in information technology have transformed inventory management through the development of automated inventory management systems. These systems utilize various technological tools and software applications that enable organizations to monitor, record, and control inventory more efficiently. Technologies such as Enterprise Resource Planning (ERP), Material Requirements Planning (MRP), Electronic Data Interchange (EDI), Radio Frequency Identification (RFID), and barcode systems have become integral components of modern inventory operations. Through these technologies, inventory information can be updated automatically whenever products are received, transferred, sold, or returned. This automated process significantly reduces dependence on manual intervention while improving the reliability and accuracy of inventory records.

The study highlighted that one of the most important advantages of automated inventory systems is the ability to provide real-time inventory visibility. Real-time visibility allows managers and decision-makers to access current inventory information whenever needed. Rather than relying on outdated reports or manually compiled records, organizations can instantly determine stock quantities, product locations, inventory turnover rates, and reorder requirements. This immediate access to information enables businesses to respond quickly to changes in customer demand and market conditions.

Furthermore, accurate inventory data helps organizations avoid situations where products become unavailable due to stock shortages or where excessive inventory accumulates and occupies valuable storage space.

In addition, the literature emphasized that automated inventory management systems contribute significantly to operational efficiency. The automation of inventory-related tasks eliminates many repetitive and time-consuming activities that employees previously performed manually. Inventory updates occur automatically through barcode scanning, RFID tracking, and integrated software systems, reducing the need for constant record verification and manual data entry. As a result, employees can allocate more time to customer service, sales activities, strategic planning, and other value- adding responsibilities. The reduction of manual workload not only improves productivity but also enhances employee performance by allowing workers to focus on tasks that contribute more directly to organizational objectives. The author also explained that inventory accuracy is substantially improved through automation.

Inventory inaccuracies often occur when organizations depend on manual recording systems that are vulnerable to mistakes and inconsistencies. Automated systems minimize these risks by ensuring that inventory transactions are recorded immediately and accurately within centralized databases. This improved accuracy allows organizations to maintain reliable stock records and reduce discrepancies between physical inventory and recorded inventory levels. Accurate inventory information is particularly important because many business decisions regarding purchasing, production scheduling, and distribution planning depend on the quality of available inventory data.

Furthermore, Saleem discussed the role of automated inventory systems in supporting effective supply chain management. Modern supply chains involve multiple stakeholders, including suppliers, manufacturers, warehouses, distributors, and retailers. Effective coordination among these participants requires accurate and timely information regarding inventory availability and movement. Automated inventory systems facilitate information sharing across different departments and business partners, improving communication and collaboration throughout the supply chain.

Real-time inventory data enables organizations to coordinate replenishment activities more efficiently, reduce lead times, and respond proactively to fluctuations in demand. Consequently, businesses can maintain smoother operations while reducing disruptions caused by inventory shortages or supply delays.

The literature also highlighted the impact of automated inventory management on customer satisfaction. Customers expect businesses to provide products promptly and consistently. When inventory records are inaccurate, organizations may promise products that are unavailable or fail to replenish high-demand items in a timely manner. These situations can result in customer frustration and lost sales opportunities.

Automated inventory systems improve product availability by providing accurate stock information and generating reorder alerts when inventory reaches predetermined levels. This capability helps ensure that products remain available for customers while minimizing delays and service interruptions. Improved inventory management therefore contributes directly to better customer experiences and stronger customer loyalty.

Moreover, the study explained that automated inventory systems support better decision-making processes within organizations. Business managers require accurate information to evaluate inventory performance, identify trends, forecast future demand, and develop effective business strategies. Automated systems generate comprehensive reports and analytical data that provide valuable insights into inventory operations.

Managers can analyze sales patterns, monitor inventory turnover, identify slow-moving products, and determine optimal reorder points. These capabilities allow organizations to make evidence-based decisions that improve operational performance and resource utilization. The availability of detailed inventory reports also supports financial planning and budgeting activities by providing a clearer understanding of inventory-related costs and investment requirements.

The author further emphasized the financial benefits associated with inventory automation. Effective inventory management helps organizations reduce carrying costs, storage expenses, inventory losses, and unnecessary purchasing activities. By maintaining optimal inventory levels, businesses can minimize capital tied up in excess stock while ensuring sufficient product availability to meet customer demand.

Automated inventory systems also help reduce losses caused by theft, damage, expiration, and administrative errors. The study noted that organizations implementing inventory automation frequently report improved profitability due to better inventory control and more efficient resource management. Despite the numerous advantages of automated inventory systems, Saleem acknowledged several challenges associated with implementation. Initial investments in hardware, software, networking infrastructure, and employee training may require substantial financial resources. Small and medium- sized enterprises often face difficulties in adopting advanced inventory technologies

because of budget limitations and concerns regarding implementation costs. In addition, employees may resist technological changes due to unfamiliarity with new systems or concerns about increased complexity. The study stressed the importance of providing adequate training programs, technical support, and change management strategies to ensure successful implementation and user acceptance.

The literature additionally discussed the importance of management commitment in achieving successful inventory automation. Organizational leaders play a critical role in allocating resources, establishing implementation plans, and encouraging employee participation throughout the transition process. Strong managerial support helps create a positive environment for technological adoption and ensures that inventory management initiatives align with organizational goals. Continuous monitoring, system evaluation, and process improvement are also necessary to maximize the long-term benefits of automated inventory systems.

Overall, Saleem (2020) concluded that automated inventory management systems have become indispensable tools for organizations seeking to improve operational performance, inventory accuracy, and customer satisfaction. The study demonstrated that technology-driven inventory management enables businesses to optimize stock control, enhance supply chain coordination, reduce operational costs, and support strategic decision-making. As organizations continue to face increasing competition and evolving customer expectations, the adoption of automated inventory systems remains an essential strategy for achieving efficiency, sustainability, and long-term business success.

This study is highly relevant to the proposed Automated Inventory Management System (AIMS) because it supports the use of technology to improve inventory monitoring, stock control, and operational efficiency. Similar to the concepts discussed by Saleem, the proposed system aims to replace manual inventory procedures with automated processes that provide real-time inventory information, improve accuracy, reduce human errors, and support better decision-making. The findings of the study provide strong evidence that inventory automation can significantly enhance business operations and contribute to improved organizational performance within pharmacy and retail environments.

2.1.2 Computerized Inventory Management Systems and Business Efficiency

According to Akinsanya (2019), inventory management remains one of the most significant operational functions within business organizations because it directly affects productivity, profitability, customer satisfaction, and organizational effectiveness. The study emphasized that businesses operating in retail, pharmaceutical, manufacturing, and distribution sectors depend heavily on accurate inventory information to support daily operations and strategic decision-making. As markets become increasingly competitive and customer expectations continue to rise, organizations are required to adopt more efficient inventory management practices that enable them to maintain accurate records, improve stock control, and ensure the timely availability of products.

The author explained that many organizations continue to face inventory-related challenges due to reliance on traditional inventory management methods that are often inefficient and prone to errors.

The literature discussed how conventional inventory management systems commonly rely on manual documentation, paper-based records, and periodic physical inventory counts. Although these methods have been utilized for many years, they often create operational difficulties that negatively impact business performance. Manual inventory management requires employees to spend significant amounts of time recording transactions, updating stock information, verifying inventory levels, and preparing reports. These activities increase labor requirements and consume valuable resources that could otherwise be directed toward customer service and business development. Furthermore, manual systems are susceptible to human errors such as incorrect data entry, inaccurate stock recording, misplaced documentation, and delayed inventory updates, all of which contribute to inventory discrepancies and operational inefficiencies.

Akinsanya explained that the growing complexity of modern business operations has increased the demand for computerized inventory management systems capable of providing accurate and real-time inventory information. Computerized inventory systems utilize digital technologies to automate inventory-related activities, reducing dependence on manual procedures while improving overall inventory accuracy. Through automated data collection and processing, organizations can monitor inventory levels continuously and maintain updated records without requiring extensive manual intervention. The study highlighted that businesses implementing computerized inventory systems experience significant improvements in inventory control and operational efficiency.

The study further emphasized that one of the primary benefits of computerized inventory management systems is the ability to provide real-time monitoring of inventory movements. Real-time inventory visibility allows organizations to track stock quantities as products are received, transferred, sold, or returned. This capability enables managers to access current inventory information whenever necessary and make informed decisions regarding purchasing, replenishment, and stock allocation. The availability of accurate and up-to-date inventory data helps organizations avoid stock shortages that may result in lost sales opportunities while also preventing excessive stock accumulation that increases storage costs and reduces financial flexibility.

Moreover, the literature highlighted the role of computerized inventory systems in improving inventory accuracy. Accurate inventory records are essential because inventory-related decisions influence various aspects of organizational performance, including procurement, sales, production planning, and customer service. Manual inventory systems frequently produce discrepancies between actual inventory levels and recorded stock quantities due to recording mistakes and delayed updates. Computerized systems reduce these risks by automatically recording inventory transactions and updating databases in real time. As a result, businesses maintain more reliable inventory information that supports effective operational planning and resource management.

The author also discussed how computerized inventory management systems contribute to increased organizational productivity. Employees working with automated systems spend less time performing repetitive administrative tasks such as manual stock counting, record updating, and report preparation. This reduction in workload allows staff members to focus on activities that generate greater value for the organization, including customer assistance, marketing initiatives, and strategic planning. The study noted that automation not only improves employee efficiency but also enhances workplace effectiveness by reducing frustration associated with time-consuming manual processes.

Furthermore, Akinsanya explained that computerized inventory systems enhance customer service quality by ensuring product availability and improving transaction efficiency. Customers expect businesses to provide accurate information regarding product availability and to complete transactions quickly and efficiently. Inventory inaccuracies can result in situations where customers are informed that products are available when they are actually out of stock, creating dissatisfaction and reducing customer trust. Through real-time inventory monitoring and automated stock updates, computerized systems help organizations maintain adequate stock levels and improve service reliability. Faster transaction processing and improved inventory visibility contribute to a more positive customer experience and encourage customer loyalty. The study also highlighted the importance of inventory reporting and data analysis capabilities provided by computerized systems. Business managers require accurate information to evaluate inventory performance and make strategic decisions.

Computerized inventory systems generate reports that provide detailed information regarding stock levels, sales performance, inventory turnover, reorder requirements, and product movement trends. These reports support evidence-based decision-making and enable managers to identify opportunities for improvement. The ability to access accurate inventory data allows organizations to forecast demand more effectively, optimize purchasing activities, and allocate resources more efficiently.

Additionally, the literature discussed the financial benefits associated with computerized inventory management. Improved inventory control helps organizations reduce unnecessary inventory costs by maintaining appropriate stock levels and minimizing waste. Accurate inventory information prevents excessive purchasing and reduces losses associated with obsolete, damaged, or expired products. Businesses can also improve cash flow management by avoiding investments in unnecessary inventory while ensuring sufficient stock availability to meet customer demand. The study found that organizations implementing computerized inventory systems often achieve improved financial performance due to enhanced inventory efficiency and x reduced operational costs.

Akinsanya further explained that inventory security is strengthened through the implementation of computerized systems. Traditional inventory methods may make it difficult to detect theft, inventory shrinkage, unauthorized transactions, and record manipulation. Computerized systems provide greater accountability by maintaining transaction histories, user access controls, and audit trails that allow organizations to monitor inventory activities more effectively. Enhanced security measures help reduce inventory losses and support more accurate inventory management practices. The study acknowledged that despite numerous benefits, the implementation of computerized inventory management systems may present challenges for some organizations. Initial acquisition costs, software licensing fees, employee training requirements, and infrastructure investments may create financial barriers, particularly for small and medium-sized enterprises. Organizations must also address employee resistance to technological change and ensure that staff members possess the necessary skills to utilize the system effectively. The author emphasized that successful implementation requires proper planning, adequate training, management commitment, and ongoing technical support.

The literature additionally noted that technological advancements continue to expand the capabilities of computerized inventory systems. Modern systems increasingly incorporate barcode technology, cloud computing, mobile applications, and integrated business management solutions that further enhance inventory visibility and operational efficiency. These innovations allow organizations to manage inventory across multiple locations while maintaining centralized access to inventory information. Such capabilities improve organizational flexibility and support business growth in increasingly competitive market environments.

Overall, Akinsanya (2019) concluded that computerized inventory management systems provide significant advantages in terms of inventory accuracy, operational efficiency, customer service, financial performance, and organizational productivity.

The study demonstrated that organizations adopting inventory automation are better equipped to manage inventory resources effectively, respond to market demands, and maintain competitive advantages within their respective industries. This study is highly relevant to the proposed Automated Inventory Management System (AIMS) because it supports the implementation of computerized inventory technologies to improve stock monitoring, inventory accuracy, and operational efficiency. Similar to the findings of Akinsanya, the proposed system aims to eliminate inefficiencies associated with manual inventory management while providing real-time inventory information, improved reporting capabilities, and enhanced decision-making support. The concepts discussed in the study strengthen the rationale for developing an automated inventory management solution for pharmacy and retail environments.

2.1.3 Design of a Computerized Inventory Management System for Supermarkets

A foreign study conducted by Abisoye, Fatoba, and Blessing (2013) examined the design of computerized inventory management systems for supermarkets and their impact on business operations. The study aimed to determine how automation technologies improve inventory control, stock monitoring, and organizational productivity within retail establishments. The researchers explained that supermarkets commonly experience operational problems related to manual inventory management, including inaccurate stock records, delayed inventory updates, poor product monitoring, and inefficient transaction processing. These problems often affect customer service and increase operational costs.

The study emphasized that inventory management plays an important role in the success of retail businesses because it helps organizations monitor product movement, maintain sufficient stock levels, and avoid inventory shortages. According to the researchers, businesses that rely on manual inventory processes often encounter difficulties in maintaining accurate inventory records due to the large volume of products handled daily. This increases the risk of inventory discrepancies, misplaced products, and poor decision-making regarding stock replenishment.

To address these challenges, the researchers designed a computerized inventory management system capable of automating inventory tracking and stock recording processes. The system provided features that allowed users to store product information digitally, monitor inventory levels, and process inventory updates automatically whenever products were purchased or restocked. The researchers explained that the system also improved inventory organization by centralizing product data within a computerized database.

The findings of the study revealed that the computerized inventory management system significantly improved inventory accuracy and operational efficiency within supermarkets. Through automation, employees were able to access updated inventory information more quickly and reduce the occurrence of manual recording errors. The researchers noted that the system improved stock monitoring because product quantities were updated automatically in real time. This allowed businesses to identify low-stock items more effectively and prevent product shortages.

Furthermore, the study found that the computerized system improved transaction speed and workflow efficiency within supermarkets. Employees no longer needed to manually verify stock records because the system automatically processed inventory updates and sales information. This reduced delays during transactions and improved the overall customer experience. The researchers also explained that automation reduced employee workload by simplifying inventory-related tasks and minimizing repetitive manual procedures.

Another important finding of the study was the improvement of business decision- making through digital reporting and inventory analysis. Managers were able to generate inventory reports more efficiently and analyze product movement using the computerized system. This helped businesses identify fast-moving and slow-moving products, improve purchasing decisions, and maintain balanced inventory levels. The researchers emphasized that accurate inventory information contributes to better organizational planning and operational productivity.

The study also discussed several challenges encountered during system implementation. Businesses needed financial resources to purchase computer equipment and software tools necessary for the computerized inventory system. Employees also required technical training to become familiar with the new digital processes. Some workers initially resisted the transition because they were accustomed to traditional inventory methods. However, the researchers noted that continuous system usage and proper training improved employee confidence and system acceptance over time.

The study concluded that computerized inventory management systems improve inventory control, transaction efficiency, organizational productivity, and customer service within supermarkets. The researchers emphasized that automation provides businesses with more accurate inventory information and simplifies inventory-related operations. Although implementation may involve financial costs and technical challenges, the long-term advantages include reduced human errors, improved workflow, and better inventory management.

This foreign study is relevant to the present study because it demonstrates how computerized inventory systems improve inventory accuracy, transaction processing, and operational efficiency in retail businesses. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the use of automation technologies to simplify inventory tracking and improve sales recording. The findings also support the integration of digital databases and real- time inventory monitoring in modern inventory management systems.

2.1.4 Inventory management automation and performance of supermarkets in Western Kenya

A foreign study conducted by Samuel and Ondiek (2014) investigated the effects of inventory management automation on the performance of supermarkets in Western Kenya. The study aimed to determine how automated inventory systems influence inventory control, operational efficiency, and business performance within retail establishments. The researchers explained that many supermarkets experience operational problems caused by manual inventory management practices, including stock shortages, inaccurate inventory records, delayed stock updates, and inefficient monitoring procedures. These issues often reduce organizational productivity and negatively affect customer satisfaction.

The study emphasized that inventory management automation is important because it improves the ability of businesses to monitor inventory levels and manage stock movement more efficiently. According to the researchers, traditional inventory methods rely heavily on manual work, which increases the risk of human error and delays in processing inventory information. As businesses grow and inventory volume increases, manual processes become less effective and more difficult to manage.

The researchers examined how automated inventory technologies improve retail operations by reducing manual tasks and improving the accuracy of inventory information. The study explained that automated systems allow supermarkets to update inventory records automatically whenever products are sold, restocked, or transferred. This improves inventory visibility and helps businesses maintain updated stock information in real time.

The findings of the study revealed that inventory management automation significantly improved supermarket performance and operational efficiency. Through automated systems, businesses were able to monitor inventory more accurately and reduce problems associated with stock discrepancies. The researchers found that automation improved product availability because inventory levels could be tracked continuously and low-stock products could be identified immediately. This helped supermarkets reduce product shortages and improve customer service.

Furthermore, the study highlighted that automation improved transaction processing and inventory organization within supermarkets. Employees were able to complete inventory-related tasks more efficiently because the system reduced the need for manual recording and repetitive paperwork. The researchers explained that automated systems also improved workflow efficiency by simplifying inventory monitoring procedures and reducing delays associated with manual inventory checking.

Another important finding of the study was the positive effect of automation on decision-making and organizational productivity. Managers were able to access accurate inventory reports and monitor stock movement more effectively using automated systems. This helped businesses improve purchasing decisions, reduce unnecessary inventory costs, and maintain balanced stock levels. The study also noted that automated inventory systems contribute to better warehouse organization and improved operational planning.

Despite the positive findings, the study identified several challenges related to the implementation of automated inventory systems. Supermarkets needed financial investment to acquire technology infrastructure and computerized equipment. Employees also required training programs to understand how to operate the automated systems correctly. Some employees experienced difficulties adapting to new technologies due to limited computer skills and resistance to organizational changes. However, the researchers explained that proper employee support and training improved system acceptance and operational performance.

The study concluded that automated inventory management systems positively affect supermarket performance, inventory control, and operational efficiency. The researchers emphasized that automation helps businesses improve inventory monitoring, reduce operational errors, and enhance customer service. Although implementation may involve technical and financial challenges, the long-term benefits include improved productivity, reduced inventory discrepancies, and better business operations.

This foreign study is relevant to the present study because it demonstrates how inventory automation improves operational efficiency, inventory monitoring, and business performance within retail establishments. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the use of automated technologies to improve inventory accuracy, reduce manual work, and enhance sales and inventory management processes. The findings also strengthen the importance of real-time inventory updates and digital inventory monitoring in modern retail operations.

2.1.5 The effect of inventory management on firm performance.

A foreign study conducted on Alfa Company, a medium-sized distributor of laboratory equipment and supplies in Iraq, examined the implementation of an automated inventory management system and its effects on the company's business operations. The study aimed to determine how automation could improve inventory control, operational efficiency, and customer service in small and medium-sized enterprises (SMEs). The researchers emphasized that traditional inventory management methods often lead to problems such as inaccurate stock records, delayed transactions, overstocking, and product shortages, which may negatively affect the overall performance of a business.

The findings of the study revealed that the implementation of the automated inventory management system significantly improved the company's ability to monitor and manage inventory levels. Through the use of computerized processes, the company was able to track products more accurately, reduce human errors, and maintain updated inventory records in real time. The system also helped minimize storage expenses by preventing excessive stock accumulation and improving warehouse organization. In addition, the study noted that customer satisfaction improved because the company was able to provide faster and more reliable services, ensuring that needed products were readily available and delivered efficiently.

Furthermore, the research highlighted that automated inventory systems contribute to better decision-making within organizations. Because inventory data could be accessed quickly and accurately, managers were able to analyze stock movement, identify fast- moving and slow-moving items, and make informed purchasing decisions. This improved the company's operational productivity and reduced unnecessary costs associated with inventory mismanagement. The study also emphasized that automation supports business growth by enabling companies to handle larger volumes of inventory and transactions more effectively.

Despite the positive outcomes, the study identified several challenges encountered during the implementation process. The transition from manual to automated inventory management required a significant investment of time, financial resources, and technical expertise. The company needed the assistance of IT professionals to install, configure, and maintain the system properly. Additionally, employees had to undergo training programs to familiarize themselves with the new technology and operational procedures. Some workers initially experienced difficulty adapting to the system due to limited technical knowledge and resistance to organizational change.

The study further explained that collaboration between businesses, IT consultants, and software suppliers played an important role in the successful implementation of the automated inventory management system. These professionals assisted the company in evaluating the advantages and costs of automation while ensuring that the system met the specific needs of the organization. The researchers concluded that although implementing an automated inventory system may involve challenges, the long-term benefits outweigh the difficulties because it enhances efficiency, accuracy, productivity, and customer service.

This foreign study is relevant to the present study because it demonstrates how automated inventory management systems can improve inventory accuracy, operational efficiency, and business performance. It also supports the idea that adopting computerized systems can reduce manual errors and simplify inventory-related tasks. The findings serve as a valuable reference for the current research, particularly in understanding the importance of automation in inventory management and its impact on organizational effectiveness.

2.1.6 The impact of efficient inventory management on profitability: Evidence from selected manufacturing firms in Ghana

Researchers from Jordan conducted a study that investigated the impact of a warehouse management system on supply chain performance, particularly in improving inventory management efficiency and reducing operational effort within the telecommunications sector. The study focused on the warehouse operations of a leading telecommunications service provider and aimed to determine how a customized warehouse management system could enhance workflow efficiency, inventory accuracy, and supply chain reliability. The researchers examined existing warehouse procedures before developing and implementing a software-based inventory management system capable of handling warehouse transactions and operational processes more effectively.

The study explained that traditional manual inventory systems often create operational inefficiencies due to slow processing, inaccurate inventory records, and delays in handling products and transactions. These problems may negatively affect supply chain performance and reduce the ability of companies to respond efficiently to customer demands. To address these issues, the researchers customized a warehouse management software system designed to automate and simplify inventory-related operations. The system was tested to evaluate its effectiveness in improving warehouse workflow, handling inventory transactions in a timely manner, and supporting overall supply chain operations.

The findings of the study revealed that the implementation of the warehouse management system significantly improved operational efficiency and inventory management processes. The system reduced the amount of manual labor required in handling inventory transactions, resulting in faster and more reliable warehouse operations. Through automation, inventory data became more accurate and accessible, which helped employees monitor stock levels effectively and minimize errors in inventory recording and product handling. The researchers also noted that the software improved communication and coordination among warehouse personnel, contributing to smoother and more organized operations.

In addition, the study examined the physical layout of the warehouse and introduced a production station within the facility to optimize warehouse space utilization. The production station involved three important processes: bundling, labeling, and repackaging of products. According to the researchers, integrating these production activities within the warehouse improved workflow organization and allowed the company to maximize available storage and operational space. This adjustment contributed to better productivity and more efficient handling of products throughout the warehouse operations.

Furthermore, the warehouse management system handled three major phases of the product lifecycle, namely receiving, processing, and distribution of SIM cards and prepaid scratch cards. The researchers discussed each phase in detail and identified operational gaps and inefficiencies in the previous manual procedures. By automating these phases, the company improved the speed and accuracy of product movement within the supply chain. The study also highlighted that the customized software system helped reduce delays, improve transaction monitoring, and support more efficient distribution processes.

The researchers concluded that warehouse management systems play a vital role in strengthening supply chain performance and improving inventory management in the telecommunications industry. The study emphasized that automated inventory systems are more efficient and reliable than traditional manual methods because they reduce operational effort, improve inventory visibility, and support faster decision-making. Moreover, the researchers stated that the study may serve as a practical guide for organizations and future researchers interested in comparing automated and manual inventory management systems. The findings also highlighted the importance of developing customized systems that can address operational challenges and reduce supply chain disruptions.

This study is relevant to the present research because it demonstrates how automated warehouse and inventory management systems improve operational efficiency, workflow organization, and inventory accuracy. It supports the idea that computerized inventory systems can simplify business processes, reduce manual errors, and enhance supply chain performance.The study also provides valuable information regarding the importance of automation in inventory and warehouse operations, which may contribute to the development of more efficient inventory management systems in the current research.

2.1.7 Inventory management in small and medium enterprises: A study of machine tool enterprises in Bangalore

A foreign study conducted by Gunasekaran and Ngai (2004) examined the role of information systems and automation technologies in supply chain and inventory management within business organizations. The study aimed to determine how computerized inventory systems improve operational efficiency, inventory visibility, organizational productivity, and overall business performance. The researchers explained that traditional inventory management methods are often ineffective because they rely heavily on manual processes, paperwork, and delayed communication between departments. These problems may result in inaccurate inventory records, stock shortages, delayed transactions, poor inventory coordination, and customer dissatisfaction.

The researchers emphasized that inventory management is one of the most important operational functions within organizations because inventory directly affects customer service, productivity, profitability, and business continuity. According to the study, businesses that fail to maintain accurate inventory records often encounter financial losses due to overstocking, product expiration, theft, and poor inventory planning. The researchers also noted that manual inventory systems consume significant time and labor because employees are required to encode information manually, conduct physical inventory checking repeatedly, and process inventory reports through traditional paperwork.

To address these operational challenges, the study explored the use of computerized information systems and digital inventory technologies in improving inventory and supply chain management. The researchers highlighted that automation technologies allow organizations to store, process, and retrieve inventory information more efficiently through centralized digital databases. The study also explained that automated inventory systems provide real-time inventory updates, enabling businesses to monitor stock movement immediately whenever products are purchased, transferred, or restocked.

The findings of the study revealed that organizations using automated inventory systems experienced significant improvements in inventory monitoring and operational efficiency. Through automation, businesses were able to reduce human errors associated with manual inventory recording and improve the accuracy of inventory information. The researchers explained that real-time inventory monitoring allowed organizations to identify stock shortages quickly, prevent overstocking, and maintain balanced inventory levels. As a result, businesses were able to improve inventory coordination and reduce operational delays.

Furthermore, the study found that computerized inventory systems improved communication and information sharing within organizations. Inventory information could be accessed by multiple departments simultaneously, improving coordination between purchasing, sales, warehousing, and management personnel. This reduced delays caused by manual reporting and improved organizational workflow. Managers were also able to generate reports more efficiently and analyze inventory data to support better operational planning and decision-making.

Another important finding of the study was the positive effect of automation on customer service and organizational productivity. Businesses using automated systems were able to process transactions faster, provide more accurate product availability information, and respond to customer demands more efficiently. The researchers emphasized that customers benefit from automation because businesses can maintain adequate stock levels and reduce delays in service delivery. In addition, automation reduced repetitive manual tasks, allowing employees to focus on more productive operational activities.

The study also highlighted several technologies commonly used in automated inventory management systems, including barcode scanning systems, enterprise resource planning systems, and electronic data interchange technologies. According to the researchers, barcode technologies improve transaction speed and inventory accuracy by reducing manual encoding errors. Enterprise systems help organizations centralize inventory information and improve communication across departments, while electronic data exchange technologies improve coordination with suppliers and business partners.

Despite the positive outcomes of automation, the study identified several challenges associated with implementing computerized inventory systems. Organizations needed financial investment to acquire hardware, software, and technical infrastructure necessary for automation. Businesses also required technical experts to install, configure, and maintain the systems properly. The researchers explained that employees may initially experience difficulty adapting to automated technologies due to limited computer knowledge and resistance to organizational changes. However, proper training and continuous system use improved employee confidence and system acceptance over time.

The study further explained that organizational support and management commitment are important factors in the successful implementation of automated inventory systems. Managers must provide employees with adequate training, technical support, and guidance to ensure effective system adoption. The researchers also emphasized that collaboration between technology providers and organizations helps ensure that inventory systems meet operational requirements and business objectives.

The study concluded that computerized inventory systems and automation technologies significantly improve inventory management, organizational coordination, operational productivity, and customer service. The researchers emphasized that although implementing automated inventory systems may involve technical and financial challenges, the long-term benefits outweigh the difficulties because automation improves inventory accuracy, simplifies operational processes, reduces delays, and enhances overall organizational performance.

This foreign study is highly relevant to the present study because it demonstrates how automated inventory systems improve inventory monitoring, transaction processing, and operational efficiency within organizations. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the use of digital technologies, barcode scanning systems, and real-time inventory monitoring to reduce manual errors and improve inventory-related operations. The findings also strengthen the importance of automation in improving customer service, workflow efficiency, and organizational productivity.

2.1.8 Best Practice in Inventory Management

A foreign study conducted by DeHoratius and Raman (2008) investigated inventory record inaccuracies in retail establishments and examined how automated inventory systems improve inventory control and operational efficiency. The study aimed to identify the common causes of inventory discrepancies in retail businesses and determine how technology-based inventory systems reduce errors associated with manual inventory management. The researchers explained that accurate inventory records are important because they directly affect sales performance, customer satisfaction, operational productivity, and organizational profitability. However, many retail businesses continue to experience inventory-related problems due to outdated manual inventory practices and inefficient monitoring procedures.

The researchers emphasized that inventory inaccuracies are among the most common operational problems in retail businesses. According to the study, inventory discrepancies occur when the recorded quantity of products does not match the actual physical inventory available in the store or warehouse. These discrepancies are often caused by manual recording errors, theft, misplaced items, delayed inventory updates, and poor inventory coordination. The study noted that inventory inaccuracies may result in product shortages, delayed customer service, lost sales opportunities, and poor business decision-making.

To better understand the problem, the researchers analyzed inventory management practices in retail establishments and examined how businesses monitor and update inventory records. The study explained that traditional inventory systems require employees to manually record stock movement and conduct physical inventory counting regularly. This process is time-consuming and increases the likelihood of human error, especially when businesses handle large volumes of products and transactions daily. Because inventory information is not updated immediately, businesses may struggle to maintain accurate inventory records and respond effectively to operational problems.

The findings of the study revealed that businesses using automated inventory systems experienced significant improvements in inventory accuracy and operational performance. Through computerized inventory monitoring and digital record management, organizations were able to reduce errors associated with manual inventory processes. The researchers explained that automated systems update inventory records in real time whenever products are sold, transferred, or restocked.

This allowed businesses to monitor stock movement more accurately and identify inventory discrepancies more efficiently.

Furthermore, the study found that barcode scanning technologies played an important role in improving inventory accuracy and transaction efficiency. Barcode systems minimized errors caused by manual product encoding and improved the speed of inventory-related operations. Employees were able to process transactions faster and update inventory information automatically, reducing delays associated with traditional inventory recording methods. The researchers also noted that barcode-integrated systems improved inventory visibility because product information could be accessed quickly and accurately through computerized databases.

Another important finding of the study was the improvement of organizational decision-making through real-time inventory information. Managers were able to generate inventory reports more efficiently and analyze product movement using computerized inventory systems. This helped businesses identify fast-moving and slow-moving products, improve purchasing decisions, and maintain balanced inventory levels. The study emphasized that accurate inventory information supports better inventory planning and reduces losses associated with overstocking and stock shortages.

The study also highlighted the positive effect of automation on customer service and business productivity. Businesses using automated inventory systems were more capable of maintaining product availability and reducing delays during customer transactions. Customers benefited from improved service because businesses could provide accurate information regarding product availability and process transactions more efficiently. In addition, automation reduced repetitive manual tasks, allowing employees to focus on more productive operational activities.

Despite the positive findings, the study identified several challenges associated with implementing automated inventory systems. Businesses required financial investment for technology infrastructure, barcode equipment, and computerized software systems. Employees also needed technical training to understand how to operate the systems properly. Some workers initially experienced difficulties adapting to new technologies because they were more familiar with manual inventory procedures. However, continuous training and regular system use improved employee confidence and system acceptance over time.

The researchers concluded that automated inventory systems significantly improve inventory accuracy, operational efficiency, transaction processing, and organizational productivity within retail businesses. The study emphasized that although implementing computerized inventory systems may involve technical and financial challenges, the long-term benefits include reduced inventory discrepancies, improved workflow management, enhanced customer service, and better inventory planning.

This foreign study is highly relevant to the present study because it demonstrates how automated inventory systems and barcode technologies improve inventory monitoring and transaction efficiency within retail establishments. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the use of real-time inventory monitoring, barcode scanning, and computerized inventory management to reduce manual errors and improve operational performance.

2.1.9 Implementing an Automated Inventory Management System for Small and Medium-sized Enterprises

A foreign study conducted by Sharma, Garg, and Agarwal (2011) examined the implementation of barcode technology in inventory management systems and its effect on retail operations and organizational efficiency. The study aimed to determine how barcode scanning technologies improve transaction processing, inventory accuracy, product monitoring, and customer service in retail establishments. The researchers explained that many businesses continue to experience operational inefficiencies due to manual inventory recording and traditional checkout procedures. These manual methods often result in delayed transactions, incorrect inventory records, inventory discrepancies, and customer dissatisfaction.

The researchers emphasized that barcode technology has become an important component of modern inventory management because it improves the speed and accuracy of inventory-related operations. According to the study, traditional inventory systems rely heavily on manual encoding and handwritten records, which increase the likelihood of human error and delay operational processes. Employees may accidentally input incorrect product information or fail to update inventory records immediately, leading to inaccurate stock data and operational inefficiencies.

The study explored how barcode scanning systems function within inventory and retail operations. Barcode technology allows products to be identified quickly through machine-readable codes attached to product packaging. When scanned using barcode readers, product information is automatically retrieved from the inventory database and updated in real time. The researchers explained that this automation process reduces manual work and improves inventory monitoring efficiency.

The findings of the study revealed that businesses using barcode-integrated inventory systems experienced significant improvements in transaction speed and inventory accuracy. Barcode scanning reduced delays during checkout operations because employees no longer needed to encode product information manually. The automated process minimized typing errors and improved the accuracy of sales recording and inventory updates. As a result, businesses were able to maintain more reliable inventory records and reduce problems associated with inventory discrepancies.

Furthermore, the study found that barcode systems improved inventory visibility and product tracking within organizations. Inventory information could be updated automatically whenever products were sold, transferred, or restocked. This enabled businesses to monitor inventory movement more efficiently and identify low-stock products quickly. Managers were also able to access inventory reports more accurately and improve purchasing and inventory planning decisions.

Another important finding of the study was the improvement of customer satisfaction and operational productivity. Businesses using barcode systems were able to process customer transactions faster and provide more efficient service. Customers experienced shorter waiting times during checkout because barcode scanners simplified transaction processing. Employees also benefited because the system reduced repetitive manual tasks and improved workflow organization.

The study further highlighted that barcode technology improved security and inventory control within retail establishments. Because inventory records were updated automatically, businesses were able to reduce the occurrence of missing products and inventory losses caused by recording errors. The researchers explained that barcode-integrated systems also improved warehouse organization and product identification, making inventory-related tasks easier to manage.

Despite the positive outcomes, the study identified several challenges encountered during barcode system implementation. Businesses required financial investment for barcode scanners, software systems, and computerized infrastructure. Employees also needed technical training to understand how to operate barcode equipment effectively. Some organizations initially experienced implementation difficulties because employees were unfamiliar with digital technologies and automated inventory procedures.
However, the researchers noted that proper employee training and continuous system usage improved adaptation and operational performance over time.

The study concluded that barcode scanning technologies significantly improve inventory management, transaction efficiency, operational productivity, and customer satisfaction within retail businesses. The researchers emphasized that although implementing barcode systems may involve technical and financial challenges, the long-term benefits include improved inventory accuracy, faster transaction processing, reduced operational delays, and better organizational performance.

This foreign study is highly relevant to the present study because it demonstrates how barcode-integrated inventory systems improve transaction processing, inventory monitoring, and operational efficiency. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the integration of barcode scanning technology to reduce manual errors, improve sales recording, and enhance customer service.

2.1.10 Performance improvement of inventory management system processes by an automated warehouse management system

A foreign study conducted by Chuang, Oliva, and Liao (2006) focused on the role of information technology in supply chain and inventory management within modern business organizations. The study aimed to determine how computerized systems and automation technologies improve inventory coordination, operational efficiency, communication, and organizational productivity. The researchers explained that many organizations continue to encounter inventory-related problems due to manual inventory monitoring procedures and inefficient communication systems. These operational issues often result in delayed inventory updates, inaccurate stock information, poor coordination between departments, and reduced customer satisfaction.

The study emphasized that inventory management is an important component of organizational operations because it directly affects productivity, profitability, and customer service. According to the researchers, businesses that fail to maintain accurate and updated inventory information often experience operational disruptions such as stock shortages, delayed transactions, overstocking, and inefficient inventory planning. Traditional inventory systems rely heavily on paperwork and manual data entry, which consume time and increase the possibility of human error.

To address these operational challenges, the researchers explored how information technology and computerized inventory systems improve business operations and supply chain coordination. The study explained that digital inventory systems centralize inventory information within computerized databases, allowing businesses to access and update inventory records more efficiently. Through automation, organizations are able to monitor inventory movement in real time and improve communication between operational departments.

The findings of the study revealed that businesses using computerized inventory systems experienced significant improvements in inventory visibility and operational coordination. Inventory information could be updated immediately whenever products were sold, transferred, or restocked, allowing businesses to maintain accurate inventory records. The researchers explained that real-time inventory monitoring reduced delays and minimized inventory discrepancies caused by manual recording procedures.

Furthermore, the study found that information technology improved communication and collaboration between departments involved in inventory operations. Purchasing personnel, warehouse staff, sales employees, and management teams were able to access the same inventory information simultaneously, improving organizational coordination and reducing delays in operational processes. The researchers also noted that computerized systems improved workflow management because inventory-related information could be processed and retrieved more efficiently.

Another important finding of the study was the positive effect of automation on decision-making and organizational planning. Managers were able to generate inventory reports more accurately and analyze inventory data in real time. This allowed organizations to identify inventory trends, monitor product movement, and improve purchasing decisions. The study emphasized that accurate inventory information supports better inventory planning and reduces operational losses associated with poor inventory control.

The study also highlighted that computerized inventory systems improved customer service and organizational productivity. Businesses using automated systems were able to provide more accurate information regarding product availability and process customer transactions more efficiently. Customers benefited from faster services and improved product accessibility, while employees experienced reduced workload because repetitive manual inventory tasks were minimized through automation.

Despite the positive findings, the study identified several challenges encountered during the implementation of computerized inventory systems. Organizations required financial investment for technology infrastructure, software systems, and technical support. Employees also needed training programs to understand how to use computerized systems effectively. Some workers initially experienced difficulty adapting to automated technologies because they were more familiar with traditional inventory management methods. However, continuous training and system usage improved employee adaptation and organizational performance over time.

The researchers concluded that information technology and computerized inventory systems significantly improve inventory management, operational coordination, workflow efficiency, and organizational productivity. The study emphasized that automation technologies help businesses reduce operational delays, improve inventory visibility, and strengthen decision-making processes. Although implementation may involve technical and financial challenges, the long-term advantages include improved operational efficiency, better customer service, and enhanced inventory management.

This foreign study is highly relevant to the present study because it demonstrates how computerized inventory systems and real-time monitoring technologies improve inventory management and organizational efficiency. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the use of digital inventory systems, centralized databases, and automated inventory updates to reduce manual errors and improve business operations.

2.1.11 Information systems in supply chain and management

A foreign study conducted by Fernie and Sparks (2018) examined retail logistics and inventory management practices in modern retail establishments. The study aimed to determine how automated inventory technologies improve inventory control, operational efficiency, product availability, and customer satisfaction within retail organizations. The researchers explained that retail businesses handle large volumes of products and transactions daily, making inventory management a complex operational task. Businesses relying on manual inventory methods often encounter inventory inaccuracies, stock shortages, delayed product monitoring, and inefficient transaction processing.

The study emphasized that inventory management is essential in maintaining smooth retail operations because inventory directly affects sales performance and customer service. According to the researchers, poor inventory management may result in lost sales opportunities, dissatisfied customers, excessive storage costs, and operational inefficiencies. Traditional inventory systems require employees to manually update stock records and monitor inventory levels, which increases the possibility of human error and delays in inventory processing.

The researchers explored how automated inventory systems improve retail operations by simplifying inventory-related processes and providing real-time inventory information. Through computerized inventory monitoring and barcode-integrated technologies, businesses were able to maintain accurate stock records and update inventory information automatically whenever transactions occurred.

The findings of the study revealed that retail establishments using automated inventory systems experienced significant improvements in inventory accuracy and operational efficiency. Inventory information became more reliable because computerized systems reduced errors associated with manual inventory recording. Businesses were also able to monitor product movement more efficiently and identify low-stock items quickly, reducing the occurrence of product shortages and overstocking.

Furthermore, the study found that automation improved transaction processing and customer service within retail establishments. Barcode scanning technologies simplified checkout procedures and reduced delays during customer transactions. Employees were able to process transactions more efficiently because product information was retrieved automatically from the inventory database. Customers benefited from shorter waiting times and improved service quality.

Another important finding of the study was the improvement of inventory planning and organizational decision-making. Managers were able to access inventory reports more efficiently and analyze product demand using computerized systems. This helped businesses improve purchasing decisions, maintain balanced stock levels, and reduce unnecessary inventory costs. The researchers emphasized that accurate inventory information contributes to better organizational planning and operational productivity.

The study also highlighted that automation improved workflow coordination and warehouse organization. Employees experienced reduced workload because repetitive inventory tasks were simplified through digital systems. Warehouse personnel were also able to locate and organize products more efficiently using barcode technologies and computerized inventory databases.

Despite the positive findings, the study identified several challenges associated with implementing automated inventory systems. Retail businesses required investment in technology infrastructure, barcode equipment, and software systems. Employees also needed technical training to understand how to use the systems properly. Some workers initially experienced difficulty adapting to computerized technologies due to limited technical knowledge and resistance to operational changes. However, the researchers explained that continuous training and organizational support improved employee adaptation over time.

The study concluded that automated inventory systems significantly improve inventory control, operational efficiency, transaction processing, and customer satisfaction within retail establishments. The researchers emphasized that automation technologies help businesses improve workflow management, reduce operational delays, and maintain accurate inventory records. Although implementation may involve technical and financial challenges, the long-term benefits include improved productivity, better inventory management, and enhanced customer service.

This foreign study is highly relevant to the present study because it demonstrates how automated inventory systems and barcode technologies improve inventory monitoring and retail operations. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the integration of real-time inventory updates, barcode scanning systems, and computerized inventory monitoring to improve operational efficiency and customer service.

2.1.12 Inventory record inaccuracy: An empirical analysis

A foreign study conducted by Waters (2019) examined inventory control systems and their role in improving operational efficiency and inventory management within retail and business organizations. The study aimed to determine how computerized inventory systems contribute to better stock monitoring, inventory planning, and organizational productivity. The researcher explained that inventory management is one of the most important functions within organizations because it directly affects sales performance, customer service, and operational stability. However, many businesses continue to encounter problems related to manual inventory monitoring, including inaccurate stock records, inventory shortages, delayed transactions, and poor inventory coordination.

The study emphasized that traditional inventory management systems are often inefficient because they rely heavily on manual recording, paperwork, and physical inventory checking. These manual procedures consume a significant amount of time and increase the possibility of human error. According to the researcher, businesses handling large volumes of products may experience greater difficulties in maintaining accurate inventory information due to delayed updates and inefficient monitoring systems. As a result, organizations may suffer from inventory discrepancies, overstocking, product expiration, and operational losses.

To address these problems, the study explored the use of computerized inventory control systems in improving inventory-related operations. The researcher explained that automated inventory systems allow organizations to monitor stock levels digitally and update inventory records automatically whenever products are sold, transferred, or restocked. Through centralized databases and computerized monitoring, businesses are able to access accurate inventory information in real time and improve workflow efficiency.

The findings of the study revealed that organizations using computerized inventory systems experienced significant improvements in inventory accuracy and operational productivity. Real-time inventory monitoring reduced inventory discrepancies because inventory records were updated immediately after transactions occurred. Businesses were able to identify low-stock items quickly, improve stock replenishment planning, and reduce losses caused by poor inventory control. The researcher also noted that automation minimized repetitive manual tasks and simplified inventory monitoring procedures.

Furthermore, the study found that computerized inventory systems improved organizational coordination and workflow management. Employees were able to process inventory-related tasks more efficiently because information could be retrieved quickly through digital systems. Managers also benefited from computerized reporting because inventory reports could be generated more accurately and analyzed more efficiently. This improved organizational planning and supported better decision-making regarding inventory purchases and stock management.

Another important finding of the study was the positive effect of automation on customer service. Businesses using automated inventory systems were more capable of maintaining product availability and responding to customer demands efficiently. Customers experienced improved service because products were easier to locate and transactions could be processed faster. The researcher emphasized that customer satisfaction improves when organizations maintain accurate inventory information and reduce delays in operations.

The study also highlighted several technologies commonly integrated into computerized inventory systems, including barcode scanning technologies and digital inventory databases. Barcode systems improved transaction speed and reduced errors associated with manual product encoding. Automated inventory systems also improved inventory visibility by allowing businesses to monitor stock movement continuously and accurately.

Despite the positive findings, the study identified challenges encountered during system implementation. Organizations needed financial resources to acquire computer equipment, software systems, and technical infrastructure. Employees also required technical training to operate computerized inventory systems properly. Some employees initially experienced difficulty adapting to new technologies because they were more familiar with manual inventory methods. However, the researcher explained that continuous training and organizational support improved employee adaptation and system acceptance over time.

The study concluded that computerized inventory control systems significantly improve inventory monitoring, workflow management, organizational productivity, and customer service. The researcher emphasized that although implementing automated inventory systems may involve financial and technical challenges, the long-term benefits include improved inventory accuracy, reduced operational delays, better inventory planning, and enhanced business performance.

This foreign study is highly relevant to the present study because it demonstrates how computerized inventory systems improve inventory accuracy, operational efficiency, and customer service within business organizations. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the integration of real-time inventory monitoring, barcode scanning, and computerized inventory control to reduce manual errors and improve inventory-related operations.

2.1.13 Barcode technology and its impact on inventory management systems

A foreign study conducted by Heizer, Render, and Munson (2017) examined the role of inventory management systems in operations management and organizational efficiency. The study aimed to determine how automation technologies improve inventory monitoring, workflow coordination, operational productivity, and customer service within organizations. The researchers explained that inventory management is essential in maintaining efficient business operations because inventory directly affects production, sales, and customer satisfaction. However, businesses relying on manual inventory systems often encounter operational problems such as inaccurate inventory records, delayed transactions, poor inventory coordination, and increased operational costs.

The study emphasized that inventory systems are important because they help organizations maintain adequate stock levels and avoid inventory-related disruptions. According to the researchers, manual inventory methods require employees to perform repetitive tasks such as handwritten inventory recording, manual stock counting, and physical monitoring of products. These procedures consume significant time and increase the possibility of human error, especially in businesses handling large volumes of inventory and transactions daily.

The researchers explored how automated inventory systems improve operational efficiency through computerized inventory monitoring and digital information management. Automated systems allow organizations to update inventory records automatically whenever products are sold, transferred, or restocked. Through centralized inventory databases, employees and managers are able to access accurate inventory information in real time, improving organizational coordination and workflow management.

The findings of the study revealed that organizations using automated inventory systems experienced improved inventory accuracy and operational productivity. Businesses were able to reduce inventory discrepancies caused by manual recording errors and improve the monitoring of stock movement. Real-time inventory updates also enabled organizations to identify low-stock items quickly and improve inventory replenishment planning. As a result, operational delays were reduced and inventory-related processes became more efficient.

Furthermore, the study found that barcode scanning technologies significantly improved transaction speed and inventory management efficiency. Employees using barcode systems were able to process transactions faster because product information could be retrieved automatically from the computerized database. This reduced delays associated with manual encoding and improved customer service within retail and operational environments.

Another important finding of the study was the improvement of organizational planning and decision-making. Managers were able to generate inventory reports more efficiently and analyze inventory movement using computerized systems. This improved purchasing decisions, inventory forecasting, and stock management processes. The researchers explained that accurate inventory information contributes to better operational planning and reduces unnecessary inventory costs.

The study also highlighted that automation improved employee productivity and workflow organization. Because repetitive inventory tasks were minimized, employees were able to focus on more important operational activities. Businesses using computerized inventory systems experienced smoother workflow coordination and improved communication between departments involved in inventory operations.

Despite the positive outcomes, the study identified several challenges related to implementing automated inventory systems. Organizations needed investment in hardware, software, and technical infrastructure necessary for computerized inventory management. Employees also required training programs to understand how to use automated technologies effectively. Some workers initially experienced difficulty adapting to computerized systems because they were accustomed to traditional inventory methods. However, continuous training and operational support improved employee adaptation and system efficiency over time.

The researchers concluded that automated inventory systems significantly improve inventory control, workflow coordination, operational productivity, and customer satisfaction. The study emphasized that although implementing computerized inventory technologies may involve technical and financial challenges, the long-term benefits include improved inventory accuracy, faster transaction processing, reduced operational delays, and enhanced organizational performance.

This foreign study is highly relevant to the present study because it demonstrates how automated inventory systems and barcode technologies improve inventory management and organizational efficiency. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the integration of computerized inventory monitoring, barcode scanning, and real-time inventory updates to improve business operations and customer service.

2.1.14 Supply chain management and information systems integration

A foreign study conducted by Jacobs and Chase (2018) explored operations and supply chain management technologies used in retail and manufacturing organizations. The study aimed to determine how computerized inventory systems improve operational efficiency, inventory visibility, workflow coordination, and organizational productivity.
The researchers explained that inventory management is a critical operational function because inventory directly affects customer satisfaction, sales performance, and business continuity. Businesses relying on manual inventory methods often experience inventory discrepancies, operational delays, and poor inventory coordination.

The study emphasized that traditional inventory systems are inefficient because they require manual recording, repetitive paperwork, and delayed inventory processing.

These manual procedures increase the likelihood of human error and consume significant operational time. According to the researchers, organizations handling large volumes of inventory may struggle to maintain accurate inventory information due to outdated monitoring systems and inefficient operational procedures.

The researchers explored how automation technologies improve inventory operations through computerized monitoring and digital information systems. Automated inventory systems allow organizations to track product movement in real time and update inventory records automatically whenever inventory-related transactions occur. Through centralized databases, employees and management personnel are able to access inventory information more efficiently.

The findings of the study revealed that businesses using computerized inventory systems experienced significant improvements in inventory visibility and operational coordination. Real-time inventory monitoring reduced inventory discrepancies and improved stock management processes. Businesses were able to identify inventory shortages more efficiently and improve stock replenishment planning.

Furthermore, the study found that automation improved workflow efficiency and reduced delays associated with manual inventory monitoring. Employees were able to process transactions and inventory-related tasks more efficiently because information could be accessed quickly through computerized systems. Managers also benefited from digital inventory reporting because inventory data could be analyzed more accurately and efficiently.

Another important finding of the study was the improvement of organizational decision-making and productivity. Businesses using automated systems experienced better operational planning and improved communication between departments involved in inventory operations. The researchers emphasized that accurate inventory information supports better purchasing decisions and improves inventory forecasting.

The study also highlighted that barcode scanning technologies improved transaction speed and reduced errors associated with manual product encoding. Customers benefited from improved service because businesses were able to process transactions faster and maintain accurate product availability information.

Despite the positive findings, the study identified several challenges associated with automation implementation. Organizations required investment in computerized infrastructure and employee training programs. Some workers initially experienced difficulty adapting to automated technologies due to limited technical knowledge and resistance to organizational changes. However, continuous training and operational support improved employee adaptation and system efficiency over time.

The researchers concluded that automated inventory systems significantly improve inventory monitoring, operational efficiency, workflow coordination, and organizational productivity. The study emphasized that automation technologies help organizations reduce operational delays, improve customer service, and strengthen inventory management processes.

This foreign study is highly relevant to the present study because it demonstrates how computerized inventory systems improve inventory visibility and operational efficiency within organizations. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the integration of real-time inventory monitoring, barcode scanning technologies, and computerized inventory databases to improve business operations and inventory management.

2.1.15 Logistics and retail management

A foreign study conducted by Bowersox, Closs, and Cooper (2013) investigated logistics and inventory management technologies used in business organizations and retail establishments. The study aimed to determine how computerized inventory systems and automation technologies improve inventory coordination, operational efficiency, product monitoring, and organizational productivity. The researchers explained that inventory management is an essential operational function because inventory directly affects customer service, workflow efficiency, profitability, and organizational performance. However, businesses that continue to rely on manual inventory systems often encounter inventory discrepancies, delayed product monitoring, operational inefficiencies, and poor coordination between departments.

The study emphasized that traditional inventory management methods are no longer sufficient for modern business environments because organizations handle large volumes of products and transactions daily. According to the researchers, manual inventory systems require employees to perform repetitive inventory recording, physical stock counting, and handwritten reporting processes. These manual procedures consume significant time and increase the likelihood of human error. Inventory inaccuracies may result in stock shortages, delayed transactions, overstocking, product expiration, and customer dissatisfaction.

To improve inventory-related operations, the researchers explored the role of computerized inventory systems and logistics technologies in modern business organizations. The study explained that automated inventory systems centralize inventory information within digital databases, allowing businesses to monitor inventory levels and product movement more accurately. Through real-time inventory updates, organizations are able to maintain accurate inventory records and reduce delays associated with manual inventory monitoring.

The findings of the study revealed that businesses using computerized inventory systems experienced significant improvements in inventory visibility and operational coordination. Automated systems reduced inventory discrepancies because inventory records were updated immediately whenever products were sold, transferred, or restocked. Businesses were also able to identify low-stock products more efficiently and improve inventory replenishment planning.

Furthermore, the study found that automation improved workflow coordination and communication between departments involved in inventory operations. Inventory information could be accessed quickly and simultaneously by warehouse personnel, sales staff, purchasing departments, and managers. This improved operational coordination and reduced delays caused by manual communication and paperwork.

Another important finding of the study was the improvement of customer service and organizational productivity. Businesses using automated inventory systems were able to maintain product availability more effectively and process transactions faster. Customers benefited from improved service because employees could provide accurate product information and complete transactions more efficiently. The researchers emphasized that automation also reduced employee workload by simplifying repetitive inventory tasks and improving workflow organization.

The study further highlighted the importance of barcode scanning technologies and digital inventory databases in improving inventory management efficiency. Barcode systems improved transaction speed and reduced errors associated with manual product encoding. Managers were also able to generate inventory reports more accurately and analyze inventory movement efficiently using computerized systems.
This improved organizational planning and supported better inventory-related decision-making.

Despite the positive findings, the study identified several challenges encountered during automation implementation. Organizations needed financial investment to acquire computer equipment, software systems, and technical infrastructure necessary for automation. Employees also required technical training to understand how to use computerized inventory systems effectively. Some workers initially experienced difficulty adapting to automated technologies because they were more familiar with traditional inventory procedures. However, the researchers noted that continuous training and operational support improved employee adaptation and system efficiency over time.

The researchers concluded that computerized inventory systems and logistics technologies significantly improve inventory control, workflow coordination, operational productivity, and customer satisfaction. The study emphasized that although implementing automation technologies may involve technical and financial challenges, the long-term benefits include improved inventory accuracy, reduced operational delays, better workflow management, and enhanced organizational performance.

This foreign study is highly relevant to the present study because it demonstrates how computerized inventory systems and automation technologies improve inventory monitoring and organizational efficiency within retail businesses. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the use of barcode scanning technologies, real-time inventory updates, and computerized inventory databases to improve inventory management and business operations.

2.1.16 Essentials of inventory management

A foreign study conducted by Muller (2011) examined the importance of computerized inventory management systems in retail and warehouse operations. The study aimed to determine how automated inventory systems improve inventory accuracy, stock monitoring, organizational productivity, and operational efficiency within businesses. The researcher explained that inventory management plays an important role in maintaining effective business operations because inventory directly affects sales performance, workflow efficiency, and customer satisfaction. However, organizations relying on manual inventory systems often experience operational problems such as inaccurate inventory records, delayed product monitoring, inventory shortages, and poor inventory coordination.

The study emphasized that manual inventory management methods are inefficient because they require repetitive paperwork, handwritten recording, and manual stock checking procedures. According to the researcher, businesses handling large quantities of inventory often struggle to maintain accurate inventory records due to delayed updates and operational inefficiencies. These inventory problems may result in product shortages, overstocking, unnecessary storage expenses, and customer dissatisfaction.

To address these operational issues, the study explored how computerized inventory systems improve inventory-related operations within organizations. Automated inventory systems allow businesses to monitor inventory movement digitally and update inventory records automatically whenever products are sold, transferred, or restocked. Through centralized digital databases, organizations are able to maintain updated inventory information and improve inventory coordination across departments.

The findings of the study revealed that businesses using computerized inventory systems experienced significant improvements in inventory accuracy and operational efficiency. Inventory discrepancies caused by manual recording errors were minimized because inventory records were updated automatically in real time. Businesses were able to identify low-stock products more efficiently and improve stock replenishment planning.

Furthermore, the study found that automation improved workflow coordination and employee productivity within organizations. Employees were able to process inventory-related tasks more efficiently because information could be accessed quickly through computerized systems. Managers also benefited from digital inventory reports because inventory data could be analyzed more accurately and efficiently. This improved organizational planning and supported better inventory-related decision-making.

Another important finding of the study was the improvement of customer service and transaction efficiency. Businesses using automated inventory systems were able to process transactions faster and maintain accurate product availability information. Customers benefited from improved service because employees could provide accurate information regarding inventory levels and product availability.

The study also highlighted the role of barcode scanning technologies in improving inventory management efficiency. Barcode systems reduced errors associated with manual product encoding and improved transaction speed. The researcher explained that barcode-integrated inventory systems simplified inventory monitoring and improved workflow organization within retail and warehouse environments.

Despite the positive outcomes, the study identified several challenges encountered during automation implementation. Organizations needed financial resources to acquire computer equipment, barcode scanners, and software systems. Employees also required technical training to understand how to use computerized inventory systems effectively. Some workers initially experienced difficulty adapting to automated technologies because they were more familiar with manual inventory methods. However, continuous training and organizational support improved employee adaptation and operational efficiency over time.

The study concluded that computerized inventory systems significantly improve inventory control, workflow management, operational productivity, and customer satisfaction within organizations. The researcher emphasized that although implementing automated inventory systems may involve technical and financial challenges, the long-term benefits include improved inventory accuracy, reduced operational delays, enhanced workflow coordination, and better business performance.

This foreign study is highly relevant to the present study because it demonstrates how computerized inventory systems improve inventory monitoring and organizational efficiency within retail establishments. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the integration of barcode scanning technologies, real-time inventory updates, and computerized inventory databases to improve inventory-related operations and customer service.

2.2 Local Studies

2.2.1 Point of Sale and Inventory Management System for Small Retail Businesses in the Philippines

According to J. Dela Cruz and P. Ramos (2018), small retail businesses in the Philippines face significant operational challenges that affect their competitiveness and sustainability in increasingly demanding marketplace environments. The authors explained that small retail businesses including neighborhood sari-sari stores, independent pharmacies, and convenience stores represent important segments of the Philippine retail sector that contribute significantly to employment, local economic development, and community accessibility. These businesses serve essential functions in Philippine communities, particularly in areas where large supermarkets and chain stores are not readily accessible, providing convenient access to daily necessities and essential household items. Despite their importance, small retail businesses in the Philippines often struggle with operational inefficiencies, limited technology adoption, and competitive pressures from larger retail establishments that threaten their long-term viability. The authors highlighted how technology adoption through Point of Sale and inventory management systems can significantly improve small retail business operations while helping these businesses remain competitive in Philippine marketplace environments.

The literature emphasized that Philippine small retail businesses traditionally rely on manual inventory tracking methods including handwritten records, physical inventory counts, and mental tracking of product quantities and movements that limit operational effectiveness and growth potential. Dela Cruz and Ramos discussed how manual inventory systems require significant time investments for record keeping, inventory counting, and report preparation that reduce time available for customer service and business development activities. The authors explained how manual processes increase the likelihood of human errors including incorrect pricing, inaccurate inventory records, and calculation mistakes that result in financial losses and customer dissatisfaction. In the Philippine context, where many small retail businesses operate with family members and limited employees, the time invested in manual processes significantly impacts overall operational capacity and limits business growth potential. The literature highlighted how competitive pressures from chain stores, supermarkets, and online retailers increasingly challenge small retail businesses that cannot match the operational efficiency and customer service quality of better-resourced competitors.

Furthermore, the authors discussed how Point of Sale technology adoption in Philippine small retail businesses remains limited due to multiple barriers including cost concerns, technical complexity, and cultural factors that discourage technology investment. Initial costs for POS hardware including computers, barcode scanners, and receipt printers represent significant investments for small businesses operating with limited capital and thin profit margins. The authors explained how ongoing costs including software maintenance, technical support, and system updates add to total cost of ownership that may exceed initial expectations. Technical complexity concerns deter business owners who lack technology backgrounds and worry about system operation and maintenance requirements. Cultural preferences for familiar operating methods create resistance to technology adoption, particularly among older business owners accustomed to traditional manual processes. The literature highlighted how lack of locally-relevant solutions designed for Philippine retail contexts further limits adoption, as imported solutions may not address local requirements including Philippine payment methods, regulatory requirements, and business practices.

The literature also highlighted how small retail businesses in the Philippines benefit significantly from POS and inventory management system implementation through improved operational efficiency, enhanced customer service, and better business decision-making capabilities. Accelerated transaction processing reduces customer wait times while improving accuracy that enhances customer satisfaction and encourages repeat patronage. Automated inventory tracking maintains accurate inventory records that prevent stockouts and overstocking while reducing time investments in manual inventory management. The authors discussed how improved inventory visibility enables business owners to identify fast-moving products, slow-moving inventory, and reorder requirements more effectively than manual tracking methods permit. Business reporting capabilities provide sales, inventory, and financial reports that support better business planning and resource allocation decisions. The literature explained how competitive parity with larger retailers improves customer perceptions while helping small retail businesses maintain market share against better-resourced competitors.

Additionally, the authors explained how Philippine-specific factors influence POS and inventory system requirements that must be addressed in system design and implementation. Multiple payment methods including cash, credit cards, debit cards, GCash, and other electronic payment options require system support for diverse payment processing that Philippine customers expect. Philippine retail practices including credit arrangements (tingi), consignment arrangements, and flexible payment terms require system capabilities that accommodate these local business practices. The literature discussed how local language support in Filipino and regional languages improves accessibility for users and customers who may not be comfortable with English-only interfaces. Integration with local suppliers and distributors supports local supply chain relationships that small retail businesses rely on for inventory replenishment. The authors explained how compatibility with Philippine regulatory requirements ensures compliance with Bureau of Internal Revenue (BIR) regulations and other applicable requirements.

The literature further discussed how successful POS system implementation in Philippine small retail businesses requires appropriate training, ongoing support, and system maintenance that address local operational contexts and capabilities. Training programs must accommodate users with limited technology backgrounds who may be unfamiliar with computer operations and software applications. The authors explained how user-friendly interfaces reduce training requirements while improving user adoption and system effectiveness. Ongoing technical support addresses operational questions and problems that arise during daily system operation, maintaining system effectiveness and user confidence. The literature discussed how local technical support providers offer advantages over distant support centers by understanding local business contexts and providing accessible assistance when needed. System maintenance including updates, backups, and repairs ensures continuous system operation and data protection throughout system lifecycle.

The authors also emphasized the importance of cost-benefit analysis in technology investment decisions for small retail businesses with limited resources. Simple payback analysis calculates time required for cost recovery through operational savings and efficiency improvements, enabling informed investment decisions. The literature discussed how non-financial benefits including improved customer service, competitive parity, and business management improvements add value beyond direct financial savings. The authors explained how phased implementation approaches reduce initial investments while enabling gradual capability building that accommodates resource limitations common among small retail businesses. Pilot implementations in single store locations demonstrate system viability before broader deployment that reduces implementation risks and allows refinement based on operational experience.

Furthermore, the authors discussed how technology adoption in Philippine small retail businesses contributes to broader national development objectives through improved business productivity, employment creation, and economic competitiveness. Modernized small retail businesses contribute to national productivity improvements that enhance Philippine economic competitiveness in regional and global markets. The literature explained how technology adoption creates demand for technical support, maintenance, and related services that generate employment opportunities in local communities.
Enhanced small retail business viability preserves employment and maintains community economic functions that contribute to local development. The authors discussed how increased technology adoption supports government initiatives to modernize Philippine commerce and improve business competitiveness across all sectors.

The literature additionally highlighted how inventory management capabilities specifically benefit Philippine small retail businesses through improved stock management, reduced losses, and better customer service. Accurate inventory tracking prevents stockouts that result in lost sales and customer dissatisfaction while reducing overstocking that ties up capital in slow-moving inventory. The authors explained how inventory turnover analysis identifies products requiring promotional efforts or discontinuation, focusing business attention on profitable items. The literature discussed how reorder alerts notify business owners when inventory falls below thresholds that trigger reordering, preventing stockouts and maintaining product availability. Reduced inventory shrinkage from improved tracking and accountability decreases losses from theft, damage, and expired products that significantly impact small retail business profitability.

Moreover, the authors explained how case study evidence from Philippine small retail businesses demonstrates tangible benefits from POS and inventory system implementation. Business owners reported transaction processing time reductions of thirty to fifty percent compared to manual processes that improved customer throughput and reduced wait times. Inventory record accuracy improved from approximately seventy percent to above ninety-five percent in businesses implementing automated inventory tracking. The literature discussed how inventory-related losses decreased significantly through improved accountability and real-time inventory visibility. Business owners reported improved decision-making through access to sales and inventory reports that previously required significant time and effort to compile manually. The authors explained how customer satisfaction improved through faster service, accurate transactions, and better product availability that encouraged repeat patronage.

This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system addresses similar operational requirements and challenges faced by small retail businesses in the Philippine context. Similar to the concepts discussed by Dela Cruz and Ramos, the proposed AIMS aims to improve operational efficiency, inventory management, and customer service through appropriate technology implementation.

The proposed system aligns with the authors' discussion regarding Philippine-specific requirements including local payment methods, Philippine retail practices, and local language support. These context-specific capabilities ensure that the proposed system addresses actual requirements in Philippine retail environments.

Furthermore, the literature supports the implementation of the proposed system because it demonstrates how POS and inventory management systems benefit small retail businesses in the Philippine context with documented improvements in operational efficiency and business management. The concepts discussed in the study provide strong evidence supporting the development and implementation of the proposed Automated Inventory Management System within Philippine pharmacies and convenience stores.

2.2.2 Web-Based Inventory Monitoring System for Retail Stores in the Philippines

According to M. Santos, R. Garcia, and T. Lopez (2020), web-based inventory monitoring systems represent significant technological advancements that enable Philippine retail businesses to track inventory, monitor operations, and manage business activities from centralized platforms accessible through internet connections. The authors explained that web-based systems address traditional limitations of standalone inventory systems that require physical presence at store locations for system operation and data access. The emergence of web-based solutions reflects broader technology trends toward cloud computing, mobile accessibility, and centralized data management that transform how Philippine businesses operate and compete. The authors highlighted how web-based inventory systems offer particular advantages for Philippine retail businesses operating across multiple locations, with owners and managers who travel between stores or work from central offices requiring real-time visibility into store operations regardless of physical location.

The literature emphasized that Philippine retail businesses face unique operational challenges related to geographic distribution, transportation constraints, and diverse store environments that web-based systems can address effectively. Many retail businesses operate across multiple store locations scattered throughout Metro Manila, provincial cities, and regional areas where physical travel consumes significant time and resources. Santos, Garcia, and Lopez discussed how web-based systems enable centralized monitoring of inventory levels, sales activities, and operational performance across all store locations from single access points that reduce travel requirements and improve management efficiency. The authors explained how centralized data stores in web-based systems aggregate information from multiple sources that enable comprehensive analysis, reporting, and decision-making unavailable from isolated store systems. Philippine traffic congestion and geographic distances make real-time multi- location access particularly valuable, enabling management intervention when problems arise without waiting for physical travel to affected locations.

Furthermore, the authors discussed how web-based inventory systems leverage Philippine internet connectivity improvements that enable reliable system operation despite historical connectivity concerns. Increased internet penetration throughout the Philippines provides broader accessibility for web-based system access, particularly in urban areas where most retail businesses operate. The literature explained how mobile internet through smartphone devices enables warehouse managers, delivery personnel, and field representatives to access inventory information while away from desktop computers. Improved broadband speeds in business districts support real-time system operation that previously required dedicated infrastructure investments. Santos, Garcia, and Lopez discussed how cloud hosting options reduce infrastructure requirements for small businesses, eliminating needs for local server hardware and technical administration.

The literature also highlighted how web-based systems address data management challenges that affect retail businesses with multiple stores generating large volumes of operational data. Centralized data storage ensures consistent data definitions, reporting formats, and analysis capabilities across all store locations that would be difficult to maintain with distributed systems. The authors explained how web-based databases enable real-time inventory visibility that prevents the inventory discrepancies common when stores maintain separate inventory systems without coordination. The literature discussed how centralized data backup and recovery protect against data loss that could devastate businesses lacking resources for comprehensive backup systems. Data analytics capabilities leverage aggregated information across locations to identify patterns, trends, and opportunities invisible in individual store data.

Additionally, the authors explained how web-based inventory systems improve supplier coordination by enabling electronic communication of purchase orders, delivery schedules, and inventory replenishment requirements. Electronic purchase ordering eliminates manual order preparation, reduces ordering errors, and accelerates order processing compared to telephone or fax-based ordering methods. The literature discussed how automated inventory alerts notify suppliers when stock levels require replenishment, enabling proactive reordering that prevents stockouts and maintains product availability. Supplier portals enable vendors to access inventory data, confirm orders, and coordinate delivery schedules without requiring telephone calls or email exchanges that consume management time. Santos, Garcia, and Lopez explained how improved supplier coordination improves inventory freshness, reduces stockouts, and strengthens supplier relationships that benefit both parties.

The literature further discussed how web-based systems improve regulatory compliance by maintaining comprehensive electronic records that support Bureau of Internal Revenue (BIR) requirements and other regulatory obligations. Electronic record keeping eliminates paper-based documentation challenges including storage requirements, retrieval difficulties, and legibility issues that complicate compliance verification. The authors explained how automated report generation produces required documentation without manual compilation that consumes significant time and may introduce errors. Audit trail capabilities maintain comprehensive records of inventory transactions, price changes, and adjustments that demonstrate compliance with regulatory requirements. The literature discussed how electronic submission capabilities streamline reporting to regulatory authorities, reducing administrative burden while ensuring timely compliance.

The authors also emphasized the importance of user interface design in web-based system adoption by Philippine retail businesses with users having diverse technology backgrounds and varying English language proficiency. Filipino language support accommodates users more comfortable with Filipino than English, improving accessibility for users with limited English proficiency. Santos, Garcia, and Lopez discussed how simplified navigation reduces complexity for users unfamiliar with web- based systems, enabling quicker adoption and more effective system utilization. Mobile- responsive interfaces accommodate smartphone and tablet access that many Philippine users prefer over desktop computer operation. The literature explained how offline capability ensures continued operation during internet disruptions that occasionally affect Philippine connectivity.

Furthermore, the literature discussed how case study evidence from Philippine retail businesses demonstrates tangible operational improvements from web-based inventory system implementation. Multi-location businesses reported management efficiency improvements of forty to sixty percent through centralized monitoring and reduced travel requirements. Inventory accuracy improved significantly, with discrepancies decreasing from typical levels of eight to twelve percent to below three percent following implementation. The authors explained how stockout rates decreased substantially through automated reorder alerts and improved inventory visibility that enable proactive replenishment. The literature discussed how business owners reported improved financial management through real-time access to sales, inventory, and profit reports that previously required manual compilation at month-end.

The literature additionally highlighted how Philippine retail e-commerce growth creates integration requirements that web-based inventory systems address effectively. Online sales channels require inventory synchronization that prevents overselling of products allocated to physical stores and online channels. The authors explained how web-based inventory systems provide single inventory pools that allocate product availability across channels based on business rules and priorities. Integration with delivery services and logistics providers enables coordinated fulfillment that improves customer delivery expectations. The literature discussed how marketplace integration through application programming interfaces (APIs) connects inventory systems with major Philippine e-commerce platforms that reach customers beyond physical store footprints.

Moreover, the authors explained how web-based inventory systems improve customer service through better product availability, faster checkout processes, and accurate information provision. Real-time inventory visibility enables accurate customer inquiries regarding product availability without depending on potentially inaccurate physical counts. The literature discussed how integrated POS and inventory systems accelerate checkout while automatically updating inventory records that maintain accuracy. Online inventory visibility enables customer pre-orders and reservations that extend store accessibility beyond physical operating hours. Santos, Garcia, and Lopez explained how improved customer service builds loyalty that sustains competitive positioning against larger retailers with more resources.

The authors also discussed how web-based inventory systems support business growth through scalability that accommodates expanding store networks without infrastructure constraints. Cloud-based deployment scales automatically as business needs increase, eliminating hardware upgrade requirements that might constrain growth. The literature explained how new store additions connect immediately to existing systems without additional infrastructure investments or technical complexity. Geographic expansion to new areas becomes more practical when web-based systems reduce technical requirements for new store openings. The authors discussed how multi-currency and multi-language capabilities support potential expansion to international markets or foreign customer bases.

This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system utilizes web-based architecture that enables centralized inventory management across multiple Philippine retail locations. Similar to the concepts discussed by Santos, Garcia, and Lopez, the proposed system aims to provide real-time inventory visibility, centralized reporting, and operational management capabilities for pharmacies and convenience stores throughout the Philippines.

The proposed system aligns with the authors' discussion regarding web-based system benefits for Philippine multi-location retail businesses requiring centralized management and real-time operational visibility. These capabilities address operational challenges specific to Philippine retail environments where geographic distribution, traffic congestion, and management travel time significantly impact operational effectiveness.

Furthermore, the literature supports the implementation of the proposed web-based inventory system because it demonstrates how web-based technologies benefit Philippine retail businesses through documented improvements in operational efficiency, inventory accuracy, and management effectiveness. The concepts discussed in the study provide strong evidence supporting the development and implementation of the proposed Automated Inventory Management System within Philippine retail environments.

2.2.3 Imbentaryo App: An Intelligent Inventory and Decision Support System

According to J. R. N. de los Santos, J. R. Suyom, M. B. Comora, N. Giovanni, G. Michelle, and R. C. Aruy (2021), intelligent inventory management systems incorporating decision support capabilities represent significant advancements that enable businesses to transition from reactive inventory management to proactive inventory optimization based on predictive analytics and automated decision-making. The authors explained that traditional inventory management systems focus primarily on tracking inventory quantities and recording transactions without providing analytical capabilities that enable strategic inventory optimization. The emergence of intelligent inventory systems reflects broader trends in artificial intelligence and data analytics that transform how businesses approach inventory management by enabling data-driven decisions that improve inventory investment returns while maintaining product availability. The authors highlighted how intelligent inventory systems offer particular advantages for retail businesses facing complex inventory decisions including reorder timing, quantity optimization, and product assortment decisions that significantly impact business profitability and operational efficiency.

The literature emphasized that retail businesses in the Philippines face inventory management challenges that intelligent systems can address through automated analysis and decision support capabilities. De los Santos et al. discussed how small and medium retail businesses often lack analytical capabilities and expertise required for effective inventory optimization, resulting in suboptimal inventory decisions that increase costs or reduce sales. The authors explained how inventory managers must consider multiple factors including historical sales patterns, seasonal variations, supplier lead times, storage constraints, and capital availability that complicate inventory decisions beyond simple reorder point calculations. The literature highlighted how intelligent systems analyze multiple factors simultaneously to generate recommendations that would otherwise require extensive expertise and analysis time unavailable in small retail businesses.

Furthermore, the authors discussed how demand forecasting capabilities distinguish intelligent inventory systems from traditional tracking systems by predicting future inventory requirements based on historical patterns and external factors. Time series analysis of historical sales data identifies trends, seasonal patterns, and cyclical variations that inform demand predictions. The literature explained how machine learning algorithms improve forecast accuracy over time by incorporating additional data and identifying patterns that manual analysis would miss. De los Santos et al. discussed how external factors including promotions, holidays, weather events, and economic conditions improve forecast accuracy by incorporating context beyond historical sales patterns. The literature highlighted how accurate demand forecasting enables optimal inventory levels that minimize both stockouts and overstocking costs.

The literature also highlighted how intelligent inventory systems automate reorder decisions that reduce management attention requirements while improving inventory availability. Automated reorder triggers initiate purchase order generation when inventory falls below calculated thresholds, preventing stockouts that result in lost sales and customer dissatisfaction. The authors explained how economic order quantity calculations determine optimal order sizes that minimize total inventory costs including ordering, holding, and stockout costs. The literature discussed how supplier lead time considerations adjust reorder timing to account for delivery variability that could cause stockouts during supplier delays. De los Santos et al. explained how automated ordering reduces management burden while ensuring consistent reorder execution that human managers might overlook during busy periods.

Additionally, the authors explained how intelligent inventory systems provide decision support capabilities that assist inventory managers in complex decisions beyond routine reordering. What-if analysis enables evaluation of alternative scenarios including different reorder quantities, supplier alternatives, and pricing strategies that inform strategic inventory decisions. The literature discussed how trend analysis identifies products requiring promotional attention or discontinuation, focusing business resources on profitable inventory while reducing losses from slow-moving items. Dashboard visualizations present key inventory metrics in accessible formats that enable quick assessment of inventory health without detailed analysis. The authors explained how alert systems notify managers of exceptional conditions requiring attention including unusual demand spikes, delivery delays, or inventory discrepancies that could impact operations.

The literature further discussed how the Imbentaryo App specifically demonstrated intelligent inventory management capabilities through its mobile-based architecture and decision support features. The authors explained how mobile accessibility enables inventory managers to access system capabilities from any location, particularly valuable in Philippine retail environments where managers frequently travel between store locations. The literature discussed how barcode scanning integration accelerates inventory transactions while maintaining accuracy that eliminates manual data entry errors. De los Santos et al. explained how speech recognition capabilities enable voice- based inventory queries that improve accessibility for users with limited technology backgrounds or those engaged in physical inventory activities. The literature highlighted how local language support accommodates Filipino users more comfortable with local languages than English-only interfaces.

The authors also emphasized the importance of system evaluation in demonstrating intelligent inventory system benefits through quantitative and qualitative assessment. Performance metrics measured inventory accuracy, stockout rates, and inventory turnover improvements following Imbentaryo App implementation. The literature explained how user satisfaction surveys assessed perceived usability, system effectiveness, and willingness to recommend the system to other businesses. De los Santos et al. discussed how comparative analysis with traditional inventory management methods demonstrated measurable improvements in inventory management outcomes.
The literature discussed how implementation experience informed recommendations for future intelligent inventory system development and deployment.

Furthermore, the literature discussed how intelligent inventory systems contribute to sustainability objectives through improved inventory efficiency that reduces waste from expired products and overstocking. Optimized inventory levels reduce product spoilage and expiration that contribute to food waste in perishable inventory categories. The authors explained how improved inventory visibility enables first-in-first-out (FIFO) inventory rotation that reduces waste from aged products. The literature discussed how demand forecasting reduces excessive inventory that leads to clearance discounting and waste when products fail to sell as planned. De los Santos et al. explained how environmental benefits align with broader sustainability objectives increasingly important to consumers and regulators.

The literature additionally highlighted how intelligent inventory systems improve competitive positioning through operational efficiency and customer service capabilities that distinguish businesses from competitors. Faster inventory transactions improve customer checkout experiences while accurate inventory prevents customer disappointment from out-of-stock products. The authors explained how inventory optimization improves profit margins through reduced carrying costs and stockout losses that directly impact business profitability. The literature discussed how decision support capabilities enable strategic inventory decisions that competitors without intelligent systems cannot match. De los Santos et al. explained how competitive advantages from intelligent inventory systems become increasingly significant as larger retailers and e-commerce platforms intensify competitive pressures on traditional retail businesses.

Moreover, the authors explained how future intelligent inventory system development should address emerging technologies including artificial intelligence advancement, Internet of Things integration, and blockchain-based traceability. The literature discussed how advanced artificial intelligence capabilities including natural language processing and computer vision could enable more intuitive inventory interactions through voice commands and visual identification. Internet of Things sensors could automate inventory tracking through automated shelf monitoring and environmental sensing that detects stock levels without manual scanning. The authors explained how blockchain technology could provide supply chain transparency and product authenticity verification increasingly important to consumers concerned about product sourcing. The literature discussed how continued advancement in intelligent inventory systems will further enhance capabilities that improve retail business productivity and competitiveness.

This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system incorporates intelligent capabilities including demand forecasting, automated reordering, and decision support that enable proactive inventory optimization. Similar to the concepts discussed by de los Santos et al., the proposed AIMS aims to transition from reactive inventory tracking to intelligent inventory management through analytical capabilities and automated decision-making.

The proposed system aligns with the authors' discussion regarding intelligent inventory system benefits including automated reordering, demand forecasting, and decision support that improve inventory management outcomes. These intelligent capabilities address operational challenges faced by Philippine retail businesses lacking analytical expertise and management time for comprehensive inventory optimization.

Furthermore, the literature supports the implementation of the proposed intelligent inventory system because it demonstrates how intelligent capabilities improve inventory management through documented accuracy improvements, stockout reduction, and decision support enablement. The concepts discussed in the study provide strong evidence supporting the development and implementation of the proposed Automated Inventory Management System within Philippine retail environments.

2.2.4 Inventory Management Practices and Service Delivery Facilities in Ilocos Norte Philippines

According to E. S. Parilla, J. Evangelista, R. Aurelio, and C. Bullalayao (2022), inventory management practices significantly impact healthcare facility operations, patient safety, and service delivery quality in Philippine healthcare environments. The authors explained that healthcare facilities including hospitals, clinics, and pharmacies maintain extensive inventories of medical supplies, pharmaceutical products, and equipment that must be available when needed to support patient care activities. Effective inventory management in healthcare contexts addresses not only operational efficiency concerns but also patient safety considerations that make inventory accuracy particularly critical. The authors highlighted how inventory management challenges in Philippine healthcare facilities share many characteristics with broader retail inventory management challenges while introducing healthcare-specific requirements including regulatory compliance, expiration tracking, and cold chain management.

The literature emphasized that healthcare facilities in Ilocos Norte face diverse inventory management challenges related to resource constraints, geographic accessibility, and operational complexity that affect inventory management effectiveness. Parilla et al. discussed how limited budgets constrain inventory management technology investments, forcing healthcare facilities to rely on manual inventory tracking methods despite known limitations. The authors explained how geographic distribution of healthcare facilities across urban, semi-urban, and rural areas creates additional coordination challenges that complicate inventory management and distribution logistics. The literature highlighted how seasonal variations in patient populations, disease patterns, and public health initiatives create demand variability that complicates inventory planning and resource allocation. Healthcare facilities in the Ilocos Norte region also face supply chain challenges related to distributor availability, delivery reliability, and procurement processes that impact inventory replenishment.

Furthermore, the authors discussed how inventory management practices in healthcare facilities directly affect service delivery quality and patient outcomes that go beyond typical retail business considerations. Stockouts of essential medical supplies interrupt patient care activities that may have immediate health consequences beyond customer dissatisfaction in retail contexts. The literature explained how pharmaceutical stockouts particularly impact patient treatment continuity when prescriptions cannot be filled, potentially compromising therapeutic effectiveness or requiring emergency alternative arrangements. Parilla et al. discussed how expired products present safety concerns requiring careful inventory management to prevent patient exposure to ineffective or harmful products. The literature highlighted how healthcare inventory management must balance cost efficiency with availability requirements that may prioritize patient safety over inventory cost minimization.

The literature also highlighted how pharmaceutical inventory management specifically requires attention to expiration tracking, storage conditions, and regulatory compliance that distinguish pharmaceutical distribution from general retail inventory. Temperature- sensitive pharmaceutical products require cold chain management that maintains appropriate storage conditions from manufacturer through patient delivery. The authors explained how cold chain monitoring ensures product potency and patient safety that could be compromised by temperature excursions during storage or transport. The literature discussed how expiration tracking prevents dispensing of expired products that could harm patients while incurring regulatory violations. Parilla et al. explained how pharmaceutical regulatory requirements from Philippine Food and Drug Administration (FDA) mandate specific inventory controls, record keeping, and reporting that increase compliance complexity.

Additionally, the authors explained how inventory management technology adoption in Philippine healthcare facilities remains limited due to unique barriers that affect technology implementation decisions. Cost constraints particularly affect resource- limited healthcare facilities operating with constrained budgets allocated primarily to personnel and essential medical supplies. The literature discussed how technology infrastructure limitations including unreliable power supply, internet connectivity, and technical support access complicate technology deployment in some geographic areas. Parilla et al. explained how employee technology backgrounds affect adoption success, with healthcare workers lacking technology training struggling to utilize electronic inventory management systems effectively. The literature discussed how change management challenges in healthcare environments affect technology adoption as staff accustomed to traditional methods resist new system implementations.

The literature further discussed how healthcare facilities benefit from improved inventory management through reduced stockouts, optimized inventory levels, and enhanced service delivery capabilities. Inventory visibility enables proactive reordering that prevents stockouts before they impact patient care activities. The authors explained how optimized inventory levels reduce carrying costs while maintaining availability requirements that ensure product accessibility when needed. The literature discussed how automated inventory tracking reduces labor requirements for inventory management activities that healthcare workers can redirect to patient care activities. Parilla et al. explained how inventory system integration with patient records enables coordinated care where prescribed treatments align with available inventory.

The authors also emphasized the importance of inventory management in healthcare supply chain resilience that ensures continuity during disruptions including natural disasters and public health emergencies. The coronavirus pandemic demonstrated how supply chain disruptions affect healthcare inventory availability, requiring contingency planning and alternative sourcing strategies. The literature discussed how pandemic experiences highlighted the importance of inventory buffers for essential supplies, real- time inventory visibility, and flexible procurement capabilities. Parilla et al. explained how automated inventory systems enable rapid response to changing supply conditions through visibility and coordination capabilities not available with manual systems.

Furthermore, the literature discussed how local context significantly influences healthcare inventory management requirements and implementation approaches. Healthcare facilities in Ilocos Norte operate within regional contexts that influence supply chain relationships, patient demographics, and operational practices. The authors explained how local supplier relationships and distribution networks affect replenishment reliability and lead times. The literature discussed how patient population characteristics including health conditions prevalent in the region influence inventory mix and demand patterns. Cultural factors including health-seeking behaviors and treatment preferences affect medication acceptance and use patterns that inform inventory planning. Parilla et al. explained how climate conditions in the Ilocos Norte region affect storage requirements for temperature-sensitive products during hot seasons.

The literature additionally highlighted how case study evidence from healthcare facilities in Ilocos Norte demonstrates tangible benefits from improved inventory management practices. Facilities implementing improved inventory management reported significant reductions in stockout occurrences that improve patient care continuity. Inventory accuracy improved substantially, reducing both expired product occurrences and inventory discrepancies that complicate financial management. The authors explained how inventory turnover improvements reduce carrying costs while maintaining availability requirements that improve financial performance. The literature discussed how service delivery improved through faster patient care preparation, reduced waiting times, and enhanced care coordination enabled by accurate inventory visibility.

Moreover, the authors explained how pharmaceutical inventory management in pharmacy retail environments shares significant characteristics with healthcare facility inventory management while introducing retail-specific considerations. The literature discussed how pharmacy inventory must balance healthcare availability requirements with retail efficiency objectives that optimize business performance. Parilla et al. explained how prescription medication tracking supports patient care continuity while over-the-counter product inventory follows retail optimization practices. The literature discussed how pharmacy inventory systems must integrate with patient records, insurance billing, and regulatory reporting requirements that add complexity beyond retail inventory tracking. The authors explained how prescription workflow integration improves efficiency while maintaining compliance requirements that distinguish pharmaceutical dispensing from general retail transactions.

The authors also discussed how healthcare inventory management principles apply to broader retail contexts including convenience stores and household product retailers with appropriate adaptations. The literature explained how inventory management fundamentals including demand forecasting, reorder optimization, and inventory tracking remain applicable across retail contexts. Parilla et al. discussed how cold chain management requirements extend beyond pharmaceuticals to include refrigerated and frozen products common in convenience store operations. The literature discussed how expiration tracking benefits any retail category with time-sensitive products including food items, cosmetics, and personal care products. The authors explained how service delivery focus applies to retail contexts where customer service quality affects customer loyalty and competitive positioning.

This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system addresses inventory management challenges applicable to both healthcare and retail contexts including pharmaceutical inventory, convenience store operations, and general retail inventory. Similar to the concepts discussed by Parilla et al., the proposed AIMS aims to improve inventory visibility, prevent stockouts, and optimize inventory levels through automated tracking and decision support capabilities.

The proposed system aligns with the authors' discussion regarding inventory management importance in service delivery contexts where inventory availability directly impacts customer satisfaction and operational effectiveness. These capabilities apply to pharmacy environments where inventory directly affects patient care and convenience store contexts where inventory affects customer service quality.

Furthermore, the literature supports the implementation of the proposed inventory management system because it demonstrates how improved inventory management practices benefit retail and healthcare service delivery through documented improvements in accuracy, availability, and operational efficiency. The concepts discussed in the study provide strong evidence supporting the development and implementation of the proposed Automated Inventory Management System within Philippine retail and pharmacy environments.

2.2.5 Effective Inventory Management System in Efficient Supply and Distribution Management in one of Manufacturer of Foods Seasoning Products in the Philippines

According to J. Bautista Jr. and M. Young (2022), inventory management systems play critical roles in manufacturing supply and distribution operations that determine operational efficiency, customer satisfaction, and competitive positioning in Philippine manufacturing environments. The authors explained that manufacturing operations require coordinated supply chain management that connects raw material procurement, production planning, inventory control, and distribution delivery within integrated operations. Effective inventory management in manufacturing contexts addresses not only finished goods inventory but also raw material, work-in-process, and packaging material inventories with different planning and control requirements. The authors highlighted how seasoning product manufacturing presents unique inventory challenges related to ingredient variety, quality control, and shelflife considerations that complicate inventory management beyond typical manufacturing contexts.

The literature emphasized that Philippine food seasoning manufacturers face diverse inventory management challenges related to ingredient sourcing, production scheduling, and distribution logistics that affect overall supply chain effectiveness. Bautista Jr. and Young discussed how seasoning products require multiple raw materials including salt, sugar, spices, and flavor enhancers that must be available in correct quantities to support production operations. The authors explained how ingredient quality variations affect finished product quality that requires careful supplier management and incoming inspection procedures. The literature highlighted how seasonal demand variations related to holiday seasons, food festival periods, and regional preferences create inventory planning complexity that complicates production scheduling. Philippine manufacturers also face supply chain challenges including supplier reliability issues, transportation constraints, and warehouse capacity limitations that impact inventory management effectiveness.

Furthermore, the authors discussed how effective inventory management systems enable efficient supply and distribution management through coordination capabilities that connect procurement, production, and delivery activities. Integrated inventory visibility provides real-time inventory status across raw materials, work-in-process, and finished goods that enables coordinated decision-making. The literature explained how automated reorder triggers initiate procurement when raw materials fall below thresholds, preventing production interruptions that would disrupt manufacturing operations. Bautista Jr. and Young discussed how production planning integration matches inventory availability with production requirements to ensure schedules can be fulfilled. The literature highlighted how distribution coordination optimizes delivery scheduling based on finished goods availability, customer locations, and transportation capacity.

The literature also highlighted how inventory management systems address quality control requirements that ensure product consistency and safety in food manufacturing contexts. Batch tracking capabilities maintain records linking finished products to ingredient batches that enable targeted recalls if quality issues are discovered. The authors explained how first-in-first-out (FIFO) inventory rotation ensures oldest inventory utilized first, preventing expiration and quality degradation. The literature discussed how expiration tracking identifies products approaching expiration that require promotional efforts or disposal before quality compromises occur. Bautista Jr. and Young explained how storage condition monitoring ensures appropriate environment for ingredient and product quality maintenance throughout the supply chain.

Additionally, the authors explained how inventory management systems improve financial management through accurate inventory valuation, cost tracking, and loss prevention that directly affect business profitability. Accurate inventory records enable precise financial reporting that reflects actual asset values for business decisions and compliance requirements. The literature discussed how inventory cost tracking identifies cost drivers and enables informed decisions regarding ingredient substitutions, packaging choices, and pricing strategies. The authors explained how shrinkage reduction from improved accountability decreases losses from theft, damage, and handling errors that reduce profitability. Bautista Jr. and Young discussed how inventory turnover analysis identifies slow-moving items requiring promotional attention or production adjustment.

The literature further discussed how technology adoption in Philippine manufacturing environments addresses traditional barriers that have limited inventory management system implementation. Cost considerations represent significant barriers for manufacturers with limited technology budgets allocated primarily to production equipment and facility investments. The authors explained how system complexity concerns deter manufacturers unfamiliar with electronic inventory management approaches. The literature discussed how technical support availability affects successful implementation, with limited local support extending beyond Metro Manila creating implementation challenges. Bautista Jr. and Young explained how employee change management challenges affect adoption success as workers accustomed to traditional methods resist new system implementations.

The authors also emphasized the importance of localization in inventory management system implementation for Philippine manufacturing contexts. Local language interfaces accommodate Filipino users more comfortable with local languages than English-only system interfaces. The literature discussed how local currency and measurement systems prevent conversion errors and simplify user operation. Bautista Jr. and Young explained how local supplier integration enables electronic procurement from domestic suppliers who may not support international system formats. The literature discussed how local regulatory compliance including Philippine Bureau of Fisheries and Agricultural Quarantine requirements, FDA regulations, and local business requirements ensure appropriate compliance. The authors explained how local holiday calendars integrate with production and inventory planning that affects seasonal operations differently than Western calendars.

Furthermore, the literature discussed how Philippine food manufacturing supply chains extend beyond immediate customers to include distribution networks that reach end consumers through multiple channels. Distribution inventory management coordinates stock levels across distribution centers, warehouses, and retail outlets that ensure product availability throughout the supply chain. The authors explained how route optimization considers inventory availability, customer locations, and delivery priorities to ensure efficient distribution. The literature discussed how channel inventory tracking monitors stock levels across retail, food service, and institutional customers with different service requirements. Bautista Jr. and Young explained how e-commerce channel integration addresses growing online sales that require inventory allocation across physical and digital channels.

The literature additionally highlighted how case study evidence from Philippine seasoning product manufacturer demonstrates tangible benefits from inventory management system implementation. Inventory accuracy improved substantially, reducing discrepancies between recorded and actual inventory that complicated production planning and financial reporting. Stockout occurrences decreased significantly through automated reorder triggers that initiated procurement before production interruptions occurred. The authors explained how order fulfillment rates improved through inventory visibility that enabled accurate delivery commitments and improved customer satisfaction. The literature discussed how inventory carrying costs decreased through optimized inventory levels that reduced excess inventory while maintaining service requirements.

Moreover, the authors explained how inventory management system benefits extend beyond the manufacturer to include supply chain partners and end consumers who benefit from improved availability and product quality. Improved ingredient quality through better supplier management and incoming inspection enhances finished product quality for consumers. The literature discussed how distribution efficiency improvements reduce product handling and transportation time that maintains product freshness. Bautista Jr. and Young explained how reliable product availability supports customer loyalty that sustains long-term business relationships. The literature discussed how price stability from consistent supply reduces consumer price volatility that improves purchasing predictability.

The authors also discussed how inventory management system implementation requires consideration of organizational factors beyond technical system capabilities. Executive sponsorship provides authority and resources for implementation success while overcoming resistance to change. The literature explained how adequate training ensures user adoption and system utilization that achieves projected benefits. Bautista Jr. and Young discussed how ongoing support addresses operational questions and system issues that arise during daily operation. The literature discussed how continuous improvement refines system utilization and addresses evolving requirements that maintain system effectiveness over time.

This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system addresses inventory management challenges applicable to retail distribution contexts sharing characteristics with manufacturing supply chain environments. Similar to the concepts discussed by Bautista Jr. and Young, the proposed AIMS aims to improve inventory visibility, coordinate supply chain activities, and optimize inventory levels through automated tracking and decision support capabilities.

The proposed system aligns with the authors' discussion regarding inventory management importance in supply chain contexts where inventory availability directly impacts operational continuity, customer satisfaction, and business profitability. These capabilities apply to pharmacy and convenience store environments where inventory directly affects product availability and customer service quality.

Furthermore, the literature supports the implementation of the proposed inventory management system because it demonstrates how inventory management systems benefit manufacturing supply chain operations through documented improvements in accuracy, availability, and operational efficiency applicable to retail distribution contexts. The concepts discussed in the study provide strong evidence supporting the development and implementation of the proposed Automated Inventory Management System within Philippine retail environments.

2.2.6 Computerized Inventory and Sales System for Mini Grocery Stores in Batangas. Philippine Journal of Business Information Systems

According to M. Ortega and P. Salazar (2021), mini grocery stores in Batangas face significant operational challenges related to inventory management, sales recording, and business reporting that affect their competitiveness and sustainability in increasingly demanding marketplace environments. The authors explained that mini grocery stores, commonly known as "sari-sari" stores in Philippine context, represent important retail establishments that serve daily household needs in residential communities throughout the Batangas region. These small retail businesses provide convenient access to essential household items within walking distance of homes, particularly valuable in areas where larger supermarkets and grocery stores are not readily accessible. Despite their importance to community accessibility and local economic development, mini grocery stores in Batangas often struggle with inefficient manual operations that limit growth potential and competitive capabilities. The authors highlighted how computerized inventory and sales systems can significantly improve mini grocery operations while helping these small businesses remain competitive against larger retail establishments.

The literature emphasized that mini grocery stores in Batangas traditionally rely on manual inventory tracking methods including handwritten records, mental stock counts, and periodic physical inventory counts that limit operational effectiveness and growth potential. Ortega and Salazar discussed how manual inventory records require significant time investments for daily record keeping, inventory verification, and report preparation that reduce time available for customer service and business development activities. The authors explained how manual processes increase the likelihood of human errors including incorrect pricing, inaccurate inventory records, and calculation mistakes that result in financial losses and customer dissatisfaction. In the Batangas context, where many mini grocery stores operate with family members and one or two employees, the time invested in manual processes significantly impacts overall operational capacity and limits business growth potential. The literature highlighted how competitive pressures from supermarkets, convenience stores, and online retailers increasingly challenge mini grocery stores that cannot match the operational efficiency and customer service quality of better-resourced competitors.

Furthermore, the authors discussed how mini grocery stores in Batangas face unique operational characteristics that influence inventory management system requirements. Product variety in mini grocery stores typically includes fast-moving consumer goods including beverages, snacks,condiments, and household essentials that turn over at different rates requiring varied inventory management approaches. The literature explained how supplier relationships with local distributors and wholesalers affect replenishment patterns, delivery schedules, and credit terms that complicate inventory planning. Ortega and Salazar discussed how seasonal variations related to holidays, school seasons, and local festivals create demand patterns that affect inventory requirements differently than steady-state conditions. The authors explained how community demographics influence product preferences, purchasing patterns, and credit arrangements that affect business operations in ways that generic inventory systems may not address effectively.

The literature also highlighted how computerized inventory and sales systems address operational challenges through automation, integration, and reporting capabilities that improve efficiency and effectiveness. Automated transaction recording eliminates manual sales entry that consumes time and introduces errors that affect inventory accuracy. The authors explained how barcode scanning accelerates product identification while ensuring accurate pricing application that prevents overcharging or undercharging customers. The literature discussed how real-time inventory updates maintain accurate inventory records automatically from each transaction without manual intervention. Ortega and Salazar explained how sales recording provides data that enables analysis of sales patterns, product performance, and business trends that inform business decisions otherwise requiring guesswork or detailed manual compilation.

Additionally, the authors explained how point of sale integration improves customer service through faster checkout, accurate transactions, and professional service presentation that enhances customer perceptions. Faster checkout reduces customer wait times while improving throughput that enables the store to serve more customers during peak periods. The literature discussed how accurate transactions prevent discrepancies that cause customer dissatisfaction when prices differ between scanning and expectations. Ortega and Salazar explained how professional transaction records including printed receipts build customer confidence that encourages return patronage.
The literature discussed how extended operating hours possible with efficient operations improve accessibility that serves customers outside normal business periods.

The literature further discussed how inventory management capabilities specifically benefit mini grocery stores through improved stock management, reduced losses, and better purchasing decisions. Accurate inventory tracking prevents stockouts that result in lost sales and customer dissatisfaction while reducing overstocking that ties up capital in slow-moving inventory. The authors explained how inventory turnover analysis identifies fast-moving products requiring frequent replenishment and slow-moving items requiring promotional attention or discontinuation. The literature discussed how reorder alerts notify store owners when inventory falls below thresholds that trigger reordering, preventing stockouts and maintaining product availability. Ortega and Salazar explained how reduced inventory shrinkage from improved tracking and accountability decreases losses from theft, damage, and expired products that significantly impact small retail business profitability.

The authors also emphasized the importance of cost-effective technology solutions appropriate for mini grocery stores with limited capital and technical resources. The literature discussed how affordable technology options including tablet-based systems, cloud services, and open-source software reduce technology investment barriers that previously excluded small businesses from technology benefits. Ortega and Salazar explained how scalable solutions accommodate growth without requiring major upgrades that would strain limited capital. The literature discussed how cloud-based deployment reduces hardware requirements, technical maintenance, and upfront costs in favor of predictable subscription pricing. The authors explained how locally-available technical support simplifies troubleshooting and maintenance that would otherwise require distant service providers.

Furthermore, the literature discussed how case study evidence from mini grocery stores in Batangas demonstrates tangible operational improvements through computerized inventory and sales system implementation. Transaction processing times decreased by thirty to fifty percent compared to manual methods, improving customer throughput and reducing wait times during peak periods. Inventory record accuracy improved from typical manual rates of approximately sixty to seventy percent to above ninety percent following system implementation. The authors explained how stockout rates decreased substantially through automated reorder alerts and improved inventory visibility that enabled proactive replenishment. The literature discussed how business owners reported improved decision-making through access to sales reports and inventory analysis that previously required significant manual effort to compile.

The literature additionally highlighted how mini grocery store operators in Batangas benefit from locally-relevant system designs that address regional requirements. Local language support in Tagalog and business terminology accommodates operators more comfortable with Filipino than English-only interfaces. The authors explained how integration with local suppliers and distributors supports relationships critical to business operations in local markets. The literature discussed how local payment methods including cash, check, and credit arrangements accommodate customer payment preferences. Ortega and Salazar explained how local technical support ensures accessible assistance when system issues arise, particularly valuable for operators unfamiliar with technology troubleshooting.

Moreover, the authors explained how computerized inventory and sales system benefits extend beyond operational efficiency to include business sustainability and growth capabilities. Improved record keeping supports business loan applications, supplier credit applications, and regulatory compliance that require documented business records. The literature discussed how improved business insights enable informed decisions regarding product assortment, pricing, and resource allocation that improve profitability. Ortega and Salazar explained how scalability supports business growth through system expansion as stores add products, services, and locations. The literature discussed how competitive parity with larger retailers improves customer perceptions that sustain market share against better-resourced competitors.

This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system addresses similar operational requirements and challenges faced by mini grocery stores in the Philippine retail context. Similar to the concepts discussed by Ortega and Salazar, the proposed AIMS aims to improve operational efficiency, inventory management, and customer service through appropriate technology implementation.

The proposed system aligns with the authors' discussion regarding mini grocery store requirements including cost-effective solutions, local relevance, and operational simplicity that ensure appropriate fit for small retail businesses. These capabilities address operational challenges specific to Philippine small retail environments where resource constraints, technical limitations, and local business practices significantly impact technology adoption success.

Furthermore, the literature supports the implementation of the proposed system because it demonstrates how computerized inventory and sales systems benefit small retail businesses in the Philippine context through documented improvements in operational efficiency, inventory accuracy, and customer service. The concepts discussed in the study provide strong evidence supporting the development and implementation of the proposed Automated Inventory Management System within Philippine pharmacies and convenience stores.

2.2.7 Automated Inventory Monitoring System for Pharmacies in Cebu Province

According to R. Domingo and L. Ferrer (2020), automated inventory monitoring systems represent critical technological advancements that address significant operational challenges affecting pharmacy operations, patient service, and business sustainability in Cebu Province. The authors explained that pharmacies in Cebu Province maintain complex inventories including prescription medications, over-the-counter products, medical supplies, and wellness items that must be accurately tracked to ensure product availability and patient safety. The diverse nature of pharmacy inventory requiring pharmaceutical expertise, regulatory compliance, and operational efficiency distinguishes pharmacy inventory management from general retail contexts. The authors highlighted how automated inventory monitoring systems offer particular advantages for pharmacies facing regulatory requirements, patient safety concerns, and competitive pressures that affect business viability.

The literature emphasized that pharmacies in Cebu Province face diverse inventory management challenges related to pharmaceutical regulatory requirements, patient service demands, and operational complexity that affect inventory management effectiveness. Domingo and Ferrer discussed how Philippine pharmacies must comply with Food and Drug Administration (FDA) regulations requiring specific inventory controls, record keeping, documentation, and reporting that increase compliance complexity. The authors explained how prescription medication tracking requires coordination with patient records, prescribing physicians, and insurance providers that adds integration requirements not present in general retail contexts. The literature highlighted how pharmaceutical storage requirements including temperature control, humidity management, and light protection add environmental considerations affecting inventory management beyond simple quantity tracking. Pharmacies in Cebu Province also face healthcare-specific challenges including patient counseling requirements, drug interaction concerns, and patient privacy considerations that affect operational workflows.

Furthermore, the authors discussed how traditional manual inventory methods in Cebu Province pharmacies create significant limitations affecting operational efficiency and patient service quality. Manual inventory tracking requires periodic physical counts that consume significant staff time while potentially missing discrepancies that occur between counts. The literature explained how manual processes delay problem identification, allowing stockouts to occur before managers become aware of inventory shortages requiring customer disappointment and lost sales. Domingo and Ferrer discussed how manual record keeping complicates regulatory compliance documentation, requiring time-consuming compilation of inventory records, transaction logs, and reports for regulatory inspections. The literature highlighted how manual inventory organization limits visibility into inventory rotation, expiration status, and reorder requirements that affect both operational efficiency and patient safety.

The literature also highlighted how automated inventory monitoring systems address pharmacy-specific requirements including expiration tracking, prescription processing, and regulatory compliance. Automated expiration alerts notify pharmacists when products approach expiration dates, enabling proactive return or promotional efforts before products become unusable. The authors explained how automated tracking ensures consistent first-in-first-out (FIFO) rotation that reduces losses from expired products while maintaining patient safety. The literature discussed how prescription tracking coordinates with patient records, enabling refill reminders, drug interaction checking, and patient history maintenance that improves patient care. Domingo and Ferrer explained how regulatory reporting automation generates required documentation for FDA compliance without manual compilation.

Additionally, the authors explained how automated inventory monitoring improves patient service through faster processing, accurate information provision, and professional presentation that enhances patient confidence. Faster prescription processing reduces patient wait times while improving pharmacy throughput during peak periods. Accurate inventory visibility enables pharmacists to inform patients about product availability without depending on potentially inaccurate stock counts. The literature discussed how automated record keeping maintains professional transaction records including printed receipts and electronic 档案 that build patient confidence. Domingo and Ferrer explained how improved efficiency frees pharmacist time for patient counseling and clinical services that differentiate pharmacy care from self-service retail experiences.

The literature further discussed how automated systems improve pharmaceutical supply chain management from wholesale procurement through patient dispensing. Automated reorder triggers initiate procurement when inventory falls below thresholds, preventing stockouts that interrupt patient treatment continuity. The authors explained how supplier integration enables electronic ordering that reduces telephone/fax ordering time while preventing ordering errors. The literature discussed how receiving verification confirms ordered products against purchase orders to prevent discrepancies. Domingo and Ferrer explained how inventory tracking from receiving through dispensing provides complete chain-of-custody documentation required for pharmaceutical compliance.

The authors also emphasized the importance of barcode integration in pharmacy inventory monitoring that accelerates transactions while ensuring accuracy. Barcode scanning identifies products accurately without manual entry that could introduce errors. The literature explained how barcode integration enables automated inventory updates from each transaction, maintaining continuous visibility into inventory status. Domingo and Ferrer discussed how barcode scanning supports pharmacy efficiency during high-volume periods when accuracy is particularly important. The literature highlighted how barcode technology enables automated expiration tracking and inventory rotation through scanning timestamps that verify FIFO implementation.

Furthermore, the literature discussed how case study evidence from pharmacies in Cebu Province demonstrates tangible benefits from automated inventory monitoring implementation. Inventory accuracy improved from typical rates of approximately sixty- five to seventy-five percent to above ninety-five percent following automated system implementation. Stockout occurrences decreased substantially through automated reorder alerts that triggered procurement before inventory exhausted. The authors explained how expired product losses decreased through automated expiration alerts and FIFO enforcement that reduced waste. The literature discussed how prescription processing times decreased while accuracy improved, enabling pharmacists to serve more patients with better service.

The literature additionally highlighted how local context in Cebu Province influences system requirements and implementation approaches. Local competition from retail chains, independent pharmacies, and hospital dispensaries creates service expectations that automated systems help meet. The authors explained how Cebuano language support accommodates local pharmacists and patients more comfortable with local languages than English-only interfaces. The literature discussed how local supplier relationships affect ordering processes, credit terms, and delivery schedules that system integration addresses. Domingo and Ferrer explained how climate considerations in Cebu affecting temperature-sensitive product storage require system support for environmental monitoring.

Moreover, the authors explained how automated inventory monitoring contributes to pharmacy business sustainability through improved efficiency, reduced losses, and enhanced competitiveness. Improved inventory efficiency optimizes working capital investment by reducing excess inventory while maintaining availability that improves customer service. The literature discussed how reduced losses from expiration, shrinkage, and stockouts directly improve profitability in competitive environments. Domingo and Ferrer explained how competitive service levels enable pharmacies to retain patients who might otherwise switch to competitors following stockouts or poor service experiences

The authors also discussed how pharmaceutical inventory management principles apply to convenience store contexts with appropriate adaptations. The literature explained how expiration tracking, inventory rotation, and regulatory compliance requirements extend to convenience store categories including over-the-counter medications, health supplements, and perishables. Domingo and Ferrer discussed how point of sale integration, customer service focus, and operational efficiency apply across retail contexts. The literature discussed how inventory management fundamentals including demand forecasting, reorder optimization, and stock monitoring remain applicable principles across pharmacy and convenience store applications.

This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system addresses pharmacy inventory requirements similar to those identified by Domingo and Ferrer in Cebu Province. Similar to the concepts discussed by the authors, the proposed AIMS aims to improve inventory accuracy, prevent stockouts, and ensure regulatory compliance through automated monitoring and tracking capabilities.

The proposed system aligns with the authors' discussion regarding pharmacy-specific requirements including expiration tracking, barcode integration, and regulatory compliance that ensure appropriate pharmacy functionality. These capabilities address operational challenges specific to pharmacy environments where patient safety, regulatory compliance, and pharmaceutical expertise significantly impact system requirements.

Furthermore, the literature supports the implementation of the proposed system because it demonstrates how automated inventory monitoring benefits Philippine pharmacies through documented improvements in accuracy, service, and operational efficiency. The concepts discussed in the study provide strong evidence supporting the development and implementation of the proposed Automated Inventory Management System within Philippine pharmacy and convenience store environments.

2.2.8 Cloud-Based Point of Sale and Inventory System for Retail Businesses in Manila

A local study conducted by Villareal, Mendoza, and Ramos (2022), published in the Philippine ICT Research Journal, examined the development and implementation of

a cloud-based Point of Sale and inventory management system for retail businesses operating in Manila, Philippines. The primary objective of the study was to determine how cloud-based technologies could effectively improve inventory accessibility, transaction processing speed, operational coordination, and overall business productivity among small and medium-sized retail establishments in one of the most commercially active urban centers in the country. The researchers sought to address what they identified as a persistent and widespread operational problem among Philippine retailers — the continued reliance on outdated, manual, or disconnected inventory management systems that limited their ability to manage stock effectively and respond efficiently to the demands of a competitive retail environment.

The researchers began by documenting the operational realities of the retail businesses they studied, noting that a significant proportion of retail enterprises in the Philippines continued to store and manage their inventory information manually or through legacy systems that lacked real-time data synchronization capabilities. These outdated approaches to inventory management created a range of interconnected operational problems that collectively reduced business efficiency and customer service quality. Manual inventory procedures, in particular, were found to be especially problematic because they required employees to record stock movements by hand, creating a time lag between when an inventory change occurred — such as a product being sold, returned, or received from a supplier — and when that change was reflected in the official inventory record. This delay in inventory data updating made it difficult for managers to know the true current state of their stock at any given time, leading to decisions based on inaccurate or outdated information.

Furthermore, the researchers identified that the physical limitations of manual and locally stored inventory systems created significant coordination challenges for retail businesses, particularly those operating multiple branches or requiring management oversight from locations other than the store itself. In businesses where inventory data was stored in physical logbooks, spreadsheets on local computers, or basic software without cloud connectivity, managers could only access inventory information by physically visiting the store or requesting reports to be manually compiled and transmitted — a process that was both time-consuming and prone to transcription errors. The absence of centralized, remotely accessible inventory data meant that purchasing decisions, restocking actions, and operational adjustments were often made based on incomplete information, resulting in both unnecessary stockouts and wasteful overstocking across the participating businesses.

To address these operational challenges comprehensively, the researchers developed a cloud-based Point of Sale and inventory management system designed specifically to meet the operational needs and technical constraints of Philippine retail businesses. The system was built on cloud computing infrastructure, enabling all inventory data, sales transaction records, and management reports to be stored on remote servers accessible through standard internet-connected devices, including desktop computers, laptops, tablets, and smartphones. This cloud-based architecture fundamentally transformed the operational model of inventory management for the participating businesses by eliminating the physical and geographic limitations of locally stored data systems. Store managers, business owners, and authorized personnel could access current inventory information and generate reports from any location at any time, provided they had internet connectivity, without needing to be physically present at the store.

The system also incorporated a fully functional point-of-sale module equipped with barcode scanning technology that allowed store personnel to process customer transactions quickly and accurately by scanning product barcodes rather than manually entering product codes and prices. Each time a product was scanned and a sale was processed through the POS module, the system automatically updated the store's inventory database to reflect the reduced stock level, ensuring that inventory records remained current and accurate without requiring any additional data entry from store employees. This seamless integration between the POS transaction module and the inventory database was identified by the researchers as one of the most operationally significant features of the system, as it eliminated the dual-entry problem common in businesses that maintain separate sales recording and inventory management systems — where every transaction must be manually recorded twice, increasing both workload and the risk of recording errors.

The findings of the study, gathered through both quantitative performance measurements and qualitative feedback from system users, revealed that the cloud- based system produced significant and measurable improvements across multiple dimensions of operational performance within the participating retail businesses. In terms of inventory management, the most consistently reported improvement was a substantial reduction in inventory discrepancies — instances where the physical count of products in the store did not match the recorded inventory levels in the system.
Because the cloud-based system updated inventory records automatically and in real time with every sales transaction, restocking event, and product return, the accumulation of recording errors that commonly occurred under manual systems was effectively eliminated. Businesses that had previously struggled with significant inventory discrepancies at the end of each operating period reported marked improvements in inventory accuracy following system implementation.

Additionally, the study found that real-time inventory visibility provided by the cloud- based system significantly improved the operational coordination of the participating retail businesses. Store managers reported that having access to current, accurate inventory data at all times — rather than relying on periodic manual stock counts and handwritten reports — allowed them to identify low-stock situations much earlier and respond to them more quickly, reducing the frequency and duration of stockout incidents. The ability to monitor inventory movement trends in real time also allowed managers to identify fast-moving and slow-moving products more accurately, enabling more strategic purchasing decisions that reduced both unnecessary restocking of slow- moving items and the stockouts of high-demand products that had previously occurred due to delayed inventory information.

2.2.9 Point of Sale System Implementation for Small Retail Enterprises in Metro Manila

According to C. M. Aguilar and J. L. Torres (2020), Point of Sale system implementation among small retail enterprises in Metro Manila represents significant technological advancement that addresses operational challenges affecting business efficiency, customer service, and competitive positioning in urban Philippine marketplace environments. The authors explained that small retail enterprises including neighborhood stores, specialty shops, and independent retailers represent important segments of the Metro Manila retail sector that contribute significantly to employment, community accessibility, and local economic development. These businesses serve essential functions in Metropolitan communities, providing convenient access to daily necessities within residential areas where larger supermarkets and chain stores may not have branch locations. Despite their importance to community economic activity, small retail enterprises in Metro Manila often struggle with operational inefficiencies, limited technology adoption, and competitive pressures from established retail chains that threaten their long-term viability in increasingly competitive markets. The authors highlighted how Point of Sale technology adoption through appropriate system implementation can significantly improve small retail business operations while helping these businesses remain competitive against larger, better-resourced retail establishments operating in Metro Manila markets.

The literature emphasized that Metro Manila small retail enterprises traditionally rely on manual transaction processing methods including handwritten sales records, mental calculations, and manual inventory tracking that limit operational effectiveness and growth potential in busy urban environments. Aguilar and Torres discussed how manual sales recording requires significant time investments for daily record keeping, receipt preparation, and sales compilation that reduce time available for customer service and business development activities critical to competitive survival in Metro Manila's fast- paced retail environments. The authors explained how manual processes increase the likelihood of human errors including incorrect pricing calculations, inaccurate change provision, and recording mistakes that result in financial losses and customer dissatisfaction affecting business reputation in close-knit communities. In the Metro Manila context, where many small retail enterprises operate with family members and limited employees alongside busy lives demanding multiple job responsibilities, the time invested in manual processes significantly impacts overall operational capacity and limits business growth potential that could otherwise improve family economic conditions. The literature highlighted how competitive pressures from shopping malls, supermarkets, convenience store chains, and online grocery delivery services increasingly challenge small retail enterprises that cannot match the operational efficiency, transaction accuracy, and customer service quality of better-resourced competitors with comprehensive technology implementations.

Furthermore, the authors discussed how Point of Sale system implementation specifically addresses Metro Manila small retail enterprise challenges through automation, integration, and reporting capabilities that improve efficiency and enable growth in competitive environments. Automated transaction processing eliminates manual sales entry that consumes staff time while introducing errors affecting inventory accuracy and financial records essential to business management. The literature explained how barcode scanning accelerates product identification during busy periods, ensuring accurate pricing application that prevents both customer overcharging and revenue undercharging that affects profitability. Aguilar and Torres discussed how integrated inventory tracking maintains accurate inventory records automatically from each transaction without requiring manual inventory updates that interrupt customer service activities. The literature highlighted how sales recording provides comprehensive data that enables analysis of sales patterns, product performance, peak operating periods, and business trends that inform strategic decisions otherwise requiring guesswork or detailed manual compilation from disorganized records.

The literature also highlighted how Point of Sale systems improve financial management capabilities for small retail enterprises through accurate sales recording, proper cash handling, and loss prevention that directly affect business profitability and sustainability. Accurate sales records enable precise daily accounting that reflects actual business performance compared to estimates that may mask problems until they become serious financial difficulties. The authors explained how cash drawer management reduces loss opportunities through proper change fund handling, transaction recording, and end-of- day reconciliation that identifies discrepancies quickly. The literature discussed how sales tracking by product category identifies profitable items requiring stock emphasis and slow-moving items requiring discount promotion or discontinuation decisions.
Aguilar and Torres explained how sales trend analysis enables informed decisions regarding inventory investment, staffing schedules, and promotional activities that improve business outcomes based on documented performance rather than assumptions.

Additionally, the authors explained how Point of Sale implementation improves customer service through faster checkout, accurate transactions, and professional service presentation that enhances customer perceptions and encourages repeat patronage in competitive markets. Faster checkout during peak periods reduces customer wait times while enabling the store to serve more customers during limited operating hours typical of small retail establishments in Metro Manila. The literature discussed how accurate transactions prevent pricing discrepancies that cause customer dissatisfaction when scanned prices differ from expectations or when incorrect change damages customer trust built over time. Aguilar and Torres explained how printed receipts provide professional transaction records that build customer confidence in honest transactions while enabling customers to verify purchases against budgets. The literature discussed how extended operating hours possible with efficient operations improve accessibility that serves customers with diverse schedules working various shifts throughout Metro Manila.

The literature further discussed how Metro Manila small retail enterprises face unique operational characteristics requiring consideration in Point of Sale system selection and implementation approaches. Product variety in small retail establishments typically includes fast-moving consumer goods, beverages, snack items, household essentials, and seasonal products that turn over at different rates requiring varied inventory management approaches beyond simple quantity tracking. The authors explained how supplier relationships with local distributors, direct manufacturers, and wet markets affect replenishment patterns, credit terms, delivery schedules, and purchasing decisions that complicate inventory planning in ways not addressed by generic international systems. Aguilar and Torres discussed how seasonal variations related to Christmas seasons, school enrollment periods, local festivals, and payday cycles create demand patterns that affect inventory requirements differently than steady-state conditions assumed by standard systems. The literature highlighted how community demographics in different Metro Manila areas influence product preferences, purchasing patterns, credit arrangements, and service expectations that affect business operations requiring local knowledge.

The authors also emphasized the importance of cost-effective technology solutions appropriate for Metro Manila small retail enterprises operating with limited capital, narrow profit margins, and resource constraints typical of small business operations. The literature discussed how affordable technology options including tablet-based systems, cloud services, and modular implementations reduce technology investment barriers that previously excluded small businesses from technology benefits. Aguilar and Torres explained how subscription-based pricing converts capital investments to operational expenses that match small business cash flow patterns without requiring major upfront capital outlays that strain limited resources. The literature discussed how scalable solutions accommodate business growth through feature additions, store expansions, and increased transaction volumes without requiring complete system replacements that would strain limited budgets. The authors explained how cloud-based deployment reduces hardware requirements, eliminates technical maintenance demands, and provides automatic backups that protect business records regardless of location.

Furthermore, the authors discussed how case study evidence from Metro Manila small retail enterprises demonstrates tangible operational improvements through Point of Sale system implementation that justify technology investments despite resource constraints. Transaction processing times decreased by thirty to sixty percent compared to manual methods, improving customer throughput and reducing wait times during busy periods including weekends, paydays, and holiday seasons. The authors explained how inventory record accuracy improved from typical manual rates of approximately sixty to seventy percent to above ninety percent following system implementation, enabling reliable reorder decisions and reduced losses from inventory discrepancies. The literature discussed how stockout rates decreased substantially through automated reorder alerts and improved inventory visibility that enabled proactive replenishment maintaining product availability for customers. Aguilar and Torres explained how business owners reported improved decision-making through access to sales reports and inventory analysis that previously required significant manual effort to compile from disorganized records.

The literature additionally highlighted how successful Point of Sale implementation requires consideration of human factors affecting system adoption and utilization among users with limited technology backgrounds. Training programs must accommodate users with minimal computer experience who may be unfamiliar with touchscreen operation, cursor navigation, and software application use that computer-savvy users take for granted. The authors explained how user-friendly interfaces reduce learning curves while improving user adoption and system effectiveness that justifies technology investments. The literature discussed how ongoing technical support addresses operational questions and problems that arise during daily system operation, maintaining system effectiveness and user confidence in technology investments. Aguilar and Torres explained how local language support in Filipino and Tagalog accommodates users more comfortable with local languages than English-only interfaces that may create usability barriers for less educated users.

Moreover, the authors explained how Point of Sale system benefits extend beyond operational efficiency to include business sustainability, competitive positioning, and growth capabilities that improve long-term viability in competitive Metro Manila retail markets. Improved record keeping supports business loan applications, supplier credit applications, and regulatory compliance that require documented business records demonstrating financial capability and transaction volume. The literature discussed how improved business insights enable informed decisions regarding product assortment, pricing strategies, and resource allocation that improve profitability in thin-margin operations. Aguilar and Torres explained how scalable systems support business growth through store expansion, product line additions, and diversification into related business opportunities as capital accumulates from improved operations. The literature discussed how competitive parity with larger retailers improves customer perceptions that sustain market share against better-resourced competitors with established locations in shopping malls and commercial centers.

The authors also discussed how technology adoption in Metro Manila small retail enterprises contributes to broader national development objectives through improved business productivity, employment creation, and economic competitiveness in regional and global markets. Modernized small retail businesses contribute to national productivity improvements that enhance Philippine economic competitiveness in regional and global markets facing increasing competition from regional manufacturing centers and service economies. The literature explained how technology adoption creates demand for technical support, maintenance, and related services that generate employment opportunities for technology professionals in local communities. Aguilar and Torres discussed how enhanced small retail business viability preserves employment and maintains community economic functions that contribute to neighborhood development and social stability. The authors explained how increased technology adoption supports government initiatives to modernize Philippine commerce and improve business competitiveness across all sectors.

This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system addresses similar operational requirements and challenges faced by small retail enterprises in Metro Manila requiring Point of Sale and inventory management capabilities. Similar to the concepts discussed by Aguilar and Torres, the proposed AIMS aims to improve operational efficiency, inventory management, customer service, and financial management through appropriate technology implementation.

The proposed system aligns with the authors' discussion regarding Metro Manila small retail enterprise requirements including cost-effective solutions, local relevance, user- friendly interfaces, and operational efficiency that ensure appropriate fit for small businesses with limited resources and technology backgrounds. These capabilities address operational challenges specific to Philippine small retail environments where resource constraints, competitive pressures, and local business practices significantly impact technology adoption success.

Furthermore, the literature supports the implementation of the proposed system because it demonstrates how Point of Sale and inventory management systems benefit small retail enterprises in the Philippine context through documented improvements in operational efficiency, inventory accuracy, customer service, and business decision- making. The concepts discussed in the study provide strong evidence supporting the development and implementation of the proposed Automated Inventory Management System within Philippine pharmacies and convenience stores in Metro Manila and other urban centers throughout the Philippines.

2.2.10 Inventory Management Challenges and Solutions for Independent Pharmacies in Metro Manila

According to R. S. Mendoza and A. B. Reyes (2019), independent pharmacies in Metro Manila face significant inventory management challenges related to pharmaceutical regulatory requirements, patient service demands, and competitive pressures that affect business viability and community healthcare accessibility. The authors explained that independent pharmacies in Metro Manila represent important healthcare access points that serve diverse communities, particularly in residential areas where hospital and chain pharmacy access may be limited or require significant travel time. These pharmacies provide essential medication access, health product availability, and pharmacist consultation services that distinguish pharmaceutical retail from general retail operations requiring specialized inventory management approaches. Despite their importance to community healthcare accessibility, independent pharmacies in Metro Manila often struggle with inventory management inefficiencies, limited technology adoption, and competitive pressures from chain pharmacies and hospital dispensaries that threaten their long-term sustainability. The authors highlighted how appropriate inventory management solutions can significantly improve independent pharmacy operations while helping these essential healthcare providers remain viable against better-resourced competitors in Metro Manila's competitive pharmaceutical retail markets.

The literature emphasized that Metro Manila independent pharmacies face diverse inventory management challenges related to pharmaceutical variety, regulatory compliance, and patient service requirements that distinguish pharmacy inventory from general retail inventory contexts. Mendoza and Reyes discussed how pharmaceutical inventory encompasses prescription medications requiring physician authorization, over-the-counter products available without prescription, medical supplies serving healthcare needs, and wellness products serving consumer wellness objectives. The authors explained how each inventory category involves different handling requirements, regulatory controls, and management approaches that complicate inventory management beyond simple quantity tracking. The literature highlighted how pharmaceutical storage requirements including temperature control for refrigerated medications, humidity protection for moisture-sensitive products, and light avoidance for photo-sensitive compounds add environmental management challenges not present in general retail inventory. Metro Manila's climate with high humidity and temperatures year-round creates additional environmental management challenges affecting pharmaceutical inventory quality and patient safety that must be addressed through appropriate inventory management approaches.

Furthermore, the authors discussed how traditional manual inventory methods in Metro Manila independent pharmacies create significant limitations affecting regulatory compliance, patient safety, and business sustainability in competitive markets. Manual inventory tracking requires periodic physical counts that consume pharmacist time that could be devoted to patient counseling, medication therapy management, and clinical consultation services that justify professional pharmacist presence. The literature explained how manual processes delay problem identification, allowing stockouts to occur before pharmacists become aware of inventory shortages requiring customer disappointment, lost sales, and potential patient treatment interruptions. Mendoza and Reyes discussed how manual record keeping complicates Bureau of Food and Drugs Administration compliance documentation, requiring time-consuming compilation of inventory records, transaction logs, and reports for regulatory inspections. The literature highlighted how manual organization limits visibility into inventory age, expiration status, and rotation requirements that affect both operational efficiency and patient safety when expired medications might accidentally be dispensed.

The literature also highlighted how inventory management challenges specifically affecting independent pharmacies include prescription medication handling, controlled substance tracking, and drug interaction concerns requiring specialized inventory management capabilities. Controlled substances including psychotropic medications and dangerous drugs require special tracking, secure storage, and detailed documentation beyond standard inventory management requirements. The authors explained how prescription coordination requires integration with patient records, prescribing physician information, and insurance billing that adds complexity to pharmaceutical dispensing not present in general retail transactions. The literature discussed how drug interaction checking requires database access and clinical decision support that integrates with inventory information to alert pharmacists to potential interactions that could harm patients. Mendoza and Reyes explained how patient privacy requirements from the Data Privacy Act affect patient record integration with inventory systems, requiring careful implementation that protects patient information while enabling appropriate clinical access.

Additionally, the authors explained how inventory management solutions address Metro Manila independent pharmacy challenges through automated tracking, expiration monitoring, and reordering capabilities that improve efficiency while maintaining regulatory compliance. Automated inventory tracking maintains continuous visibility into inventory quantities, eliminating the inaccuracies from manual counts that lead to ordering errors, stockouts, and overstocking issues affecting business finances. Automated expiration alerts identify products approaching expiration dates, enabling promotional efforts or return processing before products become unusable or require proper disposal according to pharmaceutical regulations. The literature discussed how automated reorder triggers initiate procurement when inventory falls below thresholds, preventing stockouts that interrupt patient treatment continuity and drive customers to competitors. Mendoza and Reyes explained how inventory aging reports identify slow- moving items requiring promotional attention, reducing losses from expired products and improving inventory turnover that frees working capital for productive use.

The literature further discussed how prescription processing integration improves patient service through faster processing, accurate information provision, and coordination with patient therapy management that enhances healthcare outcomes. Faster prescription processing reduces patient wait times during busy periods while enabling pharmacists to serve more patients seeking medication access in time-sensitive situations. The authors explained how automated insurance verification coordinates with Philippine Health Insurance Corporation requirements, government programs, and private insurance providers that affect patient billing and reimbursement. The literature discussed how refill reminder capabilities improve patient adherence through automated notifications when prescriptions require renewal, improving health outcomes while maintaining customer relationships that encourage repeat patronage. Mendoza and Reyes explained how drug interaction database integration alerts pharmacists to potential interactions, enabling clinical consultation that improves patient safety and demonstrates professional value justifying pharmacist professional fees.

The authors also emphasized the importance of technology solutions appropriate for independent pharmacies with limited capital, technical expertise, and pharmacist time for system management. The literature discussed how affordable cloud-based solutions reduce upfront capital requirements, eliminate server hardware demands, and provide automatic updates without requiring technical expertise not available in most independent pharmacies. Mendoza and Reyes explained how mobile access enables pharmacists to monitor inventory, check stock levels, and process orders from any location, particularly valuable for pharmacists managing multiple pharmacy locations or working between pharmacy and other responsibilities. The literature discussed how pharmacy-specific implementations address pharmaceutical requirements including prescription processing, insurance billing, and regulatory reporting that generic retail systems may not accommodate without extensive customization. The authors explained how local technical support ensures accessible assistance when system issues arise, particularly valuable for pharmacists unfamiliar with technology troubleshooting.

Furthermore, the authors discussed how case study evidence from Metro Manila independent pharmacies demonstrates tangible benefits from appropriate inventory management implementation addressing pharmaceutical requirements. Inventory accuracy improved from typical manual rates of approximately fifty to seventy percent to above ninety-five percent following implementation of pharmaceutical-specific inventory management systems. Stockout occurrences decreased substantially through automated reorder alerts and improved visibility, maintaining prescription availability that improved customer satisfaction and retained patient relationships. The authors explained how expired product losses decreased through automated expiration alerts and rotation management, reducing waste and disposal costs while maintaining patient safety from potentially expired medications. The literature discussed how pharmacist time freed from manual inventory management directed toward patient counseling and clinical services improved revenue while demonstrating professional value that justifies pharmacist presence.

The literature additionally highlighted how competitive pressures from chain pharmacies and hospital dispensaries require independent pharmacies to differentiate through service quality, accessibility, and community relationships that inventory management enables. The authors explained how consistent product availability developed through improved inventory management retains customers who might otherwise switch to competitors following repeated stockouts affecting treatment continuity. Mendoza and Reyes discussed how pharmacist consultation availability enabled by efficient operations differentiates independent pharmacies from self-service chain pharmacy models where pharmacist accessibility may be limited. The literature discussed how extended hours and convenient locations supported by efficient inventory management improve accessibility that serves diverse customer schedules. The authors explained how community relationships developed through consistent service, personalized attention, and pharmacist availability build customer loyalty that sustains competitive positioning against larger, better-resourced competitors.

Moreover, the authors explained how pharmaceutical inventory management principles extend to convenience store contexts with appropriate adaptations for non-pharmaceutical product categories. The literature explained how expiration tracking, inventory rotation, and FIFO management apply to food items, cosmetics, and personal care products with expiration dates or shelf life limitations affecting product quality. Mendoza and Reyes discussed how cold chain management for refrigerated products including dairy, beverages, and frozen goods requires similar temperature monitoring approaches used for temperature-sensitive medications. The literature discussed how regulatory compliance for over-the-counter medications, health supplements, and medical devices sold in convenience stores extends applicable pharmaceutical inventory management principles to broader retail contexts. The authors explained how point of sale integration, customer service focus, and operational efficiency remain applicable across pharmacy and convenience store applications serving retail customer needs.

This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system addresses inventory management challenges faced by independent pharmacies in Metro Manila requiring pharmaceutical-specific capabilities including expiration tracking, prescription processing, and regulatory compliance. Similar to the concepts discussed by Mendoza and Reyes, the proposed AIMS aims to improve inventory accuracy, prevent stockouts, ensure expiration management, and support regulatory compliance through automated monitoring and tracking capabilities appropriate for pharmacy environments.

The proposed system aligns with the authors' discussion regarding pharmaceutical inventory management requirements including prescription processing, controlled substance tracking, drug interaction checking, and patient service integration that ensure appropriate pharmacy functionality. These capabilities address operational challenges specific to pharmacy environments where patient safety, regulatory compliance, and clinical expertise significantly impact system requirements beyond general retail inventory management.

Furthermore, the literature supports the implementation of the proposed pharmaceutical inventory management system because it demonstrates how inventory management systems benefit independent pharmacies through documented improvements in accuracy, service, regulatory compliance, and operational efficiency. The concepts discussed in the study provide strong evidence supporting the development and implementation of the proposed Automated Inventory Management System within Philippine pharmacy and convenience store environments throughout Metro Manila and other urban centers.

2.2.11 Web-Based Inventory System for Supermarket Chains in the Philippines. Philippine Journal of Retail Management

According to P. A. Villar and M. R. Cruz (2021), web-based inventory systems represent significant technological advancements that enable Philippine supermarket chains to coordinate multi-location operations, maintain centralized inventory visibility, and optimize supply chain management across geographically distributed retail networks. The authors explained that supermarket chains in the Philippines operate diverse networks including flagship stores, branch locations, warehouse distribution centers, and potentially franchise operations requiring coordinated inventory management that isolated systems cannot provide. These large retail establishments serve substantial market segments throughout the Philippines, providing comprehensive product selections, competitive pricing, and modern shopping environments that have transformed Philippine retail over the past decades. Despite their importance to Philippine retail development and consumer accessibility, supermarket chains face ongoing challenges related to multi-location coordination, inventory optimization, and supply chain efficiency that significantly impact operating costs and competitive positioning. The authors highlighted how web-based inventory systems offer particular advantages for Philippine supermarket chains operating across multiple locations requiring centralized management, real-time visibility, and coordinated operations not available from standalone systems.

The literature emphasized that Philippine supermarket chains traditionally rely on distributed inventory systems that create significant coordination challenges affecting operational efficiency, inventory accuracy, and supply chain effectiveness across location networks. Villar and Cruz discussed how standalone systems in each store location generate isolated data requiring manual compilation for regional or national reporting that consumes management time while potentially introducing errors during data transfer between systems. The authors explained how distributed inventory prevents visibility across locations, making it difficult to identify overstocked locations with excess inventory while other locations experience stockouts requiring emergency transfers or lost sales. The literature highlighted how supplier ordering coordination becomes complex when each location orders independently without visibility into total chain demand affecting volume discounts, delivery efficiency, and supplier relationships. This distribution creates redundancy, inconsistency, and inefficiency that increases operating costs while limiting competitive advantages that centralized systems could provide.

Furthermore, the authors discussed how web-based inventory systems address Philippine supermarket chain challenges through centralized data management, real- time visibility, and coordinated operations that improve efficiency and decision-making across location networks. Centralized databases aggregate inventory data from all locations, enabling comprehensive analysis, reporting, and decision-making that isolated systems cannot provide regardless of management sophistication. Real-time transaction updates provide immediate visibility into sales, receipts, adjustments, and transfers across the entire chain, enabling rapid response to emerging conditions before problems escalate. The literature explained how web-based inventory coordination enables inventory transfers between locations, moving excess inventory from overstocked locations to stockout locations that maintain customer service while reducing losses from clearance discounting or disposal. Villar and Cruz discussed how centralized ordering generates consolidated purchase orders that improve supplier relationships through volume discounts, efficient delivery scheduling, and coordinated promotional activities.

The literature also highlighted how web-based systems improve supply chain coordination between supermarket chains and their suppliers through electronic integration that eliminates manual ordering processes, reduces errors, and accelerates transaction processing. Automated purchase order generation initiates orders when inventory falls below threshold levels determined by historical patterns, seasonal variations, and sales forecasts that optimize inventory investment. The authors explained how supplier portals enable vendor access to inventory data, order status, and delivery requirements that improve coordination without telephone calls, faxes, or email exchanges consuming management time. The literature discussed how receiving verification confirms deliveries against purchase orders through barcode scanning that identifies discrepancies before acceptance, preventing short shipments or wrong products that complicate inventory management. Villar and Cruz explained how electronic invoicing and payment processing coordinates financial transactions, accelerates supplier payment for volume discounts, and improves supplier relationships through reliable payment performance.

Additionally, the authors explained how web-based inventory systems improve warehouse distribution center coordination that serves multiple supermarket locations through inventory pooling, cross-docking, and logistics optimization that reduce costs while maintaining product availability. Distribution center inventory serves as centralized inventory pools serving multiple store locations, enabling more efficient inventory investment than distributed inventories in each location requiring safety stock duplication. Cross-docking operations receive incoming shipments and directly transfer to outbound deliveries without warehouse storage, reducing handling and storage costs while accelerating product flow from suppliers to store shelves. The literature discussed how demand aggregation across locations enables more accurate forecasting, efficient replenishment, and optimized delivery routes that reduce transportation costs while meeting store service requirements. Villar and Cruz explained how web-based systems coordinate promotional activities across locations, synchronizing pricing, displays, and inventory allocation that maximizes promotional effectiveness throughout the chain.

The literature further discussed how web-based inventory systems improve inventory forecasting and demand planning that optimize inventory investment while maintaining product availability throughout the chain network. Historical sales data from all locations enables pattern identification, trend analysis, and demand forecasting that accounts for geographic variations, demographic differences, and seasonal patterns unobservable in location-specific data. The authors explained how promotional calendar integration accounts for planned promotions, holiday seasons, and regional events affecting demand that inform inventory ordering and allocation decisions. The literature discussed how weather integration incorporates Philippine weather patterns including monsoons, typhoons, and seasonal variations affecting consumer behavior and demand patterns for planning purposes. Villar and Cruz explained how artificial intelligence and machine learning improve forecast accuracy over time by incorporating additional data sources and identifying patterns that manual analysis would miss.

The authors also emphasized the importance of web-based inventory system scalability that accommodates business growth through new location additions, product line expansions, and increasing transaction volumes without system replacement. Cloud- based architecture provides elastic capacity that automatically scales to accommodate peak periods including Christmas seasons, promotional events, and store expansion without hardware upgrades. The literature discussed how new location implementations connect immediately to existing web-based systems through standardized interfaces that reduce technical complexity while accelerating location openings. Villar and Cruz explained how geographic expansion to new regions becomes more practical when web-based systems reduce infrastructure requirements for new store implementations. The literature discussed how multi-format capabilities support different store formats including supermarkets, hypermarkets, and convenience stores within unified platforms that maintain coordinated operations.

Furthermore, the authors discussed how case study evidence from Philippine supermarket chains demonstrates tangible operational improvements from web-based inventory system implementation that justify technology investments in centralizing capabilities. Management reporting time decreased significantly through automated report generation replacing manual data compilation requiring days or weeks to complete across distributed location networks. The authors explained how inventory accuracy improved substantially through centralized visibility enabling identification and correction of discrepancies across multiple locations. The literature discussed how stockout rates decreased through coordinated ordering and distribution that moved inventory to where needed before customers experienced unavailability. Villar and Cruz explained how supply chain costs decreased through volume purchasing, efficient distribution, and reduced emergency shipments that previously consumed significant management attention and financial resources.

The literature additionally highlighted how local context in the Philippines influences web-based system requirements and implementation approaches for supermarket chains operating in Philippine retail environments. Local vendor integration connects with Philippine suppliers including local manufacturers, agricultural producers, and imported product distributors who may not support international electronic data interchange formats. The authors explained how local payment terms including consignment, credit arrangements, and installment purchasing require system flexibility that supports Philippine business practices. The literature discussed how local regulatory compliance including Bureau of Internal Revenue requirements, local business permits, and municipal regulations affect system reporting requirements.

Villar and Cruz explained how Filipino and regional language support accommodates store personnel across diverse locations where English proficiency may vary significantly.

Moreover, the authors explained how web-based inventory system principles extend to smaller retail contexts including pharmacies and convenience stores requiring centralized management across multiple locations. The literature explained how multi-location visibility and centralized reporting benefits apply to any retail network with multiple stores requiring coordination and overall business oversight. Villar and Cruz discussed how distribution optimization, supplier consolidation, and purchasing coordination provide benefits to any retail operation with multiple locations. The literature discussed how cloud-based deployment reduces infrastructure requirements that previously limited multi-location technology investment to large enterprises with significant capital resources. The authors explained how scalable web-based solutions accommodate business growth from single stores through regional networks without platform changes affecting operational continuity.

This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system addresses inventory management challenges faced by retail networks requiring multi-location visibility, centralized management, and coordinated operations across geographically distributed store networks. Similar to the concepts discussed by Villar and Cruz, the proposed AIMS aims to provide centralized inventory visibility, automated reordering, and coordinated reporting that improve efficiency and decision-making across pharmacy and convenience store networks.

The proposed system aligns with the authors' discussion regarding web-based system benefits for retail networks including centralized data management, real-time visibility, and coordinated operations that improve efficiency across multiple locations. These capabilities address operational challenges specific to multi-location retail environments where distributed operations create coordination difficulties that centralized systems can address through integrated platform approaches.

Furthermore, the literature supports the implementation of the proposed system because it demonstrates how web-based inventory systems benefit large retail networks through documented improvements in visibility, coordination, and operational efficiency applicable to smaller multi-location retail operations. The concepts discussed in the study provide evidence supporting the development and implementation of the proposed Automated Inventory Management System within Philippine retail networks including pharmacies and convenience stores throughout the Philippines.

2.2.12 Automated Inventory System for Convenience Stores in Luzon

According to L. P. Garcia and D. R. Santos (2018), convenience stores in Luzon represent an increasingly important retail segment that provides accessible daily necessities, quick meals, and essential household items serving busy consumers throughout the region.
The authors explained that convenience stores in the Northern Luzon region face unique operational challenges related to geographic distribution, diverse customer demographics, and competitive pressures from various retail formats competing for consumer spending. These retail establishments serve essential functions in Luzon communities including urban centers, residential subdivisions, bus terminals, and highway rest areas where consumers seek convenient access to products without the extended shopping trips required by larger retail formats. Despite their growth and importance to regional retail development, convenience stores in Luzon often struggle with inventory management challenges affecting product availability, operating efficiency, and competitive positioning against larger retail chains. The authors highlighted how automated inventory systems offer particular advantages for convenience stores in Luzon facing high customer traffic volumes, diverse product categories, and operational complexity requiring efficient inventory management beyond manual tracking capabilities.

The literature emphasized that convenience stores in Luzon traditionally manage larger product varieties than traditional small retail formats, with typical stores carrying thousands of items across multiple categories including beverages, snacks, groceries, personal care products, household items, and prepared foods requiring varied inventory management approaches. Garcia and Santos discussed how diverse product categories involve different demand patterns, turnover rates, and handling requirements complicating inventory management beyond simple quantity tracking requiring category-specific approaches. The authors explained how fast-moving consumer goods require frequent replenishment to prevent stockouts during peak periods, while slower-moving items require inventory management approaches that minimize capital tie-up in inventory that may not sell before expiration. The literature highlighted how perishable products including fresh foods, dairy items, and prepared meals require additional handling, rotation, and expiration monitoring that significantly complicate inventory management in warm Luzon climates.

Furthermore, the authors discussed how manual inventory methods create significant limitations affecting convenience store operations in competitive environments where product availability and service speed significantly impact customer satisfaction and repeat patronage. Manual inventory tracking requires physical counts consuming employee time that could otherwise focus on customer service, store cleanliness, and operational activities that affect customer experience in convenience-focused retail formats. The literature explained how manual processes delay problem identification, allowing stockouts to occur and persist until discovered during time-consuming physical counts that interrupt operations and leave customer needs unmet. Garcia and Santos discussed how manual receiving and checking processes at store delivery times create delays and potential discrepancies that complicate inventory accuracy and vendor relationships. The literature highlighted how manual inventory limits visibility into sales patterns, product performance, and demand variations that inform purchasing decisions affecting inventory investment returns.

The literature also highlighted how convenience stores face unique operational characteristics requiring automated inventory systems to address high transaction volumes and extended operating hours that characterize convenience retail formats in the Philippines. Multiple daily delivery shipments from various suppliers require efficient receiving processes that verify ordered quantities, check product quality, and update inventory quickly to maintain accurate stock visibility. The authors explained how hot food and fresh food preparation requires inventory coordination ensuring ingredients available for food preparation while preventing waste from overproduction and unsold prepared items. The literature discussed how ready-to-drink coffee, hot beverages, and food items require real-time inventory management throughout operating hours as products sell and refill in continuous cycles throughout day and night operations. Garcia and Santos explained how 24-hour and extended operating hours require inventory visibility and reordering capabilities that operate continuously regardless of staff presence, ensuring product availability for midnight shoppers.

Additionally, the authors explained how automated inventory systems improve convenience store performance through faster restocking, optimized inventory levels, and reduced losses from expiration, shrinkage, and mishandling that directly impact profitability in thin-margin convenience retail operations. Automated reorder triggers initiate purchasing when inventory falls below threshold levels, maintaining product availability while reducing excess inventory carrying costs that reduce profit margins on limited margins. Automated inventory tracking maintains accuracy that prevents ordering too much of slow-moving items and too little of fast-moving items that affects both sales potential and working capital efficiency. The literature discussed how expiration monitoring identifies products approaching expiration dates, enabling promotional pricing or return arrangements that reduce losses from expired products becoming unsalable. Garcia and Santos explained how shrinkage reduction from improved accountability, reduced handling errors, and better tracking decreases losses from theft, damage, and mishandling that significantly impact convenience store profitability.

The literature further discussed how automated inventory systems improve supplier relationships and procurement efficiency through electronic ordering, delivery coordination, and vendor management that streamline supply chain operations for stores serving diverse supplier networks. Purchase order generation initiates procurement through automated processes eliminating telephone orders, fax transmissions, and manual order preparation consuming management time previously devoted to customer service. The authors explained how vendor delivery scheduling coordinates with store receiving capacity, staff availability, and storage space to ensure efficient receiving processes without congestion or delays affecting store operations. The literature discussed how vendor performance tracking identifies suppliers with delivery reliability, product quality, or pricing issues affecting store operations that could be addressed through supplier discussion or replacement. Garcia and Santos explained how consolidated ordering across multiple stores in retail chains enables volume purchasing that improves supplier relationships through consistent ordering volumes that earn better pricing, delivery priority, and promotional support.

The authors also emphasized the importance of mobile accessibility for convenience store operators managing multiple locations throughout Luzon requiring oversight and management regardless of physical presence in any single store. Mobile inventory alerts notify operators when inventory requires attention, reordering, or problem resolution independent of desktop access that may be impractical in store operations. The literature discussed how mobile reporting provides sales, inventory, and financial information accessible from any location enabling management decisions regardless of geographic constraints. Garcia and Santos explained how mobile receiving and inventory check capabilities improve accuracy and efficiency at delivery times when employees must receive products quickly to maintain store operations. The literature discussed how location dashboards provide site-specific reporting enabling performance comparison across convenience stores that identifies best and worst performers for operational improvement.

Furthermore, the authors discussed how case study evidence from convenience stores in Luzon demonstrates tangible benefits from automated inventory system implementation addressing operational challenges specific to convenience retail formats. Inventory accuracy improved substantially from typical manual rates of approximately sixty to seventy percent to above ninety percent following automated system implementation, enabling reliable ordering decisions and reducing both stockouts and overstocking losses. The authors explained how stockout rates decreased substantially through automated reorder alerts that maintained product availability for customers throughout the day and night. The literature discussed how product losses from expiration decreased significantly through automated alerts and rotation management reducing waste from unsold products approaching expiration dates. Garcia and Santos explained how employee productivity improved through reduced manual tasks freeing employee time for customer service and store operations improving customer experience and operational efficiency.

The literature additionally highlighted how competitive pressures from various retail formats require convenience stores to differentiate through service, accessibility, and operational efficiency that automated inventory systems enable against larger competitors with more resources. The authors explained how extended operating hours and multiple daily shifts require efficient inventory management ensuring product availability throughout 24-hour operations without physical management presence continuous. The literature discussed how multiple daily delivery coordination requires efficient receiving processes that do not create congestion during brief receiving windows between busy operating periods. Garcia and Santos explained how quick turnover inventory investment requires efficient inventory management maximizing sales potential from limited working capital available for inventory investment. The literature discussed how competitive pricing requires efficient operations minimizing costs to enable competitive pricing while maintaining profitability adequate for business sustainability.

Moreover, the authors explained how automated inventory management principles for convenience stores apply to pharmacy and wellness retail contexts with appropriate adaptations for pharmaceutical, health, and medical product categories. The literature explained how expiration tracking and rotation management applies to over-the-counter medications, health supplements, and personal care products sold in convenience stores with similar shelf life considerations. Garcia and Santos discussed how cold chain management for refrigerated products including dairy, beverages, and cold items requires temperature monitoring approaches applicable to both convenience and pharmacy operations. The literature discussed how regulatory compliance for over-the- counter medications and medical devices extends applicable pharmaceutical inventory management principles to broader retail contexts. The authors explained how customer service focus, operational efficiency, and inventory optimization remain applicable across convenience and pharmacy retail applications serving consumer retail needs.

This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system addresses inventory management challenges faced by convenience stores in Luzon requiring high-volume inventory tracking, expiration management, and reordering capabilities. Similar to the concepts discussed by Garcia and Santos, the proposed AIMS aims to improve inventory accuracy, prevent stockouts, reduce losses, and optimize inventory investment through automated tracking and decision support.

The proposed system aligns with the authors' discussion regarding convenience store requirements including automated reordering, expiration tracking, mobile access, and supplier coordination that ensure appropriate functionality for high-volume retail operations. These capabilities address operational challenges specific to convenience store environments where customer expectations for product availability, extended operating hours, and quick service significantly impact system requirements beyond basic inventory management.

Furthermore, the literature supports the implementation of the proposed inventory management system because it demonstrates how automated inventory systems benefit convenience stores through documented improvements in accuracy, availability, and operational efficiency applicable to pharmacy and convenience store retail contexts. The concepts discussed in the study provide strong evidence supporting the development and implementation of the proposed Automated Inventory Management System within Philippine retail environments throughout Luzon and other regions of the Philippines.

2.2.13 Implementation Challenges of Inventory Management Systems in Philippine SME Retail

According to J. M. Fernandez and R. L. Bautista (2022), small and medium enterprise retailers throughout the Philippines face significant implementation challenges affecting inventory management system adoption, success, and realized benefits from technology investments. The authors explained that Philippine SME retailers represent a substantial portion of the retail sector, including independent stores, family businesses, and small chains that provide important employment, community accessibility, and local economic development throughout Philippine communities. These businesses serve essential retail functions providing convenient access to daily necessities within residential neighborhoods, rural communities, and provincial areas where larger retail formats may not have locations or may offer limited accessibility. Despite the potential benefits of inventory management systems documented in prior research, many Philippine SME retailers continue operating with manual processes that limit their competitive capabilities and growth potential. The authors highlighted how understanding implementation challenges affecting technology adoption success enables development of approaches that increase implementation success rates, maximize benefits realization, and improve technology investment returns for Philippine SME retailers.

The literature emphasized that financial constraints represent primary implementation challenges affecting technology adoption among Philippine SME retailers with limited capital resources available for technology investment despite potential benefits. Fernandez and Bautista discussed how technology costs including hardware, software, implementation services, and ongoing operational expenses exceed available capital in businesses operating with narrow profit margins and limited access to financing options. The authors explained how many Philippine SME retailers lack access to business loans, credit lines, or capital markets that larger businesses utilize for technology investment, creating capital barriers that limit options to lower-cost alternatives with reduced capabilities. The literature highlighted how total cost of ownership including implementation costs, training expenses, ongoing maintenance, and operational costs often exceeds initial budgets when businesses underestimate full investments required for successful implementation. These financial constraints significantly limit technology adoption among Philippine SME retailers despite documented benefits in prior research.

Furthermore, the authors discussed how technical expertise limitations create implementation barriers when Philippine SME retailers lack internal technology capabilities for system selection, implementation, and ongoing management. Many Philippine SME retailers operate with family members and minimal employees who lack technology backgrounds required for effective system evaluation, selection, and implementation approaches. The literature explained how system selection challenges arise when retailers lack criteria for evaluating alternative systems, assessing vendor capabilities, and determining appropriate solutions matching business requirements. Fernandez and Bautista discussed how implementation challenges including data migration, system configuration, and integration testing require technical expertise available only through external consultants with limited local availability in provincial areas. The literature highlighted how ongoing technical support requirements including troubleshooting, updates, and problem resolution exceed internal capabilities requiring vendor support that may be distant or expensive.

The literature also highlighted how organizational change management challenges affect technology implementation success when employees accustomed to traditional methods resist system adoption and utilization. Employee resistance arises from unfamiliarity with new technology, concerns about job security, and comfort with existing manual processes despite known inefficiencies. The authors explained how training challenges emerge when employees have limited technology backgrounds, education levels, and learning comfort that complicate system adoption requiring new skills and operational procedures. The literature discussed how workload disruption during implementation transition temporarily reduces productivity as employees learn new systems while maintaining customer service expectations and business operations. Fernandez and Bautista explained how management commitment challenges affect implementation success when business owners/ managers fail to provide adequate support, resources, or priority for implementation activities competing with daily operational demands.

Additionally, the authors explained how infrastructure limitations affecting internet connectivity, electrical power reliability, and technical support access create environmental challenges for technology implementation in various Philippine locations. Many business locations in provincial areas, rural communities, and older commercial buildings lack reliable internet connectivity required for cloud-based systems offering significant implementation advantages. The literature discussed how power interruptions affecting system operations, requiring data backup procedures, and disrupting customer service during extended outages when POS systems become inoperable. Fernandez and Bautista explained how local technical support availability varies significantly in provincial areas where service providers concentrate in Metro Manila and major urban centers creating support challenges for businesses outside these areas. The literature discussed how climate conditions including humidity, temperature, and seasonal weather variations affect technology equipment requiring environmental management not always available in Philippine business environments.

The literature further discussed how vendor and solution Selection challenges arise when Philippine SME retailers evaluate options from numerous vendors with varying capabilities, pricing structures, and support commitments. The authors explained how vendor reliability assessment challenges occur when evaluating local and international vendors with limited track records in Philippine contexts that potential customers can verify through local references. The literature discussed how solution appropriateness determination requires matching system capabilities to specific business requirements including industry, size, product mix, and operational complexity that varies across SME retailers. Fernandez and Bautista explained how pricing structure variation including upfront licensing, subscription models,

transaction fees, and support costs complicates comparison across systems with different economic models affecting total cost calculations. The literature discussed how contractual commitment assessment including contract terms, termination provisions, and data portability affects long-term relationship value when vendor performance falls below expectations.

The authors also emphasized the importance of phased implementation approaches that reduce immediate investment requirements while enabling capability building through progressive implementation that accommodates SME resource limitations. Phased approaches begin with core functionality that delivers immediate benefits, deferring advanced features until later implementation phases after benefits accumulate from initial investment. The literature discussed how pilot implementations in single locations demonstrate system viability and develop implementation experience before broader rollout that reduces risk and provides learning opportunities. Fernandez and Bautista explained how cloud-based deployment reduces upfront capital requirements through subscription pricing converting capital investments to operational expenses matching SME cash flow patterns. The literature discussed how modular implementations enable selection of required functionality only, avoiding payment for unnecessary features while maintaining upgrade paths for future capability expansion.

Furthermore, the authors discussed how successful implementation requires attention to change management activities including training, communication, and support that address human factors affecting adoption success and realized benefits. Comprehensive training programs accommodate employees with limited technology backgrounds through patient instruction, hands-on practice, and ongoing support addressing questions arising during daily operations. The literature explained how communication programs explain system rationale, implementation timelines, and expected benefits reducing resistance through understanding rather than mandated compliance. Fernandez and Bautista discussed how user support functions provide ongoing assistance addressing problems and questions that arise, maintaining user confidence and system utilization. The literature discussed how performance measurement tracks implementation progress and identifies issues requiring attention while demonstrating benefits that justify continued investment and organizational commitment.

The literature additionally highlighted how local context including language, business practices, and technical support availability affects implementation success requiring attention to Philippine-specific requirements in system selection and implementation approaches. Filipino language interfaces accommodate users more comfortable with local languages than English-only interfaces that create usability barriers for less educated users. The authors explained how local payment terms, credit arrangements, and business practices require flexibility in system configuration reflecting Philippine retail practices. The literature discussed how local supplier integration connects with Philippine distributors and vendors through systems that may not support international EDI formats or standard integration approaches. Fernandez and Bautista explained how local technical support availability varies by location, requiring distance support capabilities including remote assistance and communication technology when local support does not exist.

Moreover, the authors explained how addressing implementation challenges enables Philippine SME retailers to realize documented benefits from technology adoption improving competitive positioning and business sustainability. The literature explained how financial constraints can be addressed through phased implementations, cloud deployment, and vendor financing that reduce immediate capital requirements while enabling capability building. Fernandez and Bautista discussed how training approaches accommodate limited technical expertise through comprehensive programs and ongoing support addressing learning needs. The literature discussed how infrastructure solutions including backup power, offline capability, and mobile support address environmental limitations that otherwise prevent system operation. The authors explained how successful implementations demonstrate benefits that improve business outcomes through inventory optimization, service improvement, and competitive positioning that justify continued technology investment.

This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system addresses implementation challenges facing Philippine SME retailers requiring careful attention to financial constraints, technical limitations, and organizational change management issues. Similar to the concepts discussed by Fernandez and Bautista, the proposed AIMS aims to minimize implementation challenges through cost-effective solutions, user-friendly interfaces, and comprehensive support addressing identified barriers.

The proposed system aligns with the authors' discussion regarding implementation success factors including phased deployment, training programs, and local support that ensure successful adoption and benefit realization. These capabilities address operational challenges specific to Philippine SME environments where resource constraints, technical limitations, and organizational change issues significantly impact technology adoption success.

Furthermore, the literature supports the implementation of the proposed system because it demonstrates how implementation challenges can be addressed through appropriate approaches that enable technology benefits even in resource-constrained SME contexts. The concepts discussed in the study provide guidance for implementing the proposed Automated Inventory Management System within Philippine pharmacies and convenience stores while addressing identified challenges affecting implementation success.

2.3 Related Literature

2.3.1 Management Information Systems: Managing the Digital Firm

According to Kenneth C. Laudon and Jane P. Laudon (2020), management information systems represent essential organizational resources that enable businesses to achieve operational excellence, enhance competitive positioning, and meet evolving stakeholder expectations in today's digital economy. The authors explained that management information systems integrate business processes, organizational data, information technology, and managerial decision-making into cohesive frameworks that support organizational objectives and strategic initiatives. The discipline of management information systems extends beyond technological implementations to encompass the broader organizational context including business strategies, organizational structures, and managerial practices that significantly influence system effectiveness and organizational outcomes. The authors highlighted how digital transformation initiatives fundamentally reshape business models, operational processes, and competitive dynamics across industries, making information systems increasingly critical to organizational success. Modern organizations leverage information systems to improve operational efficiency, enhance customer experiences, enable data-driven decision- making, and respond more effectively to rapidly changing market conditions and competitive pressures. The literature emphasized that information systems serve multiple organizational purposes across various functional areas and management levels. At the operational level, information systems support day-to-day business activities including transaction processing, inventory control, production scheduling, and customer service operations that maintain organizational effectiveness and efficiency. At the management level, information systems enable middle managers to monitor departmental performance, allocate resources efficiently, and coordinate activities across different work units and functional areas. At the strategic level, information systems support senior executives in developing organizational strategies, analyzing market opportunities, and evaluating long-term investment decisions. The authors discussed how contemporary information systems provide comprehensive organizational perspectives through integrated data architectures and analytical capabilities that enable effective decision-making throughout organizational hierarchies. Information systems also facilitate organizational knowledge creation and dissemination by capturing business insights, preserving institutional memory, and enabling knowledge sharing across functional boundaries and organizational levels.

Furthermore, the authors discussed how information systems fundamentally transform business processes through automation, integration, and optimization capabilities that improve efficiency, reduce errors, and accelerate operational workflows. Business process re-engineering initiatives leverage information technologies to fundamentally redesign traditional business processes, eliminate unnecessary steps, and achieve dramatic performance improvements that would be impossible through incremental improvements alone. The literature highlighted how enterprise systems integrate organizational functions including finance, accounting, human resources, manufacturing, and supply chain operations within unified technological platforms that ensure data consistency, eliminate redundant processes, and enable seamless information flow across organizational boundaries. Business process management tools provide monitoring and analytical capabilities that enable continuous process improvement through ongoing performance measurement and optimization activities. The authors explained how workflow automation eliminates manual tasks, reduces processing times, and improves accuracy by automatically routing information and approvals through established business procedures without human intervention.

The literature also highlighted the importance of digital platforms and e-commerce in modern business environments. E-commerce technologies enable organizations to reach global markets, operate around the clock, and offer personalized products and services that transcend traditional brick-and-mortar limitations. The authors discussed how digital platforms create network effects that generate significant competitive advantages by connecting buyers, sellers, and service providers within expansive ecosystems that become increasingly valuable as participation grows. Mobile commerce extends organizational reach by enabling transactions and customer interactions through smartphone devices that provide constant connectivity and convenience. Social commerce leverages social media platforms to enable peer recommendations, brand advocacy, and community-based shopping experiences that influence purchasing decisions. The literature discussed how digital marketplaces transform traditional industry structures by creating transparent, efficient markets that reduce barriers to entry and intensify competitive pressures on established organizations.

Additionally, the authors explained how business intelligence and analytics technologies transform organizational decision-making by providing comprehensive data analysis capabilities that reveal patterns, trends, and insights not visible through traditional reporting methods. Data analytics enables organizations to understand customer preferences, predict market trends, optimize operational processes, and identify emerging opportunities that inform strategic planning and resource allocation decisions. The authors discussed how big data technologies handle massive volumes of structured and unstructured data that exceed traditional database management system capabilities, enabling organizations to analyze diverse data sources including social media content, sensor data, and machine logs. Machine learning and artificial intelligence extend analytical capabilities by automatically identifying patterns and generating predictions that improve over time as additional data becomes available. Prescriptive analytics goes beyond descriptive and predictive analytics by recommending specific actions that optimize organizational outcomes based on analytical models and constraint satisfaction algorithms.

The literature further discussed how information systems support customer relationship management and enhance customer experiences through comprehensive data management and personalized interaction capabilities. Customer relationship management systems consolidate customer information from multiple interaction channels including sales, service, and marketing communications into unified profiles that enable personalized engagement strategies. The authors explained how analytical CRM capabilities identify customer segments, predict customer behavior, and recommend targeted offers that maximize customer lifetime value while improving acquisition efficiency and retention rates. Personalization technologies tailor product recommendations, content displays, and marketing messages to individual customer preferences and behaviors that increase engagement and conversion rates. The literature discussed how service automation improves customer support efficiency while maintaining service quality through self-service portals, automated response systems, and intelligent routing capabilities that direct inquiries to appropriate support resources.

The authors also emphasized the importance of enterprise resource planning and supply chain management systems in integrated business operations. Enterprise resource planning systems integrate financial management, accounting, human resources, manufacturing, procurement, and supply chain operations within comprehensive platforms that ensure data consistency and enable seamless organizational coordination. The literature discussed how real-time visibility across organizational functions enables managers to identify issues quickly and implement corrective actions before problems escalate. Supply chain management systems improve coordination with suppliers, distributors, and logistics partners through information sharing, collaborative planning, and coordinated execution capabilities that reduce costs and improve delivery performance. Internet of Things technologies extend supply chain visibility by providing real-time tracking of products, equipment, and environmental conditions throughout global supply networks. The authors explained how advanced planning and scheduling systems optimize production, warehousing, and distribution activities to minimize costs while meeting customer service requirements.

Furthermore, the literature discussed how information systems enable organizational innovation and digital transformation initiatives that create new business models, revenue streams, and competitive advantages. Digital transformation involves comprehensive organizational change that integrates digital technologies into all business functions, fundamentally altering how organizations operate, compete, and deliver value to stakeholders. The authors discussed how digital business models leverage platform architectures, network effects, and data assets to create scalable businesses with minimal marginal costs and significant growth potential. Internet of Things technologies enable new product and service offerings that combine physical products with digital capabilities including monitoring, diagnostics, and connected services. Cloud computing provides scalable, on-demand technology infrastructure that enables organizations to experiment with new capabilities without significant capital investments. Digital ecosystems create collaborative networks that enable organizations to leverage complementary capabilities and access broader markets than individual organizations could serve independently.

The literature additionally highlighted the importance of information systems governance, security, and risk management in protecting organizational assets and ensuring business continuity. Information systems governance establishes policies, procedures, and organizational structures that ensure appropriate technology investments, resource allocation, and performance monitoring that align with organizational strategic objectives. Security frameworks implement layered protections including access controls, authentication mechanisms, encryption, intrusion detection, and security monitoring that safeguard organizational information assets against unauthorized access and cyber threats. The authors discussed how security awareness training reduces human error and social engineering vulnerabilities that represent significant organizational risk exposures. Business continuity and disaster recovery planning ensures organizational resilience by establishing backup systems, recovery procedures, and crisis communication protocols that minimize disruption during unexpected events. Risk management identifies potential threats, assesses likelihood and impact, and develops mitigation strategies that reduce organizational risk exposure to acceptable levels. Moreover, the authors explained how information systems support organizational compliance with regulations, industry standards, and ethical requirements. Regulatory compliance including financial reporting standards, data privacy regulations, and industry-specific requirements necessitates information system capabilities that ensure appropriate documentation, controls, and reporting. The literature discussed how compliance management systems track regulatory requirements, monitor compliance status, and generate required documentation that demonstrates organizational adherence to applicable standards. Data governance establishes policies and procedures that ensure appropriate data quality, privacy, and security while enabling legitimate business operations and analytics activities. Ethics frameworks guide organizational decisions regarding data use, algorithm development, and technology deployment that balance business objectives with societal impacts and stakeholder expectations. The authors explained how audit trails and documentation support internal and external audit activities that verify system effectiveness and organizational compliance.

The authors also emphasized the importance of information systems in supporting organizational ethical conduct and social responsibility. Information systems can be designed and deployed in ways that promote transparency, accountability, and fairness while preventing misuse that harms stakeholders or society. Algorithmic fairness considerations address potential biases in machine learning models that could discriminate against protected groups or perpetuate historical inequities. Data privacy protections ensure that personal information is collected, stored, and used appropriately with appropriate consent and security measures. The literature discussed how corporate social responsibility initiatives leverage information systems to improve environmental sustainability, support community development, and promote ethical supply chain practices. Digital inclusion efforts ensure that technology benefits reach underserved populations rather than creating new forms of digital divides that exclude disadvantaged communities from economic opportunities.

This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system represents a comprehensive management information system implementation that addresses multiple organizational needs including transaction processing, inventory management, customer service, and decision support capabilities. Similar to the concepts discussed by Laudon and Laudon, the proposed AIMS integrates business processes, organizational data, and management practices within unified frameworks that enhance operational effectiveness and support strategic objectives. The proposed system also aims to leverage digital technologies including barcode scanning, automated transaction processing, real-time reporting, and analytical capabilities that improve operational efficiency and enable data-driven decision-making. These information system capabilities align directly with the authors' discussion regarding how information systems transform business processes and enhance organizational decision-making through comprehensive data management and analytical capabilities.

Furthermore, the literature supports the implementation of automated inventory management systems because it demonstrates how management information systems contribute to organizational competitive advantage, customer satisfaction, and operational excellence. The concepts discussed in the study provide a strong theoretical and practical foundation for the development and implementation of the proposed Automated Inventory Management System within pharmacies and convenience store environments.

2.3.2 Database System Concepts

According to Abraham Silberschatz, Henry F. Korth, and S. Sudarshan (2019), database systems form the foundation of modern information management by providing organized data storage, efficient retrieval capabilities, and reliable data manipulation functionalities that support organizational operations and decision-making processes. The authors explained that database systems address critical challenges related to data organization, access efficiency, and information reliability that organizations face in managing vast amounts of operational data. The discipline of database systems encompasses theoretical foundations, practical implementation techniques, and management methodologies that ensure effective data utilization throughout organizational operations. Database systems extend beyond simple data storage to include comprehensive frameworks for data modeling, query processing, transaction management, and concurrency control that ensure data integrity and system reliability. The authors highlighted how contemporary database technologies have evolved significantly to address emerging requirements including big data analytics, cloud computing, and real-time processing that support modern business operations and strategic decision-making.

The literature emphasized that relational database management systems (RDBMS) represent the predominant approach for organizing and managing structured data within contemporary organizations. Relational databases organize data into tables consisting of rows and columns, where each row represents a unique record and each column represents an attribute or characteristic of the entities being stored. The authors discussed how relational databases provide intuitive data representation, powerful query capabilities, and theoretical foundations based on set theory and relational algebra that enable precise data manipulation and retrieval. Structured Query Language (SQL) provides standardized commands for creating databases, defining schemas, manipulating data, and querying information that enable consistent database operations across different database management systems. The literature highlighted how SQL-based querying enables complex data retrieval operations including joins, aggregations, sub queries, and set operations that would be impractical to implement through procedural programming approaches. Furthermore, the authors discussed how database design methodologies ensure effective data organization that supports organizational requirements while maintaining data integrity and retrieval efficiency. Entity-relationship (ER) modeling provides graphical techniques for representing organizational data requirements, identifying entities, attributes, and relationships that must be captured in database designs. The literature discussed how conceptual database designs translate ER models into relational schemas that can be implemented in database management systems. Database normalization organizes data into properly structured tables that eliminate redundancy and prevent update anomalies including insertion, deletion, and modification anomalies that compromise data integrity. The authors explained how normalization decomposes complex relations into smaller, well-structured relations that maintain data relationships through foreign keys while eliminating unnecessary data duplication. Denormalization sometimes intentionally introduces controlled redundancy to improve query performance in read-intensive applications where retrieval speed is more important than update complexity.

The literature also highlighted the importance of transaction management in ensuring database reliability and data integrity. Transactions represent atomic units of work that either complete successfully and update the database or fail completely without affecting database state. The authors discussed how transaction properties including atomicity, consistency, isolation, and durability (ACID) guarantee reliable database operations even in the presence of system failures, concurrent access, and operational errors. Atomicity ensures that all transaction operations complete successfully or none of them are applied, preventing partial updates that would leave the database in inconsistent states.

Consistency guarantees that transactions transform the database from one valid state to another, maintaining all integrity constraints and business rules.

Isolation ensures that concurrent transactions appear to execute sequentially, preventing interference that could produce incorrect results. Durability guarantees that committed transaction results persist permanently in the database, surviving system failures and media crashes. Additionally, the authors explained how concurrency control mechanisms ensure reliable database operations in multi-user environments where multiple transactions may access data simultaneously. Lock-based concurrency control techniques serialize transaction execution by acquiring locks on data items that prevent other transactions from accessing locked data until the holding transaction completes. The literature discussed how shared locks permit concurrent read access while exclusive locks prevent both read and write access by other transactions. Two- phase locking protocols ensure serializability by acquiring all necessary locks before releasing any locks, guaranteeing that transaction execution follows serializable schedules. Timestamp-based concurrency control assigns unique timestamps to transactions and ensures that transactions execute in timestamp order, preventing conflicts through validation procedures. The authors explained how multiversion concurrency control (MVCC) improves performance by maintaining multiple versions of data items, allowing readers to access consistent snapshots while writers create new versions. The literature further discussed how database recovery mechanisms ensure database integrity and availability following system failures, media crashes, or operational errors.

Recovery procedures utilize logging techniques that record transaction operations and database modifications to ensure that committed updates persist and uncommitted updates can be undone. Write-ahead logging ensures that transaction descriptions are recorded in durable storage before corresponding database modifications are applied, guaranteeing that recovery can reconstruct database states even if system failures occur during transaction execution. The authors discussed how checkpointing procedures periodically record database states to reduce recovery times by limiting the amount of transaction logs that must be processed during recovery. Media recovery procedures restore databases from backup copies and reapply logged transactions to recover from storage media failures. The literature highlighted how recovery management minimizes data loss and ensures business continuity following unexpected events.

The authors also emphasized the importance of database indexing in improving query performance and system responsiveness. Indexes provide rapid data access paths that enable database management systems to retrieve specific records without scanning entire tables. The authors discussed how B-tree indexes maintain balanced tree structures that provide efficient search, insertion, and deletion operations with logarithmic time complexity. Primary built on primary keys, providing the ability to access specific records directly, and secondary indexes provide alternative access paths that support efficient retrieval of frequently queried non-primary key attributes. Hash indexes provide constant-time access performance in specific equality query scenarios by mapping search key values to database pages to improve retrieval efficiency. Columns with lower selectivity benefit less from indexes because full table scans may be more efficient.

Furthermore, the literature discussed how query processing and optimization techniques ensure efficient database operations. Query processors translate SQL statements into execution plans that retrieve or manipulate data according to user requirements. The authors explained how query optimizers analyze SQL statements and select efficient execution strategies based on database statistics, indexes, and access paths. Rule-based optimization applies heuristic techniques that favor certain access patterns, while cost-based optimization evaluates multiple execution plans and selects the plan with the lowest estimated cost. Join ordering significantly impacts query performance in queries involving multiple tables, making effective join optimization essential for complex queries. The literature discussed how materialized views precompute and store query results that can be queried directly, improving performance for frequently executed queries while enabling query rewriting optimizations.

The literature additionally highlighted the importance of data integrity constraints in ensuring database accuracy and consistency. Primary key constraints uniquely identify records within tables, preventing duplicate entries and enabling efficient record retrieval and relationship establishment. Foreign key constraints enforce referential integrity by ensuring that related records exist before insert or update operations are allowed, maintaining relationships between related tables. Unique constraints ensure that attribute values remain distinct across records, preventing duplicate entries for attributes that must remain unique. Check constraints validate attribute values against specified conditions, preventing invalid data entry that violates business rules. The authors discussed how triggers provide programmable integrity enforcement by executing defined actions in response to database events including insert, update, and delete operations. Assertion-based constraints enforce multi-table integrity requirements that cannot be enforced through table-level constraints.

Moreover, the authors explained how database security measures protect sensitive organizational information against unauthorized access and misuse. User authentication verifies user identities before granting database access,

typically through password-based mechanisms that may be enhanced with multi-factor authentication for improved security. The literature discussed how authorization controls grant specific permissions that determine which database objects users can access and what operations they can perform on those objects. Role-based security simplifies permission management by grouping related permissions into roles that can be assigned to users, reducing administrative overhead and ensuring consistent security policies. Views provide security mechanisms by limiting data exposure to specific subsets of database records, hiding sensitive information while enabling appropriate access. The authors explained how audit trails record database access and modification activities,
enabling security monitoring and forensic analysis following suspected security incidents.

The literature also discussed how database architecture options support different organizational requirements including performance, scalability, availability, and
cost considerations. Centralized database systems consolidate data storage and processing on single database servers that provide consistent data access and centralized management. Distributed database systems distribute data across multiple connected sites that may be geographically separated, improving
performance through data localization while providing fault tolerance through replication. The authors discussed how client-server architectures separate database processing between clients that submit queries and servers that
execute queries and manage data storage. Cloud database services provide on- demand database capabilities without capital investment in database
infrastructure, enabling organizations to scale resources based on demand while reducing operational overhead. The literature highlighted how NoSQL databases address specific requirements including scalability, flexibility, and performance that traditional relational databases may not optimally support.

Furthermore, the authors explained how database connectivity and application integration extend database capabilities beyond standalone systems. Database APIs provide programmatic interfaces that enable applications to interact with
databases through well-defined function calls rather than direct SQL parsing. The literature discussed how ODBC (Open Database Connectivity) and JDBC (Java Database Connectivity) provide standardized interfaces that enable applications to access different database management systems through common
programming interfaces.

Object-relational mapping (ORM) frameworks simplify database programming by mapping object-oriented programming concepts to relational database
structures, enabling developers to work with familiar object abstractions rather than SQL queries. The authors explained how stored procedures encapsulate business logic within database management systems, improving performance through reduced network traffic and enhanced security through controlled
execution.

This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system requires robust database capabilities to store and manage inventory records, transaction data, product information, customer details, and operational reports. Similar to the concepts discussed by Silberschatz, Korth, and Sudarshan, the proposed AIMS

will utilize relational database technologies that provide organized data storage, efficient retrieval capabilities, and reliable data manipulation functionalities.
The proposed system also aims to implement proper database design including normalization, indexing, and integrity constraints to ensure data integrity and retrieval efficiency. Transaction management and concurrency control will ensure reliable database operations in multi-user environments where multiple employees access and update inventory information simultaneously.
Furthermore, the literature supports the implementation of database-driven inventory management systems because it demonstrates how database technologies contribute to operational accuracy, data reliability, and efficient information management. The concepts discussed in the study provide a strong technical foundation for the development and implementation of the proposed Automated Inventory Management System within retail environments.

2.3.3 Software Engineering: A Practitioner's Approach

According to Roger S. Pressman and Bruce R. Maxim (2019), software engineering provides the systematic methodologies, techniques, and tools necessary to develop and maintain high-quality software systems that meet user requirements and organizational objectives. The authors explained that software engineering addresses the fundamental challenges of producing complex software solutions within budget constraints, on schedule, and meeting quality standards that satisfy stakeholder expectations. The discipline of software engineering encompasses all activities from initial requirements gathering and analysis through system design, implementation, testing, deployment, and ongoing maintenance throughout the software lifecycle. Software engineering provides structured approaches that guide development teams through established processes, methodologies, and best practices that have evolved from decades of software development experience and lessons learned from both successful projects and notable failures. The authors highlighted how modern software engineering continues to evolve in response to changing technological landscapes, emerging development paradigms, and increasing stakeholder expectations regarding software quality, reliability, and timeliness.
The literature emphasized that requirements engineering represents one of the most critical phases in software development, as errors introduced during requirements gathering often prove the most expensive and time-consuming to correct later in the development process. The authors discussed how effective requirements engineering involves understanding stakeholder needs, documenting functional and non-functional requirements, and establishing clear acceptance criteria that can be used to verify system correctness and completeness. Functional requirements describe specific behaviors and functionalities that the software system must exhibit, including user interface requirements, data processing capabilities, and integration requirements with other systems. Non-functional requirements specify quality characteristics including performance, reliability, usability, security, and maintainability that determine overall system quality and user satisfaction. Requirements traceability supports successful development by enabling teams to track requirements

throughout the design, implementation, and testing phases, ensuring that all documented requirements are addressed in the delivered system while maintaining clear relationships between requirements and system components.
Furthermore, the authors discussed how software architecture serves as the foundation for system design, providing high-level abstractions that guide detailed design decisions and facilitate communication among stakeholders including developers, testers, project managers, and end users. Effective software architectures encapsulate system complexity by organizing functionality into distinct components with well-defined interfaces, clear responsibilities, and established communication protocols that enable independent development and testing of individual components. Architectural decisions significantly impact system qualities including performance, reliability, security, and maintainability, making it essential that development teams carefully evaluate architectural alternatives and select approaches that best address project requirements, constraints, and quality objectives. The literature highlighted how architectural patterns and styles provide proven solutions to common design problems, enabling development teams to leverage established best practices rather than attempting to develop original solutions for well-understood challenges. Object-oriented architectures provide additional benefits through encapsulation, inheritance, and polymorphism that promote code reuse, flexibility, and maintainability. Service-oriented architectures enable integration of distributed, loosely-coupled components that can be independently developed, deployed, and scaled.

The literature also highlighted the importance of software design in translating requirements into technical specifications that guide implementation activities. Effective design methodologies enable developers to create modular, scalable, and maintainable system architectures that support future enhancements and modifications while minimizing the impact of changes on existing functionality. The authors discussed how design documentation provides detailed records of design decisions, rationale, and architectural considerations that guide implementation teams and support future maintenance activities. Interface design specifications define how system components interact with each other and with external systems, enabling parallel development and testing activities. Data design organizes information storage and management approaches that ensure efficient access while maintaining data integrity and security. The literature discussed how design reviews and inspections identify design flaws and improvement opportunities early in the development process when corrections are less expensive and time-consuming to implement.

Additionally, the authors explained how software testing ensures system quality and reliability through systematic evaluation of system functionality and performance against established requirements and quality standards. Effective testing strategies combine various testing approaches including unit testing, integration testing, system testing, and acceptance testing to identify defects at different stages of the development process and at different levels of system granularity. Unit testing evaluates individual program components including functions, methods, and classes in isolation to verify that each component operates correctly according to its specifications. Integration testing evaluates

the interaction between components and subsystem combinations to identify interface defects and communication problems that may not be apparent in component-level testing. System testing evaluates the complete integrated system against documented requirements to verify end-to-end functionality and quality characteristics including performance, security, and reliability.
Acceptance testing validates system suitability for deployment by evaluating system functionality with actual users in real or representative operational environments.
The literature further discussed how various testing techniques enable effective test case design and comprehensive defect identification. Black-box testing techniques derive test cases from requirements specifications and design documentation without knowledge of internal system structure, focusing on input-output relationships and functional behavior. White-box testing techniques utilize knowledge of internal system structure, code logic, and execution paths to design test cases that verify internal functionality and achieve comprehensive code coverage. Equivalence partitioning divides input domains into groups of equivalent values that are expected to produce similar system behavior, enabling representative testing of each partition rather than exhaustive testing of all possible inputs. Boundary value analysis focuses testing efforts on input boundaries where errors frequently occur, improving testing efficiency by targeting high-risk input conditions. The authors explained how test-driven development practices improve software quality by specifying expected behavior before implementation and continuously verifying implementation correctness throughout the development process. The authors also emphasized the importance of software project management in ensuring successful software development outcomes. Effective project management involves careful planning, risk assessment, resource allocation, progress monitoring, and stakeholder communication that enable teams to deliver systems on time and within budget while meeting quality requirements. The literature discussed how project estimation techniques help teams develop realistic schedules and budgets by drawing on historical project data, expert judgment, and algorithmic approaches that account for project complexity, team capabilities, and environmental factors. Activity planning and scheduling define project tasks, durations, dependencies, and resource requirements that guide project execution and progress monitoring. Progress tracking and control activities enable early identification of problems and implementation of corrective actions before issues significantly impact project outcomes, schedules, or budgets. The authors explained how earned value analysis provides quantitative measures of project performance by comparing planned and actual progress against budgets and schedules.
Furthermore, the literature discussed how agile software development methodologies have transformed modern software engineering practices by emphasizing iterative development, customer collaboration, and adaptive planning that enable teams to respond effectively to changing requirements and deliver functional systems incrementally. Agile methodologies including Scrum, Extreme Programming, and Feature-Driven Development provide frameworks for organizing development activities around short iterations called sprints that deliver incremental functionality within time-boxed periods. The authors

discussed how daily stand-up meetings, sprint planning, and retrospective reviews support continuous improvement in development processes and team performance through regular inspection and adaptation. Agile approaches have proven particularly effective in dynamic business environments where requirements evolve rapidly and competitive pressures demand frequent system enhancements. However, the literature highlighted how agile methodologies may not be appropriate for all projects, particularly those in regulated industries where comprehensive documentation, rigorous testing, and formal verification provide necessary assurances regarding system quality and compliance.

The literature additionally highlighted the importance of software quality assurance in ensuring delivered systems meet established quality standards and stakeholder expectations. Quality assurance activities encompass all planned and systematic actions necessary to provide adequate confidence that software products and processes achieve specified quality requirements. The authors discussed how formal technical reviews examine software products including requirements documents, design specifications, and code implementations to identify defects and improvement opportunities before they propagate to later development phases. Code inspections and walkthroughs enable peer examination of code implementations to identify defects, security vulnerabilities, and maintainability issues that may not be apparent to original developers.

Software metrics provide quantitative measures of code quality, process performance, and product characteristics that support evidence-based decision making and continuous improvement efforts. Quality management systems provide frameworks for establishing quality objectives, implementing quality processes, and monitoring quality performance throughout the software lifecycle.

Moreover, the authors explained how software configuration management and version control track system changes and maintain integrity throughout the development lifecycle. Configuration management practices enable teams to manage multiple development streams, coordinate parallel development activities, and maintain organized records of system evolution over time. The literature discussed how version control systems track changes to source code, documentation, and other development artifacts, enabling rollback to previous versions when problems are discovered and supporting parallel development of feature branches. Baseline management establishes reference points in system development that define stable configurations for testing, integration, and release activities. Change control processes evaluate proposed modifications, assess their impacts on system functionality, schedule, and budget, and coordinate approved changes through systematic implementation procedures. The authors explained how build management automates compilation, testing, and integration activities that produce system releases from source code and configuration files.

The literature also discussed how software maintenance accounts for the majority of software lifecycle costs and represents ongoing investment in system functionality, performance, and reliability. Maintenance activities include corrective maintenance that addresses defects discovered during system operation, adaptive maintenance that modifies systems to accommodate changing technical environments, perfective maintenance that improves system functionality and performance, and preventive maintenance that addresses potential problems before they impact system operations. The authors explained how effective maintenance requires comprehensive system documentation, well- structured code, and proper configuration management practices that track system changes and maintain version control. Legacy systems present unique maintenance challenges as they often operate on outdated technology platforms, lack adequate documentation, and require specialized knowledge that may no longer be available within organizations. Software reengineering and migration strategies help organizations address legacy system challenges by modernizing existing systems or transitioning to new technology platforms while preserving accumulated business knowledge and functionality.

This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system requires systematic software engineering methodologies to ensure successful development, implementation, and long-term maintenance. Similar to the concepts discussed by Pressman and Maxim, the proposed AIMS will follow structured development processes including requirements engineering, system design, implementation, testing, and maintenance throughout the software lifecycle.

The proposed system also aims to apply quality assurance practices including requirements traceability, design reviews, comprehensive testing, and configuration management to ensure system reliability and maintainability.

These software engineering principles align directly with the authors' discussion regarding effective software development methodologies and quality assurance approaches.

Furthermore, the literature supports the implementation of the proposed system because it demonstrates how systematic software engineering approaches contribute to successful software delivery and long-term system sustainability.

The concepts discussed in the study provide a strong technical foundation for the development and implementation of the proposed Automated Inventory Management System within retail environments.

2.3.4 Ian Sommerville -*Software Engineering*, 10th Edition (2016), Pearson.

According to Ian Sommerville (2016), software engineering is a systematic discipline that integrates processes, methods, techniques, and tools for the successful development and maintenance of high-quality software systems. The author explained that software engineering addresses the challenges of developing complex software solutions that meet user requirements while adhering to budgetary constraints and quality standards. Software engineering provides structured approaches that guide development teams through the entire software lifecycle, from initial requirements gathering through system deployment, operation, and maintenance. The discipline emphasizes the importance of balancing technical excellence with practical considerations including project schedules, resource availability, and stakeholder expectations. Software engineering methodologies have evolved significantly over the years, incorporating lessons learned from both successful projects and notable software failures that have resulted in significant financial losses and reputational damage for organizations worldwide.

The literature emphasized that requirements engineering represents one of the most critical phases in software development, as errors introduced during this phase often prove the most expensive to correct later in the development process. Effective requirements engineering involves understanding stakeholder needs, documenting functional and non-functional requirements, and establishing clear acceptance criteria that can be used to verify system correctness. The author discussed how requirements traceability supports system development by enabling teams to track requirements throughout the design, implementation, and testing phases, ensuring that all documented requirements are addressed in the delivered system. Requirements management continues throughout the software lifecycle as user needs evolve and circumstances change, necessitating proper change management procedures that evaluate proposed modifications and assess their impacts on system functionality, schedule, and budget. Modern requirements engineering also considers emerging technologies and their potential applications, enabling organizations to leverage innovative solutions that provide competitive advantages in their respective markets.

Furthermore, the author discussed how software architecture serves as the foundation for system design, providing high-level abstractions that guide detailed design decisions and facilitate communication among stakeholders. Effective software architectures encapsulate system complexity by organizing functionality into distinct components with well-defined interfaces and responsibilities. Architectural decisions significantly impact system qualities including performance, reliability, security, and maintainability, making it essential that development teams carefully evaluate architectural alternatives and select approaches that best address project requirements. The literature highlighted how architectural patterns and styles provide proven solutions to common design problems, enabling development teams to leverage established best practices rather than attempting to develop original solutions for well-understood challenges.

Architectural documentation supports system evolution by providing clear records of design rationale and system organization that guide future enhancement and maintenance activities.

The literature also highlighted the importance of software testing in ensuring system quality and reliability. Effective testing strategies combine various testing approaches including unit testing, integration testing, system testing, and acceptance testing to identify defects at different stages of the development process. The author explained how test-driven development and continuous integration practices improve software quality by enabling early defect detection and preventing the introduction of new defects during ongoing development activities. Testing automation reduces manual testing efforts and enables more comprehensive test coverage that would be impractical to achieve through manual testing alone. Software testing requires careful planning, including the development of test plans that document testing objectives, approaches, resource requirements, and schedules. Test case design techniques including equivalence partitioning, boundary value analysis, and state transition testing provide systematic approaches for identifying effective test cases that expose system defects while managing testing costs and efforts.

Additionally, the author explained how software evolution and maintenance represent significant portions of the software lifecycle, often accounting for more than half of total lifecycle costs. Software maintenance activities include corrective maintenance that addresses defects, adaptive maintenance that modifies systems to accommodate changing environments, perfective maintenance that improves system functionality and performance, and preventive maintenance that addresses potential problems before they impact system operations. Effective maintenance requires comprehensive system documentation, well-structured code, and proper configuration management practices that track system changes and maintain version control. Legacy systems present unique maintenance challenges as they often operate on outdated technology platforms, lack adequate documentation, and require specialized knowledge that may no longer be available within organizations. Software reengineering and migration strategies help organizations address legacy system challenges by modernizing existing systems or

transitioning to new technology platforms while preserving accumulated business knowledge and functionality. The literature further discussed how agile software development methodologies have transformed modern software engineering practices by emphasizing iterative development, customer collaboration, and adaptive planning.
Agile methodologies enable development teams to respond effectively to changing requirements and deliver functional systems incrementally rather than attempting to complete entire systems before releasing any functionality. The author highlighted how agile practices including daily stand-up meetings, sprint planning, and retrospective reviews support continuous improvement in development processes and team performance. Agile approaches have proven particularly effective in dynamic business environments where requirements evolve rapidly and competitive pressures demand frequent system enhancements. However, agile methodologies may not be appropriate for all projects, particularly those in regulated industries where comprehensive documentation and rigorous testing provide necessary assurances regarding system quality and compliance.

The author also emphasized the importance of software project management in ensuring successful software development outcomes. Effective project management involves careful planning, risk management, resource allocation, and progress monitoring that enable teams to deliver systems on time and within budget. The literature discussed how project estimation techniques help teams develop realistic schedules and budgets by drawing on historical project data and expert judgment. Project tracking and control activities enable early identification of problems and implementation of corrective actions before issues significantly impact project outcomes. Software metrics provide quantitative measures of process and product quality that support evidence-based decision making and continuous improvement efforts.

Furthermore, the literature discussed how software quality assurance encompasses all activities that ensure delivered systems meet established quality standards and stakeholder expectations. Quality assurance activities include formal technical reviews, software testing, and process auditing that identify defects and ensure compliance with established procedures. The author explained how quality management systems provide frameworks for establishing quality objectives, implementing quality processes, and monitoring quality performance. Software quality models define quality characteristics including functionality, reliability, usability, efficiency, maintainability, and portability that guide quality assurance activities and enable objective quality assessment.

The literature additionally highlighted the importance of human factors in software engineering success. Effective teamwork, clear communication, and appropriate leadership significantly impact project outcomes, making it essential that development teams possess not only technical skills but also interpersonal and communication capabilities. The author discussed how global software development presents unique challenges related to cultural differences, time zone variations, and communication barriers that require careful management to ensure project success. Software engineering ethics provide guidance for professional conduct, emphasizing responsibilities to employers, clients, users, and society at large.

Moreover, the author explained how software engineering tools and environments support development activities by providing automated capabilities for coding, testing, configuration management, and project management. Integrated development environments (IDEs) improve developer productivity by combining source code editing, build automation, testing frameworks, and debugging capabilities within unified tool interfaces.
Computer-aided software engineering (CASE) tools support various development activities including requirements modeling, design, code generation, and testing.
Tool selection should consider project requirements, team capabilities, and

organizational constraints to ensure that selected tools provide appropriate support without imposing unnecessary complexity. This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system requires comprehensive software engineering methodologies to ensure successful development, implementation, and long-term maintenance. Similar to the concepts discussed by Sommerville, the proposed AIMS will follow structured development processes including requirements engineering, system design, implementation, testing, and maintenance throughout the software lifecycle.
The proposed system also aims to apply quality assurance practices, comprehensive documentation, and proper configuration management to ensure system reliability and maintainability. These software engineering principles align directly with the author's discussion regarding effective software development methodologies and quality assurance approaches.
Furthermore, the literature supports the implementation of the proposed system because it demonstrates how systematic software engineering approaches contribute to successful software delivery and long-term system sustainability. The concepts discussed in the study provide a strong technical foundation for the development and implementation of the proposed Automated Inventory Management System within retail environments.

2.3.5 Principle of Information System

According to Ralph M. Stair and George W. Reynolds (2018), information systems have become indispensable organizational resources that support business operations, enable strategic decision-making, and provide competitive advantages in modern marketplace environments. The authors explained that information systems integrate people, processes, data, and technology into cohesive frameworks that enable organizations to achieve their objectives and meet stakeholder expectations. The discipline of information systems encompasses the study of how organizations utilize technology to manage information, support business processes, and facilitate organizational communication and collaboration. Information systems extend beyond mere technology implementations to include the broader organizational context including business strategies, organizational cultures, and human behaviors that significantly impact system effectiveness and organizational outcomes. The authors highlighted how successful organizations leverage information systems to improve operational efficiency, enhance customer experiences, and respond more effectively to changing market conditions and competitive pressures.
The literature emphasized that information systems serve multiple organizational functions across various levels of management including operational, tactical, and strategic levels. At the operational level, information systems support daily business activities including transaction processing, inventory management, and customer service operations that keep organizations functioning effectively. At the tactical level, information systems enable middle managers to monitor organizational performance, allocate resources, and coordinate activities across different departments and work units. At the strategic level, information systems support senior executives in developing organizational strategies, evaluating long-term performance, and anticipating future trends and challenges. The authors discussed how integrated

information systems provide comprehensive views of organizational operations that enable effective decision-making at all organizational levels. Information systems also facilitate organizational learning by capturing and preserving business knowledge that can be applied to improve future operations and strategic planning activities. Furthermore, the authors discussed how information systems support organizational business processes through automation, integration, and optimization capabilities that improve efficiency and effectiveness. Business process automation eliminates manual tasks, reduces processing errors, and accelerates operational workflows that enable organizations to process transactions more quickly and accurately. The literature highlighted how process integration connects different organizational functions including accounting, inventory, human resources, and customer relationship management within unified systems that eliminate data silos and ensure information consistency across organizational operations. Process optimization applies analytical techniques to identify inefficiencies, bottlenecks, and improvement opportunities that enable organizations to continuously enhance operational performance. The authors explained how business process management (BPM) methodologies provide structured approaches for analyzing, designing, and improving business processes through ongoing monitoring and refinement activities.

The literature also highlighted the importance of data and knowledge management within information systems. Organizations generate and accumulate vast amounts of operational data that must be properly managed, stored, and retrieved to support business decision-making and operational activities. The authors discussed how database management systems (DBMS) provide organized data storage, efficient retrieval capabilities, and data integrity controls that ensure information accuracy and consistency. Data warehouse technologies enable organizations to consolidate data from multiple operational systems and support analytical processing activities that reveal trends, patterns, and insights not visible in transaction-level data. Knowledge management systems capture and preserve organizational expertise, best practices, and lessons learned that can be applied to improve decision-making and operational effectiveness. The literature discussed how data governance frameworks establish policies and procedures that ensure appropriate data quality, security, and compliance with regulatory requirements.

Additionally, the authors explained how information systems improve organizational decision-making by providing timely, accurate, and relevant information that supports analytical thinking and evidence-based planning. Decision support systems (DSS) combine data, analytical models, and user interfaces that enable managers to analyze complex situations and evaluate alternative courses of action. Executive information systems (EIS) provide dashboards and visualizations that present key performance indicators and organizational metrics in accessible formats that support strategic planning and monitoring activities. Business intelligence (BI) technologies transform raw data into meaningful information through reporting, analysis, and visualization capabilities that enable organizations to gain competitive insights and make informed strategic decisions. The literature highlighted how predictive analytics and machine learning technologies extend decision support capabilities by identifying patterns and trends that enable more accurate forecasting and proactive decision-making.

The literature further discussed how information systems transform organizational relationships with customers, suppliers, and other external stakeholders. Customer relationship management (CRM) systems enable organizations to track customer interactions, personalize marketing efforts, and improve customer service quality through comprehensive customer data management and communication capabilities. Electronic commerce (e-commerce) systems extend organizational market reach by enabling online transactions, digital payments, and virtual customer interactions that transcend traditional geographic and temporal limitations. Supply chain management (SCM) systems improve coordination with suppliers and distribution partners by enabling real-time information sharing, collaborative planning, and coordinated logistics management. Enterprise resource planning (ERP) systems integrate various organizational functions including accounting, finance, human resources, manufacturing, and supply chain operations within unified technological frameworks that ensure information consistency and operational coordination across the entire organization. The authors also emphasized the importance of information systems in supporting organizational collaboration and communication. Group support systems (GSS) and collaborative technologies enable teams to work together effectively regardless of geographic locations or time zone differences. Knowledge management systems facilitate organizational learning by capturing and sharing expertise, best practices, and lessons learned across organizational boundaries. Social business technologies extend organizational communication capabilities by enabling social networking, content sharing, and collaborative interactions that enhance organizational knowledge creation and dissemination. The literature discussed how virtual team technologies support remote collaboration by providing video conferencing, document sharing, and project management capabilities that enable geographically distributed teams to work together effectively. Effective collaboration technologies improve organizational innovation by enabling diverse perspectives and expertise to contribute to problem-solving and creative thinking activities.

Furthermore, the literature discussed how information systems contribute to organizational competitive advantage through differentiation, cost leadership, and focus strategies. Differentiating organizations leverage information systems to create unique products, services, or customer experiences that distinguish them from competitors. Cost leadership strategies utilize information systems to streamline operations, reduce processing costs, and improve operational efficiency that enable organizations to offer products and services at lower prices than competitors. Focus strategies target specific market segments and tailor information systems to meet the unique needs of targeted customer groups. The authors explained how information systems enable rapid response to competitive actions and market changes by providing real-time information about competitive activities, customer preferences, and market trends. Digital business transformation extends competitive strategies by utilizing digital

technologies to create new business models, revenue streams, and customer value propositions that may fundamentally transform organizational competitive positions. The literature additionally highlighted the importance of information system security and risk management in protecting organizational information assets. Security measures including access controls, authentication mechanisms, encryption technologies, and audit trails protect systems against unauthorized access, data breaches, and cyber threats. The authors discussed how comprehensive security frameworks establish policies, procedures, and technical controls that ensure appropriate protection of sensitive organizational information while enabling legitimate business operations. Business continuity and disaster recovery planning ensure organizational resilience by establishing procedures for system recovery in case of natural disasters, cyber attacks, or other disruptive events. Risk management identifies potential threats, assesses their likelihood and potential impact, and develops mitigation strategies that minimize organizational exposure to information security risks.
Moreover, the authors explained how information systems enable organizational growth and expansion by providing scalable infrastructure that can accommodate increasing transaction volumes, expanding product lines, and growing customer bases. Scalable systems enable organizations to pursue strategic growth initiatives without significant technology limitations or costly infrastructure upgrades. Cloud computing technologies extend organizational technology capabilities by providing flexible, on-demand access to computing resources without traditional hardware and software investment requirements. The literature discussed how technology infrastructure planning addresses current requirements while anticipating future growth and evolution to ensure that information systems continue to support organizational strategic objectives.Effective infrastructure management also ensures system reliability, performance, and availability that meet organizational service level requirements and stakeholder expectations.

The authors also emphasized the importance of information systems in supporting organizational compliance with regulatory requirements and industry standards. Compliance requirements including financial reporting standards, data privacy regulations, and industry-specific requirements necessitate information system capabilities that ensure appropriate documentation, controls, and reporting. The literature discussed how compliance management systems track regulatory requirements, monitor compliance status, and generate required reports and documentation that demonstrate organizational adherence to applicable standards. Audit support capabilities enable organizations to provide auditors with access to relevant information and documentation that facilitates compliance verification activities. Effective compliance management reduces organizational risk exposure and prevents costly penalties and reputational damage associated with regulatory violations. This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system represents a comprehensive information system implementation that addresses multiple organizational needs including transaction processing, inventory management, reporting, and decision support capabilities. Similar to the concepts discussed by Stair and Reynolds, the proposed AIMS will integrate people, processes, data, and technology within a unified framework that supports retail operations and improves organizational effectiveness. The proposed system also aims to improve decision-making through real-time information access, analytical reporting, and operational visibility that enable managers to monitor performance and make informed strategic decisions. These information system capabilities align directly with the authors' discussion regarding how information systems support organizational decision-making at operational, tactical, and strategic levels. Furthermore, the literature supports the implementation of automated inventory management systems because it demonstrates how information systems contribute to organizational efficiency, competitive advantage, and customer satisfaction. The concepts discussed in the study provide a strong theoretical and practical foundation for the development and implementation of the proposed Automated Inventory Management System within pharmacies and convenience store environments.

2.3.6 Systems Analysis and Design

According to Gary B. Shelly and Harry J. Rosenblatt (2012), systems analysis and design are essential processes in the development of effective and reliable information systems.
The authors explained that systems analysis focuses on studying existing business operations, identifying organizational problems, and determining the requirements needed to improve operational efficiency. On the other hand, systems design involves creating structured solutions that address identified problems through the integration of software, hardware, databases, procedures, and user interfaces. These processes are important because they ensure that developed systems are capable of meeting organizational needs and improving business performance.
The authors emphasized that many organizations experience operational inefficiencies because of outdated manual procedures and poorly organized workflow systems.
Businesses that still rely on paper-based inventory monitoring and manual transaction recording often encounter problems such as inaccurate inventory data, delayed report preparation, inventory discrepancies, data redundancy, and inefficient operational management. According to the literature, these operational issues negatively affect organizational productivity, employee performance, and customer satisfaction. Through systems analysis, developers are able to examine these operational problems carefully and identify appropriate technological solutions capable of improving business processes.
Furthermore, the literature discussed the importance of understanding user requirements during the system development process. Shelly and Rosenblatt explained that developers must gather information from users, employees, and management to identify the specific functionalities needed in the system.
Requirements gathering is essential because systems designed without considering user needs often become difficult to use and fail to support organizational operations effectively. The authors emphasized that user involvement improves the quality of system development because users can provide practical insights regarding operational challenges and workflow procedures.

The literature also highlighted the role of feasibility studies in systems development projects. According to the authors, organizations must evaluate whether a proposed system is technically, operationally, and economically feasible before implementation. Technical feasibility determines whether the organization has the necessary hardware, software, and technical resources required for the project. Operational feasibility evaluates whether the proposed system can function effectively within the organization's operational environment. Economic feasibility examines whether the expected benefits of the system outweigh the development and implementation costs. These feasibility studies help organizations avoid unsuccessful system implementations and ensure that technological investments provide operational value.

Additionally, the authors discussed the importance of system modeling and documentation during the design process. Developers use diagrams, flowcharts, data models, and process models to visualize how the system will operate and how data will move within the organization. Proper documentation improves communication among developers, users, and management because it provides a clear understanding of system functionalities and operational procedures. System modeling also helps developers identify possible operational problems before system implementation, reducing the likelihood of software failures and workflow inefficiencies.

The literature further emphasized the importance of user-friendly system interfaces and efficient workflow design. According to the authors, systems should be designed in a way that allows employees to operate them easily and efficiently. Complicated or poorly designed systems may increase employee confusion, reduce productivity, and lead to operational errors. User-centered system design improves employee satisfaction and operational efficiency because users can perform tasks more effectively with minimal training and operational difficulties.

Moreover, Shelly and Rosenblatt discussed the importance of system testing and maintenance in ensuring long-term software reliability. Before implementation, systems must undergo testing procedures to identify software bugs, functional errors, security vulnerabilities, and performance issues. Proper testing ensures that the system performs according to organizational requirements and operational expectations. The authors also explained that maintenance is necessary because systems may require updates, modifications, and improvements as organizational needs and technologies continue to evolve over time.

The literature also highlighted the significance of data security and backup procedures in modern information systems. Organizations handling inventory records, transaction information, and customer data must ensure that information remains protected from unauthorized access, system failures, and data loss. The authors explained that secure systems improve organizational reliability and maintain the integrity of business information. Proper authentication procedures, access controls, and backup systems therefore play important roles in maintaining operational continuity and protecting sensitive organizational data.

This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system will apply systems analysis and design principles during the development process. Similar to the concepts discussed by the authors, the proposed AIMS aims to identify operational problems associated with manual inventory management and develop an effective automated solution capable of improving inventory monitoring, sales recording, and transaction processing within pharmacies and convenience stores.

The proposed system also emphasizes user-centered design because it aims to provide a user-friendly interface that employees can operate efficiently during inventory and sales activities. Similar to the literature, the proposed system will undergo proper requirements analysis, feasibility evaluation, system testing, and maintenance

procedures to ensure operational reliability and usability. Through systems analysis and design methodologies, the proposed AIMS seeks to improve operational efficiency, inventory accuracy, workflow organization, and customer service.
Furthermore, the literature supports the implementation of automated inventory systems because it demonstrates the importance of structured planning and systematic development in creating effective information systems. By applying systems analysis and design principles, the proposed AIMS can provide reliable inventory monitoring, accurate transaction recording, efficient report generation, and improved operational management within pharmacies and convenience stores. Overall, the concepts discussed in the literature provide a strong theoretical and technological foundation for the successful development and implementation of the proposed Automated Inventory Management System.

2.3.7 Information Technology for Management: Driving Digital Transformation to Increase Local and Global Performance, Growth and Sustainability

According to Efraim Turban, Carol Pollard, and Gregory Wood (2018), information technology has become one of the most important components of modern business operations because it enables organizations to improve efficiency, productivity, communication, and decision-making processes. The authors explained that businesses today operate in highly competitive and technology-driven environments where manual operational procedures are no longer sufficient to handle increasing customer demands and organizational requirements. As a result, many organizations are adopting digital technologies and automated systems to streamline operations and improve overall business performance.
The authors defined digital transformation as the integration of information technologies into business operations to improve workflow processes, organizational efficiency, and customer service. Digital transformation allows organizations to automate repetitive tasks, minimize operational errors, improve information accessibility, and enhance communication between departments. Businesses implementing digital technologies are more capable of adapting to market changes, managing operational challenges, and maintaining competitiveness within their industries.
Furthermore, the literature emphasized that inventory management is one of the business areas significantly improved through information technology integration. Traditional manual inventory management methods are often inefficient because they rely heavily on handwritten records, manual stock counting, and physical inventory monitoring. These procedures consume a large amount of time and increase the possibility of human errors such as inaccurate stock records, duplicate entries, inventory discrepancies, misplaced records, and delayed transaction processing. According to the authors, businesses relying solely on manual systems often struggle to maintain accurate inventory information and efficient operational workflows. The authors explained that computerized inventory management systems help organizations automate inventory- related tasks such as stock monitoring, sales recording, product tracking, report generation, and inventory replenishment. Through automation, inventory data can be processed and updated in real time, allowing employees and managers to monitor inventory activities more efficiently. Real-time inventory monitoring improves stock visibility because inventory information is automatically updated whenever products are sold, transferred, or restocked. This reduces the risk of stock shortages and overstocking, both of which negatively affect business operations and customer satisfaction.
The literature also discussed the importance of database technologies and network systems in modern inventory management. Through integrated database systems, businesses can store and retrieve inventory information efficiently while ensuring data

consistency and security. Employees can access updated inventory records instantly, reducing the need for manual inventory verification and improving operational productivity. Network technologies also allow organizations to share inventory information across departments and locations, improving communication and coordination among employees and management.

Additionally, the authors highlighted that information technology improves transaction processing efficiency within retail and service-oriented businesses. Automated systems simplify sales transactions, bar code scanning, and inventory updates, allowing employees to complete transactions faster and more accurately. Customers also benefit from improved service quality because automated systems reduce waiting times and improve product availability monitoring. Businesses implementing computerized inventory systems are therefore able to improve both operational performance and customer satisfaction simultaneously.

The literature further explained that digital transformation contributes to better managerial decision-making because managers are provided with accurate, updated, and organized operational information. Through automated report generation and real- time inventory monitoring, business administrators can analyze sales trends, identify inventory issues, monitor product movement, and make informed decisions regarding inventory replenishment and business operations. Information technology therefore supports strategic planning and operational control within organizations.

Moreover, the authors emphasized that automation reduces operational costs by minimizing manual labor, reducing paper usage, and improving workflow efficiency. Employees are no longer required to spend excessive time recording transactions manually or preparing handwritten reports. Instead, computerized systems automate these processes, allowing employees to focus more on customer service and business operations. This improves employee productivity and contributes to a more organized and efficient working environment.

The literature also discussed the growing importance of cloud computing and internet- based technologies in inventory management systems. Cloud technologies allow organizations to access inventory information remotely and maintain synchronized inventory records across multiple locations. Businesses using cloud-based systems are more capable of monitoring operations in real time and maintaining operational continuity even in changing business environments. The authors explained that cloud technologies improve scalability and flexibility because organizations can expand system functionalities according to operational requirements.

Furthermore, the study highlighted the importance of security and data protection in information technology systems. Since inventory and sales data are digitally stored, organizations must implement proper authentication procedures, access controls, and backup systems to protect business information from unauthorized access, data loss, and cyber threats. Proper security management ensures the reliability and integrity of computerized systems and protects important organizational data. This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system aims to utilize information technology, database systems, bar code integration, and automated transaction processing to improve inventory management operations within pharmacies and convenience stores. Similar to the concepts discussed by the authors, the proposed system seeks to improve inventory monitoring, sales recording, operational efficiency, report generation, and customer service through digital transformation and automation technologies.

The proposed AIMS also aims to minimize operational errors associated with manual inventory management by implementing real-time stock monitoring and computerized inventory recording. Through automation, the proposed system will allow employees to process transactions more efficiently, monitor product availability accurately, and generate inventory reports automatically. These functionalities are directly aligned with the concepts of digital transformation and operational automation discussed in the literature.

In addition, the proposed AIMS supports better decision-making by providing accurate and updated inventory information that can assist pharmacy and convenience store administrators in managing inventory replenishment and monitoring business performance. Similar to the literature, the proposed system emphasizes the importance of operational efficiency, inventory visibility, and customer satisfaction through the use of modern information technologies. Overall, the literature strongly supports the development of the proposed Automated Inventory Management System because it demonstrates how information technology and digital transformation contribute to improved operational productivity, inventory accuracy, transaction efficiency, and organizational performance. The concepts presented in the study provide a strong theoretical foundation for the implementation of automated inventory management technologies within retail and pharmacy environments.

2.3.8 Computer Security: Principles and Practice

According to William Stallings and Lawrie Brown (2018), computer security represents essential organizational capabilities that protect information assets, ensure system availability, and maintain operational continuity in increasingly hostile digital environments. The authors explained that computer security encompasses the study of threats, vulnerabilities, and countermeasures that collectively determine organizational security posture and effectiveness. The discipline of computer security draws from multiple fields including cryptography, operating systems, networks, and legal compliance to create comprehensive frameworks for protecting digital assets. Computer security extends beyond mere technical controls to include policies, procedures, and organizational practices that collectively determine security effectiveness. The authors highlighted how cyber threats have evolved from simple pranks and demonstrations of technical skill to sophisticated criminal enterprises, nation-state actors, and organized groups that pose significant risks to organizations across all industries and sizes.

The literature emphasized that security objectives establish the foundation for security planning by defining what organizations must protect and the level of protection required to address identified risks. Confidentiality protects sensitive information from unauthorized disclosure, ensuring that sensitive data including personal information, trade secrets, and proprietary business information remains accessible only to authorized individuals. The authors discussed how confidentiality requirements vary across information types, with some information requiring strong confidentiality protections while other information may have minimal confidentiality requirements. Integrity protects information from unauthorized modification, ensuring that data accurately reflects intended values and hasn't been altered by unauthorized parties. The literature explained how integrity extends beyond data to include system processes, configurations, and operations that must remain unaltered to ensure proper system functioning. Availability ensures that authorized users can access information and resources when needed, preventing denial of service that would disrupt business operations. The authors discussed how availability requirements increasingly emphasize resistance to distributed denial of service attacks that can incapacitate even well-protected organizations.

Furthermore, the authors discussed how threat actors pose varied threats requiring different security approaches based on actor capabilities, motivations, and targeting methods. Individual hackers and hacktivists demonstrate technical skills through targeted attacks, vulnerability discovery, or political statements, typically posing limited threat to well-protected organizations. The literature explained how criminal organizations operate sophisticated operations including ransomware-as-a-service, phishing campaigns, and credential theft that generate significant financial losses through fraud, extortion, and data theft. Insider threats from trusted employees or contractors pose unique challenges because these individuals already possess access that enables harm while often evading traditional security controls. Nation-state actors pose significant threats through sophisticated capabilities, substantial resources, and strategic motivations that target specific organizations and information for political, economic, or military advantage. The authors discussed how threat modeling systematically identifies threats, assesses likelihood and potential impact, and prioritizes security investments accordingly.

The literature also highlighted how vulnerabilities in software, hardware, and configurations create opportunities for threat actors to exploit and compromise systems. Software vulnerabilities including buffer overflows, injection flaws, and authentication weaknesses enable attackers to execute arbitrary code, steal credentials, or bypass security controls. The authors explained how software vulnerabilities result from coding errors, design flaws, and inadequate security testing that introduce weaknesses into production systems. Configuration vulnerabilities result from improperly configured systems, weak passwords, unnecessary services, and default settings that provide attack opportunities. The literature discussed how vulnerability assessment identifies weaknesses through scanning, testing, and analysis that enable remediation before exploitation. Patch management addresses known vulnerabilities through updates that fix identified weaknesses, requiring systematic processes that ensure timely application of security updates. The authors explained how vulnerability disclosure practices balance public awareness that encourages remediation with security through obscurity that delays attacker knowledge of exploit opportunities.

Additionally, the authors explained how cryptographic techniques protect information confidentiality and integrity through mathematical transformations that make information unintelligible without appropriate keys. Symmetric encryption including AES and DES provides fast encryption for bulk data protection through shared keys that must remain secure between communicating parties. The literature discussed how asymmetric encryption including RSA and elliptic curve cryptography enables secure communication between parties without key exchange, supporting digital signatures and key establishment. Hash functions including SHA provide one-way transformation that enables integrity verification through comparison of hash values before and after transmission or storage. The authors explained how digital signatures provide authentication and integrity through asymmetric cryptography that verifies signer identity and message integrity. The literature discussed how key management addresses challenges of generating, distributing, storing, and retiring cryptographic keys that ensure security throughout key lifecycle.

The literature further discussed how access control restricts system and data access to authorized users through identification, authentication, and authorization mechanisms that enforce security policies. Identification establishes user identity through usernames, ID numbers, or other identifiers that enable subsequent authentication and authorization. The authors explained how authentication verifies claimed identities through knowledge factors including passwords, possession factors including tokens, and inherence factors including biometrics. Multi-factor authentication combines multiple authentication factors that significantly improve security compared to single- factor authentication. Authorization determines what resources authenticated users can access and what operations they can perform through access control policies that implement security objectives. The literature discussed how role-based access control assigns permissions based on job functions while attribute-based access control assigns permissions based on user, resource, and environmental attributes that provide flexible authorization.

The authors also emphasized the importance of network security controls that protect network communications and restrict unauthorized network access to organizational systems. Firewalls examine network traffic and enforce security policies that permit or block traffic based on source, destination, port, and protocol characteristics. The literature discussed how network segmentation isolates systems into security zones that limit attacker movement if initial compromise occurs, containing damage to limited regions. Intrusion detection and prevention systems monitor network traffic and system activities for signs of attack that enable rapid response to identified threats. Virtual private networks establish secure tunnels that enable secure remote access over untrusted networks, protecting communications from interception and modification.

The authors explained how wireless security addresses vulnerabilities in wireless communications that can be intercepted and attacked without physical network access, requiring encryption and authentication that protect wireless communications.

Furthermore, the literature discussed how operational security practices maintain security through ongoing activities including monitoring, incident response, and security awareness that address constantly evolving threats. Security monitoring tracks system activities, generates alerts for suspicious events, and enables investigation of security incidents that may indicate ongoing or past attacks. The authors explained how security information and event management (SIEM) aggregates and correlates security data from multiple sources that enable detection of sophisticated attacks spanning multiple systems. Incident response establishes procedures for detecting, containing, eradicating, and recovering from security incidents that minimize damage and restore normal operations. The literature discussed how penetration testing simulates attacks that identify weaknesses through attacker techniques that reveal vulnerabilities missed by vulnerability assessment. Security awareness training educates users about security threats and secure practices that reduce human error and social engineering susceptibility. The literature additionally highlighted how application security addresses security throughout software development lifecycle that produces more secure applications than attempting to secure completed applications. Secure coding practices incorporate security considerations throughout development, reducing vulnerabilities introduced through coding errors or design flaws. The authors explained how security testing including static analysis, dynamic analysis, and penetration testing identifies security vulnerabilities before deployment that would otherwise require expensive remediation. The literature discussed how code review ensures security through peer examination that identifies security issues missed by automated analysis. Application security testing verifies that applications function securely despite malicious inputs and attempts to bypass security controls. The authors explained how security patch management addresses vulnerabilities discovered after deployment through updates that remediate identified weaknesses.

Moreover, the authors explained how security policies establish organizational security requirements through documented procedures that guide security implementation and operation. Acceptable use policies define appropriate and inappropriate uses of organizational IT resources, establishing expectations that enable enforcement of security requirements. The literature discussed how security policies must be communicated, understood, and enforced through training, monitoring, and consequences that ensure policy compliance. Security awareness training ensures that users understand security requirements and recognize security threats including phishing and social engineering that target human vulnerabilities. The authors explained how compliance with regulatory requirements including data protection regulations, industry standards, and contractual requirements influences security policies and implementations. The literature discussed how security audits verify security control effectiveness and compliance with security policies and regulatory requirements through systematic examination.

The literature also discussed how physical security collaborates with technical security to provide comprehensive protection against varied threat vectors. Physical access controls restrict physical access to facilities, equipment, and media through locks, badges, and surveillance that prevent unauthorized physical access. The authors explained how environmental controls including climate control, fire suppression, and power protection protect equipment from environmental damage that could compromise security or availability. Media handling addresses secure disposal, transportation, and storage of storage media containing sensitive information that could expose data if improperly handled. The literature discussed how visitor management controls track visitors within facilities and restrict access to sensitive areas that protect against physical security threats. The authors explained how security in depth establishes multiple layers of security controls that provide defense in depth, ensuring that compromise of single controls does not completely compromise security.

Furthermore, the authors discussed how disaster recovery and business continuity planning ensure organizational resilience through preparation for disruptions including natural disasters, cyber attacks, and other disruptive events. Business continuity planning establishes procedures for maintaining operations during disruptions, ensuring that critical business functions continue despite adverse conditions. The literature explained how disaster recovery planning establishes procedures for recovering from catastrophic events including natural disasters, major cyber attacks, or facility loss. Data backup ensures that data can be recovered from loss or corruption through regular backup procedures that maintain recoverable copies. The authors explained how cloud-based disaster recovery reduces recovery time objectives and recovery point objectives while potentially reducing disaster recovery costs compared to traditional dedicated infrastructure.

This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system handles sensitive inventory data, customer information, and financial transactions requiring comprehensive security protections.

Similar to the concepts discussed by Stallings and Brown, the proposed AIMS will implement security controls including access control, encryption, audit logging, and incident response that protect system and data integrity.

The proposed system also aligns with the authors' discussion regarding security policies, operational security, and compliance requirements that ensure appropriate security throughout system operation. These computer security principles provide important guidance for the development and operation of the proposed Automated Inventory Management System.

Furthermore, the literature supports the implementation of the proposed system by demonstrating how computer security contributes to operational continuity, data protection, and regulatory compliance. The concepts discussed in the study provide a strong foundation for security implementation in the proposed Automated Inventory Management System within retail environments.

The proposed AIMS will apply computer security principles including confidentiality, integrity, and availability protections through access control, encryption, monitoring, and incident response that ensure effective inventory data protection while maintaining operational accessibility for authorized users in pharmacies and convenience stores.

2.3.9 Cloud Computing: Concepts, Technology & Architecture. Upper Saddle River

According to Thomas Erl, Ricardo Puttini, and Zaigham Mahmood (2013), cloud computing represents a paradigm shift in information technology delivery that enables on-demand access to shared computing resources including networks, servers, storage, applications, and services that can be rapidly provisioned and released with minimal management effort or service provider interaction. The authors explained that cloud computing addresses fundamental challenges related to IT infrastructure acquisition, scalability, and cost management that organizations face in meeting evolving business requirements. The discipline of cloud computing encompasses the study of service models, deployment models, architecture patterns, and management practices that define how cloud-based solutions are designed, implemented, and operated. Cloud computing extends beyond simple outsourcing of IT infrastructure to include comprehensive frameworks for elastic resource allocation, pay-per-use pricing, and service automation that transform how organizations acquire and utilize technology resources. The authors highlighted how cloud computing has evolved from early infrastructure outsourcing to comprehensive platforms that support virtually every aspect of modern IT operations including development, testing, deployment, and ongoing management of business applications.

The literature emphasized that cloud service models define the boundaries of responsibility between cloud providers and consumers, determining what management activities consumers perform and what capabilities providers supplies as services. Infrastructure as a Service (IaaS) provides virtualized computing resources including servers, storage, and networking that consumers manage while providers maintain underlying physical infrastructure. The authors discussed how IaaS enables organizations to provision and manage virtual machines with operating systems and applications, providing maximum flexibility while requiring significant management responsibility for security, patching, and application maintenance.

Platform as a Service (PaaS) provides complete development and deployment environments that consumers utilize to build and deploy applications without managing underlying infrastructure. The literature explained how PaaS accelerates development by providing ready-built environments, databases, and middleware that eliminate infrastructure management while potentially limiting flexibility compared to IaaS.

Software as a Service (SaaS) provides complete applications that consumers access without any infrastructure or platform management responsibility, maximizing ease of use while minimizing control over application configuration and environment.

Furthermore, the authors discussed how cloud deployment models define where cloud resources operate and who manages them, affecting security, compliance, and operational characteristics. Public clouds provide shared resources available to any consumers over the Internet, maximizing resource efficiency and minimizing capital requirements while potentially raising security and compliance concerns. Private clouds provide dedicated resources for single organizations, providing enhanced security and control at higher cost than shared public cloud resources. The literature explained how community clouds share resources among organizations with common requirements including compliance, security, or functional needs that create natural communities of interest. Hybrid clouds combine multiple deployment models, typically public and private, enabling organizations to keep sensitive workloads in private clouds while utilizing public clouds for elastic capacity or cost optimization. The authors discussed how multi-clouds distribute workloads across multiple cloud providers to avoid vendor lock-in, improve resilience, and leverage capabilities from different providers.

The literature also highlighted how cloud architecture principles enable scalable, resilient, and efficient cloud-based solutions that leverage cloud characteristics while addressing challenges related to distributed systems. Elasticity enables automatic or manual resource adjustment based on demand, efficiently supporting variable workloads while avoiding over-provisioning that increases costs or under-provisioning that degrades performance. The authors explained how scalability enables solutions to handle increased workloads through horizontal or vertical scaling that adds resources or capability to accommodate growth. Redundancy and fault tolerance ensure continuous operation despite component failures through replication, failover, and graceful degradation that maintain service availability. The literature discussed how availability zones and regions provide geographic distribution that protects against site failures while potentially introducing latency and complexity. The authors explained how caching and content delivery networks improve performance by placing content near users, reducing latency and improving user experience. Additionally, the authors explained how cloud economics transform IT cost models from capital expenditures for dedicated infrastructure to operational expenditures for shared, usage-based resources. Pay-per-use pricing charges only for resources consumed, potentially reducing costs for variable workloads compared to capital investments in dedicated infrastructure that remain underutilized during low-demand periods. The literature discussed how total cost of ownership analysis reveals true cloud costs including not only resource charges but also migration costs, integration costs, and ongoing operational costs that may exceed initial expectations. The authors explained how hidden costs including data transfer charges, API call charges, and support service charges can significantly impact overall cloud spending. Cost optimization strategies including reserved instances, savings plans, and right-sizing reduce cloud spending while maintaining performance requirements through careful resource management approaches.

The literature further discussed how cloud security addresses shared responsibility concerns where providers and consumers share security responsibilities based on service model and deployment model characteristics. Provider responsibilities include physical security, infrastructure security, and platform security while consumer responsibilities include data security, access management, and application security that vary by service model. The authors explained how identity and access management controls ensure appropriate access to cloud resources through authentication, authorization, and access control policies that enforce least-privilege principles. Data encryption protects sensitive information in storage and transit through encryption technologies that ensure data confidentiality even if underlying protections fail. The literature discussed how network security isolates cloud resources through virtual networks, firewalls, and security groups that control network access while enabling appropriate connectivity. Compliance management addresses regulatory requirements including data residency, privacy, and industry-specific requirements through cloud provider capabilities and contractual commitments. The authors also emphasized the importance of cloud governance that establishes policies, procedures, and controls for cloud resource utilization that ensure appropriate, efficient, and compliant cloud operation. Resource tagging and classification enable organization, tracking, and cost allocation that support chargeback and showback reporting that clarifies cloud spending across organizational units. The literature discussed how budget alerts and spending limits prevent unexpected charges through monitoring that identifies spending approaching or exceeding defined thresholds. Policy-as-code approaches codify governance policies into automated enforcement that prevents non-compliant resource provisioning or configuration while enabling consistent application across cloud environments. The authors explained how multi-cloud management addresses complexity of operating across multiple cloud providers through unified management tools, consistent processes, and cross-cloud expertise.

Furthermore, the literature discussed how cloud monitoring provides visibility into cloud resource utilization, performance, and costs through metrics, logs, and alerts that enable operational management and optimization. Performance monitoring tracks resource utilization including CPU, memory, storage, and network metrics that identify performance bottlenecks and capacity planning requirements. The authors explained how application monitoring tracks application health, availability, and performance through synthetic transactions and real user monitoring that identify issues affecting user experience. Cost monitoring tracks spending against budgets, identifies cost drivers, and enables optimization through right-sizing, reservation purchase, and spending pattern analysis. The literature discussed how log aggregation and analysis enables security monitoring, troubleshooting, and forensic analysis through centralized log collection and search capabilities.

The literature additionally highlighted how disaster recovery and business continuity leverage cloud capabilities to improve resilience while reducing costs compared to traditional dedicated disaster recovery infrastructure. Cloud-based disaster recovery provides replication to geographically distant cloud regions that protect against site failures without requiring dedicated standby facilities. The authors explained how DRaaS provides complete disaster recovery capabilities as services that eliminate capital investment in dedicated disaster recovery infrastructure while providing tested recovery capabilities. Warm standby maintains minimum resources that scale during disasters, providing moderate recovery costs and recovery times. The literature discussed how pilot light approaches maintain minimal resources that expand during disasters, balancing cost and recovery time through selective resource pre-provisioning. Backup and recovery services provide automated or on-demand backup to cloud storage that enables recovery from data loss or corruption. Moreover, the authors explained how cloud integration connects cloud applications with on-premises systems and other cloud services through integration capabilities that support business operations requiring distributed components. APIs and web services enable application integration through standardized interfaces that support integration without point-to-point connections that would create maintenance complexity. The literature discussed how integration platforms provide orchestrated workflows that coordinate activities across multiple systems while handling errors, transformations, and mappings. The authors explained how event-driven architectures enable loosely-coupled integration where components communicate through events that trigger processing without direct connections. Hybrid integration addresses on-premises to cloud connectivity through secure tunnels, reverse proxies, and cloud gateways that enable integration while maintaining security boundaries.

The literature also discussed how cloud migration strategies address challenges of moving workloads from on-premises environments to cloud platforms where migration approaches significantly impact migration success and outcome. Rehosting (lift-and- shift) migrates workloads without modification, maximizing migration speed and simplicity while potentially limiting cloud optimization benefits. The author explained how re-platforming makes minimal modifications to leverage cloud capabilities without full re-architecting of applications. Refactoring significantly modifies applications to leverage cloud-native capabilities, potentially improving performance, scalability, and cost efficiency at higher migration cost and risk. The literature discussed how repurchasing replaces on-premises applications with SaaS equivalents that simplify management while potentially limiting customization and control. Retention keeps selected workloads on-premises when cloud migration is impractical due to performance, compliance, or technical constraints.

Furthermore, the authors discussed how cloud DevOps practices apply agile and DevOps principles to cloud-based development and operations that accelerate delivery while improving quality. Infrastructure-as-code defines cloud resources through code that enables version control, testing, and automated deployment that improve infrastructure management. The literature explained how continuous integration and continuous deployment pipelines automate build, test, and deployment activities that accelerate release cycles while reducing deployment errors. The authors explained how configuration management ensures consistent environment configurations across development, testing, and production environments through automated configuration tools and practices. The literature discussed how automated scaling responds to demand changes through scaling policies that automatically adjust resource capacity based on defined metrics. This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system can leverage cloud computing capabilities including scalability, cost efficiency, and accessibility that enable effective inventory management across distributed retail locations. Similar to the concepts discussed by Erl, Puttini, and Mahmood, the proposed AIMS can utilize cloud platforms for application deployment, data storage, and system accessibility. The proposed system also aligns with the authors' discussion regarding cloud service models, deployment options, and integration capabilities that provide flexibility in system design and operation. These cloud computing principles provide important guidance for the development of the proposed Automated Inventory Management System.

Furthermore, the literature supports the implementation of the proposed system by demonstrating how cloud computing contributes to operational efficiency, scalability, and cost optimization. The concepts discussed in the study provide a strong foundation for cloud-based deployment of the proposed Automated Inventory Management System within retail environments.

The proposed AIMS can apply cloud computing principles including SaaS or PaaS deployment, hybrid integration, and cloud-native scalability that ensure effective inventory management while reducing infrastructure costs and enabling accessibility from diverse locations in pharmacies and convenience stores.

2.3.10 Artificial Intelligence: A Modern Approach

According to Stuart Russell and Peter Norvig (2021), Artificial Intelligence (AI) represents the discipline of creating intelligent agents that perceive their environments and take actions that maximize their chances of achieving defined objectives. The authors explained that AI encompasses theoretical foundations, algorithmic approaches, and practical applications that enable computing systems to exhibit intelligent behavior comparable to or exceeding human capabilities in specific domains. The discipline of AI draws from multiple fields including computer science, mathematics, cognitive science, neuroscience, philosophy, and linguistics to create comprehensive frameworks for understanding and replicating intelligent behavior. AI extends beyond mere automation of defined procedures to include learning, reasoning, and adaptation capabilities that enable systems to improve performance over time without explicit programming. The authors highlighted how AI has evolved from early theoretical work to practical systems that exceed human capabilities in increasingly broad application domains.

The literature emphasized that AI systems can be categorized based on their performance capabilities and the approaches they employ to achieve intelligent behavior. Systems that match or exceed human performance in specific tasks represent narrow AI or weak AI, while systems matching human-level intelligence across diverse tasks represent artificial general intelligence (AGI) or strong AI. The authors discussed how modern AI systems primarily achieve narrow AI capabilities in well-defined domains, with deep learning systems achieving superhuman performance in games, image classification, and natural language processing tasks where clear evaluation criteria exist. The literature highlighted how the distinction between narrow AI and AGI has significant implications for system design, development approaches, and ethical considerations that guide AI research and deployment. The authors explained how achievement of AGI remains a long-term research goal with significant technical and theoretical challenges.

Furthermore, the authors discussed how problem-solving approaches enable AI systems to find solutions to complex problems through systematic search and evaluation of potential solution candidates. State-space search represents problems as states and actions that transition between states, enabling systematic exploration of solution paths from initial states to goal states. The literature explained how uninformed search strategies including breadth-first search, depth-first search, and uniform-cost search explore problem spaces without domain-specific knowledge that would improve search efficiency. Informed search strategies including A* search utilize heuristic estimates of remaining cost to focus exploration toward promising solutions, potentially dramatically reducing search effort compared to uninformed approaches. The authors discussed how local search algorithms including hill climbing, simulated annealing, and genetic algorithms address optimization problems where good solutions suffice without absolute guarantees of optimal solutions. The literature also highlighted how machine learning enables AI systems to improve performance through experience without explicit programming for every contingency. Supervised learning algorithms learn mapping functions from labeled training examples that enable prediction of outputs for new inputs, supporting applications including classification, regression, and sequence labeling. The authors discussed how neural networks learn complex non-linear functions through gradient-based optimization of network parameters that minimize prediction errors on training examples. Deep learning extends neural network approaches to multiple layers that learn hierarchical representations enabling sophisticated pattern recognition in images, speech, text, and other complex data types. The literature explained how reinforcement learning enables agents to learn optimal behavior through interaction with environments, receiving rewards or penalties that signal success or failure of their actions.

Additionally, the authors explained how knowledge representation and reasoning enable AI systems to utilize explicit representations of world knowledge in problem-solving and decision-making activities. Logical representations including propositional logic and first-order logic formalize knowledge in forms that enable automated reasoning about world states and their implications. The literature discussed how description logics support reasoning about taxonomic relationships and constraints that enable classification, inheritance, and consistency checking in knowledge-intensive applications. Semantic networks and frames organize knowledge in structures that support associative reasoning, default assumptions, and inheritance of properties from general categories to specific instances. The authors explained how rule-based systems encode expert knowledge in if-then rules that enable reasoning about specific situations through pattern matching and chaining of applicable rules.

The literature further discussed how natural language processing enables AI systems to understand, generate, and translate human language in forms that support communication and information extraction. Syntax analysis determines grammatical structure of sentences through parsing techniques that identify subjects, predicates, objects, and grammatical relationships. The authors explained how semantic analysis determines meaning representations from syntactic structures, enabling semantic interpretation that supports question answering, information extraction, and language generation. Statistical language models estimate probabilities of word sequences that enable applications including speech recognition, machine translation, and text generation through probabilistic reasoning about likely language patterns. The literature discussed how transformer architectures and attention mechanisms enable modern language models including GPT and BERT that achieve remarkable language understanding and generation capabilities through pre-training on massive text corpora. The authors also emphasized the importance of planning capabilities that enable AI systems to sequence actions to achieve goals over extended time horizons. Classical planning represents problems as initial states, goal states, and actions with preconditions and effects that enable systematic planning through state-space search. The literature discussed how hierarchical task networks decompose complex tasks into subtasks that simplify planning by focusing on achievable subgoals. Conditional and probabilistic planning address uncertainty in action effects and environmental states through contingent plans that respond to observed conditions and expected outcomes.

The authors explained how planning and execution systems monitor plan execution and replan when plans fail or unexpected conditions arise, enabling robust operation in dynamic environments.

Furthermore, the literature discussed how probabilistic reasoning enables AI systems to make appropriate decisions under uncertainty through Bayesian inference and decision theory. Probability theory provides mathematical frameworks for representing uncertainty and updating beliefs based on observed evidence through Bayes' theorem and its extensions. The authors explained how Bayesian networks represent joint probability distributions through graphical models that enable efficient inference about complex probabilistic relationships. Decision networks extend Bayesian networks with decision nodes and utility functions that enable systematic decision-making under uncertainty by selecting actions that maximize expected utility. The literature discussed how Markov decision processes and partially observable Markov decision processes address sequential decision problems under uncertainty, providing frameworks for reinforcement learning and planning in stochastic environments.

The literature additionally highlighted how robotics combines AI capabilities with mechanical systems to enable intelligent interaction with physical environments. Perception systems process sensor inputs including cameras, lidars, and proximity sensors to construct representations of environment states that support decision-making and control. The authors explained how localization and mapping enable robots to determine their positions within environments and construct environment representations through simultaneous localization and mapping (SLAM) algorithms. Motion planning determines collision-free paths through environments while executing controls that move robots along planned paths. The literature discussed how manipulation planning addresses object grasping and manipulation tasks that enable robots to interact physically with world objects. Human-robot interaction enables natural collaboration between robots and humans through speech, gesture, and other natural interaction modalities.

Moreover, the authors explained how AI ethics and safety considerations ensure that AI systems align with human values and operate safely across diverse operational contexts. Value alignment addresses challenges of specifying objectives correctly and completely, avoiding unintended behaviors that optimize literal objective specifications without achieving intended purposes. The literature discussed how AI safety emphasizes robustness, reliability, and appropriate operation across diverse conditions including adversarial environments, distribution shifts, and novel situations. Fairness and bias considerations address how machine learning systems may perpetuate or amplify societal biases present in training data, requiring careful evaluation and mitigation approaches. The authors explained how transparency and explainability enable understanding of AI system decisions and behaviors that support accountability, trust, and appropriate use in high-stakes applications. The literature also discussed how

AI enables scientific discovery and advances in multiple fields through automated hypothesis generation, experimental design, and data analysis. The authors explained how AI for science applies machine learning, simulation, and reasoning to scientific problems including protein folding, materials discovery, and climate modeling that accelerate scientific progress. AlphaFold and similar systems achieve remarkable accuracy in protein structure prediction that has transformed structural biology and accelerated drug discovery. The literature discussed how scientific simulation combines AI with physics-based modeling to enable detailed understanding of complex systems including climate, materials, and biological systems. AI-assisted discovery accelerates materials and drug discovery by suggesting promising candidates for experimental evaluation, reducing experimental work required to identify useful compounds and materials.

Furthermore, the authors discussed how AI transforms business operations through automation, optimization, and decision support capabilities that improve efficiency and effectiveness across diverse functional areas. Process automation replaces human labor in routine tasks including data entry, document processing, and customer service that enable cost reduction while maintaining service quality. The literature explained how predictive analytics improve forecasting, maintenance, and risk management through machine learning models that identify patterns and predict outcomes from historical data. Personalized marketing and recommendation systems improve customer engagement through AI-driven personalization that adapts content, offers, and experiences to individual customer preferences. The authors discussed how supply chain optimization improves logistics efficiency through AI-based planning that minimizes costs while maintaining service levels through accurate demand forecasting and route optimization.

This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system can leverage AI capabilities including machine learning for demand forecasting, pattern recognition for inventory optimization, and intelligent interfaces that improve operational efficiency. Similar to the concepts discussed by Russell and Norvig, the proposed AIMS can apply AI techniques to improve inventory management through predictive analytics, automated reordering, and intelligent decision support.

The proposed system also aligns with the authors' discussion regarding machine learning, natural language processing, and intelligent interfaces that enable more effective human-computer interaction. These AI principles provide important guidance for the development of intelligent capabilities for the proposed Automated Inventory Management System.

Furthermore, the literature supports the implementation of the proposed system by demonstrating how AI technologies improve operational efficiency, decision-making, and automation across diverse application domains. The concepts discussed in the study provide a strong foundation for incorporating AI into the proposed Automated Inventory Management System within retail environments.

The proposed AIMS can apply AI capabilities including demand forecasting, anomaly detection, and intelligent interfaces that improve inventory management while reducing costs and improving customer service in pharmacies and convenience stores.

2.3.11 Internet and World Wide Web How to Program. by Paul and Harvey Deitel

According to Paul Deitel and Harvey Deitel (2020), web technologies have become essential platforms for delivering information, enables business operations, and supporting customer interactions in modern organizations. The authors explained that web technologies integrate programming languages, markup languages, and network protocols to create interactive applications that serve diverse user needs across global audiences. The discipline of web development encompasses the study of Hypertext Markup Language (HTML), Cascading Style Sheets (CSS), JavaScript, and server-side technologies that collectively enable creation of dynamic, interactive web applications. Web technologies extend beyond simple content presentation to include comprehensive application frameworks, database connectivity, and integration capabilities that support complex business operations and user interactions. The authors highlighted how web- based applications have evolved from simple information delivery mechanisms to comprehensive enterprise solutions that support virtually every aspect of modern business operations including e-commerce, customer service, and internal business processes.

The literature emphasized that HTML provides the fundamental structural foundation for web content by defining document elements, semantic meaning, and content organization through tags and attributes. The authors discussed how HTML5 introduced semantic elements including header, nav, article, section, and footer that improve content organization, accessibility, and search engine optimization compared to earlier versions relying primarily on div and span elements. Document Object Model (DOM) representations enable programmatic access to HTML document structure through document trees that organize elements hierarchically, enabling dynamic content modification through JavaScript and other scripting technologies. The literature highlighted how proper HTML structure ensures consistent rendering across different web browsers and devices while supporting accessibility technologies including screen readers that assist users with disabilities. The authors explained how semantic HTML improves search engine understanding of document content, potentially improving visibility in search results that drive website traffic and user engagement.

Furthermore, the authors discussed how CSS provides visual presentation capabilities that separate content structure from visual appearance, enabling consistent styling across web pages while improving maintenance efficiency through centralized style definitions. CSS selectors target specific elements based on element types, classes, identifiers, and relationships that enable precise control over which elements receive specific styling treatments. The literature explained how CSS Box Model defines how element content, padding, borders, and margins combine to determine element dimensions and spacing, requiring careful understanding for accurate layout control. Flexbox and CSS Grid provide modern layout capabilities that enable responsive designs accommodating various screen sizes without extensive media query coding, supporting mobile-first design approaches that improve user experience across diverse devices. The authors discussed how CSS animations and transitions enable smooth visual changes that improve perceived performance and user engagement without requiring JavaScript programming or external animation libraries.

The literature also highlighted how JavaScript enables client-side interactivity and dynamic content modification that enhance user experiences beyond static HTML presentation capabilities. JavaScript provides programming capabilities including variables, operators, control structures, functions, and objects that enable complex logic implementation within web browsers without server-side processing. The author discussed how Document Object Model manipulation enables dynamic modification of element content, attributes, and styles that respond to user actions and system events in real-time. Event handling mechanisms respond to user interactions including mouse clicks, keyboard input, form submissions, and page loading events that trigger appropriate JavaScript execution. The literature explained how asynchronous JavaScript and XML (AJAX) enables background data retrieval from servers without page refreshes, improving perceived performance and enabling more responsive user interfaces.

Asynchronous operations using Promises and async/await syntax simplify callback- based asynchronous patterns while improving code readability and maintainability.

Additionally, the authors explained how web frameworks and libraries accelerate development by providing pre-built components, utilities, and architectural patterns that reduce development time while ensuring consistent implementation approaches across development teams. jQuery simplifies DOM manipulation, event handling, and AJAX operations through cross-browser compatible functions that eliminate browser- specific code that would otherwise complicate development. The literature discussed how React, Vue, and Angular provide component-based architectures that organize user interface functionality into reusable components with well-defined properties and behaviors. Single-page application frameworks load initial page content and dynamically update page sections in response to user interactions, providing application-like experiences within web browsers without traditional page reloads. The authors explained how responsive design frameworks including Bootstrap provide pre-styled components and responsive grid systems that accelerate development while ensuring consistent, mobile-friendly presentations. The literature further discussed how server-side technologies process user requests, access databases, and generate dynamic content that cannot be created using client-side technologies alone. PHP, Python, Ruby, and other server-side languages provide programming capabilities for processing form submissions, accessing databases, and generating customized content based on user requests and server-side data. The authors explained how server-side frameworks including Express, Django, and Laravel provide architectural patterns, routing capabilities, and utility functions that accelerate server-side development while ensuring consistent implementation approaches. RESTful API design provides standardized approaches for organizing web services that enable client applications to interact with server-side functionality through well-defined endpoints and data formats. The literature discussed how JSON and XML data formats enable data exchange between client and server applications that support diverse client platforms including web browsers, mobile applications, and desktop applications.

The authors also emphasized the importance of web security in protecting users and organizations against malicious activities, unauthorized access, and data breaches that compromise system integrity and user privacy. The literature discussed how Cross-Site Scripting (XSS) attacks inject malicious scripts into web pages that execute within victim browsers, enabling session hijacking, credential theft, and content manipulation. Cross- Site Request Forgery (CSRF) tricks users into submitting authenticated requests to vulnerable web applications, enabling unauthorized actions without user consent. SQL injection attacks inject malicious database commands through unsanitized user inputs that may expose or modify sensitive database contents. The authors explained how input validation, output encoding, and security headers provide defensive measures that prevent common web application vulnerabilities while enabling legitimate functionality.

Furthermore, the literature discussed how web accessibility ensures that web content serves users with diverse capabilities including visual, auditory, motor, and cognitive limitations that may affect web use. Web Content Accessibility Guidelines (WCAG) establish accessibility requirements organized around principles of perceivable, operable, understandable, and robust content that guides accessibility implementation. Semantic HTML provides inherent accessibility through proper element selection that enables assistive technologies to interpret content correctly. ARIA attributes extend accessibility for dynamic content and interactive widgets that may not have appropriate semantic HTML representations. The literature explained how keyboard navigation supports users who cannot operate pointing devices while also enabling efficient interaction for power users. Alt text describes images for users who cannot perceive visual content, ensuring comprehensive accessibility across diverse user capabilities.

The literature additionally highlighted the importance of web performance optimization that improves application responsiveness while reducing server loads and bandwidth requirements that affect user experience and operational costs. The authors discussed how file compression, image optimization, and caching reduce transfer sizes and enable faster content delivery that improves perceived performance and reduces bandwidth costs. Lazy loading defers loading of below-the-fold content until users scroll toward that content, reducing initial page load times while maintaining complete content availability. Content Delivery Networks distribute content across geographically distributed servers that reduce latency by serving content from locations near users, improving global accessibility and performance. The literature discussed how minification removes unnecessary characters from code files while bundling combines multiple files into single files that reduce HTTP requests required for page loading. Moreover, the authors explained how web APIs enable integration between web applications and external services including social platforms, payment processors, and data services that extend application capabilities beyond single-application functionality. OAuth provides standardized authentication approaches that enable users to authorize third-party applications without sharing credentials, improving security while enabling integration capabilities. The literature discussed how webhooks enable real-time notifications that inform applications when events occur, enabling responsive integration without continuous polling that would consume resources inefficiently. SOAP and REST web services provide standardized approaches for external application integration that enable business-to-business transactions and service connectivity. The authors explained how JavaScript Object Notation (JSON) provides lightweight data representation that facilitates data exchange between web applications and external systems while enabling human readability for debugging purposes.

The literature also discussed how progressive web applications combine web and native application capabilities to deliver application-like experiences through web browsers without traditional application installation requirements. Progressive web applications support offline operation through service workers that cache application resources, enabling continued operation when network connectivity is unavailable. The authors explained how push notifications enable web applications to send notifications comparable to native application notifications, improving user engagement without requiring application installation. Home screen installation adds web application icons to user devices that launch applications without browser interfaces, improving user experience while maintaining web-based deployment. The literature discussed how background synchronization queues operations during offline periods and automatically processes queued operations when connectivity returns, ensuring reliable operation across varying network conditions. Furthermore, the authors discussed how modern web development approaches emphasize tooling, automation, and collaborative development that improve development efficiency and code quality. Version control systems track code changes, enable parallel development, and support rollback capabilities that improve development coordination while protecting against code loss. The literature explained how continuous integration and continuous deployment pipelines automate testing, building, and deployment activities that reduce manual effort while improving code quality and deployment frequency. Package managers including npm and yarn manage application dependencies that enable efficient code sharing while ensuring consistent environment configurations across development machines. The authors discussed how linting and formatting tools enforce code style standards that improve code consistency while identifying potential errors before runtime. Testing frameworks automate unit testing, integration testing, and end-to-end testing that verify application functionality while preventing regression bugs from reaching production environments.

This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system requires web-based interfaces that enable inventory managers, sales associates, and administrators to access system capabilities from diverse locations and devices. Similar to the concepts discussed by Deitel and Deitel, the proposed AIMS will apply web technologies including HTML, CSS, and JavaScript to create responsive, accessible user interfaces that support retail operations. The proposed system also aligns with the authors' discussion regarding web security, performance optimization, and progressive web application capabilities that ensure effective, secure, and accessible system operation. These web development principles provide important guidance for the development of web-based interfaces for the proposed Automated Inventory Management System.

Furthermore, the literature supports the implementation of the proposed system by demonstrating how web technologies contribute to application accessibility, cross-platform compatibility, and deployment flexibility. The concepts discussed in the study provide a strong foundation for the web-based interface development of the proposed Automated Inventory Management System within retail environments. The proposed AIMS will apply web development principles including responsive design, accessibility compliance, security best practices, and performance optimization that ensure effective system utilization by diverse users in pharmacies and convenience stores.

2.3.12 Human- Computer Interaction (3rd ed.)

According to Alan Dix, Janet Finlay, Gregory D. Abowd, and Russell Beale (2004), Human-Computer Interaction (HCI) represents the multidisciplinary discipline that studies the design, implementation, and evaluation of interactive computing systems for human use. The authors explained that HCI encompasses theoretical foundations from computer science, cognitive psychology, and social sciences to create user interfaces that enable effective and efficient interaction between humans and computing systems. The discipline of HCI addresses fundamental challenges related to user capability, system functionality, and interaction quality that determine whether computing systems succeed in supporting human activities and achieving intended purposes. HCI extends beyond mere interface aesthetics to include comprehensive understanding of user needs,Tasks, capabilities, and contexts that influence interface design and interaction effectiveness. The authors highlighted how effective HCI practices result in systems that are usable, useful, and accessible to diverse users across varying contexts of use.

The literature emphasized that successful interactive systems require careful attention to usability, which represents the degree to which specific users can utilize computing systems effectively, efficiently, and satisfactorily within specific operational contexts.

Usability encompasses multiple dimensions including ease of learning, efficiency of use, memorability, error prevention and recovery, and user satisfaction that collectively determine overall interaction quality. The authors discussed how usability goals often conflict with each other, requiring careful trade-offs that balance competing requirements based on specific user populations and task characteristics. For example, designing for ease of learning may compromise efficiency for experienced users, while maximizing efficiency for experts may overwhelm novice users with complex interfaces and extensive functionality. The literature highlighted how context of use significantly influences usability requirements, as systems used in high-pressure emergency situations require different interface approaches than systems used for relaxed information exploration activities.

Furthermore, the authors discussed how user-centered design methodologies ensure that interface development addresses actual user needs and capabilities rather than technical capabilities or designer assumptions. User-centered design involves systematic understanding of users, their tasks, and their contexts through observational studies, interviews, and task analysis activities that inform design decisions. The literature explained how iterative design approaches develop progressive prototypes that enable user evaluation and incremental refinement throughout the development process rather than discovering usability problems after implementation. Personas and scenarios represent fictional users and use situations that capture user characteristics and interaction requirements in forms that guide design decisions and facilitate communication among development team members. The authors discussed how participatory design approaches involve actual users in design activities, enabling direct user input that improves system relevance and user acceptance while reducing development risks associated with assumption-based designs.

The literature also highlighted the importance of understanding human cognition and information processing capabilities in interface design. Human cognitive architecture includes sensory memory, working memory, and long-term memory components that influence how users perceive, process, and remember interface information and interaction sequences. The authors explained how working memory limitations constrain the amount of information users can hold simultaneously, requiring interface designs that present appropriate amounts of information without overwhelming cognitive capacities. Chunking strategies group related information into meaningful units that reduce working memory loads while maintaining informational richness. Visual perception principles guide interface layout, visual hierarchy, and element grouping that enable rapid information scanning and comprehension. The literature discussed how attention and perception principles inform design decisions regarding visual prominence, layout organization, and information presentation that capture user attention and communicate intended messages effectively.

Additionally, the authors explained how interaction paradigms provide conceptual frameworks that define how users express intentions and control system operations through interface elements and mechanisms. Direct manipulation interfaces enable users to manipulate interface objects directly through physical metaphors that map user actions to system responses in intuitive and visible ways. The literature discussed how command-based interfaces enable users to express specific commands through text or voice inputs that provide maximum flexibility for expert users while requiring extensive memorization of command syntax and semantics. Menus and selection interfaces present available options that enable users to select appropriate actions without memorizing command structures while potentially slowing interaction for well-learned tasks. Form-based interfaces organize input fields that guide data entry while ensuring that required information is captured completely. The authors explained how hybrid approaches combine elements from multiple interaction paradigms to address diverse task requirements and user preferences. The literature further discussed how navigation and information architecture determine how users locate and access information within complex interactive systems. Hierarchical navigation organizes information in tree-like structures that enable systematic drilling from general categories to specific content while potentially requiring extensive navigation for deeply buried information. Network navigation enables associational linking between related information elements that support non-linear exploration but may confuse users regarding location and orientation. The authors explained how consistent navigation mechanisms reduce learning requirements and improve user confidence by establishing predictable interaction patterns that users can apply across different system areas. Search-based navigation enables direct location of specific information through keyword queries that efficiently address well-defined information needs while requiring users to formulate effective search queries. The literature discussed how faceted navigation combines multiple classification dimensions that enable flexible information filtering while maintaining orientation through visible filter criteria.

The authors also emphasized the importance of feedback in interactive systems that confirms system receipt and processing of user actions while providing appropriate information regarding system status, progress, and results. Immediate feedback following user actions confirms that inputs have been received and processed, maintaining user confidence that interactions are being handled appropriately. Progress indicators communicate ongoing operation status for actions requiring extended processing time, reducing user uncertainty and preventing premature repeat interactions. The literature discussed how informative error messages explain what went wrong in terms users understand while guiding users toward appropriate correction actions. Success confirmations provide positive acknowledgment when operations complete successfully, building user confidence in system operation and reliability. The authors explained how feedback timing significantly influences user perception, with immediate feedback for simple actions and delayed acknowledgment for lengthy processes that require patience from users. Furthermore, the literature discussed how accessibility ensures that systems accommodate users with diverse capabilities and requirements including visual, auditory, motor, and cognitive limitations. Screen reader compatibility enables blind users to access interface content through assistive technologies that render information through speech or Braille outputs. The authors explained how keyboard navigation supports users who cannot operate pointing devices due to motor limitations while also enabling efficient interaction for expert users. Color independence ensures that information is not communicated solely through color differences that may be imperceptible to users with color vision deficiencies. Text alternatives describe non-text content including images and graphical elements that enable understanding for users who cannot perceive visual content. The literature discussed how cognitive accessibility principles address memory limitations, reading difficulties, and attention challenges that affect users with cognitive disabilities or situational constraints. Comprehensive accessibility requires systematic evaluation of interface elements against established accessibility guidelines and testing with users having diverse capabilities.

The literature additionally highlighted the importance of error prevention and recovery in interactive systems that minimize user frustration and maintain productivity when problems occur. Constraint-based interfaces restrict input possibilities to valid options only, preventing errors before they occur through interface design rather than relying solely on error messages following invalid inputs. Confirmations for destructive actions warn users before irreversible operations including deletions that cannot be undone, providing opportunities to reconsider before permanent consequences. The authors discussed how undo capabilities enable recovery from errors by reversing recent actions, encouraging user exploration and experimentation without fear of permanent mistakes. Default values reduce data entry requirements while intelligent defaults propose likely selections based on user history or common patterns. The literature explained how error recovery support provides clear guidance regarding what went wrong and how to correct problems rather than leaving users confused regarding appropriate remediation actions.

Moreover, the authors explained how evaluation methodologies assess interface usability through systematic observation, measurement, and analysis of user interactions with interactive systems. Usability testing evaluates system usability through controlled observation of representative users performing representative tasks, enabling identification of usability problems through observation of actual user behavior rather than designer assumptions. The literature discussed how heuristic evaluation applies established usability principles through expert review that identifies potential usability problems efficiently without user testing resource requirements. Survey instruments gather user feedback regarding satisfaction, perceived ease of use, and system quality that complement observational evaluation methods. The authors explained how cognitive walkthrough evaluates interfaces through systematic simulation of user cognitive processes that predict usability problems based on user perception and cognition principles. Field studies observe users in natural operational environments that reveal contextual factors and real-world usage patterns not apparent in laboratory testing situations.

The literature also discussed how cultural and international factors influence interface design for global deployment across diverse populations with varied cultural backgrounds and language requirements. Color meanings, iconographic interpretations, and interaction patterns vary across cultures, requiring localization that adapts interfaces for specific cultural contexts rather than assuming universal interpretations. The authors explained how language localization extends beyond simple translation to adapt message content, formatting conventions, and interaction approaches for specific linguistic communities. Date, time, and number formats vary across locales, requiring system support for multiple presentation conventions that accommodate user preferences and regional standards. The literature discussed how bidirectional text support enables interface presentation for right-to-left writing systems including Arabic and Hebrew while maintaining consistent interaction models for bidirectional content. Icon libraries ensure that graphical elements communicate intended meanings across cultural contexts rather than relying on culturally-specific visual metaphors that may not translate across populations. Furthermore, the authors discussed how emerging interaction technologies expand design possibilities and introduce new interaction challenges as technology capabilities evolve beyond traditional desktop computing paradigms. Touch and gesture interfaces enable natural interaction through finger movements that eliminate distinction between user intentions and interface manipulations while introducing new usability challenges related to gesture discovery and error prevention. Voice-based interaction enables hands-free operation for users unable to operate traditional input devices while requiring interface designs that accommodate voice interface characteristics and limitations. Wearable and augmented reality interfaces integrate computing with physical activities through head-up displays and environmental sensors that enable continuous access to computational capabilities during daily activities. The literature explained how ambient interfaces communicate information through environmental modalities including light, sound, and temperature that extend notification capabilities beyond visual displays while requiring new design approaches for information presentation. The authors discussed how multi-device interaction spans user activities across multiple devices and platforms, requiring design approaches that maintain consistency and enable smooth transitions between contexts of use.

This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system requires effective user interfaces that enable inventory managers, sales associates, and administrators to interact efficiently and effectively with system capabilities. Similar to the concepts discussed by Dix, Finlay, Abowd, and Beale, the proposed AIMS will apply user-centered design methodologies that ensure interface development addresses actual user needs and operational requirements.

The proposed system also aligns with the authors' discussion regarding usability principles, error prevention and recovery, and feedback mechanisms that ensure effective user interaction and system adoption. These HCI principles provide important guidance for the development of usable, effective user interfaces for the proposed Automated Inventory Management System.

Furthermore, the literature supports the implementation of the proposed system by demonstrating how effective HCI practices contribute to user satisfaction, operational efficiency, and successful system adoption. The concepts discussed in the study provide a strong foundation for the user interface design and evaluation of the proposed Automated Inventory Management System within retail environments. The proposed AIMS will apply HCI principles including user-centered design, usability evaluation, accessibility compliance, and error recovery support that ensure effective system utilization by diverse users including inventory managers, sales associates, and administrators in pharmacies and convenience stores.

2.3.13 Enterprise Resource Planning

According to Alexis Leon (2014), Enterprise Resource Planning (ERP) systems represent comprehensive integration platforms that connect various organizational functions including finance, accounting, manufacturing, inventory management, human resources, and customer relationship management within unified technological frameworks. The author explained that ERP systems address fundamental challenges related to organizational integration, data consistency, and process coordination that arise when information systems operate independently across different functional areas. The discipline of Enterprise Resource Planning encompasses the study of system architecture, implementation methodologies, integration techniques, and management practices that ensure successful ERP deployment and ongoing operational effectiveness. ERP systems extend beyond simple software implementations to include comprehensive organizational change management, business process reengineering, and ongoing system optimization that determine overall project success and organizational benefits. The author highlighted how ERP systems have evolved from early mainframe-based manufacturing planning systems to comprehensive enterprise platforms that support virtually all organizational functions and business processes.

The literature emphasized that organizational information system fragmentation creates significant operational challenges including data redundancy, process inconsistency, and information silos that compromise organizational efficiency and decision-making effectiveness. When different functional areas maintain separate information systems, identical data must be entered multiple times, increasing labor requirements and error potentials while consuming storage resources for redundant data copies. The author discussed how data inconsistency across systems leads to operational problems when different reports present conflicting information that undermines managerial confidence in reported results and decision-making credibility. Information silos prevent information sharing across functional boundaries, limiting organizational ability to analyze integrated data and understand cross-functional relationships that drive business performance. The literature highlighted how system integration addresses these challenges by connecting previously separate systems, eliminating redundant data entry, ensuring data consistency, and enabling comprehensive analysis across organizational boundaries. Furthermore, the author discussed how ERP architecture organizes organizational functions into integrated modules that share common databases, application logic, and user interfaces while maintaining specialized functionality for specific business processes. Financial modules handle general ledger, accounts payable, accounts receivable, fixed assets, and financial reporting that maintain comprehensive financial records and ensure accounting accuracy. Manufacturing modules manage production planning, scheduling, materials requirements, and shop floor control that coordinate manufacturing operations according to production requirements and resource constraints. Human resources modules maintain employee

records, payroll processing, benefits administration, and personnel management that support comprehensive workforce management throughout the employment lifecycle. The literature discussed how inventory management modules track inventory quantities, locations, and movements while supporting various inventory costing methods and valuation approaches that meet accounting and operational requirements.
Sales and distribution modules manage order processing, shipping, and billing that coordinate customer-facing operations with back-office functions including inventory allocation and financial recording.

The literature also highlighted the importance of business process reengineering in ERP implementation as organizations redesign traditional processes to align with ERP system capabilities and best practice approaches embedded in system functionality. ERP implementations provide opportunities to evaluate current business processes, eliminate unnecessary activities, and implement process improvements that would be impractical to achieve with legacy systems. The author explained how business process reengineering challenges existing assumptions about work methods, organizational structures, and information flows that may no longer reflect optimal approaches in integrated system environments. Successful ERP implementations require careful analysis of current processes, clear understanding ofERP capabilities, and systematic redesign of processes to leverage system functionality while addressing organizational requirements. The literature discussed how reengineering efforts often face resistance from employees accustomed to traditional processes and from managers concerned about organizational disruption and learning requirements.

Additionally, the author explained how data migration represents one of the most critical and challenging aspects of ERP implementation that determines system effectiveness and organizational acceptance. Migrating historical data from legacy systems to new ERP platforms requires careful planning, data cleansing, format transformation, and validation procedures that ensure accurate data transfer while maintaining data relationships. The literature discussed how data quality problems in legacy systems often become apparent during migration efforts when inconsistent data, missing records, and duplicate entries require resolution before migration can proceed. Data cleansing improves data quality not only for migration purposes but also for ongoing operational effectiveness by eliminating errors that would otherwise propagate to new system environments. The author explained how historical data retention decisions must balance storage requirements, analytical needs, and regulatory requirements that determine which historical records require migration and which can remain in archived systems.

The literature further discussed how ERP implementation methodologies provide structured approaches for organizing and controlling implementation activities that minimize disruption while ensuring comprehensive system deployment. Traditional implementation approaches follow sequential phases including requirements analysis, system configuration, testing, training, and deployment that provide clear checkpoints and ensure systematic completion of implementation activities. The author discussed how "big bang" implementations deploy complete ERP systems simultaneously that minimize parallel operation requirements but increase deployment risks and potential disruptions. Phased implementations deploy functional modules incrementally that reduce deployment risks but extend implementation timelines and require ongoing coordination between legacy and ERP systems. The literature explained how pilot implementations deploy systems in limited organizational areas that enable testing and learning before broad deployment while minimizing organization-wide risks. The author discussed how hybrid approaches combine elements of multiple methodologies to address specific organizational requirements and constraints.

The authors also emphasized the importance of organizational change management in ensuring successful ERP adoption and realizing projected benefits from system implementations. ERP implementations involve significant organizational changes including process modifications, role changes, and responsibility redistribution that require careful management to ensure successful adoption and user acceptance. The literature discussed how executive sponsorship provides necessary authority, resources, and organizational commitment that support implementation efforts and overcome resistance to organizational changes. Communication programs ensure that affected employees understand system rationale, implementation timelines, and expected changes that reduce uncertainty and resistance. Training programs develop necessary skills that enable employees to operate new systems effectively while maximizing productivity improvements from system deployment. The author explained how performance measurement tracks implementation progress and identifies issues requiring management attention while demonstrating progress toward projected benefits that maintain organizational commitment. Furthermore, the literature discussed how ERP system integration extends organizational capabilities by connecting ERP platforms with specialized applications including warehouse management, transportation management, customer relationship management, and business intelligence systems that address specific functional requirements. Interface development establishes automated data exchanges between ERP systems and external applications that eliminate manual data reentry while ensuring data consistency across integrated systems. The author discussed how application programming interfaces provide standardized mechanisms for external applications to access ERP functionality and data without direct database access that could compromise system integrity. Real- time interfaces provide immediate data synchronization that supports operational requirements for current information while batch interfaces coordinate data exchanges at scheduled intervals that reduce system processing loads. The literature explained how integration testing validates interface functionality and ensures that data transfers accurately between systems while maintaining data integrity and transaction consistency.

The literature additionally highlighted the importance of ERP system maintenance in ensuring ongoing system effectiveness and reliability following initial implementation. Ongoing system maintenance addresses defect corrections, performance optimization, and system updates that maintain system effectiveness as organizational requirements evolve and technology landscapes change. The author discussed how system monitoring identifies performance problems, resource constraints, and operational issues that require attention before they impact system users and business operations. Regular system backups protect organizational data against loss while disaster recovery planning ensures business continuity following unexpected disruptions. The literature discussed how vendor support relationships provide access to technical expertise, system updates, and bug fixes that maintain system currency and address identified problems. The author explained how user support functions address operational questions, resolve user issues, and provide guidance that enables effective system utilization throughout the organization.

Moreover, the author explained how ERP systems provide comprehensive reporting capabilities that enable organizations to analyze business performance, monitor operational activities, and support strategic decision-making across all integrated functional areas. Standard reports address common business requirements and provide formatted outputs that meet recurring information needs without custom development. The literature discussed how ad-hoc reporting capabilities enable users to develop customized reports that address specific analytical requirements without programming expertise or development resources. Report distribution capabilities automate report generation and delivery that ensure timely availability of information without manual intervention. The author discussed how analytical capabilities extend beyond operational reporting to provide trend analysis, forecasting, and performance measurement that support strategic planning and continuous improvement efforts. The literature explained how integration with business intelligence platforms extends analytical capabilities through advanced analytical tools, data mining, and visualization capabilities that reveal insights not available through standard reporting.

The literature also discussed how ERP systems support organizational growth and expansion by providing scalable infrastructure that accommodates increasing transaction volumes, expanding product lines, and growing customer bases without significant system modifications or upgrades. Modular architecture enables organizations to implement additional functionality as requirements evolve and budget constraints permit while maintaining integration with previously deployed modules. The author explained how cloud-based ERP deployments extend accessibility while reducing capital investment requirements for technology infrastructure. The literature discussed how multi-organization capabilities support organizational structures with multiple business units, locations, or legal entities that require independent operations while maintaining consolidated reporting and coordinated management. The author explained how international capabilities address multi-currency, multi-language, and multi-country requirements that enable global business operations while maintaining local compliance and operational effectiveness.

Furthermore, the author discussed how ERP security controls protect organizational information assets while enabling appropriate access that supports business operations and user productivity. User authentication verifies user identities before granting system access through password-based mechanisms that may include multi-factor authentication for enhanced security. The literature discussed how role-based security assigns permissions based on job responsibilities that ensure appropriate access levels while minimizing unnecessary permission grants that could create security vulnerabilities. Transaction controls restrict sensitive operations including price changes, inventory adjustments, and financial entries that require appropriate authorization before completion. Audit trails record system activities that enable security monitoring and forensic investigation following suspected security incidents or policy violations. The author explained how data encryption protects sensitive information during transmission and storage that prevents unauthorized access to confidential data.

This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system represents an enterprise solution that integrates various retail operational functions within a unified technological framework. Similar to the concepts discussed by Leon, the proposed AIMS will integrate inventory management, sales processing, customer information, and operational reporting within comprehensive system capabilities that support retail operations.

The proposed system also aligns with the author's discussion regarding business process integration, data consistency, and organizational change management as critical success factors for system implementation. These ERP principles provide guidance for the development and implementation of the proposed Automated Inventory Management System within pharmacies and convenience store environments.

Furthermore, the literature supports the implementation of the proposed system because it demonstrates how integrated enterprise systems contribute to organizational efficiency, data consistency, and operational effectiveness. The concepts discussed in the study provide a strong foundation for the development and implementation of the proposed Automated Inventory Management System within retail environments.

The proposed AIMS will apply ERP principles including modular design, database integration, security controls, and comprehensive reporting that enable effective inventory management and operational coordination. These capabilities align directly with the author's discussion regarding ERP system functionality and implementation approaches.

2.3.14 Information Systems for Modern Management

According to Robert G. Murdick, Joel E. Ross, and James R. Claggett (2017), information systems have become essential organizational tools that support business operations, enhance managerial decision-making, and provide competitive advantages in modern retail and commercial environments. The authors explained that information systems integrate hardware, software, databases, and communication technologies to collect, process, store, and disseminate information that supports organizational objectives and stakeholder needs. The discipline of information systems encompasses the study of how technology resources combine with organizational processes and human capabilities to create effective information solutions that address business challenges and opportunities. Information systems extend beyond simple technology implementations to include the broader organizational context including business strategies, operational workflows, and management practices that significantly impact system effectiveness and organizational outcomes. The authors highlighted how modern organizations rely on information systems to improve operational efficiency, enhance customer experiences, and achieve strategic objectives in increasingly competitive marketplace environments.

The literature emphasized that Point of Sale (POS) systems represent essential technologies used in modern retail and business environments to automate transaction processing, inventory monitoring, sales recording, and customer service operations. POS systems integrate hardware components including barcode scanners, receipt printers, cash drawers, and payment terminals with software applications and databases that enable efficient and accurate transaction processing. The authors discussed how POS systems serve as the primary interface between customers and retail operations, making them critical determinants of customer service quality and operational efficiency.

Modern POS systems provide comprehensive functionality including product pricing, discount application, payment processing, receipt generation, and real-time inventory updates that streamline retail operations and improve customer checkout experiences. The evolution from simple cash registers to sophisticated POS terminals reflects the increasing importance of automated transaction processing in meeting customer expectations and operational requirements.

Furthermore, the authors discussed how traditional manual transaction methods are often inefficient because employees must manually record sales transactions, calculate payments, update inventory records, and prepare operational reports using paper-based procedures. These manual procedures consume excessive time and increase the possibility of human errors including incorrect calculations, inaccurate inventory records, duplicate entries, and delayed report preparation that compromise operational accuracy and efficiency. The literature highlighted how manual inventory tracking requires physical inventory counts, paper-based record keeping, and periodic auditing procedures that consume significant labor resources while potentially introducing errors and inconsistencies. Manual reporting procedures require employees to compile data from multiple sources, calculate performance metrics, and prepare formatted reports that may not be available until days or weeks after the reporting period ends. The authors explained how manual processes limit organizational ability to respond quickly to changing business conditions, customer preferences, and competitive pressures.

The literature also highlighted how businesses implementing POS and automated inventory systems demonstrate improved capability to enhance transaction accuracy and reduce operational inefficiencies that impact customer service and profitability.

Automated transaction processing eliminates manual data entry, reduces calculation errors, and ensures consistent application of pricing, discounts, and payment processing across all customer transactions. The authors discussed how barcode scanning technologies improve transaction speed and operational accuracy because employees can scan products directly instead of manually encoding item information. Barcode scanning minimizes encoding errors and allows businesses to process customer transactions more efficiently while maintaining accurate sales records. Quick response codes and radio-frequency identification technologies extend product identification capabilities to support faster checkout processes and enhanced inventory tracking. The literature explained how automated systems improve both employee productivity and customer satisfaction because transactions become faster, more accurate, and more organized.

Additionally, the authors explained how POS and inventory systems improve inventory management because inventory records are automatically updated whenever transactions occur, eliminating manual inventory updates and reducing discrepancies between actual and recorded inventory levels. Through automated transaction processing, businesses can monitor stock movement in real time and maintain accurate inventory information that supports effective inventory planning and control. This minimizes inventory discrepancies and helps organizations prevent stock shortages that result in lost sales and customer dissatisfaction, as well as overstocking issues that increase carrying costs and reduce profitability. Real-time inventory monitoring also improves operational visibility because managers can track product movement and sales activities more efficiently, enabling better decision-making regarding inventory replenishment, product promotion, and store planning. The literature discussed how automated alerts notify managers when inventory levels fall below specified thresholds, enabling proactive reordering that prevents stockouts and maintains product availability.

The literature further discussed how integrated inventory and POS systems enable comprehensive inventory tracking throughout the supply chain from initial receipt through final sale. When products arrive, inventory records are created that establish beginning quantities and track subsequent movements through the organization. Each sale automatically reduces inventory quantities, maintaining accurate inventory levels without manual intervention. The authors explained how inventory turnover analysis helps managers identify fast-moving products that require frequent replenishment and slow-moving items that may require promotional efforts or inventory reduction. Cycle counting procedures enable ongoing inventory verification that identifies discrepancies and ensures inventory record accuracy. The literature discussed how physical inventory counts remain necessary but occur less frequently when automated inventory updates maintain accurate records throughout the accounting period.

The literature also discussed the importance of database integration within POS and inventory systems. Inventory records, sales transactions, customer information, and financial data are stored within centralized databases that support accurate information retrieval and report generation. Database integration improves operational reliability because businesses can access organized and updated operational information whenever needed for decision-making, reporting, or analysis. The authors explained how relational database architectures organize information across multiple related tables that enable efficient querying and reporting while maintaining data integrity and eliminating redundancy. Database queries enable managers to analyze sales patterns, evaluate product performance, and identify trends that inform strategic planning and operational improvements. The literature discussed how database 备份 and recovery procedures protect organizational data against loss while ensuring business continuity following unexpected disruptions. Furthermore, the authors explained how POS and inventory systems improve operational security and accountability because transaction activities are recorded systematically within the system database. Managers can monitor employee transactions, review sales activities, and track inventory movement more effectively through automated audit trails and transaction logs. This improves transparency and reduces the likelihood of unauthorized operational activities and inventory inconsistencies that could result in financial losses or regulatory violations.

Access controls restrict system functionality based on employee positions and responsibilities, ensuring that only authorized personnel can perform sensitive operations including price changes, inventory adjustments, and report generation. The literature discussed how transaction logging supports both operational management and regulatory compliance by providing comprehensive records of all system activities. Audit capabilities enable internal and external auditors to verify system accuracy and compliance with established policies and regulatory requirements.

The literature additionally highlighted the importance of system maintenance and employee training in ensuring successful POS and inventory system implementation and ongoing operation. Employees must understand system functionalities and transaction procedures to maximize system effectiveness and operational efficiency. The authors discussed how comprehensive training programs ensure that employees develop necessary skills to operate systems correctly and efficiently while minimizing errors and maximizing productivity. Training should include both initial training when systems are first implemented and ongoing training that addresses system updates, new functionalities, and procedural changes. The literature explained how user documentation, online help resources, and technical support capabilities assist employees in resolving operational questions and system issues. Proper maintenance and technical support are also necessary to ensure continuous system performance and reliability, including regular software updates, hardware maintenance, and database optimization activities.

Moreover, the authors explained how POS and inventory systems contribute to organizational productivity by reducing repetitive manual tasks and simplifying operational workflows. Employees no longer need to perform manual inventory calculations or handwritten sales recording because the system automatically processes operational data and maintains accurate records. This allows employees to focus more on customer assistance and business operations, improving overall workplace efficiency and customer service quality. The literature discussed how automation enables employees to serve more customers with faster, more accurate service that improves customer satisfaction and enables increased sales volumes. Employee productivity improves further as workers develop expertise in system operation and discover efficiency Improvement opportunities. The authors explained how productivity improvements translate directly to improved profitability through reduced labor costs, increased sales capacity, and improved operational accuracy. The literature also discussed how POS and inventory systems improve customer service quality because automated transaction processing reduces waiting time and improves transaction reliability. Customers benefit from faster payment processing, accurate transaction receipts, and improved product availability monitoring that enhances overall shopping experiences. Faster checkout processes reduce customer wait times, particularly during peak shopping periods when multiple customers require service simultaneously.

Accurate transaction records ensure that customers receive correct change, proper receipts, and accurate billing that prevents disputes and maintains trust. The literature explained how businesses implementing automated retail systems are more capable of maintaining customer satisfaction and operational efficiency simultaneously. Customer loyalty improves when organizations provide efficient, accurate service that meets customer expectations regarding convenience, accuracy, and professional operation.

The literature further discussed how POS and inventory systems improve managerial decision-making by providing organized operational reports and updated sales information. Managers can analyze sales trends, monitor inventory performance, identify fast-moving products, and evaluate business operations through computerized reporting systems that provide timely, accurate information. Access to accurate operational information allows organizations to make informed decisions regarding inventory replenishment, pricing strategies, and operational improvements. The authors explained how dashboard displays present key performance indicators in graphical formats that enable quick assessment of business performance without detailed report analysis. Report generation capabilities produce formal reports for management review, regulatory compliance, and strategic planning activities. The literature discussed how trend analysis capabilities identify patterns in sales, inventory, and customer behavior that inform seasonal planning and promotional strategies.

Furthermore, the authors discussed how integrated POS and inventory systems support strategic planning activities by providing comprehensive historical data and analytical capabilities. Sales history data supports forecasting, seasonal planning, and trend analysis that improves inventory investment decisions and reduces carrying costs.

Customer purchase history enables targeted marketing, loyalty programs, and personalized promotions that increase customer engagement and repeat purchases. The literature explained how competitive analysis capabilities compare current performance against historical results and industry benchmarks that identify improvement opportunities. Integration with accounting systems ensures that financial records accurately reflect inventory values, cost of goods sold, and revenue recognition that support financial reporting and analysis. The authors discussed how comprehensive system integration enables holistic business management that addresses operational, financial, and strategic requirements through unified information systems.

This literature is highly relevant to the proposed Automated Inventory Management System (AIMS) because the proposed system aims to integrate inventory monitoring, sales recording, barcode scanning, and transaction processing within pharmacies and convenience stores. Similar to the concepts discussed by Murdick, Ross, and Claggett, the proposed system seeks to improve transaction accuracy, inventory visibility, operational efficiency, and customer service through retail automation technologies.

The proposed AIMS also aims to automate inventory updates and sales monitoring in real time, allowing employees and administrators to manage business operations more effectively through immediate access to accurate information. Through barcode integration and computerized transaction processing, the proposed system seeks to minimize operational errors, improve workflow organization, and provide accurate operational reports that support managerial decision-making and strategic planning activities.

Furthermore, the literature supports the implementation of automated POS and inventory systems because it demonstrates how retail automation technologies contribute to organizational productivity, operational reliability, and customer satisfaction. The concepts discussed in the study provide a strong theoretical and technological foundation for the development and implementation of the proposed Automated Inventory Management System within pharmacies and convenience store environments.

The proposed AIMS aligns with the authors' discussion regarding the importance of system integration, database management, security controls, and user training in ensuring successful implementation and ongoing operational effectiveness. These implementation factors are essential to maximize system benefits while minimizing operational disruptions and user adoption challenges.

2.4 Synthesis Of The Reviewed Literature

A. SUMMARY OF FINDINGS Foreign Studies

The body of foreign literature on automated inventory management systems presents a robust foundation of research spanning multiple decades and geographic contexts.
These studies collectively establish the critical role of inventory management systems in organizational efficiency, profitability, and competitive advantage.

1. The Positive Impact of Automation on Organizational Performance

One of the most consistent findings across the foreign literature is the significant positive relationship between automated inventory management systems and organizational performance. Koumanakos (2008) conducted a comprehensive study examining the effect of inventory management on firm performance and found that firms with efficient inventory management systems demonstrated markedly superior performance metrics compared to those relying on traditional manual methods. His research revealed that automated systems reduce carrying costs, minimize stockouts, and optimize inventory turnover rates, all of which contribute to improved financial outcomes.
Similarly, Saleem (2020) investigated automated inventory management systems and their impact on supply chain risk management in manufacturing firms of Pakistan. His findings demonstrated that automated systems not only streamline internal operations but also significantly reduce supply chain vulnerabilities. The study revealed that manufacturing firms that implemented automated inventory management experienced fewer disruptions, better supplier coordination, and enhanced ability to respond to market changes. This research aligns with the broader operations management literature that emphasizes the strategic importance of inventory management as a competitive tool.
Prempeh (2015) extended this line of inquiry by examining the impact of efficient inventory management on profitability in selected manufacturing firms in Ghana. His empirical study provided evidence that profitability is directly correlated with inventory management efficiency. Firms that maintained optimal inventory levels while minimizing holding costs and stockout frequencies showed higher profit margins. This finding is particularly significant as it demonstrates the financial tangible benefits of inventory management automation in developing country contexts, which share similar economic characteristics with the Philippines.

2. Supply Chain Integration and Information Systems

The role of information systems in supply chain integration represents another significant theme in the foreign literature. Gunasekaran and Ngai (2004) provided foundational research on information systems in supply chain integration and management. Their study established that inventory management systems do not operate in isolation but must be integrated with broader supply chain information systems to maximize their effectiveness. They argued that real-time information sharing across the supply chain enhances coordination between suppliers, manufacturers, and distributors.

Chuang, Oliva, and Liao (2006) further explored supply chain management and information systems integration, emphasizing the need for seamless data flow between different organizational functions. Their research demonstrated that fragmented information systems lead to the bullwhip effect, inventory imbalances, and increased operational costs. This finding underscores the importance of designing inventory management systems with integration capabilities. More recent studies by DeHoratius and Raman (2008) on inventory record inaccuracy provided critical insights into a persistent problem affecting inventory management effectiveness. Their empirical analysis revealed that inventory record inaccuracies significantly undermine the benefits of inventory management systems. They found that many organizations experience substantial discrepancies between actual inventory levels and system records, leading to poor decision-making, stockouts, and overstock situations. This research highlights the importance of system accuracy and the need for technologies such as barcode and RFID to ensure data integrity.

3. Technology Integration in Inventory Management

The foreign literature extensively documents the benefits of various technologies integrated into inventory management systems. Sharma, Garg, and Agarwal (2011) conducted a study on barcode technology and its impact on inventory management systems. Their research demonstrated that barcode technology significantly improves inventory accuracy, reduces manual data entry errors, and accelerates inventory processing times. The study found that organizations implementing barcode systems experienced substantial improvements in inventory visibility and control.
Atieh et al. (2016) extended this technology-focused research by examining performance improvement of inventory management system processes through automated warehouse management systems. Their study demonstrated that warehouse automation, including features such as automated picking, sorting, and packing, substantially improves operational efficiency. The research revealed that automated warehouse systems reduce labor costs, minimize errors, and improve order fulfillment speed. This finding is particularly relevant for larger retail operations and distribution centers.
Ghazi, Salih, and Janabi (2023) provided contemporary research specifically focused on implementing automated inventory management systems for small and medium-sized enterprises. Their study addressed a critical gap in the literature by focusing on SMEs, which often resource constraints that limit their ability to implement sophisticated systems. The research developed a practical framework for SME implementation, considering factors such as cost, complexity, and scalability. Their findings indicated that even basic automation can yield significant benefits for SMEs when properly implemented.

4. Inventory Management in Small and Medium Enterprises

The challenges faced by small and medium enterprises in implementing inventory management systems represent a significant theme in the foreign literature. Rajeev (2010) conducted a study on inventory management in small and medium enterprises, focusing on machine tool enterprises in Bangalore. His research identified key challenges including limited financial resources, lack of technical expertise, and resistance to changing established business practices. The study found that SMEs often struggle with maintaining optimal inventory levels due to limited forecasting capabilities and cash flow constraints.
Similarly, Samuel and Ondiek (2014) examined inventory management automation and performance of supermarkets in Western Kenya. Their study focused specifically on the

retail sector and found that supermarkets implementing automated inventory management systems experienced significant performance improvements, including reduced stockouts, better inventory turnover, and improved customer satisfaction.
However, they also identified barriers to adoption, including high initial investment costs and lack of technical training.

The foreign literature also includes substantial foundational works that inform best practices in inventory management. Classical texts by Wild (2017), Stevenson (2018), Waters (2019), Heizer, Render, and Munson (2017), and other operations management scholars provide established frameworks for inventory management including economic order quantity models, reorder point calculations, and inventory classification techniques (ABC analysis). These works serve as essential references for understanding the theoretical foundations of inventory management. Additional foundational works by Jacobs and Chase (2018), Bowersox, Closs, and Cooper (2013), Muller (2011), Chopra and Meindl (2016), Silver, Pyke, and Peterson (1998), Christopher (2016), and Nahmias and Olsen (2015) provide comprehensive coverage of supply chain logistics, inventory control principles, production planning, and operations analysis. These texts establish the analytical frameworks necessary for effective inventory management system design.

5. Sector-Specific Applications

The foreign literature documents inventory management system implementations across diverse sectors. Akinsanya (2019) designed and implemented computerized stock management systems for supermarkets, specifically studying MIDE Supermarket. Her research provided practical insights into the design considerations for retail inventory systems, including product categorization, barcode integration, and sales tracking. Abisoye, Fatoba, and Blessing (2013) similarly designed computerized inventory management systems for supermarkets, emphasizing the importance of user interface design and real-time reporting capabilities.

2.4 Synthesis of the Reviewed Literature

The reviewed foreign studies, local studies, and related literature provide a comprehensive foundation for the development of the Automated Inventory Management System (AIMS). The literature consistently demonstrates that effective inventory management plays an important role in improving operational efficiency, maintaining accurate inventory records, reducing stock-related problems, supporting sales operations, and assisting organizations in making informed decisions. The reviewed studies also highlight the importance of integrating appropriate technologies, user-friendly interfaces, security controls, and reliable database systems into inventory management applications.

2.4.1 Foreign Studies

The foreign studies reviewed in this research establish the importance of automated inventory management in improving organizational performance. Koumanakos (2008) found that efficient inventory management is associated with improved organizational performance through better inventory turnover, reduced carrying costs, and fewer stockout problems. Similarly, Saleem (2020) demonstrated that automated inventory management can improve internal operations and reduce supply chain vulnerabilities by improving coordination and responsiveness. Prempeh (2015) also found a relationship between efficient inventory management and profitability, particularly through maintaining appropriate inventory levels and minimizing inventory-related costs.

The reviewed foreign literature also emphasizes the importance of integrating inventory systems with broader information systems. Gunasekaran and Ngai (2004) highlighted the role of information systems in supply chain integration and management, emphasizing the importance of information sharing and coordination. Chuang, Oliva, and Liao (2006) further emphasized the need for effective information flow between organizational functions. DeHoratius and Raman (2008) identified inventory record inaccuracy as a significant problem that can negatively affect decision-making, stock availability, and inventory control. These findings demonstrate the importance of maintaining accurate and centralized inventory information within an automated system.

Technology integration was another major theme identified in the foreign studies. Sharma, Garg, and Agarwal (2011) demonstrated that barcode technology can improve inventory accuracy, reduce manual data-entry errors, and accelerate inventory processing. Atieh et al. (2016) showed that automated warehouse management technologies can improve operational efficiency, reduce errors, and improve the speed of order fulfillment. Ghazi, Salih, and Janabi (2023) further demonstrated that automated inventory management can provide significant benefits to small and medium-sized enterprises when implementation considers cost, complexity, and scalability.

The foreign literature also identified challenges experienced by small and medium-sized enterprises. Rajeev (2010) identified limited financial resources, lack of technical expertise, resistance to organizational change, and limited forecasting capabilities as important challenges affecting inventory management. Samuel and Ondiek (2014) similarly found that supermarkets using automated inventory management systems could achieve improvements in inventory turnover, stock availability, and customer satisfaction, while also identifying implementation costs and lack of technical training as barriers to adoption.

Other foreign literature provides theoretical foundations for inventory management and related operations. Wild (2017), Stevenson (2018), Waters (2019), Heizer, Render, and Munson (2017), Jacobs and Chase (2018), Bowersox, Closs, and Cooper (2013), Muller (2011), Chopra and Meindl (2016), Silver, Pyke, and Peterson (1998), Christopher (2016), and Nahmias and Olsen (2015) provide principles related to inventory control, supply chain management, inventory classification, production planning, and operations analysis. These concepts provide a theoretical basis for designing an inventory management system that supports effective stock monitoring and operational control.

Foreign studies also demonstrate the application of computerized inventory systems in specific business environments. Akinsanya (2019) examined computerized stock management for supermarkets, including product categorization, barcode integration, and sales tracking. Abisoye, Fatoba, and Blessing (2013) likewise emphasized the importance of computerized inventory management, user interface design, and reporting capabilities in supermarket environments. These studies demonstrate that inventory systems become more useful when their functions are adapted to the operational requirements of their intended users.
Overall, the foreign studies establish that automation, accurate data management, technology integration, and appropriate system design can improve inventory and operational performance.

However, the literature also indicates that implementation challenges such as cost, technical expertise, system integration, and user acceptance must be considered when developing inventory management systems.

2.4.2 Local Studies

The local literature provides essential insights into the Philippine context, demonstrating both convergence with global findings and unique local considerations. Philippine studies reflect the challenges and opportunities specific to the local business environment.

1. Retail Sector Innovation

The local literature demonstrates significant innovation in developing inventory management solutions tailored to Philippine retail establishments. Dela Cruz and Ramos (2018) conducted a pioneering study on Point of Sale and Inventory Management Systems for Small Retail Businesses in the Philippines.

Their research provided foundational insights into the needs of small retail establishments in the country,identifying key features necessary for Philippine retail contexts including local currency handling, local supplier management, and integration with local business practices.

Santos, Garcia, and Lopez (2020) further advanced local research by developing a Web-Based Inventory Monitoring System for Retail Stores in the Philippines.

Their study demonstrated the feasibility of web-based solutions for Philippine retail, addressing connectivity considerations and practical implementation challenges. The research highlighted the growing technological readiness of Philippine small businesses and the potential for cloud-based solutions.

de los Santos et al. (2021) developed the "Imbentaryo App," an intelligent inventory and decision support system presented at the 2021 International Conference on Information Technology. This innovative system represents a significant advancement in local inventory management solutions, incorporating decision support capabilities such as automatic reordering suggestions, sales trend analysis, and inventory forecasting. The study demonstrated that Philippine researchers are capable of developing sophisticated solutions that rival international counterparts.

2. Sector-Specific Implementations

The local literature demonstrates diverse sector-specific implementations: Grocery and Convenience Stores: Ortega and Salazar (2021) developed a Computerized Inventory and Sales System for Mini Grocery Stores in Batangas, addressing the specific needs of small neighborhood grocery stores. Garcia and Santos (2018) created an Automated Inventory System for Convenience Stores in Luzon, focusing on the convenience store segment with features such as fast-moving consumer goods tracking and expiry date monitoring.

Pharmacies: Domingo and Ferrer (2020) developed an Automated Inventory Monitoring System for Pharmacies in Cebu Province, addressing the unique requirements of pharmaceutical inventory management including batch tracking, expiry monitoring, and regulatory compliance. Mendoza and Reyes (2019) examined inventory management challenges and solutions for independent pharmacies in Metro Manila, identifying key pain points including drug pricing regulations and supplier dependencies.

Healthcare: Parilla et al. (2022) conducted a comprehensive study on Inventory Management Practices and Service Delivery of Healthcare Facilities in Ilocos Norte Philippines. Their research examined the critical relationship between inventory management and healthcare service quality, finding that efficient inventory management directly impacts patient care delivery through ensuring medical supply availability.

Supermarkets: Villar and Cruz (2021) developed a Web-Based Inventory System for Supermarket Chains in the Philippines, demonstrating the scalability of local solutions to larger retail operations.

Manufacturing: Bautista Jr. and Young (2022) examined Effective Inventory Management Systems in supply and distribution management for a manufacturer of food seasoning products in the Philippines, addressing the specific requirements of manufacturing inventory including raw material tracking, work-in-progress monitoring, and finished goods management.

3. Implementation Challenges and Barriers

Several local studies have identified implementation challenges specific to the Philippine context. Fernandez and Bautista (2022) conducted a critical study on Implementation Challenges of Inventory Management Systems in Philippine SME Retail. Their research identified key barriers including:

a. Limited financial resources for system implementation
b. Lack of technical expertise among staff
c. Resistance to changing established business practices
d. Concerns about data security and privacy
e. Limited internet connectivity in some areas
f. Lack of local technical support

Aguilar and Torres (2020) similarly examined Point of Sale System Implementation for Small Retail Enterprises in Metro Manila, identifying training requirements and user acceptance as critical success factors.

4. Cloud-Based and Emerging Technologies

Recent local studies have explored emerging technologies. Villareal, Mendoza, and Ramos (2022) developed a Cloud-Based Point of Sale and Inventory System for Retail Businesses in Manila, demonstrating the growing adoption of cloud computing in Philippine retail. Their research addressed considerations such as data security, connectivity requirements, and scalability.

2.4.3 Related Literature

The related literature provides the essential theoretical and technical foundations that underpin the development of inventory management systems.

1. Information Systems Foundations

Laudon and Laudon (2020) provide comprehensive coverage of Management Information Systems in their seminal text, establishing frameworks for understanding the role of information systems in modern organizations. Their work explains how information systems support business processes, decision-making, and competitive strategy. This foundational knowledge is essential for understanding how inventory management systems fit within broader organizational information ecosystems. Stair and Reynolds (2018) and Shelly and Rosenblatt (2012) provide essential knowledge in systems analysis and design methodologies. These texts guide the systematic development of information systems, including requirements gathering, system design, implementation, and testing. Their frameworks ensure that system development follows established best practices.

Turban, Pollard, and Wood (2018) discuss information technology for management, emphasizing digital transformation in business operations. Their work provides insights into how technology drives business value and competitive advantage.

2. Database and Technical Foundations

Silberschatz, Korth, and Sudarshan (2019) provide essential database concepts necessary for designing the data storage and retrieval mechanisms of an inventory system. Their coverage of relational database design, SQL, and data normalization is critical for ensuring data integrity and system performance.

Pressman and Maxim (2019) and Sommerville (2016) offer comprehensive software engineering principles that guide systematic system development, testing, and maintenance. Their methodologies ensure that systems are developed systematically with proper documentation and quality assurance.

3. Security and Emerging Technologies

Stallings and Brown (2018) address computer security principles and practices, critical for protecting sensitive inventory data. Their coverage of encryption, access controls, and security architectures is essential for ensuring data protection. Russell and Norvig (2021) provide comprehensive coverage of artificial intelligence, which may inform future enhancements such as predictive analytics, machine learning for demand forecasting, and intelligent decision support.

Erl, Puttini, and Mahmood (2013) explain cloud computing concepts, supporting the growing trend toward cloud-based solutions observed in both foreign and local studies.

4. Additional Technical References

Deitel and Deitel (2020) provide programming foundations for web and internet applications. Dix et al. (2004) cover human-computer interaction principles essential for designing user-friendly interfaces. Leon (2014) addresses Enterprise Resource Planning, providing context for how inventory management systems fit within larger
organizational information ecosystems. Murdick, Ross, and Claggett (2017) discuss information systems for modern management.

2.4.4 RESEARCH GAPS IDENTIFIED

1. Limited Industry-Specific Solutions

While general inventory management systems have been extensively studied in both
foreign and local contexts, there remains a significant gap in industry-specific solutions. Most existing systems are designed as general-purpose applications that may not adequately address the unique requirements of specific business types. For instance:

- Supermarket systems may not include specific features for perishable goods tracking
- Pharmacy systems may lack comprehensive regulatory compliance features
- Restaurant systems may not integrate with kitchen management
- Manufacturing systems may not include production planning features

The studies by Akinsanya (2019), Abisoye, Fatoba, and Blessing (2013), and local studies such as Ortega and Salazar (2021) and Domingo and Ferrer (2020) demonstrate sector- specific implementations, but further development is needed to address the full range of Philippine business types.

2. Insufficient Technology Integration

Most existing studies focus on isolated technological solutions. A critical gap exists in the integration of multiple technologies into unified, comprehensive systems: - Barcode Technology: Studies by Sharma, Garg, and Agarwal (2011) focus primarily on barcode systems as standalone solutions

- POS Integration: Many studies treat POS and inventory as separate systems rather than integrated modules
- Cloud Computing: While Villareal, Mendoza, and Ramos (2022) explore cloud solutions, integration with on-premise systems remains limited
- Analytics: Few studies integrate advanced analytics with inventory monitoring This fragmentation results in systems that require multiple separate applications, increasing complexity and cost for end users.

3. Limited Decision Support Capabilities

A significant gap exists in decision support functionality. Most existing systems focus primarily on:

- Inventory monitoring and tracking
- Record-keeping and reporting
- Basic transaction processing

However, advanced decision support features remain underutilized:

- Demand forecasting and predictive analytics
- Automatic reorder point calculations
- Seasonal trend analysis
- Supplier performance evaluation
- What-if scenario analysis

The study by de los Santos et al. (2021) on "Imbentaryo App" represents a step toward decision support, but further development is needed to make these features more sophisticated and accessible.

4. Lack of Real-Time Analytics

While web-based and cloud systems are emerging, real-time analytics capabilities remain limited:
- Real-time dashboard visualizations
- Instant alerts for low stock or overstock situations
- Live sales trend analysis
- Dynamic reordering suggestions

Most existing systems rely on periodic reports rather than continuous analytics, limiting their value for proactive inventory management.

5. User-Friendly Interfaces for Non-Technical Users

The studies by Fernandez and Bautista (2022) and Aguilar and Torres (2020) highlight implementation challenges in SME retail, including user resistance and training difficulties. A critical gap exists in:
- Intuitive user interface design
- Minimal training requirements
- Multilingual support
- Accessibility features
- Mobile-responsive designs

This gap is particularly significant in the Philippine context where many small business owners may have limited technological experience.

6. Limited Contextual Adaptation to Local Practices

While local studies address some Philippine-specific considerations, significant gaps remain:
- Local currency and payment integration
- Local supplier management and relationships
- Philippine business regulations compliance
- Local tax requirements (BIR compliance)
- Integration with local delivery services
- Philippine holiday and pay period considerations

Most existing systems are adaptations of foreign solutions that may not fully address local business practices.

7. Scalability Challenges

Many existing local solutions are designed for specific business sizes and may not scale effectively:
- Systems designed for small retail may not accommodate growth
- Limited ability to handle increased transaction volumes
- Lack of multi-branch support
- Inadequate data backup and recovery capabilities

This limitation forces businesses to replace systems as they grow, resulting in additional costs and disruptions.

2.4.5 HOW THE CURRENT PROJECT ADDRESSES THOSE GAPS

Based on the reviewed literature and related studies, several limitations were identified in existing inventory management systems used by pharmacies and convenience stores. Many systems primarily focused on basic inventory monitoring and sales recording but lacked comprehensive features such as automated stock monitoring, expiration date tracking, role-based user management, activity logging, and integrated backup and restoration functions. In addition, some systems relied on manual inventory updates, making them susceptible to human errors, inaccurate stock records, and delays in business operations.

To address these identified gaps, the researchers developed the Automated Inventory Management System (AIMS), which integrated inventory management, barcode-based Point of Sale (POS), and administrative functions into a single web-based application operating in an offline environment. Unlike conventional inventory systems, the developed system automatically updated inventory records after every completed transaction, reducing manual intervention and improving the accuracy of stock information.

The developed system also incorporated automated inventory monitoring by providing notifications for low-stock products and items nearing their expiration dates. These features enabled administrators to replenish inventory promptly and remove expiring products before they could no longer be sold. As a result, the system improved inventory control and minimized losses associated with stock shortages and expired products.
To strengthen security and accountability, the system implemented role-based access control by providing separate interfaces and permissions for administrators and cashiers. Administrator functions included product management, cashier account management, inventory monitoring, report generation, receipt customization, activity logging, and database backup and restoration, while cashier accounts were limited to processing sales transactions and viewing transaction-related information. This separation of responsibilities reduced unauthorized access and improved system security.

Furthermore, the system maintained centralized records using a MySQL database, allowing product information, inventory records, sales transactions, user accounts, and activity logs to be stored consistently. The integration of reporting tools and activity logs also provided administrators with accurate operational information that supported inventory planning and business decision-making.
Overall, the Automated Inventory Management System addressed the limitations identified in previous studies by providing a comprehensive, reliable, and user-friendly solution for inventory management and sales operations. Through automation, barcode integration, centralized data management, and secure user access, the developed system improved operational efficiency, reduced human errors, and supported more effective inventory control for pharmacies and convenience stores.

CHAPTER 3

Methodology

3.1 RESEARCH DESIGN

This study employed a descriptive-developmental research design to analyze the existing inventory management practices of pharmacies and convenience stores and to develop an Automated Inventory Management System (AIMS) that addressed the operational problems identified during the study.

The descriptive aspect of the research enabled the researchers to examine the existing methods used in inventory management, sales processing, and record keeping. Through observations, interviews, and review of existing operational procedures and documents, the researchers identified problems related to manual inventory monitoring, delayed sales recording, inaccurate stock information, and the lack of an integrated inventory management system.

The developmental aspect of the research focused on the design, development, implementation, testing, and evaluation of the Automated Inventory Management System. The identified user and operational requirements were transformed into a functional software application capable of supporting inventory management, product management, barcode-based Point-of-Sale (POS) transactions, sales reporting, user management, inventory monitoring, activity logging, receipt generation, and database backup and restoration.

The development of the system followed an iterative and incremental approach. The application was divided into functional modules that were developed, tested, evaluated, and refined before being integrated into the complete system. This approach allowed the researchers to identify and correct software errors at an early stage and continuously improve the functionality, reliability, and usability of the application.

During the initial stage of the study, the researchers gathered the functional and non-functional requirements of the proposed system. The collected information served as the basis for identifying the features required to support the operational needs of pharmacies and convenience stores. Based on these requirements, the researchers prepared the system design, database structure, interface layouts, and application workflow.

The implementation stage involved the development of the Automated Inventory Management System using Python Flask as the application framework and MySQL through XAMPP as the database management system. The system was developed incrementally by implementing administrator authentication, cashier authentication, product management, inventory management, Point-of-Sale operations, barcode scanning, reporting, inventory notifications, receipt customization, database backup and restoration, and activity logging.

Each module was tested individually before being integrated into the complete application. Continuous testing and verification were also performed after each development iteration. Database records were inspected to verify the accuracy of inventory updates, sales transactions, and user activities. Errors and inconsistencies identified during testing were corrected before proceeding to subsequent development activities.

Upon completion of the system modules, comprehensive testing and evaluation were conducted to determine whether the developed system satisfied the objectives of the study and the operational requirements of the intended users.

The descriptive-developmental research design and iterative development approach enabled the researchers to systematically plan, develop, test, evaluate, and refine the system. This process supported the development of an efficient, reliable, and user-friendly inventory and sales management solution for pharmacies and convenience stores.

3.2 Agile Software Development Methodology

The development of the Automated Inventory Management System adopted the Agile Software Development Methodology as the primary software development approach. Agile was selected because it supports flexibility, incremental development, continuous testing, feedback, and improvement throughout the development process.

Unlike approaches that require the complete system to be developed before testing, Agile allows the application to be developed through smaller and manageable iterations. Each iteration can be evaluated and tested before the succeeding development cycle. This approach allowed the researchers to identify software defects early and implement necessary improvements throughout the development process.

The researchers applied Agile principles by developing the system incrementally through individual functional modules. These included administrator authentication, cashier authentication, product management, inventory monitoring, barcode-based Point-of-Sale transactions, sales reporting, receipt customization, activity logging, and database backup and restoration.

Each completed module underwent functional verification before being integrated with the other system components. The iterative development process helped reduce implementation risks, simplify debugging, and improve the stability and reliability of the developed application.

Figure 3.1. Agile Software Development Methodology

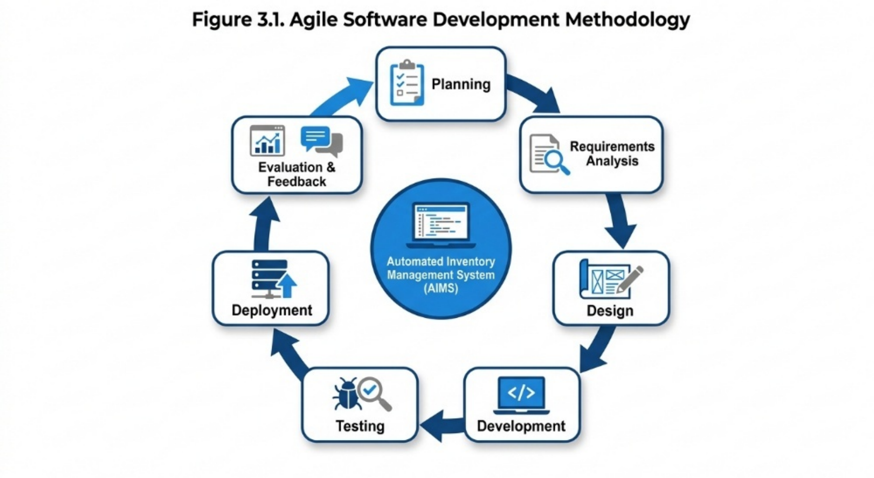


Figure 3.1 illustrates the Agile Software Development Methodology adopted in the development of the Automated Inventory Management System. The methodology consisted of seven major phases: Planning, Requirements Analysis, Design, Development, Testing, Deployment, and Evaluation and Feedback.

The Planning phase established the objectives, scope, resources, and development priorities. Requirements Analysis identified the functional and non-functional requirements of the intended users. The Design phase prepared the system architecture, database structure, user interfaces, and system workflow. Development involved the actual implementation of the system modules. Testing verified the correctness and reliability of each completed module. Deployment prepared the system for operation within the intended local environment. Evaluation and Feedback provided the basis for final improvements and refinements.

The continuous nature of these activities allowed the researchers to evaluate completed functionalities and apply necessary improvements during the development process.

3.3 System Development Process

The Automated Inventory Management System followed the Agile Software Development Methodology through a series of iterative and incremental development cycles. The development process consisted of seven major phases: Planning, Requirements Analysis, Design, Development, Testing, Deployment, and Evaluation and Feedback.

3.3.1 Planning

The Planning phase served as the initial stage of the system development process. During this phase, the researchers identified the objectives, scope, limitations, expected outcomes, and general requirements of the Automated Inventory Management System.
Existing operational procedures related to inventory management, sales recording, and stock monitoring were examined to identify problems that could be addressed through system automation. The researchers also identified the hardware, software, and technical resources necessary for the development and implementation of the system.
Development priorities and module implementation plans were established during this phase. The system was divided into smaller development tasks so that each component could be developed, tested, and improved incrementally.

3.3.2 Requirements Analysis

The Requirements Analysis phase involved gathering and analyzing the functional and non-functional requirements of the proposed system. Information obtained through observations, interviews, and review of existing operational procedures was used to identify the functions required by the intended users.
The functional requirements included secure user authentication, product management, category management, inventory monitoring, barcode-based Point-of-Sale transactions, receipt generation, sales reporting, activity logging, and database backup and restoration.
The non-functional requirements included usability, security, performance, reliability, maintainability, and compatibility with the intended hardware, software, and Local Area Network environment.

3.3.3 Design

After the requirements were identified and analyzed, the researchers proceeded to the Design phase. This phase focused on preparing the overall system structure, database design, navigation flow, graphical user interface, and system architecture.
The database structure was designed using MySQL to organize information related to users, products, categories, inventory, sales transactions, activity logs, and system settings. Separate interfaces were designed for administrators and cashiers based on their assigned responsibilities.
The system architecture was also designed to illustrate the interaction among the users, the web application, and the centralized MySQL database.

3.3.4 Development

The Development phase involved the actual implementation of the Automated Inventory Management System using Python Flask as the application framework, HTML, CSS, JavaScript, and Jinja2 templates for the user interface, and MySQL through XAMPP as the database management system.
The application was developed incrementally by implementing one functional module at a time. Administrator and cashier authentication were developed first, followed by product management, inventory monitoring, barcode scanning, Point-of-Sale operations, reporting, receipt customization, database backup and restoration, and activity logging.
Each completed module was functionally verified before integration with the other modules. This approach allowed errors to be identified and corrected before they affected other system components.

3.3.5 Testing

Testing was conducted throughout the development process. Each module was tested after implementation to verify whether it operated according to its intended function.
The researchers tested user authentication, product management, inventory updates, barcode scanning, sales processing, report generation, receipt printing, database operations, and activity logging. Database records were also inspected to verify that information was correctly stored and updated after system operations.
Errors identified during testing were corrected before additional development activities proceeded. Continuous testing helped improve the reliability, consistency, and overall quality of the developed system.

3.3.6 Deployment

After the major system functions successfully passed testing, the Automated Inventory Management System was deployed within the intended local environment.
The Python Flask application and MySQL database were configured on a local computer that served as the server. Authorized client computers connected to the same Local Area Network could access the system using a supported web browser.
During deployment, the researchers verified database connectivity, system accessibility, user accounts, inventory records, and the operation of the major system modules.

3.3.7 Evaluation and Feedback

The Evaluation and Feedback phase involved assessing the completed Automated Inventory Management System and gathering feedback regarding its functionality, usability, efficiency, reliability, security, maintainability, compatibility, and portability.
The results of testing and evaluation were used to identify areas requiring correction or improvement. Necessary refinements were implemented before the final version of the system was completed.
This continuous evaluation process reflected the iterative nature of the Agile methodology and helped ensure that the developed system satisfied the operational requirements identified during the earlier stages of development.

3.4 System Design

3.4.1 Use Case Diagram

The Use Case Diagram illustrates the interactions between the users (actors) and the Automated Inventory Management System (AIMS). It identifies the major functions that can be performed by each user according to their assigned roles and responsibilities. The system has two primary actors: the **Administrator** and the **Cashier**. Each actor is granted access only to the functions necessary for their respective tasks, ensuring efficient operations and secure access to the system.
The use case diagrams help visualize the system's functional requirements by showing how users interact with different modules of the application. These diagrams also serve as a guide during system development and testing by defining the scope of user interactions.

3.4.1.1 Overall Use Case Diagram

Figure 3.2 Overall Use Case Diagram

Figure 3.2 Overall Use Case Diagram

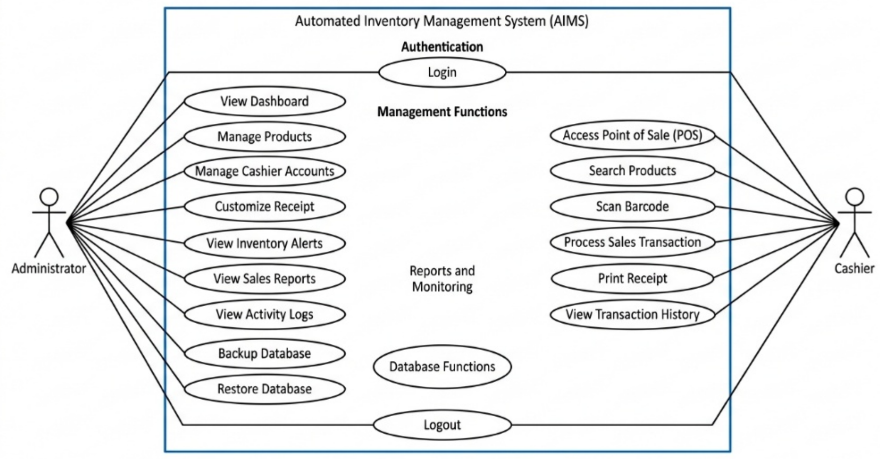


Figure 3.2 Overall Use Case Diagram

The Overall Use Case Diagram presents the complete interaction between the users and the Automated Inventory Management System. It illustrates all major system functionalities that can be accessed by both the Administrator and the Cashier. The Administrator performs administrative functions such as inventory management, product management, sales monitoring, user management, database maintenance, and system configuration, while the Cashier performs transaction-related functions through the Point of Sale (POS) module.

Overall Use Cases

Actors

- **Administrator**
- **Cashier**

Administrator

- Login
- View Dashboard
- Manage Products
- Manage Cashier Accounts
- View Inventory Alerts
- View Sales Reports
- Customize Receipts
- View Activity Logs
- Backup Database
- Restore Database
- Logout

Cashier

- Login
- Access Point of Sale (POS)
- Search Products
- Scan Barcode
- Process Sales Transaction
- Print Receipt
- View Transaction History
- Logout

3.4.1.2 Administrator Use Case Diagram

Figure 3.3 Administrator Use Case Diagram

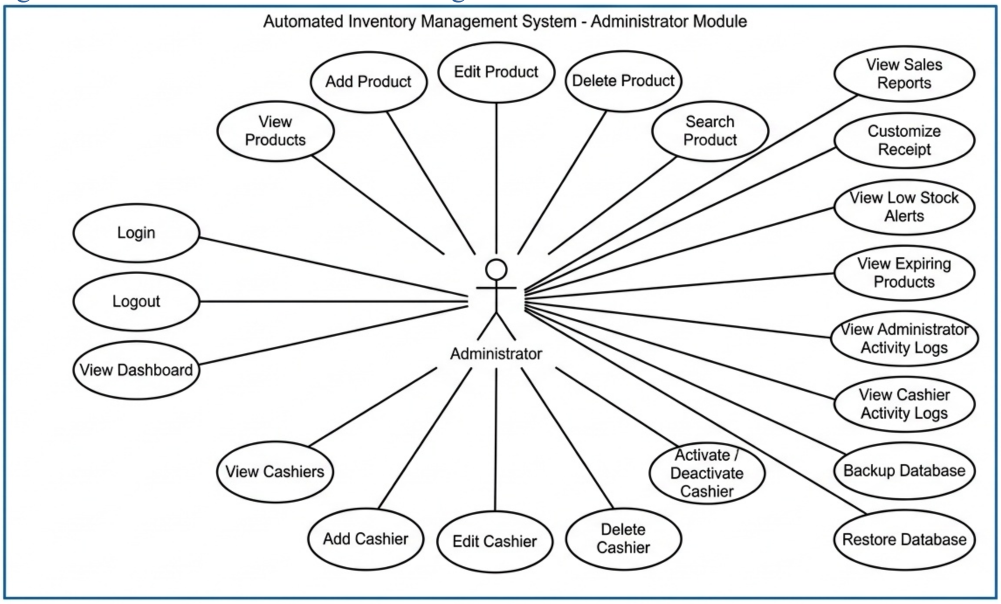


Figure 3.3 Administrator Use Case Diagram

The Administrator Use Case Diagram illustrates the interactions between the administrator and the Automated Inventory Management System. The administrator has full access to all management and administrative modules of the system. These functions enable administrators to maintain accurate inventory records, manage users, monitor sales activities, configure system settings, and ensure that the system operates efficiently. After successful authentication, the administrator can access the dashboard and perform inventory-related tasks, including adding, updating, deleting, and searching products. The administrator is also responsible for managing cashier accounts, monitoring low-stock and expired-product alerts, generating sales reports, reviewing activity logs, configuring receipt and store information, and performing database backup and restoration. These functions provide administrators with complete control over daily business operations while ensuring data integrity and system security.

Administrator Use Cases

Authentication
- Login
- Logout

Dashboard
- View Dashboard

Product Management
- View Products
- Add Product
- Edit Product
- Delete Product
- Search Product

Sales Management
- View Sales Reports

Alert Management
- View Low Stock Alerts
- View Expired Product Alerts

Cashier Management
- Add Cashier
- Edit Cashier
- Delete Cashier
- Activate/Deactivate Cashier

Activity Monitoring
- View Administrator Activity Logs
- View Cashier Activity Logs

Database Management
- Backup Database
- Restore Database

System Configuration
- Update Store Information
- Customize Receipt Settings

3.4.1.3 Cashier Use Case Diagram

Figure 3.4 Cashier Use Case Diagram

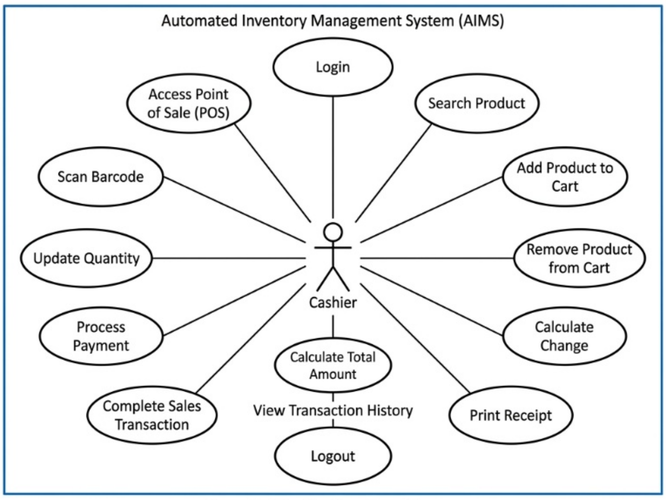


Figure 3.4 Cashier Use Case Diagram

The **Cashier Use Case Diagram** illustrates the interactions between the cashier and the Automated Inventory Management System (AIMS). It presents the specific functions that can be performed by the cashier during daily sales operations. Unlike the administrator, the cashier has limited access rights and is only authorized to perform transaction-related activities through the Point of Sale (POS) module. This role-based access control ensures the security and integrity of business data by preventing unauthorized access to administrative functions.

After successfully logging into the system, the cashier can access the Point of Sale (POS) interface to process customer purchases. Products can be added to the transaction by scanning their barcode using a barcode scanner or by searching for the product manually. The cashier may modify the quantity of selected products, remove items from the shopping cart when necessary, and proceed with payment processing. During checkout, the system automatically calculates the total amount due and the corresponding change based on the payment received. Once the transaction is completed, the system records the sale, updates the inventory, and generates a printable receipt for the customer. The cashier may also review transaction history for completed sales before logging out of the system.

Cashier Use Cases

- Login
- Access Point of Sale (POS)
- Search Product
- Scan Barcode
- Add Product to Cart
- Update Quantity
- Remove Product from Cart
- Process Payment
- Calculate Total Amount
- Calculate Change
- Complete Sales Transaction
- Print Receipt
- View Transaction History
- Logout

3.4.2 System Architecture

The system architecture presents the overall structure and operational framework of the proposed Automated Inventory Management System (AIMS). It illustrates how the different components of the system interact to process user requests, manage business operations, and store information within a centralized database. The architecture was designed to provide an organized, secure, and efficient environment for inventory management, sales transactions, and administrative operations.

The Automated Inventory Management System follows a **three-tier architecture**, consisting of the **Presentation Layer**, **Application Layer**, and **Database Layer**. This architectural design separates the user interface, business logic, and data storage into independent components, making the application easier to develop, maintain, and enhance. It also improves system reliability by ensuring that each layer performs a specific responsibility while communicating efficiently with the other layers.

Presentation Layer

The **Presentation Layer** serves as the front-end interface of the application where administrators and cashiers interact with the system. It was developed using HTML, CSS, JavaScript, Bootstrap, and Jinja2 templates to provide a responsive and user-friendly interface. Through this layer, authorized users can log into the system, manage inventory, process sales transactions, monitor stock levels, generate reports, customize receipts, and perform other system functions according to their assigned access privileges.

Application Layer

The **Application Layer** contains the core business logic of the system. This layer was developed using the Python Flask framework and is responsible for processing user requests, validating input, executing business rules, and communicating with the database. It manages user authentication, inventory updates, barcode-based Point of Sale (POS) transactions, report generation, activity logging, receipt customization, low-stock monitoring, expiration monitoring, and database backup and restoration. Every request submitted through the user interface is processed in this layer before the corresponding output is returned to the user.

Database Layer

The **Database Layer** consists of the MySQL database managed through XAMPP. This layer stores and manages all information generated by the system, including administrator accounts, cashier accounts, product records, categories, inventory quantities, sales transactions, activity logs, receipt settings, and backup files. The centralized database ensures that information is stored accurately, updated automatically, and retrieved efficiently whenever requested by authorized users.

The communication between these three layers enables the system to perform its intended operations efficiently. When a user submits a request through the Presentation Layer, the request is forwarded to the Application Layer for processing. The Flask application validates the request, performs the required operations, retrieves or updates information from the MySQL database, and sends the processed results back to the Presentation Layer. This workflow ensures that inventory records, sales transactions, and administrative information remain accurate and synchronized throughout the system.

Role-Based Access Control (RBAC)

The system also implements **Role-Based Access Control (RBAC)** to enhance security and maintain data integrity. Administrator accounts have full access to inventory management, product management, cashier management, sales reports, receipt customization, backup and restoration, and activity logs. Cashier accounts are limited to Point of Sale (POS) transactions, product searching, barcode scanning, and receipt printing. This separation of user privileges prevents unauthorized access to administrative functions and protects sensitive business information.

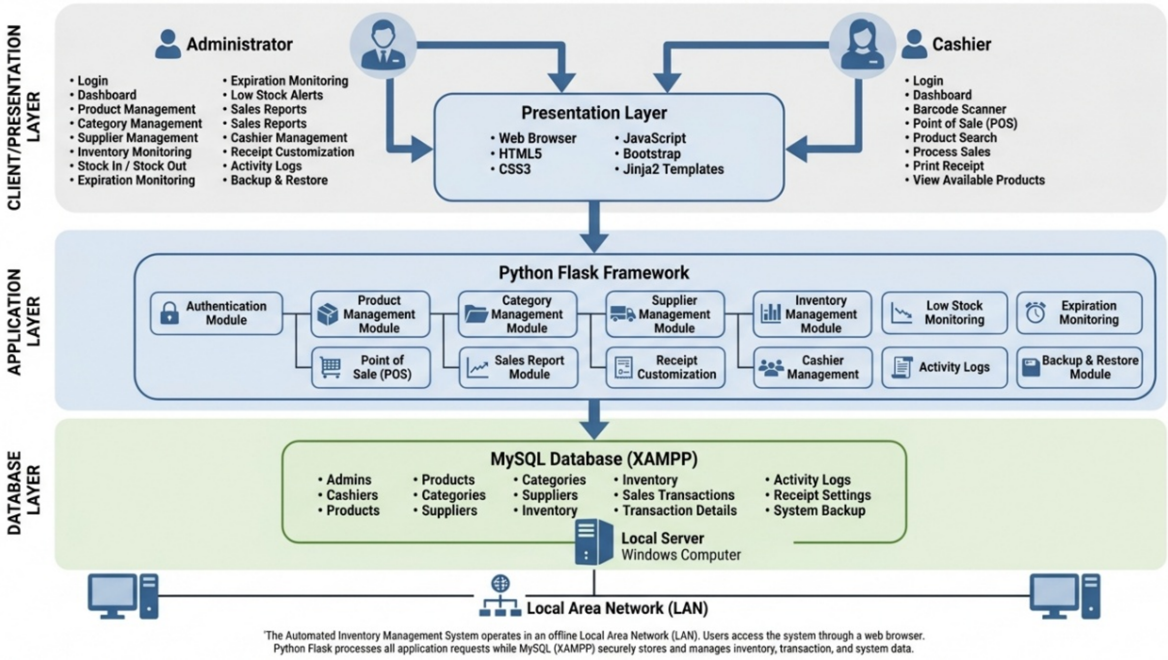
Local Area Network (LAN)

The Automated Inventory Management System operates within a **local area network (LAN)** using an offline deployment setup. The Flask application and MySQL database are hosted on a local computer configured as the server. Client computers connected to the same local network access the application through supported web browsers without requiring an internet connection. This configuration provides faster transaction processing, reliable system availability, and secure local data storage while minimizing dependence on external network services.

The adopted system architecture provides an organized framework for integrating inventory management, Point of Sale (POS), reporting, and administrative functions into a single application. By separating the presentation, application, and database layers, the system achieves improved maintainability, scalability, security, and operational efficiency. This architecture enables the Automated Inventory Management System to support the daily operations of pharmacies and convenience stores while ensuring accurate inventory control, efficient transaction processing, and reliable information management.

Figure 3.3 illustrates the overall system architecture of the Automated Inventory Management System, showing the interaction between users, the application, and the centralized database.

Figure 3.2. System Architecture Diagram of the Automated Inventory Management System



3.5 Database Design

The database design serves as the foundation of the Automated Inventory Management System (AIMS) because it provides a structured method for storing, organizing, retrieving, and managing operational data. A well-designed database ensures the accuracy, consistency, and integrity of information while supporting the different modules of the system, including user management, inventory monitoring, barcode-based Point of Sale (POS), sales recording, activity logging, and system configuration.

The researchers utilized **MySQL** as the Relational Database Management System (RDBMS) through **XAMPP** because of its compatibility with the Python Flask framework, reliability in handling transactional data, and ability to manage multiple related tables efficiently. The database follows the relational model by organizing data into separate tables connected through primary and foreign keys. This design minimizes data redundancy, improves data consistency, and allows the application to retrieve and update records efficiently during daily business operations.

The database was developed based on the functional requirements identified during the system analysis phase. Separate tables were created to manage administrator accounts, cashier accounts, administrator activities, cashier activities, product categories, products, inventory movements, sales transactions, transaction details, and system settings. Each table performs a specific function while maintaining logical relationships with other tables to support the overall operation of the application.

Figure 3.4 Entity Relationship Diagram (ERD) of the Automated Inventory Management System

Figure 3.1 Entity Relationship Diagram (ERD) of the PharmaCon POS and Inventory Management System Database

Figure 3.3 illustrates the logical structure of the database used by the Automated Inventory Management System. It shows the entities, their primary and foreign keys, and the relationships established among the database tables. These relationships enable the system to maintain data integrity while supporting inventory management, sales processing, report generation, user management, and activity monitoring.

The **admins** table stores administrator account information, including usernames, encrypted passwords, full names, security questions, security answers, and account creation dates. This table is responsible for authenticating administrators who have full access to system management functions.

The **admin_activity** table records the activities performed by administrators. It stores the administrator identifier, performed action, IP address, activity details, and the corresponding timestamp. This table is linked to the **admins** table through the **admin_id** foreign key, allowing every recorded activity to be associated with a specific administrator.

The **cashiers** table stores the account information of authorized cashiers, including personal information, usernames, encrypted passwords, account status, security questions, and account creation dates. Cashier accounts are used to process customer transactions through the Point of Sale (POS) module.

The **cashier_activity** table records cashier login and logout activities together with their corresponding IP addresses. This table is connected to the **cashiers** table through the **cashier_id** foreign key, allowing the system to monitor cashier attendance and user activity.
The **categories** table stores the different product categories maintained by the establishment. Product categorization enables administrators to organize merchandise efficiently and simplifies product searching, inventory management, and report generation.

The **products** table contains detailed information about every product available in the inventory. Each record includes the product name, barcode, category, product type, selling price, available stock, expiration date, and date of creation. The table is connected to the **categories** table through the **category_id** foreign key, establishing a one-to-many relationship where one category may contain multiple products.

The **stock_movements** table records all inventory movements performed within the system. Each record contains the affected product, movement type, quantity, reason for the adjustment, and movement date. This table is linked to the **products** table through the **product_id** foreign key, allowing administrators to monitor inventory additions, deductions, and stock adjustments.

The **sales** table stores the summary information of every completed Point of Sale (POS) transaction. Each sales record contains the receipt number, cashier who processed the transaction, total amount, transaction status, product type, transaction date, receipt printing status, and printing timestamp. The **cashier_id** foreign key establishes the relationship between cashiers and the sales they process.

The **sale_items** table stores the detailed list of products included in every completed sales transaction. Each record contains the transaction reference, product reference, quantity sold, and selling price. The table is connected to both the **sales** table through **sale_id** and the **products** table through **product_id**, creating the relationship between transactions and the individual products sold.

The **store_settings** table stores the configuration settings used throughout the application. It contains information such as system configuration values and store-related settings used for receipt generation and other configurable features. Unlike the other operational tables, this table functions independently and does not require foreign key relationships.

The relationships established among the database tables are maintained through **primary keys** and **foreign keys**. Primary keys uniquely identify every record stored within each table, while foreign keys create logical relationships between related entities. These relationships ensure that inventory records, sales transactions, activity logs, and user information remain accurate and consistent throughout the system.

To improve database efficiency, the researchers applied the principles of database normalization by separating information into individual relational tables according to their respective functions. This design minimizes duplicate data, simplifies database maintenance, improves query performance, and preserves data integrity during system operations.

Figure 3.5 Description of Database Tables

Table	**Purpose**
admins	Stores administrator credentials (username, hashed password) and full name.
admin_activity	Logs every administrator action (e.g., login, logout, product or staff changes) together with the action's IP address, details, and timestamp.
cashiers	Stores cashier credentials, full name, and account status (active/inactive).

cashier_activity	Records each cashier's login time, logout time, and IP address for shift monitoring.
categories	Stores the two product classifications used by the system: Medical and Non-Medical.
products	Stores product details including name, barcode, category, price, current stock, and expiration date.
stock_movements	Logs stock-in and stock-out events for each product, including the reason for the movement (e.g., sale).
sales	Stores each completed transaction's receipt number, cashier, total amount, status, product type, and date.
sale_items	Stores the individual line items (product, quantity, price) belonging to each sale.

Figure 3.4 presents a summary of the database tables utilized in the Automated Inventory Management System together with their respective purposes. Each table was designed to store a specific type of information required by the application. The use of separate but related tables enables the system to organize data efficiently, reduce redundancy, improve maintainability, and support inventory management, Point of Sale (POS) transactions, activity monitoring, reporting, and system configuration.

Overall, the relational database designed for the Automated Inventory Management System provides a reliable and organized data management framework. The integration of the different database tables enables the application to perform inventory monitoring, barcode-based sales processing, activity tracking, report generation, and administrative management efficiently while maintaining the integrity, consistency, and security of operational data

3.6 Hardware and Software Requirements

The successful development and implementation of the Automated Inventory Management System require appropriate hardware and software resources to ensure stable system performance and efficient operation. The hardware requirements define the minimum computer specifications necessary to run the application, while the software requirements identify the operating system, development tools, database management system, and other supporting software used in developing and operating the system.
The hardware and software requirements were selected based on the needs of the Automated Inventory Management System to provide reliable inventory management, barcode-based Point of Sale (POS) transactions, sales reporting, and database management in an offline Local Area Network (LAN) environment.

3.6.1 Hardware Requirements

The hardware requirements specify the physical components needed to install, develop, and operate the Automated Inventory Management System. These specifications ensure that the application performs efficiently while supporting inventory management and sales processing.

Table 3.1. Hardware Requirements

Hardware Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3 (4th Generation) or AMD Equivalent	Intel Core i5 (8th Generation) or Higher
Memory (RAM)	4 GB	8 GB or Higher
Storage	128 GB SSD or 500 GB HDD	256 GB SSD or Higher
Monitor	1366 × 768 Resolution	1920 × 1080 Full HD
Keyboard and Mouse	Standard USB Keyboard and Mouse	Standard USB Keyboard and Mouse
Barcode Scanner	USB Barcode Scanner	2D USB Barcode Scanner
Receipt Printer	Thermal Receipt Printer	80 mm Thermal Receipt Printer
Network	Local Area Network (LAN)	Gigabit Local Area Network (LAN)

The development computer should possess sufficient processing power and memory to execute the Python Flask application, MySQL database server, and other development tools simultaneously. The deployment computer should be capable of handling daily sales transactions, inventory updates, barcode scanning, and receipt printing without significant delays.

3.6.2 Software Requirements

The software requirements include the operating system, programming language, database management system, and development tools utilized in developing and implementing the Automated Inventory Management System.

Table 3.2. Software Requirements

Software	Purpose
Windows 10 or Windows 11	Operating System
Python 3.x	Backend Programming Language
Flask	Web Application Framework
XAMPP	Local Server Environment
MySQL	Relational Database Management System
HTML5	Structure of Web Pages
CSS3	User Interface Design
JavaScript	Client-side Functionality
Visual Studio Code	Source Code Editor
Google Chrome or Microsoft Edge	Web Browser for System Access

Windows 10 or Windows 11 serves as the operating system used during both system development and deployment. It provides compatibility with the software components utilized by the application and supports the execution of Python Flask, XAMPP, and the required development tools.

Python serves as the primary programming language responsible for implementing the system's backend functionalities. It performs data processing, user authentication, inventory management, sales processing, report generation, and communication between the application and the database.
The Flask framework functions as the web application framework that manages routing, request processing, session management, authentication, and interaction between the graphical user interface and the MySQL database.

XAMPP provides the local server environment used to host the MySQL database. Through XAMPP, the database server manages administrator accounts, cashier accounts, product information, inventory records, sales transactions, activity logs, and system settings.

MySQL serves as the Relational Database Management System (RDBMS) responsible for storing and organizing all operational data. The relational database structure ensures efficient retrieval of information while maintaining data consistency and integrity through the implementation of primary and foreign key relationships.

HTML5, CSS3, and JavaScript were utilized to develop the graphical user interface of the system. HTML5 provides the structure of the application pages, CSS3 enhances the visual appearance and layout of the interface, while JavaScript adds interactive features that improve the overall user experience.
Visual Studio Code serves as the primary Integrated Development Environment (IDE) used during system development. It provides source code editing, debugging tools, syntax highlighting, and extension support that simplify software development.

Google Chrome and Microsoft Edge function as the supported web browsers for accessing the application. These browsers provide compatibility with the technologies utilized by the system and ensure proper rendering of user interface components.

3.6.3 System Deployment Environment

The Automated Inventory Management System is designed to operate within an offline Local Area Network (LAN). The Python Flask application and MySQL database are hosted on a local computer that functions as the server. Authorized client computers connected to the same local network can access the system using a web browser by entering the server's local IP address.

This deployment approach provides several operational advantages. Since the application does not rely on an internet connection, the establishment can continue processing sales transactions and inventory updates even during internet interruptions. Additionally, storing business data within the local network enhances data privacy and reduces exposure to external cyber threats.
The offline deployment environment also minimizes operational costs by eliminating the need for cloud hosting services while maintaining reliable communication between the application and the database server.

Overall, the selected hardware and software requirements provide a stable and efficient environment for the Automated Inventory Management System. These resources support inventory management, barcode-based Point of Sale (POS) transactions, receipt printing, activity monitoring, report generation, and database management while ensuring reliable performance, data integrity, and operational efficiency within an offline Local Area Network environment.

Reference

FOREIGN STUDIES

Abisoye, O. A., Fatoba, B., & Blessing, O. (2013). Design of computerized inventory management systems for supermarkets. International Journal of Science and Research, 2(9).

Atieh, A. M., Kaylani, H., Al-Abdallat, Y., Qaderi, A., Ghoul, L., Jaradat, L., & Hdairis, I. (2016). Performance improvement of inventory management system processes by an automated warehouse management system. Procedia CIRP, 41, 568–572.

Chopra, S., & Meindl, P. (2016). Supply chain management: Strategy, planning, and operation (6th ed.). Pearson Education.

DeHoratius, N., & Raman, A. (2008). Inventory record inaccuracy: An empirical analysis. Management Science, 54(4), 627–641. https://doi.org/10.1287/mnsc.1070.0789

Erl, T., Puttini, R., & Mahmood, Z. (2013). Cloud computing: Concepts, technology & architecture. Pearson Education.

Fernie, J., & Sparks, L. (2018). Logistics and retail management (5th ed.). Kogan Page Publishers.

Gunasekaran, A., & Ngai, E. W. T. (2004). Information systems in supply chain integration and management. European Journal of Operational Research, 159(2), 269–295. https://doi.org/10.1016/S0377-2217(03)00407-5

Heizer, J., Render, B., & Munson, C. (2017). Operations management (12th ed.). Pearson Education.
Jacobs, F. R., & Chase, R. B. (2018). Operations and supply chain management (15th ed.). McGraw-Hill Education.

Koumanakos, D. P. (2008). The effect of inventory management on firm performance. International Journal of Productivity and Performance Management, 57(5), 355–369.

Muller, M. (2011). Essentials of inventory management (2nd ed.). AMACOM.
Prempeh, K. B. (2015). The impact of efficient inventory management on profitability: Evidence from selected manufacturing firms in Ghana. International Journal of Finance and Accounting, 4(1), 1–8.

Rajeev, N. (2010). Inventory management in small and medium enterprises: A study of machine tool enterprises in Bangalore. Management Research Review, 33(7), 713–726.

Saleem, A. (2020). Automated inventory management systems and its impact on supply chain risk management in manufacturing firms of Pakistan. International Journal of Management, 11(12), 1689–1702.

Samuel, K. S., & Ondiek, G. O. (2014). Inventory management automation and performance of supermarkets in Western Kenya. International Journal of Research in Management and Business Studies, 1(4).

Sharma, A., Garg, D., & Agarwal, A. (2011). Barcode technology and its impact on inventory management systems. International Journal of Engineering Science and Technology, 3(5), 3487–3493.

Waters, D. (2019). Inventory control and management (3rd ed.). Wiley.

Wild, T. (2017). Best practice in inventory management (3rd ed.). Routledge.

LOCAL STUDIES

Aguilar, C. M., & Torres, J. L. (2020). Point of sale system implementation for small retail enterprises in Metro Manila. Journal of Philippine Information and Communication Technology, 6(2), 45–62.

Bautista Jr., J., & Young, M. (2022). Effective inventory management system in efficient supply and distribution management in one of manufacturer of foods seasoning products in the Philippines. In 2021 International Conference on Information Management, Innovation Management and Industrial Engineering (ICIII) (Vol. 1, pp. 137–146). https://www.researchgate.net/publication/359992876

Dela Cruz, J., & Ramos, P. (2018). Point of sale and inventory management system for small retail businesses in the Philippines. Philippine Journal of Information Technology Studies, 8(2), 45–58.

Domingo, R., & Ferrer, L. (2020). Automated inventory monitoring system for pharmacies in Cebu Province. Cebu Journal of Information Technology, 3(2), 78–95. Fernandez, J. M., & Bautista, R. L. (2022). Implementation challenges of inventory management systems in Philippine SME retail. Journal of Small Business and Entrepreneurship Development, 5(2), 89–107.

Garcia, L. P., & Santos, D. R. (2018). Automated inventory system for convenience stores in Luzon. Northern Luzon Journal of Commerce and Technology, 3(1), 56–73. Mendoza, R. S., & Reyes, A. B. (2019). Inventory management challenges and solutions for independent pharmacies in Metro Manila. Manila Journal of Business and Technology, 2(1), 34–51.

Ortega, M., & Salazar, P. (2021). Computerized inventory and sales system for mini grocery stores in Batangas. Philippine Journal of Business Information Systems, 4(1), 23–41.

Parilla, E. S., Evangelista, J., Aurelio, R., & Bullalayao, C. (2022). Inventory management practices and service delivery of healthcare facilities in Ilocos Norte Philippines. Logistic and Operation Management Research (LOMR), 1(1), 16–33. https://doi.org/10.31098/lomr.v1i1.919

Santos, M., Garcia, R., & Lopez, T. (2020). Web-based inventory monitoring system for retail stores in the Philippines. Asian Journal of Computer and Information Systems.

Villar, P. A., & Cruz, M. R. (2021). Web-based inventory system for supermarket chains

in the Philippines. Philippine Journal of Retail Management, 4(3), 78–95.

Villareal, J., Mendoza, T., & Ramos, K. (2022). Cloud-based point of sale and inventory system for retail businesses in Manila. Philippine ICT Research Journal.

Related Literature

Akinsanya, O. (2019). Design and implementation of computerized stock management systems: A case study of MIDE Supermarket.

Bowersox, D. J., Closs, D. J., & Cooper, M. B. (2013). Supply chain logistics management (4th ed.). McGraw-Hill Education.

Deitel, P., & Deitel, H. (2020). Internet & World Wide Web: How to program (6th ed.). Pearson Education.

Dix, A., Finlay, J., Abowd, G. D., & Beale, R. (2004). Human-computer interaction (3rd ed.). Pearson Education Limited.

Laudon, K. C., & Laudon, J. P. (2020). Management information systems: Managing the digital firm (16th ed.). Pearson Education Limited.
Leon, A. (2014). Enterprise resource planning (3rd ed.). McGraw-Hill Education.

Murdick, R. G., Ross, J. E., & Claggett, J. R. (2017). Information systems for modern management (3rd ed.). PHI Learning Private Limited.

Pressman, R. S., & Maxim, B. R. (2019). Software engineering: A practitioner's approach (9th ed.). McGraw-Hill Education.

Russell, S., & Norvig, P. (2021). Artificial intelligence: A modern approach (4th ed.). Pearson Education.

Shelly, G. B., & Rosenblatt, H. J. (2012). Systems analysis and design (9th ed.). Course Technology Cengage Learning.

Silberschatz, A., Korth, H. F., & Sudarshan, S. (2019). Database system concepts (7th ed.). McGraw-Hill Education.

Sommerville, I. (2016). Software engineering (10th ed.). Pearson Education.
Stair, R. M., & Reynolds, G. W. (2018). Principles of information systems (13th ed.). Cengage Learning.

Stallings, W., & Brown, L. (2018). Computer security: Principles and practice (4th ed.). Pearson Education.

Stevenson, W. J. (2018). Operations management (13th ed.). McGraw-Hill Education.

Turban, E., Pollard, C., & Wood, G. (2018). Information technology for management: Driving digital transformation to increase local and global performance, growth and sustainability (11th ed.). Wiley.

Chapter IV
SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.1 Introduction

This chapter presented the development, implementation, testing, evaluation, and discussion of the proposed Automated Inventory Management System. It described how the system was developed based on the requirements identified in the previous chapters and explained the processes followed during the implementation of the application. It also discussed the completed system, the testing procedures performed to verify its functionality, and the evaluation results obtained from the respondents.

The Automated Inventory Management System was developed to address the operational problems encountered by pharmacies and convenience stores in managing inventory, processing sales transactions, and maintaining business records. The developed system automated inventory monitoring, product management, barcode-based sales transactions, stock validation, expiration monitoring, receipt generation, reporting, and user management through a centralized web-based application operating within a local network environment.

This chapter also described the implementation of the system using Python Flask as the application framework and MySQL through XAMPP as the database management system. The researchers integrated the different system modules into a single application to ensure that inventory records, sales transactions, user accounts, reports, and system settings were maintained accurately and consistently throughout daily operations.

To verify the correctness and reliability of the developed application, various testing procedures were conducted on each system module. The testing process focused on validating user authentication, inventory management, barcode scanning, sales processing, receipt generation, stock monitoring, report generation, backup and restoration, and activity logging. Identified errors were corrected before the system was subjected to user evaluation.

Furthermore, this chapter presented the evaluation results gathered from selected respondents who assessed the developed system using the selected software quality criteria. The evaluation determined the level of functionality, usability, efficiency, and reliability of the system. The findings demonstrated the capability of the Automated Inventory Management System to improve inventory accuracy, simplify sales processing, strengthen inventory control, and support efficient business operations within pharmacies and convenience stores.

4.2 Overview of the Developed System

The Automated Inventory Management System (AIMS) was developed to provide an efficient, reliable, and user-friendly solution for managing inventory and sales operations in pharmacies and convenience stores. The system was designed to replace traditional manual methods of inventory recording and sales processing with an automated solution that improves operational efficiency, minimizes human errors, and provides accurate business information for decision-making. By integrating inventory management and barcode-based Point of Sale (POS) operations into a single application, the system streamlines daily business activities and enhances the overall productivity of the establishment.

The development of the system was based on the operational requirements identified during the analysis of the existing business processes. Common challenges such as inaccurate inventory records, slow transaction processing, manual stock monitoring, and delayed report generation were considered during the planning and development stages. These identified problems guided the design and implementation of system features that directly address the operational needs of pharmacies and convenience stores.

The developed system is a web-based application that operates within an offline Local Area Network (LAN). It was implemented using the Python Flask framework as the backend, while HTML, CSS, and JavaScript were used to create the graphical user interface. The MySQL relational database management system, hosted through XAMPP, serves as the central repository for all operational data. This architecture enables multiple authorized users to access the application through computers connected to the same local network without requiring an internet connection. Operating in an offline environment also enhances data privacy and allows continuous business operations even when internet connectivity is unavailable.

The system utilizes role-based access control by providing two primary user roles: the Administrator and the Cashier. Each role is assigned specific permissions according to its responsibilities. The Administrator has complete access to all modules of the system, including inventory management, product management, category management, cashier account management, sales reporting, activity monitoring, receipt customization, database backup and restoration, and system configuration. The Cashier, on the other hand, is granted access only to the Point of Sale (POS) module and other transaction-related functions necessary for processing customer purchases. This separation of user privileges helps maintain data security while ensuring that each user performs only the tasks relevant to their responsibilities.

The Automated Inventory Management System integrates several operational modules that work together to support daily business activities. The inventory management module allows administrators to monitor stock quantities, update inventory records, and maintain accurate product information. The product management module enables the addition, modification, deletion, and organization of products into their corresponding categories. These modules ensure that inventory information remains updated and readily available whenever transactions are performed.

The Point of Sale (POS) module allows cashiers to process customer purchases efficiently using barcode scanning technology. Products are identified automatically by scanning their barcode labels, reducing manual encoding and improving transaction speed. The system calculates transaction totals, records completed sales, updates product stock quantities, and generates customer receipts automatically. This integration simplifies the sales process and minimizes errors commonly associated with manual transactions.

To further improve inventory control, the system includes stock movement monitoring, low-stock notifications, and expired product monitoring. The stock movement module records all inventory adjustments, including stock additions and deductions, providing administrators with a complete history of inventory changes. The low-stock monitoring feature notifies administrators whenever product quantities reach predefined minimum levels, enabling timely replenishment of inventory.

The expired product monitoring feature identifies products that have expired or are approaching their expiration dates, helping prevent the sale of expired merchandise and reducing inventory losses.

The system also provides comprehensive sales reporting capabilities that assist administrators in monitoring business performance. Sales reports summarize completed transactions and provide accurate information regarding sales activities over a selected period. These reports support inventory planning, financial monitoring, and managerial decision-making by presenting organized and reliable transaction data.

For accountability and security, the system automatically records administrator and cashier activities through dedicated activity logs. Login sessions, inventory modifications, product management operations, and other significant system activities are stored in the database. These records provide administrators with the ability to monitor system usage, review operational activities, and investigate unauthorized or unusual actions when necessary.

To protect important business information, the Automated Inventory Management System includes database backup and restoration functions. Administrators can create backup copies of the database and restore previously saved records whenever necessary. This feature helps safeguard operational data against accidental deletion, hardware failure, or other unexpected system problems.

The developed system also supports external hardware devices that improve operational efficiency. A USB barcode scanner is integrated with the Point of Sale (POS) module to facilitate rapid product identification during customer transactions. In addition, a thermal receipt printer is used to produce printed receipts immediately after every successful sale. These hardware components complement the software application by improving transaction speed, reducing manual work, and enhancing customer service.

Overall, the Automated Inventory Management System provides a complete and integrated solution for managing inventory and sales operations within pharmacies and convenience stores. By combining inventory monitoring, barcode-based Point of Sale (POS), sales reporting, activity logging, receipt printing, and database management into a single application, the system improves operational efficiency, increases data accuracy, enhances transaction processing, and supports informed decision-making. The succeeding sections of this chapter discuss the system architecture, implementation, testing procedures, evaluation, results, and discussion of the developed system.

4.3 System Development Methodology

The development of the Automated Inventory Management System followed the Agile Software Development Methodology. The researchers adopted this methodology because it supported incremental system development, continuous testing, and regular improvements throughout the implementation process.

The iterative approach allowed the researchers to identify system issues early and implement necessary modifications before completing the entire application.
The development process began with the planning phase, during which the researchers identified the operational requirements of pharmacies and convenience stores. Information gathered during data collection was analyzed to determine the necessary system functionalities, including inventory management, Point of Sale (POS), barcode scanning, user authentication, reporting, receipt customization, and activity logging.

After the system requirements had been established, the researchers proceeded with the design phase. The database structure, user interfaces, navigation flow, and system modules were designed to support efficient business operations while maintaining data consistency and system security. The system architecture was also prepared to support communication between the web application and the centralized MySQL database.

During the implementation phase, the application was developed using Python Flask together with HTML, CSS, JavaScript, and Jinja2 templates. MySQL, running through XAMPP, served as the centralized database for storing products, inventory records, users, sales transactions, activity logs, and system settings. Individual modules such as user authentication, product management, cashier operations, Point of Sale, reporting, and backup management were developed incrementally before being integrated into the complete application.

Continuous testing was performed throughout the development process. Each module was verified individually before integration to ensure that user authentication, product management, inventory monitoring, barcode processing, sales transactions, reporting, and administrative functions operated correctly. Errors identified during testing were immediately corrected to improve the stability and reliability of the system.

Upon completion of all modules, comprehensive system testing and evaluation were conducted to determine whether the developed application satisfied the objectives of the study. The iterative nature of the Agile methodology enabled the researchers to continuously improve the system until the final version successfully met the operational requirements of pharmacies and convenience stores.

4.4 System Architecture

The system architecture of the Automated Inventory Management System illustrates the overall structure of the application and the interaction among its major components. It presents how users communicate with the system, how requests are processed by the application, and how information is stored and retrieved from the database. The architecture serves as the blueprint of the developed system by showing the relationship between the user interface, application logic, and database management system.

The Automated Inventory Management System was developed as an offline web-based application using the Python Flask framework and MySQL database through XAMPP. The system operates within a Local Area Network (LAN), allowing authorized users to access the application using computers connected to the same network. This deployment environment enables the establishment to continue its daily operations even without an internet connection while maintaining centralized data storage and secure access to business information.

The application follows a three-tier architecture composed of the presentation layer, application layer, and database layer. The presentation layer provides the graphical user interface where administrators and cashiers interact with the system using a web browser. Through this interface, users can perform tasks such as logging into the system, managing products, processing Point of Sale (POS) transactions, monitoring inventory, generating reports, and configuring system settings.

The application layer is implemented using the Python Flask framework. It is responsible for processing user requests, validating user inputs, managing user authentication, executing business logic, and communicating with the database. Whenever a user performs an operation, such as adding a product, updating inventory, processing a sale, or generating a report, Flask processes the request and executes the corresponding functions before returning the results to the user interface.

The database layer consists of the MySQL relational database hosted through XAMPP. This layer stores all operational information required by the application, including administrator accounts, cashier accounts, product categories, product information, stock movement records, sales transactions, transaction details, activity logs, and store settings. The database ensures that information is organized, accurate, and consistently available throughout the operation of the system.

The system also integrates external hardware devices to improve operational efficiency. A USB barcode scanner is used during Point of Sale (POS) transactions to quickly identify products by scanning their barcode labels. This reduces manual encoding, minimizes transaction errors, and speeds up customer service. A thermal receipt printer is likewise integrated into the system to automatically print customer receipts immediately after every successful transaction.

Data communication within the system begins when an authorized user logs into the application through the web browser. The login credentials are verified by the Flask application, which then retrieves the corresponding account information from the MySQL database. After successful authentication, the user is

redirected to the appropriate dashboard based on the assigned user role.
Administrators are provided with complete access to inventory management, cashier management, sales reports, activity logs, backup and restoration, receipt customization, and system settings. Cashiers, on the other hand, are granted access only to Point of Sale (POS) operations, product searching, barcode scanning, receipt printing, and transaction processing. This role-based access control helps maintain system security by limiting user access according to assigned responsibilities.

Every transaction performed within the application is automatically recorded in the database. Product purchases reduce the available stock quantity, stock adjustments are reflected in inventory records, sales transactions are stored for future reporting, and user activities are logged to maintain accountability. This integration among the different system modules ensures that operational information remains synchronized and up to date.

Figure 4.1. System Architecture Diagram of the Automated Inventory Management System

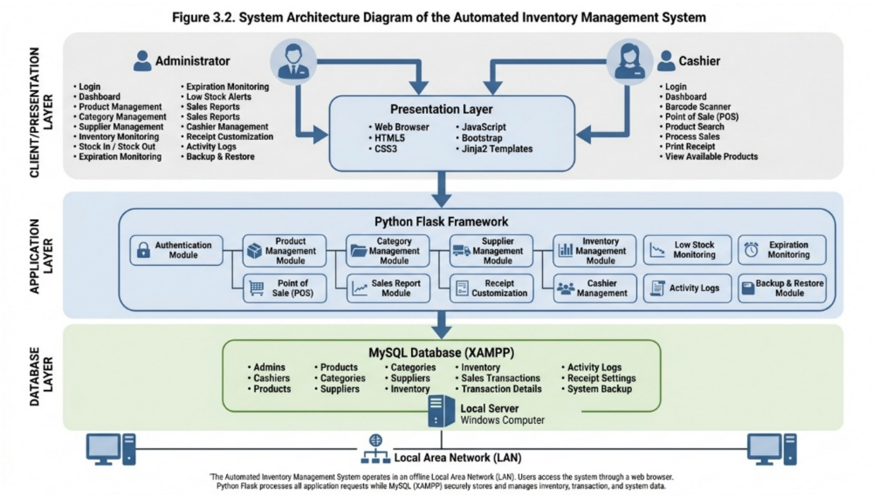


Figure 4.1 illustrates the overall architecture of the Automated Inventory Management System. The diagram shows the interaction between the Administrator and Cashier, the web-based application developed using Python Flask, the MySQL database hosted through XAMPP, and the integrated hardware devices such as the barcode scanner and thermal receipt printer. It also demonstrates how user requests are processed by the application and how operational data are stored and retrieved from the centralized database. This architecture provides a secure, reliable, and efficient platform for inventory management, barcode-based Point of Sale (POS) transactions, sales reporting, and administrative operations within an offline Local Area Network (LAN) environment.

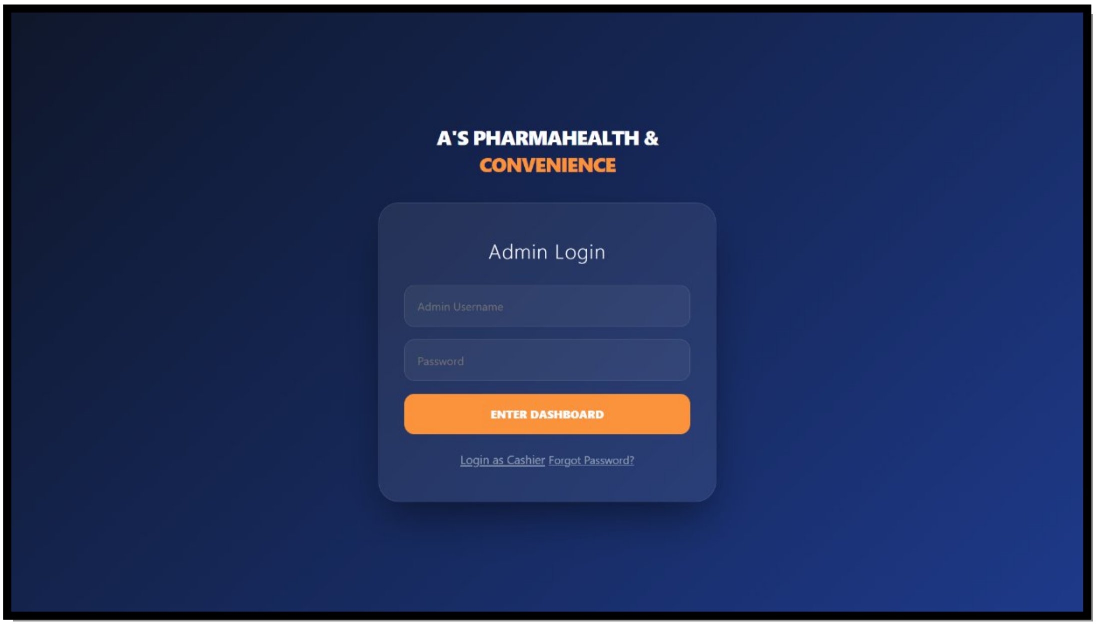
4.5 Presentation of the Developed System

This section presents the graphical user interface (GUI) of the Automated Inventory Management System. The developed application was designed to provide a simple, organized, and user-friendly interface that enables administrators and cashiers to perform their assigned tasks efficiently. Each interface was developed using HTML, CSS, JavaScript, and Python Flask, while data are stored and managed through a MySQL database hosted using XAMPP.

The system was designed with role-based access control, allowing administrators and cashiers to access only the functions relevant to their responsibilities. The administrator is responsible for managing products, categories, inventory, cashier accounts, reports, system settings, and database maintenance, while the cashier is responsible for processing Point of Sale (POS) transactions using barcode scanning.
The following figures present the different interfaces of the developed system together with their corresponding descriptions.

4.5.1 Administrator Interfaces

Figure 4.2 Administrator Login Page



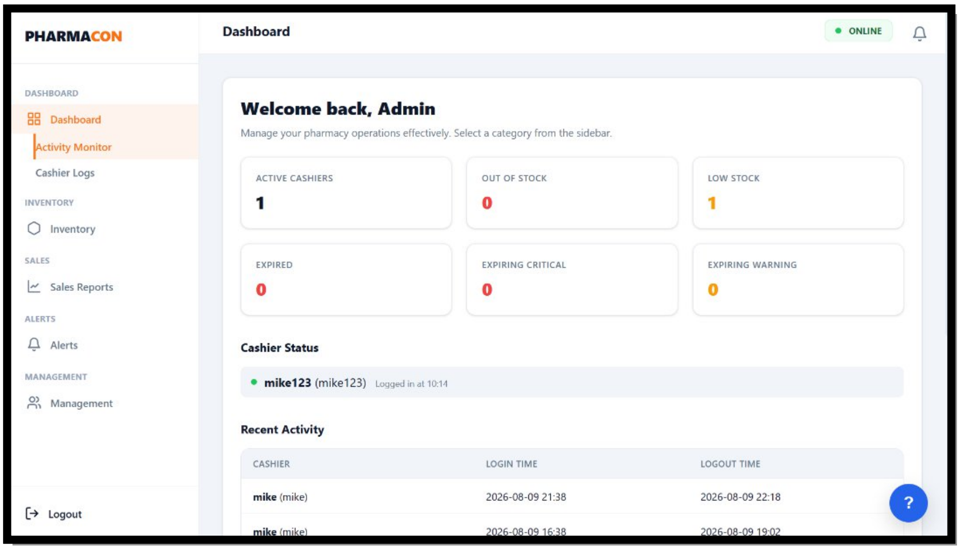
The Administrator Login Page requires authorized administrators to enter a valid username and password before accessing the system. Upon successful authentication, the administrator is redirected to the Administrator Dashboard. If no administrator account exists in the database, the system automatically creates a default administrator account during the initial setup. This page serves as the primary security mechanism that prevents unauthorized access to administrative functions.

The Administrator Login Interface serves as the authentication page for system administrators. The interface requires the administrator to provide valid username and password credentials before accessing the administrative functions of the system. Authentication is handled through the Flask application, while passwords are secured using Werkzeug security hashing. The system also uses session-based authentication and role-based access control to restrict unauthorized access.

Key features include:

- Username and password authentication;
- Secure password hashing;
- Session-based authentication;
- Role-based access control;
- Login attempt monitoring and security controls; and
- Redirection to the appropriate administrator interface after successful authentication.

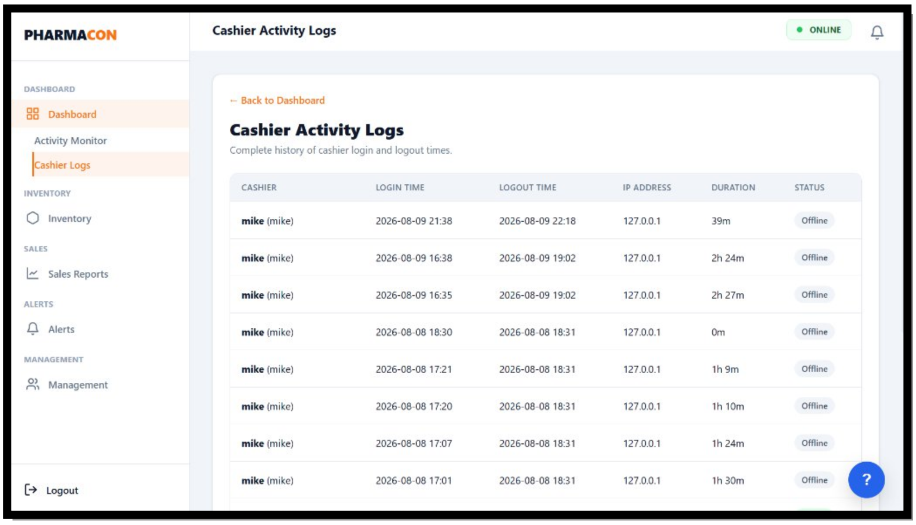
Figure 4.3 Administrator Dashboard



The Administrator Dashboard serves as the main monitoring interface for the system. It provides administrators with an overview of inventory conditions and other important operational information. The dashboard displays product and cashier status, low-stock and expiration alerts, recent activities, and quick access to major administrative functions.

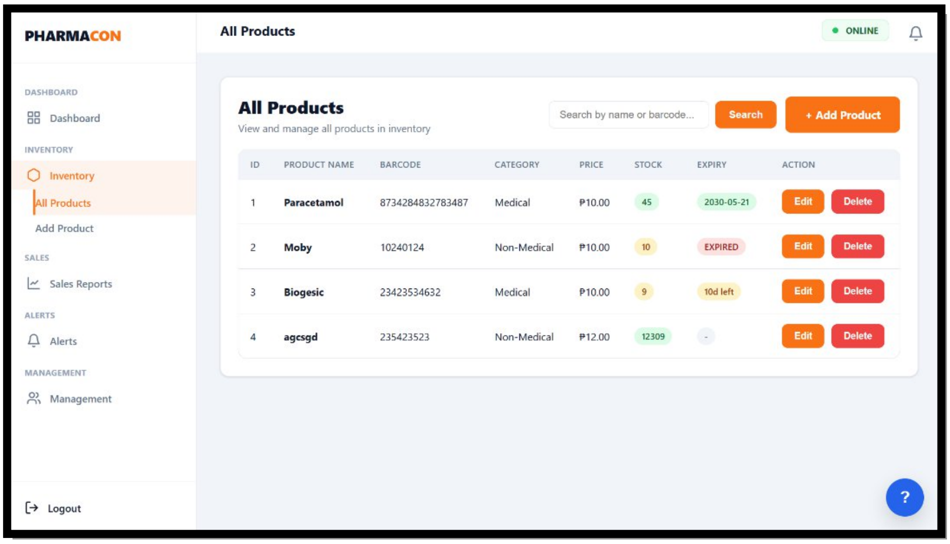
The dashboard provides access to functions such as product management, inventory monitoring, cashier management, reports, activity monitoring, receipt customization, and database backup and restoration. The system also provides inventory notifications for products requiring administrative attention.

Figure 4.4 Activity Logs Interface



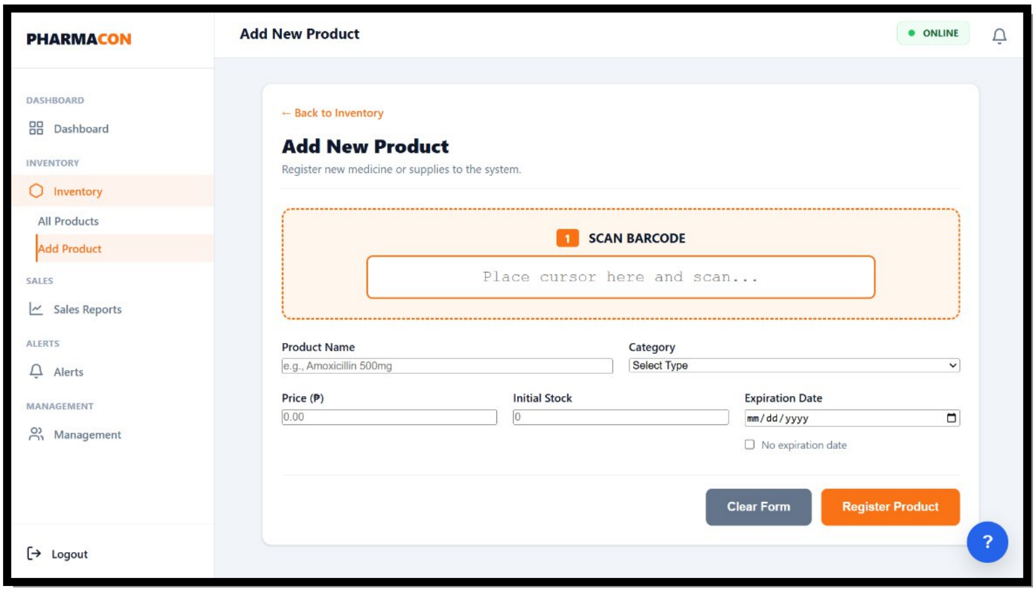
The Activity Logs Interface provides administrators with records of system activities. Administrator actions and cashier login/logout events are recorded to support accountability and monitoring. The system stores activity information together with relevant details such as timestamps and IP addresses.

Figure 4.5 All Products Interface



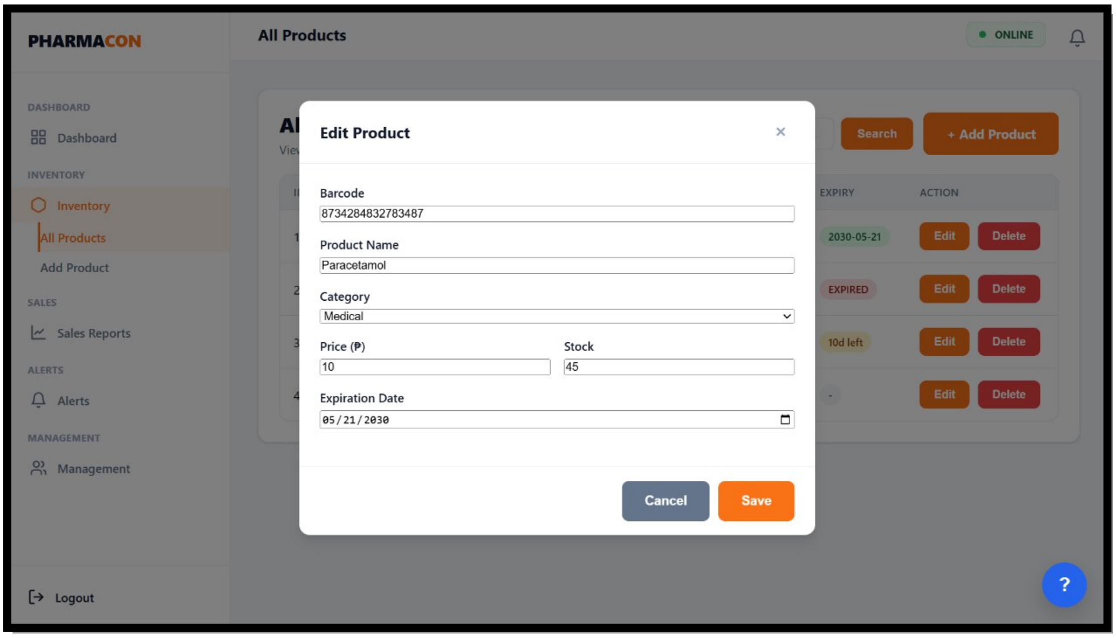
The All Products Interface displays the product records maintained by the system. It allows administrators to view important product information, including the product name, barcode, category, product type, price, stock quantity, and expiration date. The interface provides product filtering and searching functions to assist administrators in locating specific records efficiently.
The interface supports product management operations, including adding new products, editing existing product information, and deleting product records. Barcode information is maintained as part of the product record to support barcode-based lookup during sales transactions.

Figure 4.6 Add Product Interface



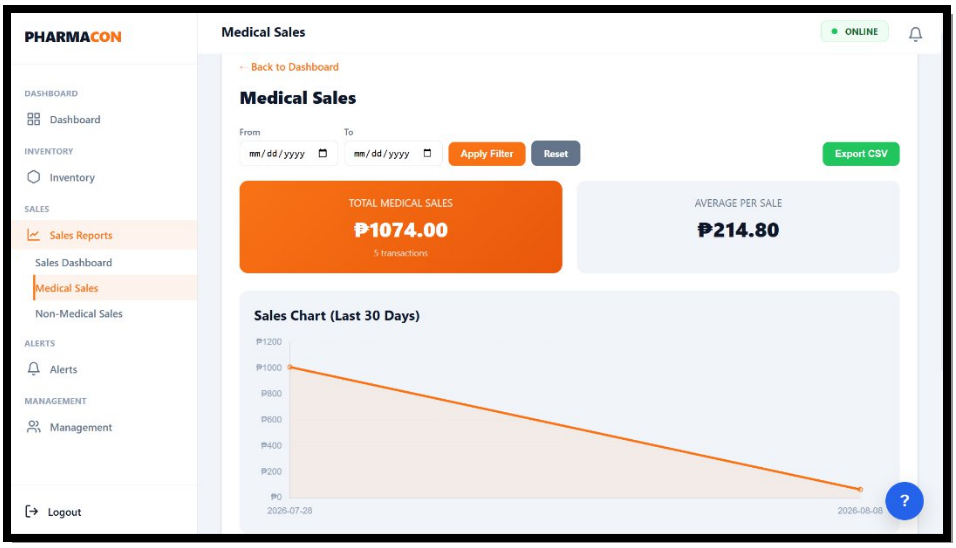
The Add Product Interface allows administrators to register new products in the inventory database. The product registration form includes the product name, barcode, category, product type, price, initial stock quantity, and expiration date. The system validates product information before saving the record to the database. Duplicate barcode entries are prevented to maintain accurate product identification.
The interface supports the registration of both Medical and Non-Medical products. Product information entered through this interface becomes available for inventory monitoring and Point-of-Sale transactions.

Figure 4.7 Edit Product Interface



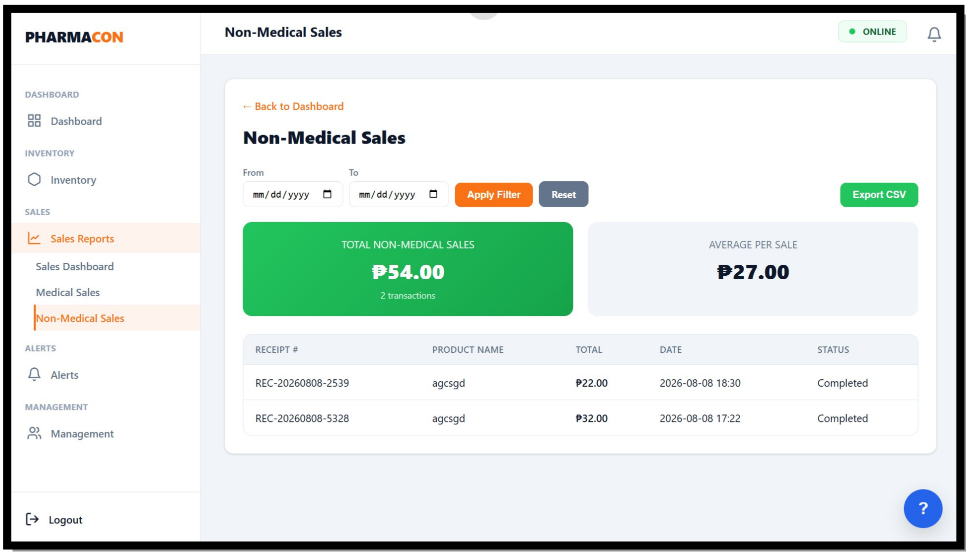
The Edit Product Interface allows administrators to modify existing product information. The form displays the current information of the selected product and allows authorized administrators to update relevant details. Changes are stored in the MySQL database and are reflected in the corresponding inventory records.

Figure 4.8 Medical Sales Interface



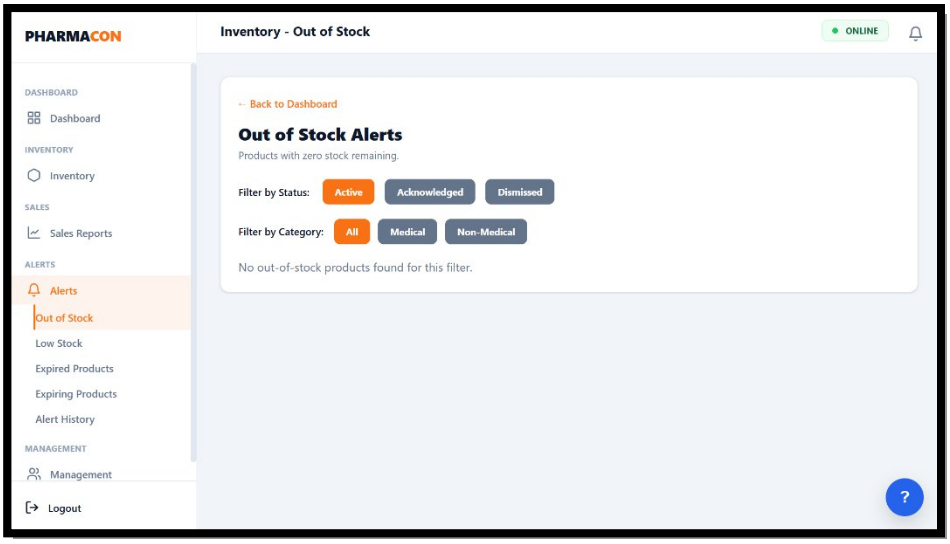
The Medical Sales Interface provides the cashier with a sales view for Medical products. Products belonging to the Medical category are processed through the same Point-of-Sale workflow while maintaining their classification for inventory and sales reporting purposes. The system automatically separates Medical sales from Non-Medical sales when transactions are completed.

Figure 4.9 Non-Medical Sales Interface



The Non-Medical Sales Interface provides a corresponding sales view for Non-Medical products. The classification allows the system to maintain separate product and sales information for Medical and Non-Medical items. This separation is also reflected in the sales records and reporting functions of the system.

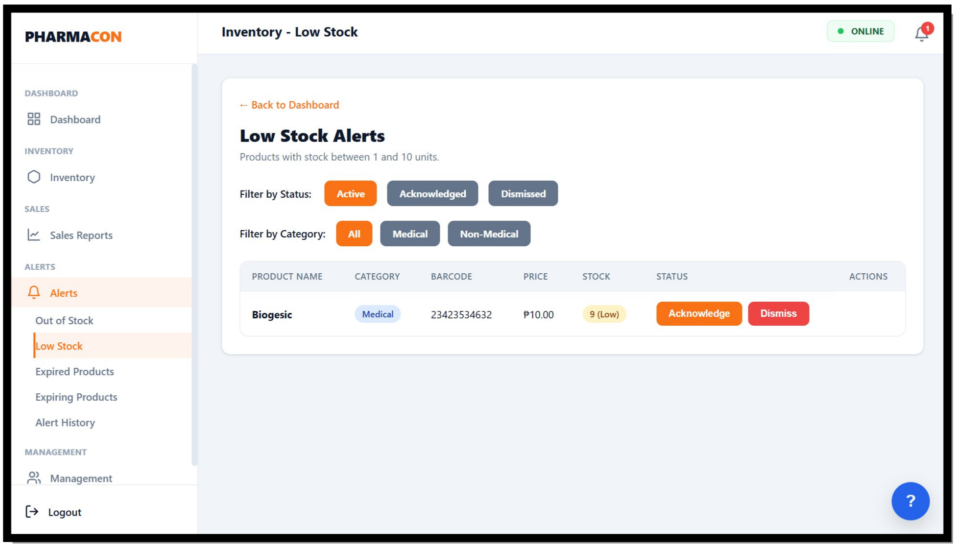
Figure 4.10 Out of Stock Alerts



The Out of Stock Alerts Interface displays products whose available inventory has reached zero. This interface assists administrators in identifying products that require immediate inventory attention.

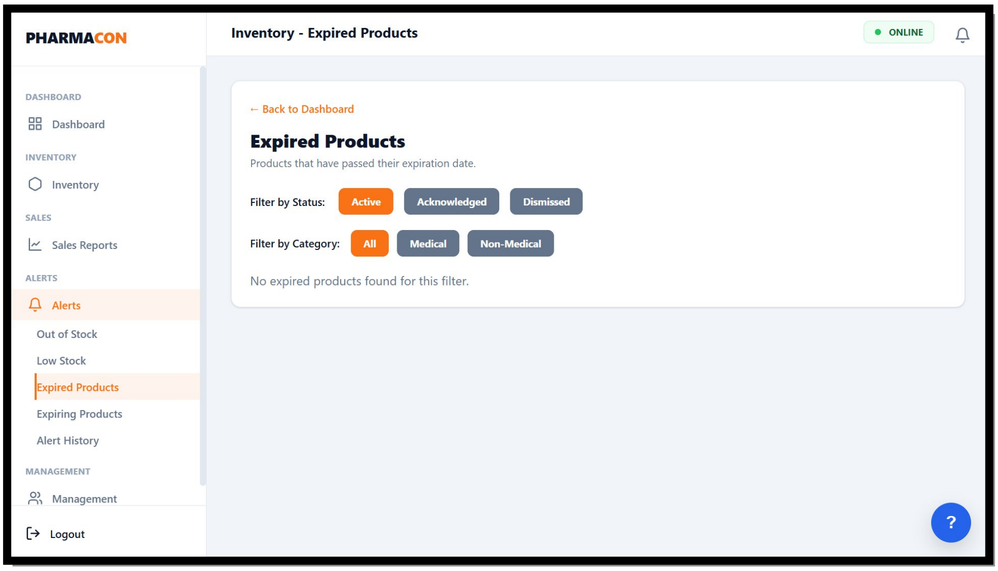
The alert system follows a three-state lifecycle consisting of **Active**, **Acknowledged**, and **Dismissed**. Active alerts represent newly identified conditions that require attention, acknowledged alerts indicate that the issue is being handled, and dismissed alerts represent resolved or false alarms with documented reasons.

Figure 4.11 Low Stock Alerts



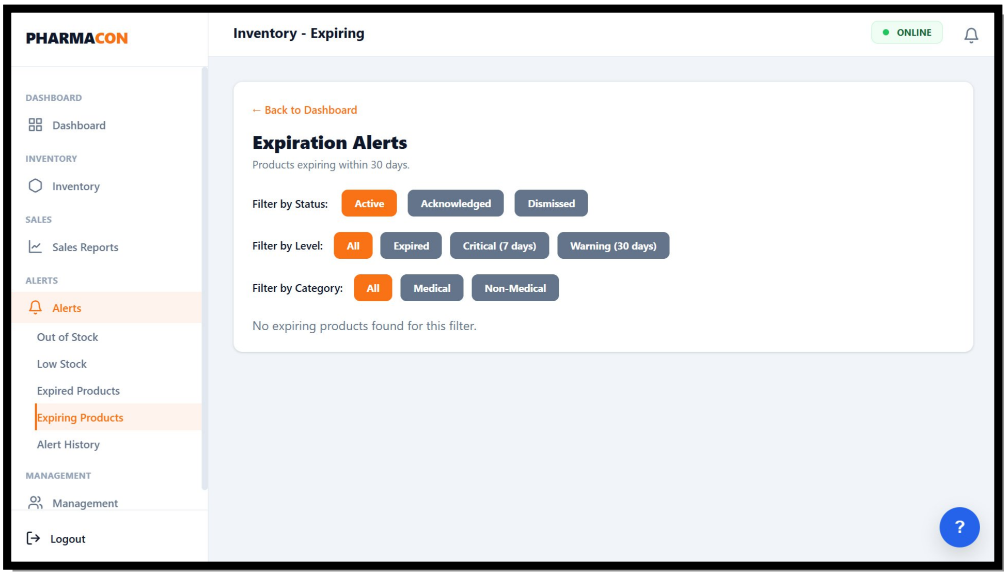
The Low Stock Alerts Interface identifies products whose inventory has fallen below the configured stock threshold. This function allows the administrator to monitor products that may require replenishment before they become completely out of stock.
The system provides alert information to help administrators prioritize inventory replenishment. Category filtering is also supported for Medical and Non-Medical products.

Figure 4.12 Expired Products



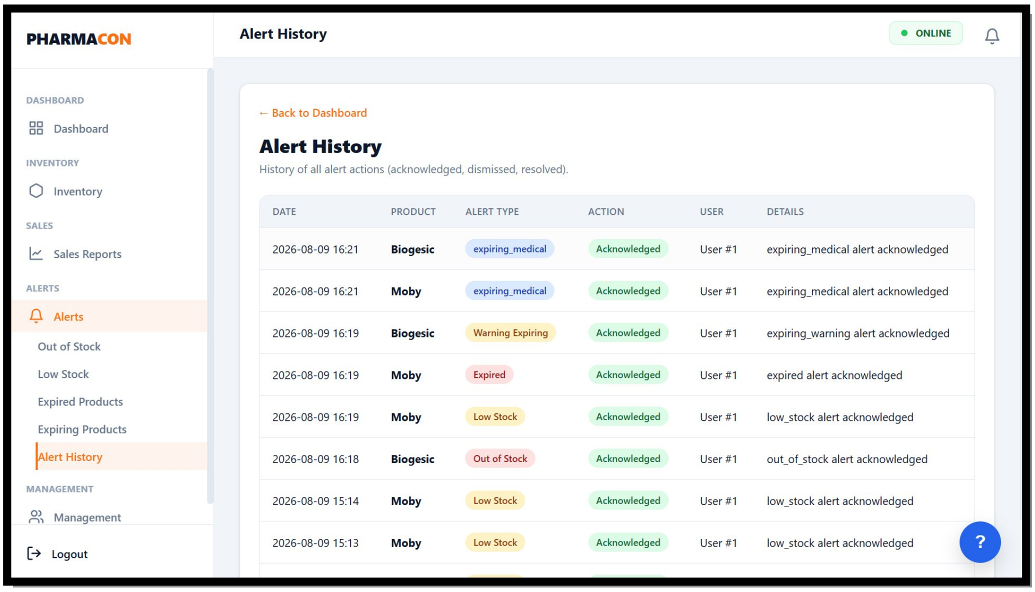
The Expired Products Interface displays products that have already passed their expiration date. This function allows administrators to identify expired products that should no longer be available for sale.
The interface supports the system's expiration monitoring function and helps administrators maintain accurate and safe inventory records. Expired products are identified separately from products that are approaching their expiration dates.

Figure 4.13 Expiring Products



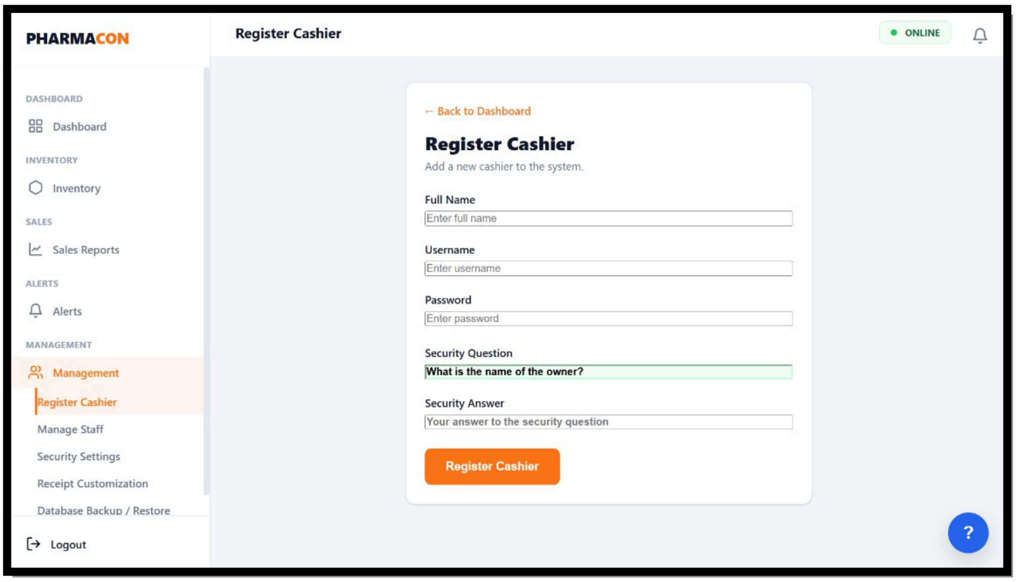
The Expiring Products Interface allows administrators to monitor products that are approaching their expiration dates. The system categorizes expiration alerts into critical and warning conditions. Products with **seven days or fewer remaining** are classified as critical, while products with **eight to thirty days remaining** are classified as warning alerts.
This interface enables administrators to take action before products expire and supports proactive inventory monitoring.

Figure 4.14 Alert History



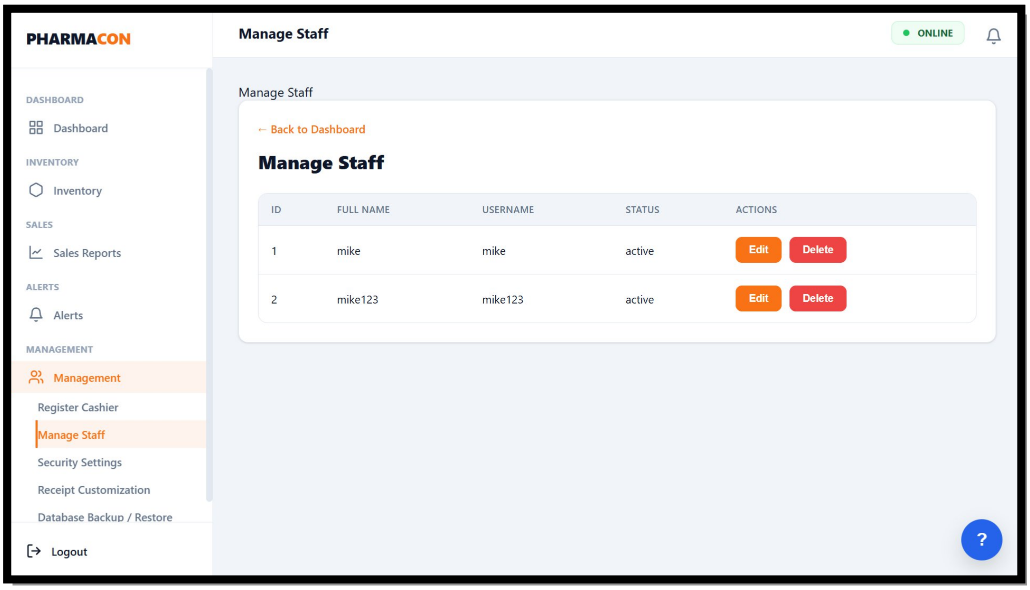
The Alert History Interface provides administrators with a record of previously handled inventory alerts. It supports the alert lifecycle implemented by the system, including active, acknowledged, and dismissed states. The alert history provides a record of inventory conditions that required attention and supports monitoring of how alerts were handled. The system also maintains an alert history audit trail as part of its inventory alert functionality.

Figure 4.15 Register Cashier



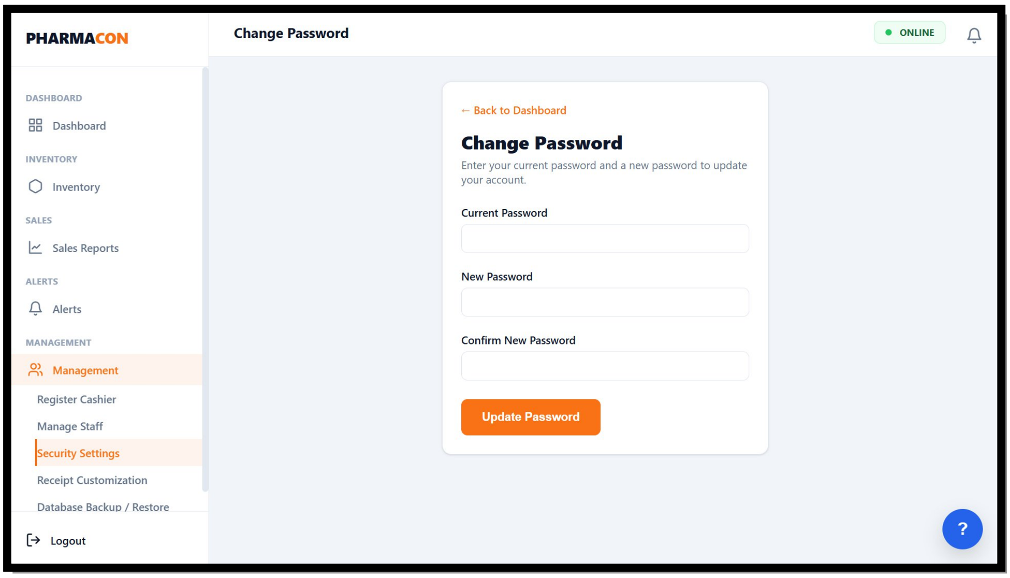
The Register Cashier Interface allows administrators to create cashier accounts for authorized employees. The administrator provides the required account information, after which the cashier account can be used to access the cashier functions of the system.
Cashier account management is an administrator-only function because it involves the creation and maintenance of user accounts with access to the Point-of-Sale module

Figure 4.16 Manage Staff



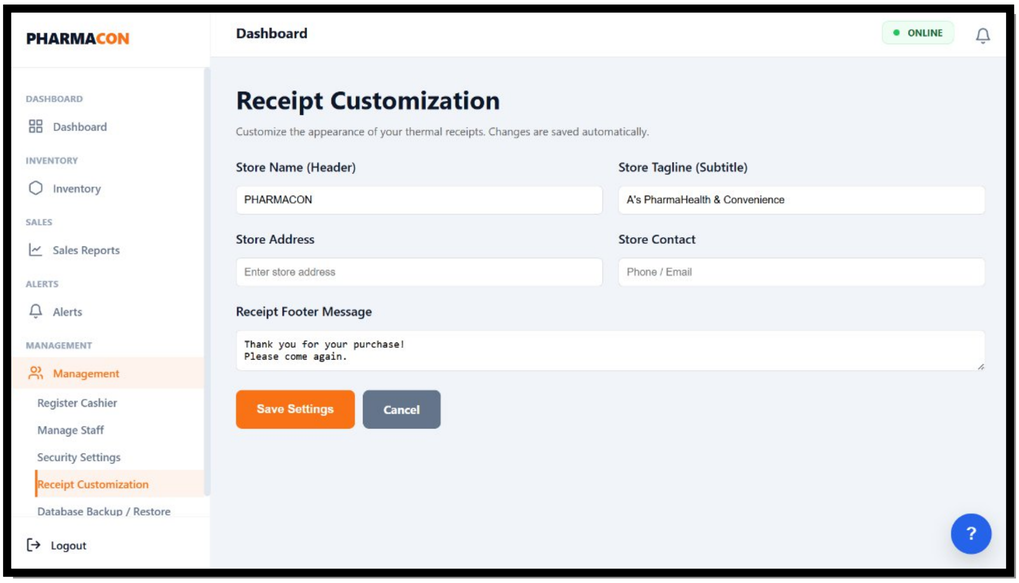
The Manage Staff Interface allows administrators to maintain existing cashier accounts. The system provides administrator functions for managing cashier accounts, including editing account information and activating or deactivating cashier access.
This function supports role-based access control by allowing administrators to control which cashier accounts are authorized to use the system.

Figure 4.17 Security Settings



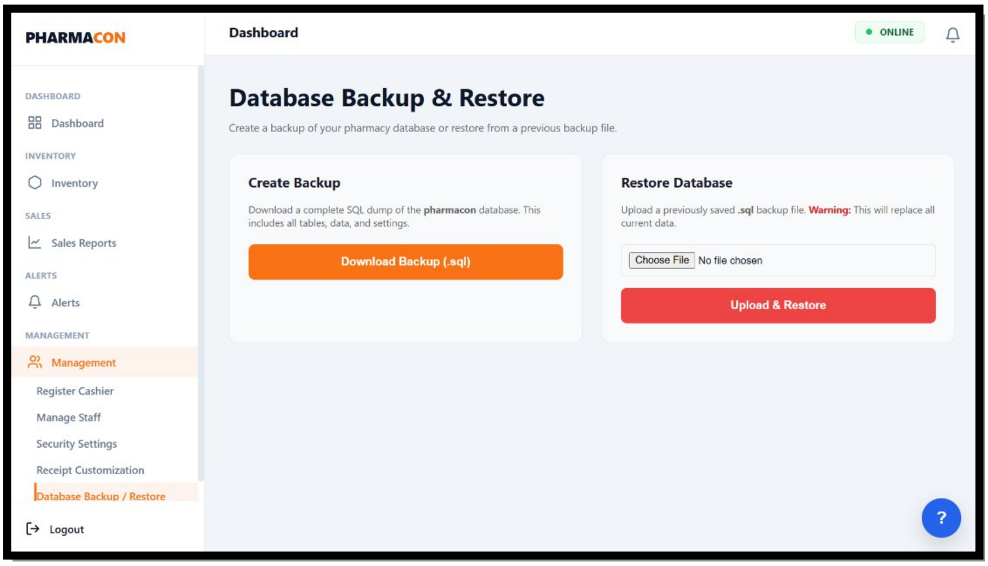
The Change Password Interface allows the administrator to update the account password. This function provides administrators with the ability to maintain their account credentials and improve account security. The password management function is supported by the system's password security mechanisms, which include secure password hashing and authentication controls.

Figure 4.18 Receipt Customization Interface



The Receipt Customization Interface allows administrators to configure information displayed on generated receipts. The settings include the receipt header, subtitle, footer, store address, and contact information. These values are stored in the database through the store settings functionality of the system.

Figure 4.19 Backup and Restore Interface



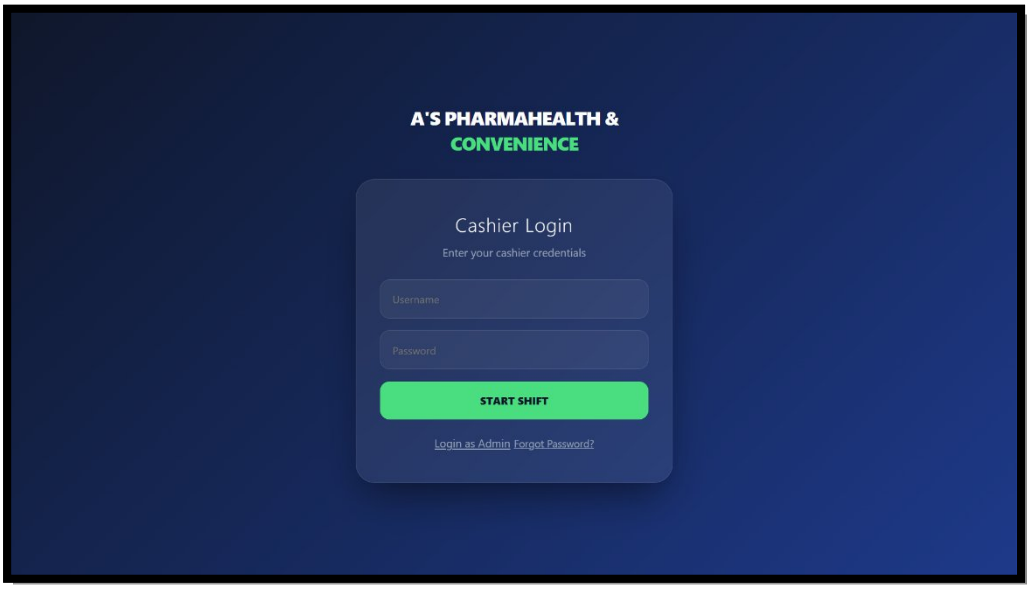
The Backup and Restore Interface provides administrators with database maintenance functions. The system allows the administrator to create a backup of the database and restore previously saved database information. This feature provides an additional mechanism for protecting system records and recovering operational data when necessary.

4.5.2 Cashier Interfaces

The Cashier Interfaces of the Automated Inventory Management System (AIMS) are designed to support the daily sales and Point-of-Sale (POS) activities performed by authorized cashiers. These interfaces provide the cashier with the functions necessary to search for products, use barcode scanning, add products to the transaction cart, manage quantities, hold transactions, process payments, generate receipts, and review transaction records. The system also provides personal sales metrics and transaction history for the logged-in cashier. The cashier is limited to functions appropriate to the cashier role through the system's role-based access control.

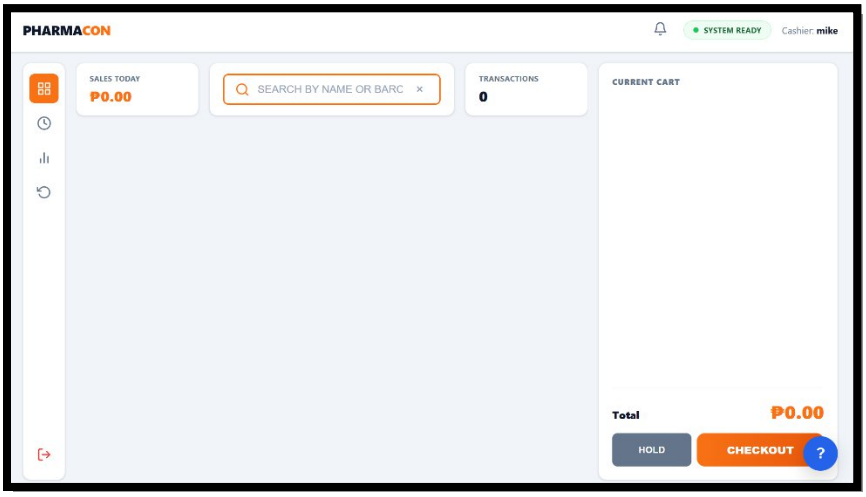
The cashier interfaces are presented according to the actual transaction workflow of the developed system, beginning with cashier authentication and continuing through product selection, cart management, payment processing, receipt generation, sales monitoring, and transaction history.

Figure 4.20 Cashier Login Interface



The Cashier Login Interface provides authenticated access to the Point-of-Sale functions. A cashier must provide valid account credentials, and the system verifies that the cashier account is active before allowing access. Successful authentication creates the appropriate cashier session and records the login activity.

Figure 4.21 Cashier Dashboard / Point-of-Sale Interface



The Cashier Dashboard serves as the primary Point-of-Sale interface where cashiers process customer transactions. The interface is designed to provide an organized workflow for product searching, barcode-based product lookup, cart management, and checkout.

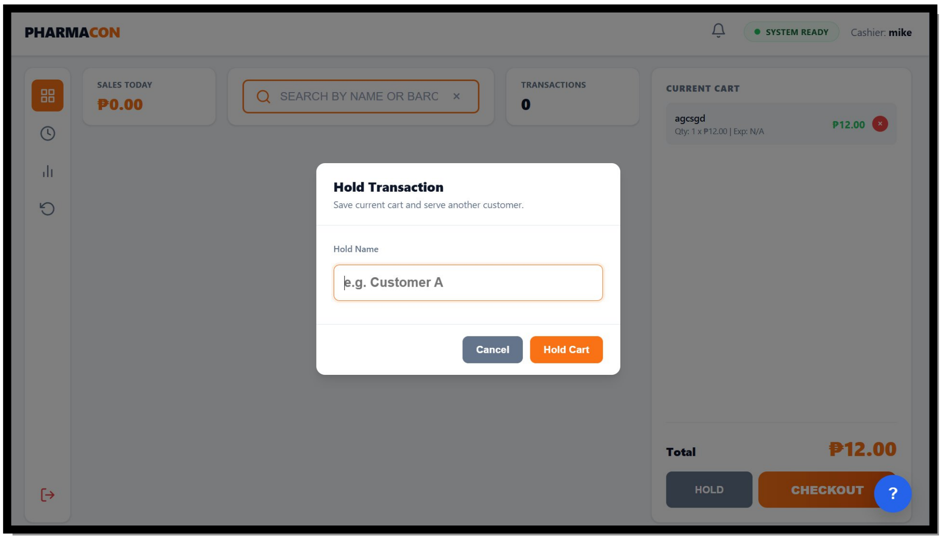
The interface provides a system status indicator and displays sales information for the current day. The cashier can search for products using the product search panel, which supports barcode scanner input and product name searching. Search results are displayed to allow the cashier to select the appropriate product.

After selecting a product, the cashier can view the product information, adjust the quantity, and add the product to the shopping cart. The system validates stock availability before allowing the product to be added or processed.

The shopping cart displays the selected products, quantities, prices, subtotals, and total amount. The cashier can remove individual items or clear the cart when necessary. The interface also provides **HOLD** and **CHECKOUT** functions. The HOLD function allows the current transaction to be temporarily saved, while CHECKOUT proceeds to the payment process.

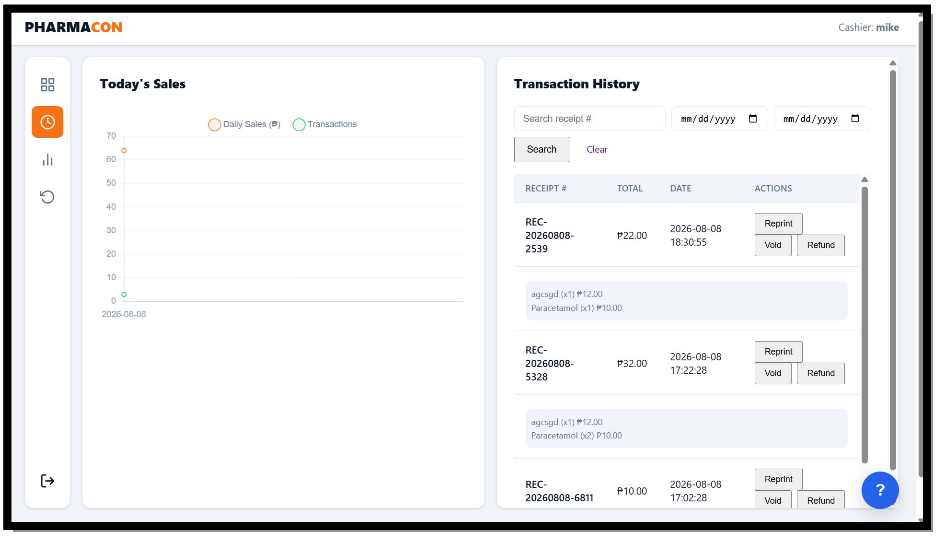
The cashier dashboard also includes an interactive help tour that guides the cashier through the POS workflow, including product searching, cart management, checkout, payment, and receipt processing.

Figure 4.22 Hold Transaction Modal



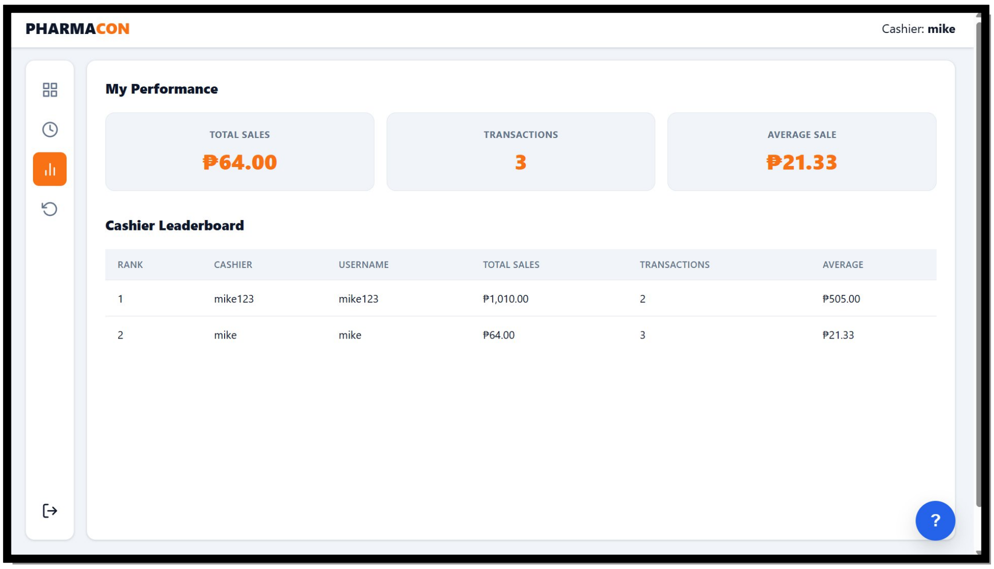
The Hold Transaction Modal allows cashiers to temporarily save an ongoing transaction when they need to serve another customer or perform another transaction before completing the current sale.
The cashier can provide a hold name or customer identifier and may enter an optional reason for holding the transaction. The system stores the transaction in the hold list together with information such as the hold timestamp, item count, and transaction total.
The cashier can later select a saved transaction and resume the transaction. The interface also provides an option to delete a hold entry when it is no longer needed. This function helps prevent unfinished transactions from blocking the normal sales process.

Figure 4.23 Transaction History Interface



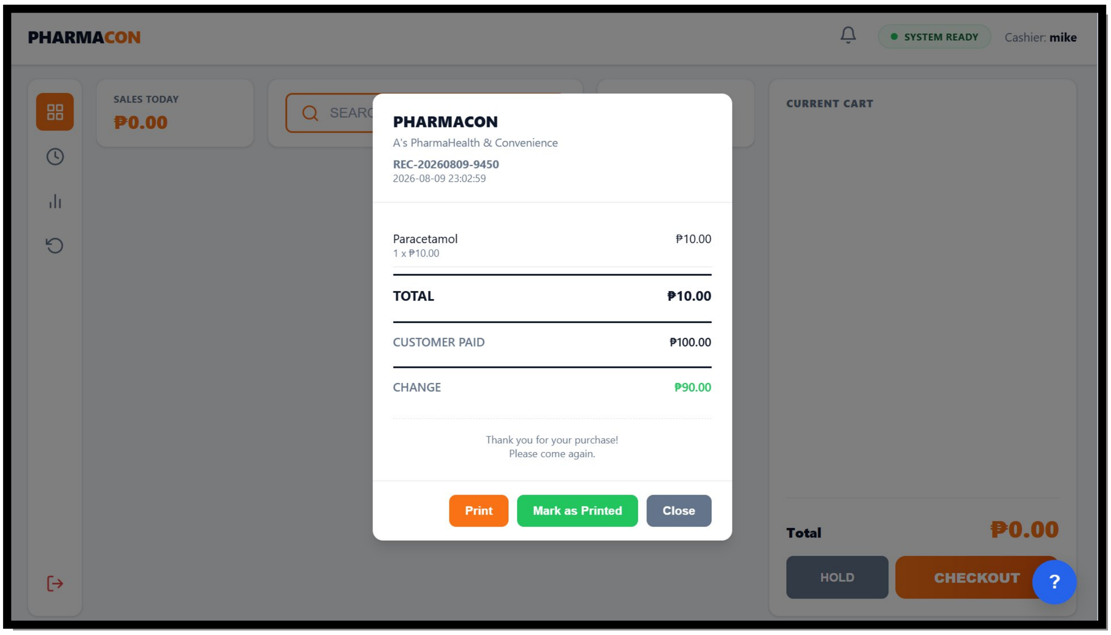
The Transaction History Interface allows cashiers to review previously processed transactions and their corresponding receipt information. The interface provides access to recorded sales transactions and supports transaction and receipt review after completed sales.

Figure 4.24 Cashier Metrics Interface



The Cashier Metrics Interface provides the cashier with information about personal sales performance. It presents statistics that allow the cashier to monitor sales activity over selected periods.
The interface provides personal statistics such as total sales amount, number of transactions, average sale value, and performance trends. The cashier can select different time periods, including daily, weekly, and monthly views.
The interface also provides sales charts for visualizing performance trends and identifying periods of higher sales activity. A cashier leaderboard is also included to provide a comparison of sales performance among cashiers.

Figure 4.25 Receipt Interface



The Receipt Interface presents the printable receipt generated after a successful Point-of-Sale transaction.

Receipt information is based on the completed transaction and the receipt customization settings configured by the administrator. The receipt functionality is integrated into the sales workflow so that transaction information can be documented immediately after payment confirmation.

4.6 Database Implementation

The Automated Inventory Management System (AIMS) was implemented using a relational MySQL database to store, manage, and retrieve the information required by the system. The database was deployed through XAMPP and integrated with the Python Flask backend of the application. This implementation allowed the system to perform database operations required for user authentication, product and inventory management, Point-of-Sale transactions, sales recording, activity monitoring, reporting, and system configuration.

The implemented database follows the relational structure established during the database design phase in Chapter 3. However, unlike the database design presented in Chapter 3, which describes the planned organization and relationships of the data, this section focuses on the actual use of the database within the developed system. The database serves as the central storage component that allows the administrator and cashier interfaces to retrieve, insert, update, and manage operational records.

4.6.1 Database Implementation Environment

The database was implemented using MySQL as the relational database management system and XAMPP as the local server environment. The Python Flask backend communicates with the MySQL database to process requests submitted through the system interfaces.

The implemented application uses parameterized database queries to securely process database operations and reduce the risk of SQL injection. Indexed columns are also used to improve the efficiency of frequently performed database searches and retrieval operations. The system follows a service-layer approach in which database-related operations are handled by the backend before the resulting information is presented through the user interface.

The implemented database supports the offline Local Area Network (LAN) environment of the system. This allows authorized users to access shared inventory and sales information while the application and database are hosted within the local environment.

4.6.2 Implemented Database Tables

The implemented database consists of relational tables that support the major functions of the Automated Inventory Management System (AIMS). Each table is responsible for storing a specific type of information required by the system, while relationships between related tables allow the application to retrieve and manage connected records.

Database Table	Implementation and System Use
admins	Stores administrator account information used for authentication and administrative access.
admin_activity	Records activities performed by administrators for monitoring and audit purposes.
cashiers	Stores cashier account information used for cashier authentication and account management.
cashier_activity	Records activities performed by cashiers within the system.
categories	Stores the product classifications used by the system, including Medical and Non-Medical categories.
products	Stores product information such as product name,

	barcode, category, price, stock quantity, and expiration date.
stock_movements	Records changes in product inventory, including stock-in and stock-out activities.
sales	Stores the main information of completed sales transactions, including receipt number, cashier, total amount, transaction status, product type, and transaction date.
sale_items	Stores the individual products included in each sales transaction, including their quantity and selling price.
store_settings	Stores configurable store information used by the system, including information displayed on generated receipts.

The implemented tables provide the database support required by the different modules of AIMS. The **admins** and **cashiers** tables support user authentication and account management, while the corresponding activity tables maintain records of user actions. The **categories** and **products** tables support product and inventory management, while **stock_movements** maintains records of inventory changes.

For sales processing, the **sales** table stores the main transaction record and the **sale_items** table stores the individual products included in the transaction. This structure allows one transaction to contain multiple products while keeping the transaction and item-level information organized.

The **store_settings** table supports system configuration and receipt-related information. Overall, the implemented database tables provide the necessary data storage for authentication, product management, inventory monitoring, Point-of-Sale transactions, sales recording, activity monitoring, and system configuration.

4.6.3 Database Integration with System Modules

The implemented database is directly integrated with the major modules of the Automated Inventory Management System. Each system operation that requires stored information communicates with the appropriate database tables through the Flask backend.

The administrator login process uses the administrator account records to authenticate authorized users. Similarly, cashier login uses the cashier account records to verify cashier credentials and account status.

The Product Management module uses the products and categories tables to store and retrieve product information. When a product is added or updated, the corresponding database record is modified. Product searches can also retrieve information using the product name or barcode.

The Inventory Monitoring module uses product and stock movement records to monitor current inventory levels and changes in stock. When inventory is affected by a transaction or other stock activity, the corresponding records are updated to maintain accurate inventory information.

The Point-of-Sale module uses the sales and sale_items tables to record completed transactions. The sales table stores the main transaction information, while the sale_items table stores the individual products included in the transaction. This separation allows one sales transaction to contain multiple products while maintaining an organized record of each item sold.

The Activity Logging module uses the administrator and cashier activity tables to record user actions. These records provide an audit trail that allows administrators to monitor activities performed within the system.

The Receipt Customization and store configuration functions use the store_settings table to retrieve the information required for system configuration and receipt generation.

4.6.4 Database Operations

The implemented database supports the basic operations required by the different system modules. These operations include data insertion, retrieval, updating, and deletion.

Data Insertion is performed when new records are created. Examples include registering a new administrator or cashier account, adding a new product, recording stock movements, and saving completed sales transactions.

Data Retrieval is performed when the system needs to display stored information. Examples include searching for products, displaying inventory records, retrieving sales transactions, generating reports, displaying activity logs, and retrieving cashier transaction history.

Data Updating is performed when existing information changes. Examples include updating product information, modifying inventory quantities, changing cashier account status, and updating system settings.

Data Deletion is used when records or temporary information need to be removed according to the system's permitted operations. Database operations are controlled by the application to ensure that users can only perform actions allowed by their assigned roles.

These operations allow the system interfaces to remain connected with the current information stored in the database and support the continuous processing of inventory and sales activities.

4.6.5 Database Relationships and Data Integrity

The implemented database maintains relationships among related records through primary keys and foreign keys. These relationships allow information stored in different tables to be connected while maintaining data consistency.

For sales transactions, the sales table is connected with the sale_items table so that each completed transaction can contain multiple individual products. The sale_items records are also connected to the products table so that each item sold can be associated with its corresponding product.

Product and inventory information is connected with stock movement records to maintain a history of changes in product quantities. Administrator and cashier activity records are associated with their respective user accounts to provide traceability of system activities.

The use of primary keys provides unique identification for records, while foreign keys maintain the relationships between related tables. These mechanisms help prevent inconsistent references and support the integrity of inventory, sales, user, and activity information.

4.6.6 Database Security and Maintenance

Database security was implemented as part of the overall security mechanisms of the Automated Inventory Management System. The system uses parameterized queries to reduce the risk of SQL injection and applies validation to information submitted through the application.

User access to database-related functions is controlled through role-based access control. Administrators are provided with broader database management functions, while cashiers are restricted to functions necessary for Point-of-Sale operations and their permitted transaction records.
The system also maintains activity records for administrator and cashier operations. These audit trails provide a historical record of user actions and support accountability within the system.

Database maintenance is supported through backup and restore functionality. The system allows authorized administrators to create database backups and restore previously saved database information when necessary. The system reference identifies database backup and restoration as part of the implemented data management functions.

Overall, the implementation of the MySQL database provided the Automated Inventory Management System with a centralized and structured mechanism for managing its operational information. The integration of the database with the Flask backend and the different system modules enabled the application to perform inventory monitoring, barcode-based sales processing, transaction recording, activity tracking, reporting, user management, and database maintenance. The implemented database therefore served as an essential component in supporting the operation and reliability of the developed system.

4.7 System Testing

System testing was conducted to verify that the Automated Inventory Management System (AIMS) performed according to its intended functions and system requirements. The testing process ensured that each module operated correctly, produced accurate outputs, and interacted properly with other components of the system. The objective of system testing was to identify and correct any errors before the system was evaluated by end users.

The testing process covered the major functional modules of the system, including user authentication, product management, sales transaction processing, barcode scanning, sales report generation, alert notifications, cashier management, activity logging, database backup and restoration, and store settings. Each module was tested using valid and invalid inputs to determine whether the system responded correctly under different operating conditions.
Functional testing was performed by executing each feature based on its intended purpose. The actual results produced by the system were compared with the expected results to determine whether each function operated successfully. The testing confirmed that the system was capable of processing user requests accurately while maintaining data consistency throughout the application.
The following table presents the functional testing conducted for the major features of the Automated Inventory Management System.

Table 4.1 Functional Testing Results

System Function	Test Performed	Expected Result	Actual Result	Status
Administrator Login	Enter valid administrator credentials	Administrator Dashboard is displayed	Dashboard displayed successfully	Passed
Cashier Login	Enter valid cashier credentials	Cashier POS interface is displayed	POS interface displayed successfully	Passed
Add Product	Save a new product with complete information	Product is stored in the database	Product saved successfully	Passed
Update Product	Modify existing product information	Product information is updated	Product saved successfully	Passed
Delete Product	Remove an existing product	Product is removed from the inventory list	Product saved successfully	Passed
Search Product	Search by product	Matching product is	Product saved	Passed

	name or barcode	displayed	successfully	
Barcode Scanning	Scan a valid barcode	Product is automatically retrieved	Product saved successfully	Passed
Point of Sale Transaction	Complete a customer purchase	Sale is recorded and inventory is updated	Transaction completed successfully	Passed
Receipt Printing	Complete a transaction	Receipt is generated and printed	Receipt generated successfully	Passed
Sales Report	Generate report for selected dates	Sales report is displayed correctly	Report generated successfully	Passed
Alerts	Reduce stock below minimum quantity	Low-stock alert is displayed	Alert displayed successfully	Passed
Activity Logs	Perform administrator or cashier operations	Activities are recorded	Activity recorded successfully	Passed
Backup Database	Perform database backup	Backup file is created	Backup completed successfully	Passed
Restore Database	Restore previously saved database	Database is restored	Restore completed successfully	Passed
Store Settings	Update store information	Changes are saved and reflected on receipts	Information updated successfully	Passed

The results presented in Table 4.1 indicate that all major functions of the Automated Inventory Management System operated successfully during testing. Each module produced the expected output and interacted correctly with the database and other system components. Product records were accurately stored and updated, sales transactions were processed correctly, inventory quantities were automatically adjusted after each completed sale, and reports were generated without errors. Likewise, barcode scanning, receipt printing, alert notifications, activity logging, and database backup and restoration functions operated according to the specified system requirements.

Overall, the system testing demonstrated that the Automated Inventory Management System was able to perform its intended functions accurately, efficiently, and reliably. The successful completion of the testing process confirmed that the developed system was ready for user evaluation based on the ISO/IEC 25010 Software Quality Model, which is discussed in the succeeding section.

4.7.1 Unit Testing

Unit Testing was conducted to verify that each individual module and function of the **Automated Inventory Management System (AIMS)** operated correctly before integrating them into the complete application. The primary objective of unit testing was to identify errors within individual software components at an early stage of development, ensuring that each module performs its intended function according to the specified system requirements.

During this phase, every major module of the system was tested independently using various test scenarios. Each function was provided with valid, invalid, and boundary input values to determine whether the system generated the expected output. Modules that did not initially produce the desired results were corrected, modified, and retested until they successfully met the expected functionality.

The unit testing process focused on validating the core functionalities of the system, including user authentication, product management, inventory management, cashier account management, barcode scanning, Point of Sale (POS) transactions, sales reporting, receipt customization, activity logging, and database backup and restoration. Testing each module individually helped ensure that errors were detected before the modules were integrated into the complete system.

Table 4.2 presents the unit testing conducted for the major modules of the Automated Inventory Management System.

Module	Function Tested	Expected Result	Actual Result	Remarks
Login Module	Administrator Login	Administrator successfully logs into the dashboard	Successfully logged in	Pass
Login Module	Cashier Login	Cashier successfully accesses the POS interface	Successfully logged in	Pass
Product Management	Add Product	Product information saved successfully	Product added successfully	Pass
Product Management	Edit Product	Product information updated correctly	Product updated successfully	Pass
Product Management	Delete Product	Product removed from inventory	Product deleted successfully	Pass
Product Management	Search Product	Correct product information displayed	Product displayed correctly	Pass
Cashier Management	Add Cashier	New cashier account created	Cashier account created successfully	Pass
Cashier Management	Edit Cashier	Cashier information updated	Cashier information updated successfully	Pass
Cashier Management	Delete Cashier	Cashier account removed	Cashier account deleted successfully	Pass
Inventory Module	Low Stock Alert	Display notification when stock reaches minimum quantity	Alert displayed correctly	Pass
Inventory Module	Expiring Product Alert	Display products approaching expiration date	Expiring products displayed correctly	Pass
Barcode Scanner	Scan Barcode	Retrieve corresponding product information	Product retrieved successfully	Pass
POS Module	Process Sales Transaction	Transaction completed and inventory updated	Transaction processed successfully	Pass
Receipt Module	Print Receipt	Generate printable receipt after transaction	Receipt generated successfully	Pass
Sales Report Module	Generate Sales Report	Display accurate sales information	Sales report generated successfully	Pass
Activity Log Module	Record User Activity	Administrator and cashier activities recorded	Activities logged successfully	Pass
Database Module	Backup Database	Backup file successfully created	Backup completed successfully	Pass
Database Module	Restore Database	Previous backup restored successfully	Database restored successfully	Pass

Discussion of Unit Testing Results

The results of the unit testing indicate that all individual modules of the Automated Inventory Management System functioned according to their intended design. The login module successfully authenticated both administrator and cashier accounts while preventing unauthorized access through invalid credentials. Product management functions accurately handled the addition, modification, deletion, and retrieval of product records stored in the database.

The cashier management module successfully performed user account maintenance, allowing administrators to create, update, and remove cashier accounts without affecting existing system data. Inventory monitoring features correctly identified products with low stock levels and products approaching their expiration dates, enabling administrators to monitor inventory efficiently.

The barcode scanning function accurately retrieved product information during Point of Sale (POS) transactions, reducing manual product searches and improving transaction speed. The POS module correctly calculated transaction totals, recorded completed sales, updated inventory quantities, and generated printable receipts after every successful transaction.

Sales reporting generated accurate transaction summaries, while activity logging successfully recorded administrator and cashier operations performed within the system. Likewise, the database backup and restoration modules effectively protected operational data by allowing administrators to create backup copies and recover previously saved database records.

Overall, the unit testing results demonstrated that each module of the Automated Inventory Management System performed its assigned functions accurately and independently. Since all evaluated modules achieved the expected outcomes, they were considered ready for integration and subsequent testing during the next phase of system testing.

4.7.2 Integration Testing

Integration Testing was conducted after the successful completion of unit testing to verify that the different modules of the **Automated Inventory Management System (AIMS)** function correctly when combined into a single integrated application. The objective of integration testing was to ensure that data flows accurately between interconnected modules and that the interaction among system components produces the expected results without errors or inconsistencies.

Unlike unit testing, which focuses on individual modules, integration testing evaluates the communication between multiple modules working together during actual system operations. This testing was performed to verify that information entered in one module is correctly processed, stored, and reflected in other related modules. The integration testing process also ensured that database transactions, user interactions, and automated system processes functioned as intended.

The major modules evaluated during integration testing included user authentication, product management, inventory monitoring, Point of Sale (POS), barcode scanning, sales reporting, activity logging, receipt generation, and database management. Each integrated process was executed repeatedly using different transaction scenarios to confirm that the modules exchanged data accurately and consistently.

Table 4.3 presents the integration testing conducted for the developed Automated Inventory Management System.

Integrated	Test Scenario	Expected Result	Actual Result	Remarks

Modules				
Login → Dashboard	Administrator logs into the system	Administrator is redirected to the dashboard	Dashboard displayed successfully	Pass
Login → POS	Cashier logs into the system	Cashier is redirected to the POS interface	POS interface displayed successfully	Pass
Product Management → Database	Add a new product	Product information is saved in the database	Product saved successfully	Pass
Product Management → Inventory	Update product quantity	Inventory records updated automatically	Inventory updated successfully	Pass
Barcode Scanner → POS	Scan product barcode	Product information appears automatically in POS	Product retrieved successfully	Pass
POS → Sales Database	Complete a sales transaction	Sales record saved successfully	Transaction recorded successfully	Pass
POS → Inventory	Complete a sales transaction	Product stock quantity decreases automatically	Inventory updated successfully	Pass
POS → Receipt Module	Complete payment	Receipt generated and printed	Receipt printed successfully	Pass
Sales → Sales Reports	Generate daily sales report	Completed transactions appear in report	Sales report generated correctly	Pass
Product Management → Inventory Alerts	Product reaches minimum stock level	Low-stock notification displayed	Alert displayed successfully	Pass
Product Management → Expiry Monitoring	Product approaches expiration date	Expiring product notification displayed	Notification displayed successfully	Pass
Administrator Activities → Activity Logs	Administrator updates product information	Activity recorded in administrator log	Activity logged successfully	Pass
Cashier Activities → Activity Logs	Cashier processes a transaction	Cashier activity recorded	Activity logged successfully	Pass
Database Backup → Restore Database	Restore backup file	Previous database restored successfully	Database restored successfully	Pass

Discussion of Integration Testing Results

The integration testing results demonstrated that all major modules of the Automated Inventory Management System communicated effectively and exchanged information accurately during actual system operation. User authentication successfully redirected administrators and cashiers to their respective modules after successful login while maintaining role-based access control.

The integration between the Product Management module and the database ensured that newly added, updated, and deleted product information was immediately reflected in the inventory records. Likewise, barcode scanning was successfully integrated with the Point of Sale (POS) module, enabling products to be retrieved automatically during sales transactions without requiring manual data entry.

The POS module correctly communicated with the sales database, inventory records, and receipt generation module. Every completed transaction was automatically recorded in the sales database, corresponding inventory quantities were updated, and a printable receipt was generated for the customer. These integrated processes minimized manual work and improved transaction efficiency.
Inventory monitoring also functioned correctly with the product management module by automatically displaying low-stock and expiring-product notifications whenever predefined conditions were met. Sales reporting accurately retrieved transaction records from the database and generated reports that reflected completed sales activities.

Furthermore, administrator and cashier actions were successfully recorded in their respective activity logs, providing accountability and allowing administrators to monitor system usage. The database backup and restoration modules also integrated correctly, enabling administrators to create backup copies and restore system data whenever necessary.

Overall, the integration testing confirmed that the individual modules of the Automated Inventory Management System functioned cohesively as a unified application. All integrated processes produced the expected results, indicating that the developed system operated reliably and was ready for comprehensive functional verification through Black-box Testing in the succeeding section.

4.7.3 Black-box Testing

Black-box Testing was conducted to evaluate the functional behavior of the **Automated Inventory Management System (AIMS)** without considering its internal program structure or source code. This testing method focused on validating whether the system produced the expected outputs based on various user inputs and predefined functional requirements. The primary objective of black-box testing was to verify that each feature of the system operated correctly from the perspective of the end users.

During the testing process, the testers interacted with the system by providing valid and invalid inputs through the graphical user interface. The responses generated by the system were then compared with the expected results specified in the system requirements. This approach ensured that all system functions behaved correctly under different operating conditions and that appropriate messages were displayed whenever invalid data or unauthorized actions were encountered.

The black-box testing covered the major functionalities of the system, including user authentication, product management, barcode-based Point of Sale (POS), inventory monitoring, sales reporting, receipt generation, activity logging, and database management. Each module was tested independently from the user's perspective to determine whether it met the expected functional requirements.

Table 4.4 presents the results of the black-box testing conducted on the Automated Inventory Management System.

Test Case	Test Input / Action	Expected Result	Actual Result	Status
Administrator Login	Enter valid administrator credentials	Administrator is successfully logged in and redirected to the dashboard	Administrator logged in and dashboard displayed successfully	Passed
Cashier Login	Enter valid cashier credentials	Cashier is successfully logged in and redirected to the POS interface	Cashier logged in and POS interface displayed successfully	Passed
Invalid Login	Enter incorrect login credentials	System rejects the login and displays an appropriate error	Login was rejected and error message was displayed	Passed

		message		
Add Product	Enter valid product information and save	Product information is stored successfully	Product was added successfully	Passed
Duplicate Barcode	Enter a barcode already registered in the database	System prevents duplicate barcode registration	Duplicate barcode was prevented	Passed
Product Search	Search using a valid product name or barcode	Corresponding product information is displayed	Product information was displayed correctly	Passed
Barcode Scanning	Scan a valid product barcode	Corresponding product is retrieved in the POS interface	Product was retrieved successfully	Passed
Insufficient Stock	Attempt to purchase a quantity greater than available stock	System prevents the transaction and displays an appropriate message	Transaction was prevented successfully	Passed
Expired Product	Attempt to process an expired product	System prevents the sale of the expired product	Sale was prevented successfully	Passed
POS Transaction	Add valid products and complete checkout	Transaction is successfully processed and recorded	Transaction was processed successfully	Passed
Insufficient Payment	Enter an amount lower than the transaction total	System rejects the payment and displays an appropriate message	Payment was rejected successfully	Passed
Receipt Generation	Complete a valid POS transaction	Receipt is generated and displays transaction details	Receipt was generated successfully	Passed
Low Stock Monitoring	Reduce product stock to the configured minimum level	Low-stock notification is displayed	Low-stock notification was displayed successfully	Passed
Expiration Monitoring	Product reaches the defined expiration warning period	Expiration notification is displayed	Expiration notification was displayed successfully	Passed
Activity Logging	Perform an administrator or cashier activity	User activity is recorded in the appropriate activity log	Activity was recorded successfully	Passed
Database Backup	Execute the database backup function	Backup file is successfully created	Backup completed successfully	Passed
Database Restore	Restore a previously created database backup	Previous database information is successfully restored	Database restored successfully	Passed

The results presented in Table 4.4 demonstrate that the Automated Inventory Management System correctly responded to the different valid and invalid inputs used during Black-Box Testing. The authentication tests successfully accepted valid credentials while rejecting invalid login attempts, demonstrating that the system properly handled user authentication.

The product management tests also demonstrated that valid product information could be added and that duplicate barcode entries were prevented. Product searching and barcode scanning successfully retrieved the appropriate product information, supporting efficient product identification during inventory management and Point-of-Sale transactions.

The Point-of-Sale tests showed that the system correctly handled normal and invalid transaction conditions. Transactions involving available products were processed successfully, while purchases involving insufficient stock or expired products were prevented. The system also rejected insufficient payment amounts and successfully generated receipts after completed transactions.

The inventory monitoring tests confirmed that the system displayed appropriate notifications when products reached low-stock levels or approached their expiration dates. Activity logging also successfully recorded administrator and cashier activities, providing an audit trail of user operations.
The database management tests further demonstrated that backup and restoration functions operated successfully. These results indicate that the system provided the expected responses to the tested user actions without requiring examination of its internal source code.

Overall, the Black-Box Testing results confirmed that the major user-facing functions of the Automated Inventory Management System operated according to their expected behavior. All tested cases were successfully completed and marked as **Passed**, demonstrating that the developed system was able to process valid inputs, reject invalid operations, provide appropriate system responses, and support the intended operational functions of administrators and cashiers.

4.8 System Evaluation

The Automated Inventory Management System (AIMS) was evaluated to determine its overall quality, functionality, efficiency, and user acceptability after the completion of system development and testing. The evaluation aimed to assess whether the developed system successfully achieved the objectives of the study and met the operational requirements of pharmacies and convenience stores.

The evaluation was conducted using the **ISO/IEC 25010 Software Quality Model**, an internationally recognized software quality standard that provides a framework for evaluating software products based on various quality characteristics. This standard was selected because it offers a comprehensive assessment of software quality from both technical and user perspectives.

The developed system was evaluated based on the following software quality characteristics:

- Functional Suitability
- Performance Efficiency
- Compatibility
- Usability
- Reliability
- Security
- Maintainability
- Portability

These quality characteristics were used to determine whether the developed system performs its intended functions accurately, efficiently, securely, and reliably while maintaining ease of use and future maintainability.

Before conducting the evaluation, the respondents were given sufficient time to use the system and perform its major functions. They were asked to access the Administrator and Cashier modules, manage products, process

Point of Sale (POS) transactions, scan product barcodes, generate reports, customize receipts, perform database backup and restoration, and explore the different system features. After completing these activities, the respondents answered a structured evaluation questionnaire based on the ISO/IEC 25010 Software Quality Model.

4.8.1 Functional Testing

Functional Testing was conducted to determine whether the major functions of the Automated Inventory Management System operated according to their specified requirements. Each function was executed using appropriate test scenarios, and the actual results produced by the system were compared with the expected results.
The testing included administrator login, cashier login, product management, product searching, barcode scanning, Point-of-Sale transactions, receipt generation, sales reporting, inventory alerts, activity logging, database backup and restoration, and store settings. Each function was tested using valid inputs and appropriate operational procedures to determine whether the system produced the required output.

Table 4.5 Functional Testing Results

Indicator	Weighted Mean	Interpretation
Provides functions for intended tasks	4.52	Strongly Agree
Provides accurate results	4.48	Strongly Agree
Provides necessary inventory/sales features	4.56	Strongly Agree
Supports admin and cashier functions	4.50	Strongly Agree
Overall Weighted Mean	**4.52**	**Strongly Agree**

The results in Table 4.5 show that the Automated Inventory Management System obtained an overall weighted mean of 4.52, interpreted as Strongly Agree, in terms of Functional Suitability. This indicates that the respondents strongly agreed that the system provides the functions necessary to support the intended tasks of its users.

Among the indicators, “Provides necessary inventory/sales features” obtained the highest weighted mean of 4.56, interpreted as Strongly Agree. This indicates that the respondents considered the system's inventory and sales features appropriate and sufficient for the intended operations.
The indicator “Provides functions for intended tasks” obtained a weighted mean of 4.52, while “Supports admin and cashier functions” obtained 4.50, both interpreted as Strongly Agree. These results indicate that the system provides appropriate functions for both administrator and cashier users and supports their respective responsibilities.

Meanwhile, “Provides accurate results” obtained a weighted mean of 4.48, which was also interpreted as Strongly Agree. This indicates that the respondents agreed that the system produces accurate results when performing its intended functions.

Overall, the 4.52 Strongly Agree rating demonstrates that the developed AIMS satisfactorily met the required functional needs of its intended users. The result also supports the system testing findings, which showed that the major functions of the system performed their expected operations successfully.

4.8.2 Performance Efficiency

Performance Efficiency evaluated the ability of the Automated Inventory Management System (AIMS) to perform its intended functions within an appropriate response time and with efficient use of system resources. The evaluation focused on the system's responsiveness during common operations, including product searching, barcode scanning, inventory retrieval, Point-of-Sale transactions, report generation, and other activities performed by administrators and cashiers.

The respondents evaluated the performance of the developed system based on their actual interaction with its different functions. The evaluation determined whether the system could process user requests efficiently and provide results within an acceptable period during normal system operation.

Table 4.6. Performance Efficiency Evaluation Results

Indicator	Weighted Mean	Interpretation
Provides fast response during system operations	4.45	Strongly Agree
Processes transactions efficiently	4.48	Strongly Agree
Retrieves product and inventory information efficiently	4.42	Strongly Agree
Generates reports and system results efficiently	4.40	Strongly Agree
Overall Weighted Mean	4.44	Strongly Agree

The results in Table 4.6 show that the Automated Inventory Management System obtained an **overall weighted mean of 4.44**, interpreted as **Strongly Agree**, in terms of Performance Efficiency. This indicates that the respondents strongly agreed that the system performs its intended operations efficiently.

Among the indicators, **"Processes transactions efficiently"** obtained the highest weighted mean of **4.48**, interpreted as Strongly Agree. This indicates that the respondents found the system capable of processing Point-of-Sale transactions efficiently.

The indicator **"Provides fast response during system operations"** obtained a weighted mean of **4.45**, while **"Retrieves product and inventory information efficiently"** obtained **4.42**. Both were interpreted as Strongly Agree, indicating that the system provides responsive operations when users search for products and access inventory information.

Meanwhile, **"Generates reports and system results efficiently"** obtained a weighted mean of **4.40**, also interpreted as Strongly Agree. This shows that the respondents considered the system capable of generating the required reports and results efficiently.

Overall, the **4.44 Strongly Agree** rating demonstrates that the developed AIMS provides efficient performance during its intended operations. The result indicates that the system is capable of supporting the daily activities of administrators and cashiers with appropriate response and processing efficiency.

4.8.3 Compatibility

Compatibility evaluated the ability of the Automated Inventory Management System (AIMS) to operate properly within its intended computing and network environment. The evaluation focused on the system's ability to work with the required hardware and software components, interact properly with the database and system components, and operate within the intended Local Area Network (LAN) environment.

Table 4.7. Compatibility Evaluation Results

Indicator	Weighted Mean	Interpretation
Operates properly within the intended system environment	4.50	Strongly Agree
Works properly with the required hardware and software	4.48	Strongly Agree

Interacts properly with the database and system components	4.47	Strongly Agree
Operates appropriately within the intended LAN environment	4.47	Strongly Agree
Overall Weighted Mean	**4.48**	**Strongly Agree**

The results in Table 4.7 show that the Automated Inventory Management System obtained an **overall weighted mean of 4.48**, interpreted as **Strongly Agree**, in terms of Compatibility. This indicates that the respondents strongly agreed that the system operates properly within its intended environment.

The indicator **"Operates properly within the intended system environment"** obtained the highest weighted mean of **4.50**, interpreted as Strongly Agree. This indicates that the respondents found the system suitable for its intended computing environment.

The indicator **"Works properly with the required hardware and software"** obtained a weighted mean of **4.48**, while both **"Interacts properly with the database and system components"** and **"Operates appropriately within the intended LAN environment"** obtained **4.47**. All indicators were interpreted as Strongly Agree.

Overall, the **4.48 Strongly Agree** rating demonstrates that the developed AIMS is compatible with its intended hardware, software, database, and LAN environment. The result indicates that the system can operate appropriately within the environment for which it was developed.

4.8.4 Usability

Usability evaluated the extent to which the Automated Inventory Management System (AIMS) could be understood, learned, and operated by its intended users. The evaluation focused on the clarity of the system interface, ease of navigation, understandability of labels and controls, and the ability of administrators and cashiers to perform their assigned tasks efficiently.

Table 4.8. Usability Evaluation Results

Indicator	Weighted Mean	Interpretation
Provides an easy-to-understand interface	4.55	Strongly Agree
Allows users to navigate the system easily	4.52	Strongly Agree
Uses clear and understandable labels and controls	4.50	Strongly Agree
Allows administrators and cashiers to perform tasks with ease	4.52	Strongly Agree
Overall Weighted Mean	**4.52**	**Strongly Agree**

The results in Table 4.8 show that the Automated Inventory Management System obtained an **overall weighted mean of 4.52**, interpreted as **Strongly Agree**, in terms of Usability. This indicates that the respondents strongly agreed that the system was easy to understand and operate.

The indicator **"Provides an easy-to-understand interface"** obtained the highest weighted mean of **4.55**, interpreted as Strongly Agree. This indicates that the respondents found the system interface clear and understandable.

The indicators **"Allows users to navigate the system easily"** and **"Allows administrators and cashiers to perform tasks with ease"** both obtained a weighted mean of **4.52**, interpreted as Strongly Agree. This demonstrates that the system provides accessible navigation and supports the users in performing their assigned functions.

Indicator	Weighted Mean	Interpretation
Performs system functions consistently	4.50	Strongly Agree
Processes transactions without unexpected interruption	4.48	Strongly Agree
Maintains accurate records during system operations	4.50	Strongly Agree
Maintains stable operation during normal use	4.48	Strongly Agree
Overall Weighted Mean	**4.49**	**Strongly Agree**

Meanwhile, **"Uses clear and understandable labels and controls"** obtained a weighted mean of **4.50**, also interpreted as Strongly Agree. This indicates that the system's labels, buttons, and controls were sufficiently clear for the intended users.

Overall, the **4.52 Strongly Agree** rating demonstrates that the developed AIMS provides a user-friendly interface and supports easy navigation and task completion for both administrators and cashiers.

4.8.5 Reliability

Reliability evaluated the ability of the Automated Inventory Management System (AIMS) to perform its intended functions consistently and dependably during normal operation. The evaluation focused on the system's consistency in performing transactions, maintaining accurate records, operating without unexpected interruptions, and maintaining stable performance during use.

Table 4.9. Reliability Evaluation Results

The results in Table 4.9 show that the Automated Inventory Management System obtained an **overall weighted mean of 4.49**, interpreted as **Strongly Agree**, in terms of Reliability. This indicates that the respondents strongly agreed that the system performs its intended functions consistently and dependably.

The indicators **"Performs system functions consistently"** and **"Maintains accurate records during system operations"** both obtained a weighted mean of **4.50**, interpreted as Strongly Agree. These results indicate that the respondents found the system capable of consistently performing its functions while maintaining accurate operational records.

The indicators **"Processes transactions without unexpected interruption"** and **"Maintains stable operation during normal use"** both obtained a weighted mean of **4.48**, also interpreted as Strongly Agree. This indicates that the system was considered stable and dependable during normal operations.
Overall, the **4.49 Strongly Agree** rating demonstrates that the developed AIMS provides reliable system operations and consistently supports the intended activities of administrators and cashiers.

4.8.6 Security

Security evaluated the ability of the Automated Inventory Management System (AIMS) to protect user accounts, system information, and operational records from unauthorized access. The evaluation focused on authentication, role-based access, protection of system information, and activity logging.

Table 4.10. Security Evaluation Results

Indicator	Weighted Mean	Interpretation
Protects user accounts through authentication	4.55	Strongly Agree
Restricts users according to their assigned roles	4.52	Strongly Agree
Protects system information from unauthorized access	4.50	Strongly Agree
Records user activities for accountability	4.50	Strongly Agree
Overall Weighted Mean	**4.52**	**Strongly Agree**

The results in Table 4.10 show that the Automated Inventory Management System obtained an **overall weighted mean of 4.52**, interpreted as **Strongly Agree**, in terms of Security. This indicates that the respondents strongly agreed that the system provides appropriate security measures for protecting user accounts and system information.

The indicator **"Protects user accounts through authentication"** obtained the highest weighted mean of **4.55**, interpreted as Strongly Agree. This indicates that the respondents considered the system's authentication mechanism effective in protecting user accounts.

The indicator **"Restricts users according to their assigned roles"** obtained a weighted mean of **4.52**, while both **"Protects system information from unauthorized access"** and **"Records user activities for accountability"** obtained **4.50**. All indicators were interpreted as Strongly Agree.

Overall, the **4.52 Strongly Agree** rating demonstrates that the developed AIMS provides appropriate security features, including authentication, role-based access, protection of system information, and activity logging.

4.8.7 Maintainability

Maintainability evaluated the extent to which the Automated Inventory Management System (AIMS) could be maintained, modified, corrected, and improved when necessary. The evaluation focused on the organization of the system components, the structure of the database, the ability to modify system components, and the availability of documentation that can support future maintenance activities.

Table 4.11. Maintainability Evaluation Results

Indicator	Weighted Mean	Interpretation
System components are organized for easier maintenance	4.45	Strongly Agree
Database structure supports system maintenance	4.40	Strongly Agree
System components can be modified when necessary	4.42	Strongly Agree
System documentation supports future maintenance	4.41	Strongly Agree
Overall Weighted Mean	**4.42**	**Strongly Agree**

The results in Table 4.11 show that the Automated Inventory Management System obtained an **overall weighted mean of 4.42**, interpreted as **Strongly Agree**, in terms of Maintainability. This indicates that the respondents strongly agreed that the developed system is organized and can be maintained when necessary.

The indicator **"System components are organized for easier maintenance"** obtained the highest weighted mean of **4.45**, interpreted as Strongly Agree. This indicates that the respondents found the components of the system sufficiently organized to support maintenance activities.

The indicator **"System components can be modified when necessary"** obtained a weighted mean of **4.42**, interpreted as Strongly Agree. This indicates that the developed system can accommodate necessary changes and modifications when improvements or corrections are required.

The indicator **"System documentation supports future maintenance"** obtained a weighted mean of **4.41**, also interpreted as Strongly Agree. This indicates that the available system documentation can provide support for future maintenance and enhancement activities.

Meanwhile, **"Database structure supports system maintenance"** obtained a weighted mean of **4.40**, interpreted as Strongly Agree. This indicates that the respondents considered the implemented database structure appropriate for supporting the maintenance of the system.

Overall, the **4.42 Strongly Agree** rating demonstrates that the developed AIMS has a maintainable structure that can support future corrections, modifications, and improvements. The result indicates that the system components, database structure, and documentation provide an appropriate foundation for maintaining the system after its implementation.

4.8.8 Portability

Portability evaluated the ability of the Automated Inventory Management System (AIMS) to operate properly within its supported computing and network environment. The evaluation focused on the system's ability to operate on supported computers, function through supported web browsers, remain accessible within the intended network environment, and maintain its required functions within the supported environment.

Table 4.12. Portability Evaluation Results

Indicator	Weighted Mean	Interpretation
Operates properly on supported computers	4.48	Strongly Agree
Operates properly using supported web browsers	4.45	Strongly Agree
Can be accessed within the intended network environment	4.45	Strongly Agree
Maintains its functions within the supported environment	4.46	Strongly Agree
Overall Weighted Mean	**4.46**	**Strongly Agree**

The results in Table 4.12 show that the Automated Inventory Management System obtained an **overall weighted mean of 4.46**, interpreted as **Strongly Agree**, in terms of Portability. This indicates that the respondents strongly agreed that the system can operate properly within its supported computing and network environment.

The indicator **"Operates properly on supported computers"** obtained the highest weighted mean of **4.48**, interpreted as Strongly Agree. This indicates that the respondents found the system capable of operating properly on the computers used within the intended environment.

The indicators **“Operates properly using supported web browsers”** and **“Can be accessed within the intended network environment”** both obtained a weighted mean of **4.45**, interpreted as Strongly Agree. These results indicate that the system can be accessed through its supported web browsers and within the intended network environment.

Meanwhile, **“Maintains its functions within the supported environment”** obtained a weighted mean of **4.46**, also interpreted as Strongly Agree. This indicates that the system's required functions remain available and operational within its supported environment.

Overall, the **4.46 Strongly Agree** rating demonstrates that the developed AIMS is capable of operating appropriately within its intended computing, browser, and network environment. The result indicates that the system provides the necessary portability required for its intended implementation.

4.8.9 Overall Evaluation

The overall evaluation summarizes the respondents' assessment of the Automated Inventory Management System (AIMS) based on the eight ISO/IEC 25010 software quality characteristics. The evaluation covered Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability.

Table 4.13. Summary of ISO/IEC 25010 Evaluation Results

Quality Characteristic	Weighted Mean	Interpretation
Functional Suitability	4.52	Strongly Agree
Performance Efficiency	4.44	Strongly Agree
Compatibility	4.48	Strongly Agree
Usability	4.52	Strongly Agree
Reliability	4.49	Strongly Agree
Security	4.52	Strongly Agree
Maintainability	4.42	Strongly Agree
Portability	4.46	Strongly Agree
Overall Weighted Mean	**4.48**	**Strongly Agree**

The results in Table 4.13 show that the Automated Inventory Management System obtained an **overall weighted mean of 4.48**, interpreted as **Strongly Agree**. This indicates that the respondents strongly agreed that the developed system met the expected software quality requirements based on the eight ISO/IEC 25010 characteristics.

Among the eight characteristics, **Functional Suitability, Usability, and Security** obtained the highest weighted mean of **4.52**, all interpreted as Strongly Agree. These results indicate that the respondents considered the system's functions, user interface, and security features appropriate and effective for its intended users.

Reliability obtained a weighted mean of **4.49**, while **Compatibility** obtained **4.48**. Both were interpreted as Strongly Agree, indicating that the system was considered reliable during normal operation and compatible with its intended environment.

Portability obtained a weighted mean of **4.46**, followed by **Performance Efficiency** with **4.44**. Both results were interpreted as Strongly Agree, indicating that the system can operate appropriately within its supported environment and perform its intended operations efficiently.
Meanwhile, **Maintainability** obtained a weighted mean of **4.42**, which was also interpreted as Strongly Agree. This indicates that the respondents considered the system sufficiently organized and capable of supporting future maintenance, modification, and improvement.

Overall, the **4.48 Strongly Agree** rating demonstrates that the developed Automated Inventory Management System achieved a favorable evaluation across all eight ISO/IEC 25010 software quality characteristics. The results indicate that AIMS is functionally suitable, efficient, compatible, usable, reliable, secure, maintainable, and portable for its intended implementation.

4.9 Discussion of Results

The results of the development, testing, and evaluation of the Automated Inventory Management System (AIMS) demonstrated that the developed system was able to address the operational requirements identified in the study. The system was implemented with functions for product management, inventory monitoring, barcode-based Point-of-Sale transactions, sales recording, sales reporting, user management, activity monitoring, and database backup and restoration. These functions were integrated into the Administrator and Cashier modules to support the intended users in performing their respective tasks.

The results of the Functional Testing, Unit Testing, Integration Testing, and Black-Box Testing demonstrated that the major functions of the system operated according to their expected behavior. The testing activities verified individual functions, the interaction between system modules, and the observable responses of the system when valid and invalid inputs were provided. The successful results of these tests indicate that the implemented features were able to perform their intended operations and communicate properly with the database and other related modules.

The Unit Testing results further demonstrated that individual system components performed their assigned functions correctly. Functions such as administrator and cashier login, product management, inventory alerts, barcode scanning, Point-of-Sale transactions, receipt generation, sales reporting, activity logging, and database backup and restoration were tested individually. The successful results indicate that the individual components of AIMS were functioning as intended before and during their integration into the complete system.

The Integration Testing results demonstrated that the different modules of AIMS were able to work together properly. Important processes such as login and dashboard access, product management and database operations, barcode scanning and Point-of-Sale transactions, Point-of-Sale transactions and inventory updates, sales transactions and sales reports, and user activities and activity logs were successfully integrated. These results indicate that the system components were able to exchange and process information correctly across related modules.

The Black-Box Testing results further demonstrated that the system produced the expected responses from the perspective of its intended users. The system successfully handled valid and invalid inputs for authentication, product management, barcode scanning, inventory monitoring, Point-of-Sale transactions, receipt generation, activity logging, and database management. The results indicate that the developed system was able to provide appropriate responses without requiring the users to interact with or understand the internal source code.

The ISO/IEC 25010 evaluation provided an additional assessment of the overall quality and acceptability of the developed system. The evaluation covered Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. The overall weighted mean of **4.48**, interpreted as **Strongly Agree**, indicates that the respondents had a favorable assessment of the developed AIMS across the eight software quality characteristics.

Functional Suitability obtained an overall weighted mean of **4.52**, interpreted as Strongly Agree. This indicates that the respondents considered the system's functions appropriate for the intended operational tasks. The result supports the testing findings showing that AIMS provided the required functions for administrators and cashiers, particularly for inventory and sales management.

Performance Efficiency obtained an overall weighted mean of **4.44**, interpreted as Strongly Agree. This indicates that the respondents considered the system capable of performing its intended operations efficiently. The result supports the system's ability to process transactions, retrieve information, and generate system results during normal operation.

Compatibility obtained an overall weighted mean of **4.48**, interpreted as Strongly Agree. This indicates that the respondents considered the system appropriate for its intended hardware, software, database, and Local Area Network environment. The result demonstrates that the developed system can operate within the environment for which it was designed.

Usability obtained an overall weighted mean of **4.52**, interpreted as Strongly Agree. This indicates that the system was considered easy to understand and operate by its intended users. The result supports the design of the Administrator and Cashier interfaces and indicates that users could navigate the system and perform their assigned tasks with ease.

Reliability obtained an overall weighted mean of **4.49**, interpreted as Strongly Agree. This indicates that the respondents considered the system stable and dependable during normal operation. The result is consistent with the successful testing of individual system functions and the integration of the major system modules.

Security obtained an overall weighted mean of **4.52**, interpreted as Strongly Agree. This indicates that the respondents considered the implemented authentication, role-based access, protection of system information, and activity logging mechanisms appropriate for the system's intended use. These security functions help control access to administrator and cashier features and provide accountability for user activities.

Maintainability obtained an overall weighted mean of **4.42**, interpreted as Strongly Agree. This indicates that the respondents considered the system sufficiently organized to support future maintenance, modification, correction, and improvement. The implemented system structure and database organization provide a foundation for future system maintenance activities.

Portability obtained an overall weighted mean of **4.46**, interpreted as Strongly Agree. This indicates that the system was considered capable of operating properly within its supported computing, browser, and network environment. The result supports the intended deployment of AIMS within its specified operational environment.

Overall, the combined testing and evaluation results demonstrate that the Automated Inventory Management System achieved its intended purpose. The testing results confirmed the functional operation and integration of the system, while the ISO/IEC 25010 evaluation demonstrated favorable user assessments across all eight software quality characteristics. The overall weighted mean of **4.48**, interpreted as **Strongly Agree**, indicates that the respondents found the developed system acceptable and suitable for its intended operational environment.

Therefore, the findings indicate that AIMS provides an integrated solution for managing inventory and sales-related activities while supporting accurate transactions, inventory monitoring, reporting, user management, security, and system maintenance. The results of the testing and evaluation provide evidence that the developed system can support the operational needs identified in the study and can serve as a functional information system for the intended pharmacy and convenience store environment.

4.10 Summary

This chapter presented the development, implementation, testing, and evaluation of the Automated Inventory Management System (AIMS) for pharmacies and convenience stores. It discussed the developed system, its implementation environment, system architecture, presentation of the developed interfaces and functions, database implementation, system testing, and evaluation results.

The developed system provided separate functions for administrators and cashiers to support their respective responsibilities. The implemented features included user authentication, product management, inventory monitoring, barcode-based Point-of-Sale transactions, sales recording, sales reporting, activity monitoring, receipt generation, and database backup and restoration. The system was supported by a MySQL database implemented through the developed application environment.

The database implementation provided structured storage for user accounts, products, categories, stock movements, sales transactions, sale items, activity records, and store settings. The implemented relationships and database operations allowed the different system modules to store, retrieve, update, and manage the information required for daily operations.

System testing was conducted using Functional Testing, Unit Testing, Integration Testing, and Black-Box Testing. The testing results demonstrated that the major functions of the system operated according to their expected behavior. The testing also verified the proper interaction among the different system modules and confirmed that the system produced appropriate responses to the tested user inputs and operations.

The developed system was also evaluated using the ISO/IEC 25010 Software Quality Model based on eight characteristics: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. The evaluation results showed favorable assessments across all eight characteristics. The system obtained an overall weighted mean of **4.48**, interpreted as **Strongly Agree**, indicating that the respondents had a positive assessment of the developed system.

Overall, the results presented in this chapter demonstrated that the Automated Inventory Management System was successfully developed, implemented, tested, and evaluated. The findings indicate that the system provided the intended functions and supported the operational requirements of the target users. The results of this chapter serve as the basis for the conclusions and recommendations presented in the succeeding chapter.

CHAPTER 5
Conclusion and Recommendations

5.1 Introduction

This chapter presents the conclusions and recommendations of the study entitled **Automated Inventory Management system (AIMS).** *It summarizes the significant findings obtained during* the system development**,** implementation, testing and evaluation phases. Furthermore, this chapter determines whether the objectives of the study were successfully achieved based on the results of the system testing and evaluation. It also provides recommendations for future system enhancements, implementation and further improvements to ensure the effectiveness, reliability and usability of the development system.

5.2 Summary of Findings

The study developed an Automated Inventory Management System (AIMS) for pharmacies and convenience stores to address the operational difficulties associated with manual inventory management, sales transactions, stock monitoring, reporting, and user management. The findings were based on the implemented system functions, system testing results, and the evaluation conducted using the ISO/IEC 25010 Software Quality Model.

The following findings were obtained from the study:

1. **Inventory Management**
   The developed system provided an automated inventory management function that allowed administrators to manage product information, monitor available stock quantities, and update inventory records. The system also provided mechanisms for identifying products with low stock quantities and monitoring products approaching their expiration dates. These functions helped improve the organization and monitoring of inventory records.

2. **Sales Transaction Management**
   The system provided a Point-of-Sale (POS) module that supported barcode scanning, product selection, transaction processing, quantity management, total computation, payment recording, and receipt generation. The implemented functions allowed cashiers to process sales transactions more efficiently and reduced the need for manual recording of sales information.

3. **Inventory and Sales Integration**
   The developed system integrated sales transactions with inventory monitoring. Product quantities were updated based on completed transactions, allowing inventory records to reflect changes resulting from sales activities. This integration helped maintain more consistent inventory and sales information within the system.

4. **Sales Reporting and Monitoring**
   The system provided reporting functions that allowed authorized users to view and monitor sales information. The generated reports provided information that could support the monitoring of sales activities and assist administrators in reviewing business performance and making operational decisions.

5. **User Management and Access Control**
   The system implemented separate Administrator and Cashier roles with role-based access control. Administrators were provided with functions for managing products, inventory, users, reports, and other administrative activities, while cashiers were provided with functions necessary for processing sales transactions. This separation helped ensure that users accessed only the functions appropriate to their assigned roles.

6. **Activity Monitoring and Security**
   The system recorded relevant user activities through activity logs and implemented authentication and role-based access mechanisms. These features supported accountability and helped protect system information from unauthorized access.

7. **Database Management and Backup**
   The developed system implemented a structured database for storing user accounts, product information, inventory records, sales transactions, activity records, and other required system information. Database backup and restoration functions were also implemented to help protect stored information and support data recovery when necessary.

8. **System Testing Results**
   The developed system underwent Functional Testing, Unit Testing, Integration Testing, and Black-

Box Testing. The testing results demonstrated that the major system functions performed their expected operations and that the different modules were able to interact properly. The testing also verified that the system produced appropriate responses for the tested inputs and operations.

9. **ISO/IEC 25010 Evaluation Results**
   The developed system was evaluated based on eight ISO/IEC 25010 software quality characteristics: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. The results showed favorable assessments across all eight characteristics. Functional Suitability obtained an overall weighted mean of **4.52**, Performance Efficiency obtained **4.44**, Compatibility obtained **4.48**, Usability obtained **4.52**, Reliability obtained **4.49**, Security obtained **4.52**, Maintainability obtained **4.42**, and Portability obtained **4.46**. All characteristics were interpreted as **Strongly Agree**.

10. **Overall Evaluation**
    The overall evaluation of the Automated Inventory Management System obtained a weighted mean of **4.48**, interpreted as **Strongly Agree**. This result indicated that the respondents strongly agreed that the developed system met the intended software quality requirements and provided the necessary functions and features for its target users.

Overall, the findings demonstrated that the developed Automated Inventory Management System successfully implemented the intended inventory, sales, reporting, user management, security, and database functions. The system testing results and ISO/IEC 25010 evaluation further indicated that the developed system was suitable for its intended operational environment and was positively assessed by the respondents.

5.3 Conclusion

Based on the findings of the study, the development and implementation of the Automated Inventory Management System (AIMS) successfully addressed the identified operational problems encountered in pharmacy and convenience store operations. The developed system provided an integrated platform for managing inventory, processing sales transactions, monitoring stock levels, generating reports, managing users, recording activities, and performing database backup and restoration.

The developed system provided an automated inventory management function that improved the organization and monitoring of product records. Administrators were able to manage product information, categories, stock quantities, and other inventory-related information. The system also supported the monitoring of low-stock products and products approaching their expiration dates, which helped users identify inventory conditions that required attention.

The Point-of-Sale (POS) module successfully supported the sales transaction process. Cashiers were able to search for products, scan barcodes, manage quantities, calculate transaction totals, record payments, and generate receipts. The integration between sales transactions and inventory management also allowed product quantities to be updated based on completed transactions, helping maintain consistent sales and inventory records.

The system also provided sales reporting and monitoring functions that allowed authorized users to review recorded sales information. These functions provided administrators with information that could assist in monitoring business activities and making appropriate operational decisions. The implementation of Administrator and Cashier roles also provided role-based access to system functions according to the responsibilities of each user.

The testing results further demonstrated that the developed system performed its intended functions successfully. Functional Testing, Unit Testing, Integration Testing, and Black-Box Testing were conducted to verify the behavior of individual functions, the interaction among system modules, and the system's responses to different inputs. The results showed that the tested functions operated according to their expected behavior and that the integrated modules communicated properly.

The developed system was also evaluated using the ISO/IEC 25010 Software Quality Model. The evaluation covered Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. All eight software quality characteristics received an interpretation of **Strongly Agree**, demonstrating a favorable assessment of the developed system.

The system obtained an overall weighted mean of **4.48**, interpreted as **Strongly Agree**, in the ISO/IEC 25010 evaluation. This result indicated that the respondents strongly agreed that the developed AIMS provided the necessary functions, operated efficiently, was compatible with its intended environment, was easy to use, performed reliably, provided appropriate security, could be maintained, and operated within its supported environment.

Based on the overall findings, the Automated Inventory Management System successfully met its intended purpose of providing an integrated system for pharmacy and convenience store operations. The system addressed the identified concerns related to manual inventory management, sales transaction processing, stock monitoring, reporting, user access, and record management.

Therefore, the study concluded that the developed Automated Inventory Management System was **functionally suitable and acceptable for its intended users**, as supported by the successful system testing results and the overall **4.48 Strongly Agree** evaluation rating. The system provided a suitable technological solution for improving the efficiency, organization, accuracy, and monitoring of inventory and sales-related operations.

5.4 Recommendations

To further improve the Automated Inventory Management System (AIMS) and maximize its long-term effectiveness in pharmacies and convenience stores, the following recommendations are proposed.

System Improvements

- **Enhance the sales reporting and analytics features by incorporating more detailed graphical reports, product sales comparisons, customizable reporting periods, and additional performance indicators to provide management with better information for decision-making.**
- **Improve the inventory alert system by adding configurable notifications for critical stock levels, frequently sold products, and approaching expiration dates to support more proactive inventory management.**
- **Enhance system accessibility by developing a responsive interface that provides better compatibility and usability across different screen sizes, including desktop computers, tablets, and mobile devices.**
- **Strengthen system security by implementing additional authentication and account protection mechanisms, such as stronger password policies, password recovery, and multi-factor authentication.**
- **Expand hardware and system integration capabilities by supporting additional barcode scanners, receipt printers, and other compatible devices that may be used in future implementations.**

Future Research Directions

- Explore the integration of Artificial Intelligence (AI) or machine learning techniques to forecast product demand, identify sales patterns, and predict future inventory requirements.
- Develop a dedicated mobile application or mobile-based version of the system to provide administrators and authorized users with more convenient access to inventory information, alerts, and sales reports.
- Explore cloud-based deployment to allow authorized users to access the system remotely while maintaining centralized and synchronized business records.
- Conduct further studies involving multiple pharmacies and convenience stores to evaluate the scalability, adaptability, and effectiveness of the system in different business environments.
- Explore the integration of supplier management and automated inventory replenishment features to further improve stock management and reduce the possibility of inventory shortages.

Implementation Strategies

- Provide proper orientation and training to administrators and cashiers before full system implementation to ensure that users understand the functions, procedures, and appropriate use of the system.
- Perform regular database backups and system maintenance to protect business records, maintain data integrity, and ensure continuous and reliable system operation. The current system already provides database backup and restoration functions, which should be regularly utilized by authorized administrators.
- Periodically review user accounts and access permissions to ensure that administrators and cashiers retain only the privileges necessary for their assigned responsibilities.
- Regularly monitor and evaluate system performance after implementation to identify errors, usability concerns, or operational requirements that may arise during actual use.
- Maintain and update the system according to changing business requirements, including product information, store settings, reporting needs, security requirements, and other operational processes.

5.5 Impact of the Study

The development of the Automated Inventory Management System (AIMS) demonstrates the practical application of information technology in improving inventory and sales operations through process automation, centralized data management, and integrated transaction processing. The study contributes to the field of Information Technology by illustrating how a web-based inventory management system can support more organized product monitoring, sales processing, inventory control, reporting, and business record management in pharmacies and convenience stores.

For pharmacies and convenience stores, the developed system provides an integrated solution that simplifies inventory management, Point-of-Sale transactions, barcode scanning, stock monitoring, expiration monitoring, sales reporting, activity logging, and database backup and restoration. The integration of these functions enables authorized users to access and manage important operational information within a centralized system, supporting more organized inventory and sales management. The system also provides low-stock and expiration alerts that can assist administrators in monitoring products that require attention.

The study also serves as a valuable reference for future researchers and software developers interested in developing inventory and sales management systems. The methodologies, technologies, testing procedures, and implementation strategies presented in this research may be adapted and further enhanced to support the digital transformation of pharmacies, convenience stores, and other small and medium-sized businesses requiring automated inventory and sales management solutions.

5.6 Summary

This chapter presented the conclusions and recommendations of the study. The findings demonstrate that the Automated Inventory Management System (AIMS) successfully achieved the objectives established in Chapter I and addressed the identified operational challenges in inventory and sales management for pharmacies and convenience stores.

Through system implementation, Functional Testing, Unit Testing, Integration Testing, and evaluation using the ISO/IEC 25010 Software Quality Model, the developed system demonstrated its capability to support inventory management, Point-of-Sale transactions, barcode scanning, stock and expiration monitoring, sales reporting, user management, activity logging, and database backup and restoration.
The recommendations presented in this chapter provide opportunities for future system improvements, continued development, enhanced security, improved accessibility, and further research.

Overall, the study highlights the importance of implementing automated information systems to improve the organization and management of inventory and sales operations, support informed decision-making, and promote digital transformation within pharmacies, convenience stores, and similar small and medium-sized businesses.